BASE24®

Release 6.0 Version 1 and 2 Implementation Guide

Version Record

06/2002, Version 2.0 Second Printing

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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol “|” in the outside margin.

Version 1.0 (J. Urbanec, C. Rink, L. Culp)-06/2001

Initial Version

Version 2.0 (L. Culp, J. Dargy, T. Engle, R. Barker, L. Watson, J. Urbanec, J. Atkinson, R. Krogstrand, D. Hatfield, C. Rink, S. Schmitt, E. Allen, M Chapman, S Naik, N. Robeson)-06/2002

Second Version

Modified the following existing subsections:

- Renamed “Dial-Up ATM Support” to “Dial-Up ATM Support for Diebold 10XX/478X and NCR 5XXX Devices”
- 6.0 General Changes subsection for sections 4 was changed for the FILE-BUFFERED field in the base segment of the ECF.
- 6.0 General Changes subsections for sections 4, 5, 6, 7, 9, and 12 were changed for the READ-PAST-INITIAL-EOF field in the base segment of the ECF.
- EMV Support includes information regarding the NCR 5XXX ATM and the Hypercom POS device.
- Appendix D was updated, since the AAREADME files have been renamed to ensure that they are unique per product. For example, the AAREADME file on BA60UC04 was changed to AAREAD04.

Added the following new subsections:

- Appendix I – Converting from Release 5.1.2/1.2 to 6.0
- BASE24-card
- CAF Updates to Monad-mfm
- Electronic Benefits Transfer (EBT)
- Dassault ATM Support
- Derived Unique Key Per Transaction (DUKPT)
- Format 2 Files
- Interactive Response Notification Fee Processing and Negative Surcharge Support
- Multiple Logical Network Routing Indicator

- Multiple Account Select by Qualifier Support
- Stored Value
- Transaction Security Services
- Triple/Single DES Translate Support

Section 1

Release 6.0 Introduction

The ***BASE24 Release 6.0 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 5.3 or 6.0 systems to the latest version of BASE24 Release 6.0.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their current BASE24 Release 5.3 or release 6.0 systems to the latest version of release 6.0. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to the latest version of release 6.0.

BASE24 Release 6.0 products include the following:

- BASE24-atm Release 6.0
- BASE24-pos Release 6.0
- BASE24-teller Release 6.0
- BASE24 Kernel/IEP Release 6.0
- BASE24-telebanking Release 6.0
- BASE24-billpay Release 6.0
- BASE24-customer service Release 6.0
- BASE24 Interchange Interfaces Release 6.0
- BASE24 Value-Added Products:
- BASE24-Infobase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager

Who should use this guide

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration from release 5.3 or a previous version of release 6.0 to the latest version of release 6.0. In either case, the reader should be experienced with BASE24 software, Compaq NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in section 3.

To implement release 6.0 from a previous release, users should be on release 5.3. Refer to the ***BASE24 Release 5.3 Implementation Guide*** to assist in planning the tasks necessary to get to release 5.3.

Using this guide, you can do the following

- Determine what ACI services are available to assist in the migration to the latest version of release 6.0.
- Develop a project plan to migrate from release 5.3 or a previous version of 6.0 to release 6.0.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 enhancements.

Release Indicator or Number

The BASE24 release indicator or number is used by several BASE24 processes to indicate which format of messages should be used; it does not indicate the release of the software module itself. The release indicator or number for release 6.0 is 60. The release number used for release 6.0 Integrated Transaction Server (ITS) architecture products (e.g., BASE24-telebanking) is 6.0. The usage of the release indicator or number for release 6.0 is summarized below:

Refresh Process

The Refresh process determines the release input format by looking at the refresh file header. All releases of the BASE24 Release 6.0 Refresh process use the same release file header and file trailer record formats.

Refresh still supports previous release (that is, BASE24-atm Release 5.x, BASE24-mail Release 5.x, BASE24-pos Release 5.x, BASE24-teller Release 5.x, and BASE24-telebanking Release 1.3) file formats for all files that can be refreshed, except the CCF and CSF.

Extract Process

The release number is used to indicate the format of the data on the extract tape for all base files. The BASE24 user enters the value in the release number field on the Extract Configuration File (ECF) screen.

When running release 6.0, the current extract release number entered on the ECF screens is 60. The Extract process still supports previous release (i.e., BASE24-atm Release 5.x, BASE24-pos Release 5.x, BASE24-teller Release 5.x, and BASE24-telebanking Release 1.3) file formats, and this is indicated by entering 50 on the ECF screens.

ISO Host Interface Process

The release indicator is used to specify the release of the messages being exchanged between the DPC and the ISO Host Interface process. The BASE24 user enters the value in the release indicator field on the Host Configuration File (HCF) screen. The release indicator translates into a release number within the ISO message.

When running release 6.0, the current BASE24 release indicator is 01 and translates to release number 60 in the ISO message. A release indicator value of 02 in the HCF maps to release number 50 for the BASE24 Base, BASE24-atm, BASE24-pos, and BASE24-mail products. The release indicator on the HCF for BASE24-teller is always mapped to 50 in the ISO message, indicating current release. The release indicator on the HCF for BASE24-telebanking is always mapped to 11 in the ISO message, indicating current release.

From Host Maintenance Process

The From Host Maintenance message format is determined by the LCONF parameter, FHM-REL-IND. When running release 6.0, the current BASE24 release indicator is 01 and maps to 60 in the message header for the ISO FHM interface, indicating a release 6.0 format. A value of 02 in the LCONF is the previous format and maps to 50 in the message header for the ISO FHM interface, indicating a release 5.x format.

The From Host Maintenance process rejects a message if the release number in the header does not correspond to the value in the LCONF.

Software Requirements

Compaq NSK Software Requirements

ACI recommends that the BASE24 user be on a current Compaq NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the Compaq NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several of the new files.

XPNET

ACI requires that the BASE24 user be on the XPNET 3.0 system.

Previous Release Conversion Planning

Previous release BASE24 users must develop their own project plans for migrating their current BASE24 networks to release 6.0. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

For previous release conversions, ACI recommends that BASE24 users read the ***BASE24 Release 5.3 Implementation Guide*** to get to release 5.3. Previous releases of BASE24 products include the following:

- BASE24-atm Release 3.1, 4.0, 5.0, 5.1, 5.1.2, 5.3
- BASE24-pos Release 3.0,3.4, and 5.0, 5.1, 5.1.2, 5.3
- BASE24-teller Release 3.0, 3.4, and 5.0, 5.1, 5.1.2, 5.3
- BASE24-atm Self-Service Banking Release 5.0, 5.1, 5.1.2, 5.3
- BASE24-from host maintenance 3.1, 4.0, 5.0, 5.1, 5.1.2, 5.3
- BASE24-telebanking Release 1.0, 1.1, 1.1.2, 1.3
- BASE24-customer service Release 1.0, 1.1.2, 1.3

Using both the ***BASE24 Release 5.3 Implementation Guide*** and this guide, the BASE24 user can create a comprehensive plan to migrate a pre-release 5.3 network to BASE24 Release 6.0.

Previous Version Conversion Planning

Users of previous BASE24 release 6.0 versions must develop their own project plans for updating their current release 6.0 systems to the latest version. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

Appendix H, provides page numbers identifying areas of this document that have been modified for enhancements specific to the latest version of release 6.0. These areas of modification are either entire new enhancement descriptions and/or notes that follow the following tag line:

“Release 6.0 Version 1 to Release 6.0 Version 2 Consideration:”

In general, all changes for the latest version of release 6.0 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the customer’s current 6.0 modules.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 software. Such manuals include the formal publications mentioned under the Publications heading in section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 Classic 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped with the BASE24 release 6.0 software.

The BASE24 Classic 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to the move the install document to your hard drive and open it with Microsoft Word.

Installation Guide for BASE24-telebanking/billpay Release 6.0 (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24-telebanking/billpay release 6.0 system from scratch, although not compilation of BASE24-billpay. Refer to the ***Installation Guide for BASE24-billpay Release 6.0*** for more information on the installation of BASE24-billpay for release 6.0. This document is included as a Microsoft Word document that will be shipped with the BASE24-telebanking release 6.0 software. This document should be used in conjunction with the ***BASE24 Classic 6.0 Installation Guide***.

Installation Guide for BASE24-billpay Release 6.0 (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24-billpay Release 6.0 system from scratch. This document is included in hard copy with all BASE24-billpay Release 6.0 tapes. This document should be used in conjunction with the ***BASE24 Classic 6.0 Installation Guide.***

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a Compaq NSK TGAL document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

User Bulletin for BASE24 Release 6.0 (Informal)

A primary reference for information about all changes made to each BASE24 release 6.0 source module. This documentation is included as a set of Compaq NSK TGAL documents and is shipped with BASE24 release 6.0 software. This documentation consists of a number of files located on the BA60UB04 and BA60UB05 subvolumes. All files ending in FH contain fix history sections for all changes made. All files ending in FP contain a list of all fixed Procedures for each change made to all modules, except those files that have been resequenced.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade from release 5.3/1.3 to release 6.0. This document is included as a Compaq NSK TGAL document and is shipped with the release 6.0 software. This document is named AAREAD04 and is located on the BA60UC04 subvolume.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration:

A separate reference document for information about data file impacts and methods that can be used to upgrade from the previous version of release 6.0 to the most current is also available. This is a Compaq NSK TGAL document shipped with the release 6.0 software as well. This document is named AAREAD05 and is located on the BA60UC05 subvolume.

ACI MAKE Release 5 User Guide (formal) - TO-MK000-06

This manual provides an explanation of the ACI Make and Makescan utilities, and an overview of their functionality. It also provides information for using the utilities, including the command syntax and input files involved. It includes guidelines for modifying the input files for customized development (i.e., CSMs) and contains a demonstration of the ACI Make utility. This document is available on the BASE24 Documentation CD-ROM.

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

ATM Service Personnel

People who service ATMs. This servicing could include fixing mechanical problems, resupplying the terminal, balancing the terminal, etc.

BASE24 Integrated Transaction Server (ITS)

A BASE24 product family that uses object-oriented design written in the TAL programming language. For example, BASE24-telebanking is a BASE24 ITS product.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Clerk

A retail clerk entering transactions at a POS terminal.

Combination (Combo) Card

A card similar to a credit card with a magnetic stripe and a microprocessor chip embedded on it.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects.

This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Customer Service Representative

The person responsible for customer contact and performing cardholder or merchant assistance.

Dynamic Key Management (DKM)

The automated distribution of encryption keys between BASE24 and an outside entity.

Enhanced Device Handler

A Device Handler process that supports the Transactions Allowed enhancement and the TDF/PTDF Expansion enhancement. For example, the Diebold 10XX/478 Device Handler in BASE24-atm and the ACI Standard POS Device Handler in BASE24-pos are enhanced Device Handler processes.

Enhanced Interchange Interface

An Interchange Interface process that supports the Transactions Allowed enhancement (i.e., it can use the new ICFE file). For example, the Visanet Interchange Interface process is an enhanced Interchange Interface process.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Host Programmer

A programmer who supports the host mainframe system linked to BASE24. Typically, responsibility includes the coding for host message exchanges with BASE24, including refresh and extract.

Initialization Program

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The initialization program can be run while the online processes are running. The program must consider records that are locked by an online process and retry reading the record until it is unlocked.

This method also requires system down time for the time it takes to roll in new objects for processes that access the new fields in the file.

Integrated Circuit Card (ICC)

An Integrated Circuit Card (ICC) is a plastic card, usually the size of a credit card, that contains an embedded microprocessor chip. This chip is capable of storing large amounts of cardholder information and can contain multiple applications. The terms chip card and smart card are sometimes used interchangeably with integrated circuit card.

Integrated Endpoint Process (IEP)

An interface between endpoints and applications in the Integrated Transaction Services (ITS) environment of BASE24.

Network Security Personnel

A person who works specifically in the area of BASE24 security and is responsible for setting up and administering any or all aspects of BASE24 security. In its usage, the term suggests that security functions are somewhat specialized BASE24 functions and would probably be handled by a separate person or group due to the high degree of confidentiality required.

Non-Enhanced Device Handler

A Device Handler process that does not support the Transactions Allowed enhancement and the TDF/PTDF Expansion enhancement. For example, the BAS24-atm Triton Device Handler process is a non-enhanced Device Handler process.

Non-Enhanced Interchange Interface

An Interchange Interface process that does not support the Transactions Allowed enhancement (i.e., it still uses the ICF file). For example, the Deluxe ISO Generic Interface process is a non-enhanced Interchange Interface process.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with log messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

Smart Card

A card similar to a credit card with a microprocessor chip embedded on it. The chip holds various types of information in electronic form with sophisticated security mechanisms. A smart card is also known as an Integrated Circuit Card (ICC). Refer to Integrated Circuit Card (ICC) for additional information.

Stored Value Card (SVC)

A smart card that contains an electronic purse application. It holds electronic money. This application is used to replace coins and bank notes for everyday purchases.

System Manager

The person ultimately responsible for the BASE24 system at the customer site. Usage of the term suggests a person with advanced system knowledge and has the ability and authority to make and implement long-term processing and operations decisions.

Systems Programmer

A Compaq NSK programmer employed by a BASE24 user who supports the customer's in-house data processing and BASE24 systems.

Teller

A cashier in a financial institution who enters transactions through a teller terminal.

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Section 2

BASE24 Release 6.0 Functionality

This section contains an overview of the BASE24 Release 6.0 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled by the BASE24 Release 6.0 user. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24 Kernel/IEP (applicable to all ITS based products)
- BASE24-telebanking
- BASE24-billpay
- BASE24 Interchange Interfaces
- BASE24 Value-Added Products
    - BASE24-InfoBase
    - BASE24 Retail Products
        - BASE24-check auth
        - BASE24-frequent shopper

- BASE24-refunds
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 offering.

BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0, they must be reviewed by all BASE24 users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

Several 6.0 general changes have been made to BASE24 Base modules.

These changes have impacts that users can learn more about within Section 4 of this document.

- The BASE24 ITS Control Server name has changed to include the logical network name. This allows BASE24 ITS to have multiple logical networks under one BASE24 AFT subsystem.
- A BASE24 Classic Control Server similar to the BASE24 ITS Control Server has been created. Several server objects (e.g., ATD, APCF) are bound into one executable server, and therefore only one server is defined in the PATHCONF for these new servers.
- The release 6.0 file conversion programs use a set of skeleton conversion files that are the basis for each file conversion program and necessary to Make the programs.
- For release 6.0, the Make files (i.e., FM files, MM files, and M files) have been sequenced. When fixes are made to these files in the future, BASE24 users will receive FIX files for these modifications instead of file replacements.
- For release 6.0, ACI is introducing a new version of the Make utility (i.e., version 5).
- Release 4.0 formats for refresh, extract, and ISO messages used by ISO Host Interface, BIC ISO Interchange Interface, and From Host Maintenance processes are no longer supported in release 6.0.
- Three user fields were added to several DDL layouts for future product enhancements, future regional enhancements, and future CSMs. Note that the new user fields were added only to the base segment of a segmented file. These fields were added now, so file conversions may be avoided with later enhancements and customer-specific changes.

- Fields were added to several DDL layouts for CSEs that had already been done in a previous release, and may be supported in product in future releases. These fields were added now, so that file conversions may be avoided with later enhancements.
- For release 6.0, the Surcharge File (SURF) has been modified for the Multiple Currency Support enhancement to include a CURRENCY CODE field in the primary key. All BASE24 users who implement the surcharge feature, regardless of whether the Multiple Currency Support enhancement is used, are impacted by this modification.
- Support for some modules has been dropped in Release 6.0. Contact your ACI Customer Manager for the list of modules that have been dropped.
- For release 6.0, the Extract Configuration File (ECF) has been modified for the Format 2 File Support enhancement to support larger values for the various extent fields in the ECF. The ECF has also been modified to allow the extract disk file to be buffered. In addition, the ECF supports the configuration of reading past the initially determined end-of-file or reading up to the initially determined end-of-file. All BASE24 users, regardless of whether format 2 files will be used in their BASE24 system, are required to run an ECF Conversion program.

EMTs In BASE24 Classic

Significant improvements have been made in the way memory-resident tables are created and managed by the BASE24 Classic applications. In previous releases, each instance of a BASE24 Classic process would create its own memory-resident tables. The “EMTs in BASE24 Classic enhancement” allows BASE24 Classic processes to share extended memory tables, eliminating redundant copies of data in memory. The enhancement also makes use of a batch process, EMT Build, to create the Extended Memory Table files in the background. This approach removes a significant burden from the BASE24 Classic applications during initialization and warmbooting.

This new architecture offers the advantages of data verification through batch processing, conserved initialization time at startup, greater efficiency during multiple-step searches, and data sharing by multiple processes within a CPU. The enhancement improves overall system efficiency and allows greater operational control in the staging of database changes.

Shared Extended Memory Tables and the EMT Build module have been used extensively in BASE24 ITS since they were introduced in the early 1990s.

LCONF File Maintenance

The Logical Network Configuration File (LCONF) is the central location of file information and processing information for BASE24. One LCONF file exists for each logical network. Records in the LCONF are classified as either assigns or params, depending on their function. Assigns link the physical names of objects in the logical network to names used in the BASE24 code. Params set processing values at the logical network level or process level.

The previous method of managing the contents of the LCONF file involved the use of a stand-alone program written using the Entry Programming language. Support for the entry language was discontinued by Compaq NSK several years ago. To use the ENTLCONF program, it was necessary to start the program from a TACL prompt to read or modify the LCONF. The main purpose of this enhancement is to upgrade the technology used to maintain the LCONF.

In Release 6.0, the LCONF files are now maintained using the BASE24 AFT subsystem. The LCONF is accessed using the LNCF selection on the BASE24 AFT virtual menu. The LNCF screens look and function like other BASE24 files maintenance screens. Online Maintenance File (OMF) auditing will track any LCONF file changes. With these changes, more than one person at a time can make changes to the LCONF file.

It is not possible to start the BASE24 AFT subsystem without the existence of an LCONF. The LCONF must contain the required assigns and params (e.g., logical network name, location of the security file (SEC), location of the OMF, etc.). If an LCONF is not available with these assigns and params, the BASE24 user utilizes the FUP CREATE command to create the LCONF file and the LCONF Load Program to load the required assigns and params. Refer to the FUP input file LCONFFUP and the obey file LCONFLDR located on the BA60MISC subvolume to aid you in accomplishing these tasks.

The new fields added to the LCONF record are located at the end of the existing record to minimize the impact to application programs that read the LCONF. A conversion program is available to change existing LCONF records to the new record layout.

The ability to maintain assigns and params is provided with the LCONF maintenance programs. The edit function for file names in LCONF assign records has been changed so that part of the file name and part of the template name will no longer be required to match.

Print options and global editing options remain as they were with the ENTLCONF program.

Multiple Logical Network Routing Indicator

This enhancement provides a new token for use by back office settlement and reconciliation systems to identify transactions that have been routed between two logical networks. The token is added to transactions that have been routed from an acquiring logical network using Sproute routing, or "LGNT Routing", based on a card prefix match from a Card Prefix File of another Logical Network. Prior to this enhancement, these transactions are logged on both logical network log files with identical content, making them appear as duplicate transactions to back office applications. The transaction log file records have no indication of which logical network acquired the transaction and no indication that a duplicate record exists in another transaction log file. With the enhancement, presence of the new token indicates that the transaction has been "LGNT Routed", and the contents of the token indicates which Logical Network acquired the transaction. By using this token, back office applications can determine that the record is duplicated on another transaction log file, and can determine if the record belongs to the acquirer or the issuer Logical Network. Duplicates can be identified as the records are read from the transaction log files, rather than merging and sorting transaction log files to identify the duplicates. Transaction Based Pricing Reports were also enhanced to use this token in identifying transactions that have been routed between two logical networks.

PI-TABLE Expansion

The PI-TABLE Expansion enhancement separates the use of product identifiers and segment identifiers and expands the number of products and file segments that can be supported by BASE24. The PI-TABLE previously contained 32 product positions. The limited entry capacity of the PI-TABLE is the primary reason for the PI-TABLE Expansion enhancement. Note that the term *segment* refers to a portion that physically resides in a product-shared segmented data file (i.e., segmented file) in BASE24, and data contained in a segment is specific to the product it represents.

The use of the PI-TABLE product positions has moved beyond the simple identification of products. As more add-on modules (e.g., Address Verification, Non-Currency Dispense) were developed with BASE24 releases 5.1, 5.1.2, and 5.3, the PI-TABLE was the mechanism that enabled these add-on features. Thus, a single product (e.g., BASE24-atm) required more than one PI-TABLE entry (e.g., ATM entry, NCD entry). Finally, the increased usage of PI-TABLE product positions also resulted from the goal to integrate ACI solutions. For example, the retail products (e.g., Frequent Shopper) were not developed as BASE24 products, yet it was desirable to have a single BASE24 AFT subsystem for BASE24 users who have both BASE24 and retail products. This resulted in PI-TABLE entries used only for BASE24

AFT menu navigation purposes.

For these reasons, the PI-TABLE has been split into two groups of tables, one for products used for BASE24 AFT menu navigation and one for segments, both of which have 256 entries. An entry in the PI-TABLE now has only one meaning instead of the three possibilities that were present prior to this enhancement -- there will be an entry on the BASE24 Virtual Menu. A new table, called the Segment Table (i.e., SEG-TABLE), has been created that defines the file segments. The SEG-TABLE can hold up to 256 segment entries. A product can have one entry on the BASE24 Virtual Menu and one segment. A product cannot have multiple Virtual Menu entries and multiple segments. A product can have a segment and not have an entry on the Virtual Menu, or a product can have an entry on the Virtual Menu but have no segment. As part of this enhancement the new version of the PI-TABLE and the new SEG-TABLE is provided with default values set in each. These tables need to be customized for each BASE24 system. Both the PI-TABLE and the new SEG-TABLE reside in the BA60AFT.PITABLE edit file.

In summary, the following benefits are a result of the PITABLE Expansion enhancement:

- Ability to define entries for only BASE24 AFT subsystem menu navigation without using an entry for file segmentation (e.g., retail products).
- Ability to define entries for new file segments without using an entry for BASE24 AFT subsystem menu navigation (e.g., Non-Currency Dispense).
- Ability to support new enhancements that require new segments.
- Ability to define 256 products.
- Ability to define 256 segments.

Transactions Allowed

Flexibility and configurability have been greatly improved with the Transactions Allowed enhancement. In previous releases, several different tables were maintained by the user to determine which transactions were allowed or disallowed. A new method for managing allowed transactions changes the way these parameters are organized and gives the user more power to manage, simplify, and extend product offerings.

The enhanced processing accomplishes the following:

- Removes the limits on the number of entries
- Consolidates the configuration screens

- Expands configuration options
- Simplifies configuration set-up

In previous releases, users enabled or disabled transactions in six different transactions allowed tables in different files (e.g., IDF, TDF, PTDF). The user indicated by transaction type (e.g., withdrawal) whether the transaction type was allowed and under what conditions. The configuration tables were hard coded and made it difficult to add new transaction types. The user had to access multiple instances of the configuration tables, and in some cases there was no way to restrict transactions by the type of accounts used in the transaction. The Transactions Allowed enhancement overcomes these limitations.

New files were created which allow acquirers, retailers, and issuers to define allowed transactions at the processing code level (i.e., transaction code, account 1 type, account 2 type) versus the transaction code level. One record exists for each processing code. This eliminates the hard-coded maximum number of transactions that can be supported by an acquirer, retailer, or issuer.

The new transactions allowed configurations are based on ISO processing codes rather than the product-specific transaction types. This provides a common representation of processing codes across products. Furthermore, by referencing the full processing code, users can allow or restrict transactions by *to* and *from* account types as well. For example, a withdrawal from checking could be allowed, but a withdrawal from savings could be restricted. This is a feature previously supported only in the IDF Transactions Allowed table. By placing these options at an ISO processing code level, the user now has a broader range of configuration options for both issuers and acquirers, as well as a simplified representation of the processing codes.

The transactions allowed configurations are organized separately for the issuer and acquirer. Device Handler and Interchange Interface processes reference the new Acquirer Processing Code File (APCF) to determine if the acquirer allows the transaction. Similarly, Authorization modules and Interchange Interface processes reference the new Issuer Processing Code File (IPCF) to determine whether the issuer allows the transaction. The alignment of processing codes by issuer and by acquirer greatly expands the options and simplifies the configuration effort for the user.

The number of transaction allowed entries is unlimited. Rather than displaying a hard coded table of transactions, users can now add and update an unlimited number of records in the configuration files. This flexibility extends configuration options, and allows new processing codes to be easily added in future enhancements.

Transactions allowed parameters are configured in a set of common files shared by multiple products. Although not all transactions are supported by all products (e.g., a transfer transaction is not a valid BASE24-pos transaction), the common files reduce the number of configuration screens and simplifies product maintenance.

To further simplify the organization of configuration records, a *profile* is used. Part of the key into the new IPCF and APCF files is the profile field, which can be used to group identical configurations into a single definition. For example, a group of 100 terminals can share one profile rather than using 100 identical table entries. The profile feature reduces set-up time, reduces entry error, and simplifies configuration management.

Also, as part of the Transactions Allowed enhancement, an authorization destination can be configured at the ISO processing code level in the APCF. This authorization destination in the APCF can override the Authorization process name defined in the terminal data files (i.e., ATD, PTD) and Enhanced Interchange Configuration File (i.e., ICFE). At this time, the APCF authorization destination is intended to be a non-BASE24 process (i.e., customer-specific process) running under XPNET.

In conjunction with the processing code files created by the Transactions Allowed enhancement, the BASE24-atm enhanced Device Handler processes use the information defined in the new file called the Terminal Receipt File (TRF). This file is used to define the approved and denied receipt print options. This file is accessed instead of the hard coded receipt tables in the TDF, and therefore giving the user a wider range of receipt configuration options.

In summary, the following benefits are a result of the Transactions Allowed enhancement:

- An unlimited number of allowed transactions can be defined for an acquirer, retailer, or issuer.
- Since part of the key into the new files is the processing code, transactions allowed can be defined at a more detailed level (i.e., processing code versus transaction code).
- Since part of the key into the new files is a profile name, repetitive configuration at the acquirer, retailer and issuer levels is eliminated. For example, if a group of 100 POS devices supports the same transaction set, each of the PTD records would contain the same profile value. Only one set of allowed transactions would be configured in the new file, because the common PTD profile allows the same transactions allowed configuration records to be read for each of the 100 devices.
- The COMPLETION REQUIRED flag is configured in the IPCF. This allows institutions to define completions required at the processing code level. Note that this replaces the COMPLETION REQUIRED flag in the ATM segment and POS segment of the IDF. Note that the flag in the ATM segment pertained to all

BASE24-atm transactions, and the flag in the POS segment pertained to each BASE24-pos transaction code.

- The new file for transactions allowed at the acquirer level (i.e., APCF) allows the authorization destination to be overridden. In BASE24-atm, all transactions are sent to the authorization process defined in the ATD. In BASE24-pos, all transactions are routed/authorized within the router/auth process that is bound in with the device handler. The new authorization destination can be used to selectively send certain transactions to a different non-BASE24 authorization entity. This functionality is useful in supporting user-defined transactions or when integrating multiple ACI solutions.

Note: Not all BASE24 modules have been enhanced to utilize the new transaction allowed processing. Refer to the product specific sections in this guide for more details about this enhancement as well as your ACI Customer Manager.

Optional Enhancements

The following BASE24 Base Release 6.0 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

ACI Proactive Risk Manager Interface Components

ACI Proactive Risk Manager is a fraud detection system that integrates with the BASE24-atm and BASE24-pos products. This enhancement provides the “hooks” that are required by the ACI Proactive Risk Manager product editions to support a “near real-time” interface between BASE24 and ACI Proactive Risk Manager. In this environment, the BASE24-atm Standard Internal Message (STM) or BASE24-pos Standard Internal Message (PSTM) are routed to the ACI Proactive Risk Manager Input Process (PIP). The Master File, is updated using BASE24-from host maintenance and contains the cardholder specific data that is used by the PIP.

To support the ACI Proactive Risk Manager interface, changes have been made to the BASE24 Base, BASE24-atm, BASE24-pos, BASE24-from host maintenance, Extract and Refresh modules. The implementation of these product “hooks” are required for all BASE24 customers, whether or not they have ACI Proactive Risk Manager implemented in their environment.

If the ACI Proactive Risk Manager functionality is being utilized, the ACI Proactive Risk Manager Routing Table (PTBL) is read during initialization by the BASE24-pos Router/Authorization and BASE24-atm Authorization processes and stored in extended memory. After each transaction is authorized, and once the response has been returned to the acquirer, the affected authorization process checks the PTBL entries to determine if the transaction needs to be sent to ACI Proactive Risk Manager. If the cardholder FIID or BIN in the transaction matches an entry in the PTBL, a copy of the transaction is sent to the PIP process for transmission to the ACI Proactive Risk Manager system.

The processing and routing of BASE24-atm Authorization and BASE24-pos Router/Authorization processes are not affected by the enhancement. Normal transaction authorization and response processing is not dependent on the ACI Proactive Risk Manager transaction scoring process.

CAF Updates to ACI Smart Chip Manager

ACI Smart Chip Manager is a card management system for multifunctional or multi-application smart cards. The ACI Smart Chip Manager application supports the issuing of multi-application cards and manages the life cycle of these cards and the life cycle of their card applications. BASE24 can be configured as a smart card application optionally managed by ACI Smart Chip Manager; therefore, when updates are made to the BASE24 Cardholder Authorization File (CAF), these updates, if applicable, are forwarded to ACI Smart Chip Manager for updating the ACI Smart Chip Manager database.

When the BASE24 CAF is altered using BASE24-from host maintenance or the BASE24 AFT subsystem, BASE24 can optionally send a notification message to the ACI Smart Chip Manager. The 0300 ISO message format is used between BASE24 and ACI Smart Chip Manager for the update messages. These messages are supported from a host application.

If the ACI Smart Chip Manager functionality is being utilized, BASE24-from host maintenance forwards applicable CAF add, update, delete, and increment messages to a Host Interface process that supports BASE24-from host maintenance and any applicable connections to ACI Smart Chip Manager. If a station and line is configured to ACI Smart Chip Manager, messages can be sent online. If a station or line are not configured, these messages are written to the SAF and can be extracted and sent to ACI Smart Chip Manager in a SAF extract tape or file format.

The CAF requester/server forwards applicable CAF add, update and delete messages to a Notification Server. The Notification Server reformats the message to an ISO message format and sends it to a Server M-PATH (Server Path) that needs to be configured. Server M-Path has a destination of a Host Interface process that supports BASE24-fhm and any applicable connections to ACI Smart Chip Manager. If a station and line is configured to ACI Smart Chip Manager, messages can be sent online. If a station and line are not configured, these messages are written to the SAF and can be extracted and sent to ACI Smart Chip Manager in a SAF extract tape or file format.

To support this enhancement, changes have been made to BASE24 Base and BASE24-from host maintenance. The implementation of these changes are required for all BASE24 customers with the exception of the Notification Server and Server M-Path.

BASE24 users interested in supporting updates to ACI Smart Card Manager via Refresh should contact their ACI Customer Manager.

EMV Support

Chip technology and its use in the payments industry was pioneered in France in the 1980s, and Europe continues to be at the forefront of the global migration to chip cards. By implementing this technology, French banks have witnessed dramatic decreases in fraud levels and significant reductions in processing costs. More recently, the UK banks also have embraced the move to chip technology as an answer to the increasing levels of fraud in their region. There also has been increased activity in the Asia/Pacific region, most notably in Korea and Taiwan.

In those regions where chip technology is being adopted, it is possible for one or more smart card applications (e.g., credit, debit, stored value, loyalty) to reside on a single multi-application card. In addition, it also is likely that multiple card associations will offer different smart card products within the same region. At the same time, merchants demand that the same piece of POS equipment process all cards and applications. In order for this interoperability to exist, the card associations needed to ensure compatibility among the different applications. Using the new EMV ´96 ICC specifications, the card associations subsequently developed EMV programs to support members wanting to add a smart card chip to their branded credit and debit cards.

EMV Transaction Processing

When a smart card is presented to a terminal, the terminal determines by reading the chip card data which applications are supported by both the card and terminal. The terminal displays all mutually supported applications to the cardholder, and the cardholder selects the appropriate application. If these applications cannot be displayed, the terminal selects the application with the highest priority, as designated by the issuer during card personalization.

If the EMV credit or debit application is selected, the terminal requests that the card identify the data to be used for that application. The terminal reads that data from the card and determines whether to perform offline static or dynamic data authentication. *Offline static data authentication* is used to ensure that the card data has not been fraudulently altered since the card was personalized. *Offline dynamic data authentication* is used to ensure that the card data has not been fraudulently altered and that the card is genuine. If offline static data authentication is performed, the terminal uses the card data read from the card along with the application's "signature," which is created from signing the card data with the private key of the RSA public key algorithm. The terminal verifies the signed card data and cardholder data using the issuer's public key (which is stored in the card signed by the card scheme's private key) and the card scheme's public key (which is stored in the terminal) and compares it to the clear data to ensure that they match.

If offline dynamic data authentication is performed, the terminal verifies signed static data using the issuer's public key and the card scheme's public key as is done for offline static data authentication. The terminal then generates a dynamic challenge and transmits it to the card, which signs the challenge with the card's private key. The terminal verifies the signed dynamic data using the smart card's public key, which is stored in the card signed by the issuer's private key, and compares it to the challenge to ensure that they match.

The terminal determines whether the cardholder should be prompted to enter a cardholder verification method (CVM) based upon the issuer's CVM data initialized in the card. Cardholder verification is used to ensure that the cardholder is legitimate and the card is not lost or stolen. If the card supports an offline personal identification number (PIN) and the terminal supports an offline PIN pad, the terminal prompts the cardholder to enter the PIN and transmits the plain text PIN to the card, which matches it against the PIN stored in the card to ensure that they match.

Both the terminal and card can perform offline risk management (e.g., floor limit checking, transaction activity accumulation checking) to determine whether the transaction should be approved, declined, or transmitted online for authorization. If both the card and the terminal indicate that the transaction satisfied the criteria for offline authorization, the transaction can be approved offline, and the card generates a DES Encryption Algorithm (DEA) based cryptogram called the transaction certificate. The transaction certificate is based on card, transaction, and terminal data. The transaction certificate and the data used to generate it are transmitted in the clearing message for future cardholder disputes or chargeback purposes. A transaction certificate may be used as a proof of transaction when a cardholder incorrectly repudiates a transaction or to verify that the transaction data has not been changed by the merchant or acquirer. A cryptogram identical to the transaction certificate is generated for a declined transaction; it is called an application authorization cryptogram (AAC).

If the criteria for offline authorization are not satisfied and if the terminal has online capabilities, the terminal transmits an online request message to the issuer (or its agent) indicating why the transaction was transmitted online. Prior to the transaction being transmitted online, the card generates a cryptogram called the authorization request cryptogram (ARQC) based on card, transaction, and terminal related data. The terminal transmits the ARQC and the data used to generate it in the request message. During online processing, the issuer performs card authentication to verify that the transmitted ARQC is valid (in other words, to ensure that the card data has not been skimmed from a genuine card to a counterfeit card). The issuer generates a reference ARQC using the clear data transmitted in the request message and compares it to the transmitted ARQC to ensure that they match. The issuer then can use the ARQC to create a second cryptogram called the authorization response cryptogram (ARPC), which is sent to the card in the response message and is used by the card to verify that the issuer is genuine. Issuer authentication is used to ensure that certain security-related parameters in the card are not reset after an online authorization unless the issuer is proven to be genuine. This prevents criminals from circumventing the card’s security features by simulating online processing and fraudulently approving a transaction. If the issuer authentication fails, the card cannot be used to generate further offline transactions. Issuer authentication is performed using a method similar to that used for card authentication.

As part of online processing, the issuer can choose to transmit certain commands in the response message, which the terminal transmits to the card to perform functions such as updating card parameters, blocking the application, unblocking or changing the offline PIN, etc. This is called issuer script processing.

To successfully complete the online transaction, the card generates a transaction certificate as described above. If the terminal transmits a clearing message subsequent to an authorization message, the transaction certificate is transmitted in the clearing message. If the terminal transmits a single message to the acquirer, the terminal may not transmit a separate message containing the transaction certificate to the acquirer.

BASE24 Base processing has been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV ‘96.

Note: Not all BASE24 modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Format 2 File Support

Format 2 File Support was released in Compaq’s NSK D46 operating system. Format 2 File Support is available for the TLF, PTLF, ITLF and the ILF of selected Switch 6.0 interchange interfaces in BASE24 Release 6.0 for D46 and higher Compaq NSK operating systems.

Format 2 File Support gives BASE24 users the ability to format Enscribe entry-sequenced disk file (i.e., TLF, PTLF, ITLF, and ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain much more data than the existing format 1 files. Files, whether format 1 or format 2, can have up to sixteen partitions. Format 1 files (i.e., entry-sequenced files) are limited to 4 gigabytes. Each partition can have up to two gigabytes for a maximum of 4 gigabytes. Format 2 files have new block and label formats and can hold as much as 1024 gigabytes of data and are currently limited only by disk volume size. Large format 1 files, limited by the capacity of a single disk volume, have used partitioning as a means to increase file capacity.

When using Compaq's NSK D46 Operating System or subsequent operating systems, key-sequenced files such as the CAF, PBF, and NEG are able to support format 2 files.

BASE24 users must calculate how adding tokens to the entry-sequenced files — TLF, PTLF, ITLF and ILF — affect the amount of disk space required. Format 1 entry-sequenced files allow a maximum logical record length of 4072 bytes. Format 2 entry-sequenced files allow a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are logged to these entry-sequenced files, the length of each record in the entry-sequenced file could increase beyond the limit that allows two records to be written for each logical record. If this situation occurs only one record per block can be logged, which causes the disk space requirements to double for these files.

Customers should test format 2 files in a test environment to ensure records do not exceed the 4048 limit. Format 1 BASE24 users are not affected, since the DDLs were not changed to restrict the size of the log files with the exception of the TLF Token. If the TLF Token is logged and extracted, the TLF, PTLF, and ILF will change in size once this enhancement is put into place regardless if you are using format 1 or format 2 files.

Refer to the appropriate product sections of this document for detailed information impacting each file.

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement.

Multiple Currency Support

The increased mobility of the world is driving changes in financial applications to provide access to accounts held in foreign currencies. To remain competitive, changes in the world economic environment require more and more financial institutions to support multiple currency processing. Reasons for this include the increasing number of people traveling and working across country borders, as well as the consolidation of processing for banks and retailers operating across borders into a single operational environment.

Traditionally, BASE24 processing has operated in a single currency. BASE24 has permitted only one currency to be operative within the system at any one time, defined within the system as the Logical Network Currency.

This is not sufficient in many countries where account holders are permitted to maintain accounts in different currencies, and where the currency used in transaction processing differs from device to device. Therefore, BASE24-atm and BASE24-pos applications have been enhanced to support any number of currencies. Currency conversions are performed as necessary providing the following capabilities:

- Transactions can be denominated in any currency that is supported in the logical network.
- Devices can be configured to operate in a currency that differs from the default currency of the owning institution, and the same as or different from currencies used at other terminals belonging to the same institution.
- Customer accounts can be denominated in any currency supported in the logical network.
- Where multiple accounts are associated with a single card, these accounts may be in different currencies.
- Transactions that are performed in one currency can be processed against an account held in a different currency.
- Limit and usage checking can be performed regardless of the currency of the transaction.
- All currencies used during processing of a single transaction are stored in a BASE24 token and can be configured to be logged in the transaction log files, including the equivalent amounts in each currency used. This information can be made available to external interfaces and extracted for host processing, if required.

The currency conversion services provided by the BASE24 Multiple Currency Support enhancement utilize the triangulation method of currency conversion and are compliant with the European Union Council Regulation (EC) No. 1103/97. Currency conversions have the following characteristics:

- Conversion between two currencies takes place using exchange rates held against a user chosen *Base Currency* and follows appropriate conversion rules. The base currency is a currency that is selected for the system at installation time and is unlikely to change. For example, in the case of a BASE24 user using the Multiple Currency Support enhancement to support the European Single Currency (Euro), the base currency will always be the Euro.
- Exchange rates for each of the currencies to be used in the system are held in the Exchange Rate File (ERF), with one rate for each currency supported. The exchange rates are held in terms of the rate at which the base currency is converted to the given specific currency unit.
- Exchange rates are held to six significant figures and rounding is performed to adhere to the conversion precision used by MasterCard, Europay, and Visa.

- BASE24 Base processing has been enhanced to allow institutions and retailers to offer accounts in different currencies, by defining each account with its own currency. In addition, cardholders can perform transactions in different currencies at different POS and ATM devices, all associated with a single financial institution.

The following should be considered when configuring the Multiple Currency Support enhancement. Refer to the particular product sections to see the considerations for the supported product set.

- Only BASE24-atm and BASE24-pos have been enhanced to process Multiple Currency transactions.
- The BASE24-atm and BASE24-pos reports have not been enhanced to process in multiple currencies.
- BASE24 Interchange Interface reports have not been modified to support reporting in multiple currencies.
- There is no provision for supporting multiple currency reconciliation or totals in the BIC ISO Interchange Interface.
- Currency codes have not been added to all BASE24 Files Maintenance screens containing amount fields.
- There is not multiple currency support for BASE24-from host maintenance CAF or PBF increment requests where the currency from the host is different than the currency in the file.
- There is currently no Refresh, Extract, or From Host Maintenance process support of the Exchange Rate File (ERF). The ERF can only be updated using BASE24 Files Maintenance.

Note: Not all BASE24 modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Transaction Security Services

With the advent of triple DES, ACI introduces a new architecture for managing transaction security. In BASE24 Release 6.0, all cryptographic functions, such as PIN verification, message authentication, chip authentication, and card verification can be managed through the new *BASE24 Transaction Security Services*. This advanced cryptographic and key management system is powered by ACI's Enterprise Services (ES) architecture and is configured and managed using ACI's new java-based GUI environment. Transaction Security Services is the foundation for all new and future cryptographic functions such as triple DES, derived unique key per transaction (DUKPT) and public key infrastructure (PKI).

Transaction Security Services was developed for BASE24 Release 6.0 consistent with the release themes: extended risk management, ease of migration, and "broadening the base" to enable new solution offerings. BASE24 Transaction Security Services provide the following features and benefits:

<u>Risk Management</u>

- Supports triple DES for stronger PIN protection and reduced potential fraud losses
- Supports longer keys for protection against key exhaustion attacks
- Supports stronger key storage techniques, such as Atalla's AKB Scheme, to protect against compromise of keys

<u>Ease of Migration</u>

- Minimizes triple DES impact to other modules, simplifying retrofit of customizations
- Eliminates impact to existing files used for key management
- Minimizes impact due to new key lengths and stronger key storage schemes
- Allows phased migration to triple DES by accommodating a mixed DES/triple DES environment

<u>Broadening the Base – Enhancements</u>*:*

- Graphical User Interface (GUI)
- Public-Key Cryptography (Automated ATM key distribution) *
- Derived Unique Key Per Transaction (DUKPT) *

- New hardware security module (HSM) enhancements and APIs
- Foundation for new cryptographic standards

* *Available for TSS in upcoming enhancements. Contact your ACI Customer Manager for availability.*

How TSS Works

In traditional BASE24 environments, the device handlers, authorization modules and interchange interface processes manage their own cryptographic services. This includes storing and retrieving keys, and making calls to the Hardware Security Module (HSM). In an environment using Transaction Security Services (TSS), the applications call the TSS process rather than the HSM. TSS manages the storing and retrieving of keys and making calls to the HSM.

With BASE24 Transaction Security Services, the application processes use “key locators” in place of actual keys. Key locators serve as tags, or indexes, representing a key. The actual keys are stored in new key files managed exclusively by Transaction Security Services. A new key file can hold single-length keys for traditional DES operations, double-length and triple-length keys for triple DES, or longer keys for public-key operations. Keys can be stored with a header and a Message Authentication Code (MAC) value as required with newer Atalla HSMs.

Rather than calling a security device directly to perform cryptographic services, BASE24 processes call Transaction Security Services. This is a direct call using interprocess I/O and does not pass through XPNET, although the Transaction Security Services process is managed by XPNET. Transaction Security Services uses the key locators to retrieve the actual key. Keys are retrieved from memory whenever possible to improve performance.

Migration Options

Several migration options are available to customers. TSS is not required for single DES implementations, or when triple DES is provided by select interfaces capable of a triple DES “translate mode”. Refer to the Triple/Single DES Translation support enhancement for more details. TSS can be implemented in a single DES environment to facilitate a phased migration to triple DES. When TSS is implemented, devices and interfaces can be migrated to triple DES on an as-needed basis (provided the interface or device handler has been upgraded for TSS and triple DES support).

TSS is required to support any one of the following items:

- Atalla AKB Support

- Triple DES within BASE24
- Triple DES Support from Device Handlers
- End-to-End triple DES support

Supported Products

All fully supported BASE24-atm Device Handlers are capable of supporting Transaction Security Services. The NCR 5XXX, Diebold 10XX/478X, and Tidel device handlers have been enhanced to support triple DES. The Triton, Fujitsu, and Cash Source Plus device handlers will be upgraded for triple DES as vendor specifications become available and customer demand warrants development. Other device handlers classified with a lower support category may require customer funding. Contact your ACI Customer Manager if interested in triple DES device handler support.

All fully supported interfaces are capable of supporting Transaction Security Services. As vendor specifications become available, and customer demand warrants development, triple DES support will also be added to these interfaces.

All fully supported BASE24-pos Device Handlers, except the BASE24-pos APACS Device Handler, are capable of supporting Transaction Security Services. Support for the APACS MACing feature will be available in an upcoming enhancement to TSS. As vendor specifications become available, and customer demand warrants development, triple DES support will also be added to these device handlers.

Derived Unique Key Per Transaction (DUKPT) will be uplifted to use TSS in a forthcoming enhancement. Customers currently using DUKPT cannot upgrade any portion of their BASE24 system to TSS until DUKPT has been uplifted. Contact your ACI Customer Manager for availability.

Atalla and Thales e-Security (Racal) security devices are supported by TSS. Contact your security device vendor or ACI Customer Manager for additional details on minimum device and version dependencies. Many older platforms and software versions will not be supported due to technology changes and the implementation of triple DES.

Unsupported Features

The following functions have previously been supported in BASE24 but are not supported in the current version of Transaction Security Services.

- Atalla Identikey PIN Verification method *
- Diebold Number Table PIN Verification Method *
- Non-ANSI PIN/PAN PIN blocks
- Certain uses of the KEYF intermediate Key

- Certain configurations for clear PIN processing

*May be reinstated based upon customer demand. Contact your ACI Customer Manager for availability.

Triple/Single DES Translation Support

Several BASE24 Release 6.0 Interchange Interfaces (Switch 6.0) have been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 interchange interfaces and the interchange itself. These BASE24 Release 6.0 interchange interfaces run in all releases of BASE24. No other BASE24 modules have been enhanced at this time to support triple DES. **Note:** ACI will only support full end-to-end Triple DES processing in release 6.0. Because of interchange mandates, BASE24 interchange interfaces may be required to support triple DES encryption before customers can implement end-to-end support or when running BASE24 interchange interfaces in a release 5.3 or earlier network.

The interchange interfaces enhanced to support triple to single DES transactions are:

- BASE24 Interface ISO (BIC ISO) interchange interface
- Banknet ISO interchange interface
- MDS/Cirrus interchange interface
- MDS Maestro Gateway interchange interface
- VisaNet ISO interchange interface
- VisaNet Interlink Gateway interchange interface
- VisaNet/Plus Gateway interchange interface
- Link ISO interchange interface
- EPS-Net ISO interchange interface

The ISO Host Interface (HISO) has been enhanced to support triple to single DES translations as well, but only for release 6.0.

Triple/Single DES Translation support invokes a new type of PIN translation. PINs may be translated from encryption under a double length key (3DEA) to encryption under a single length key (DEA) or from DEA encryption to 3DEA encryption. The special translation is only done in following three situations:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware
- External PIN encrypted in hardware to internal PIN encrypted in hardware

When an interchange interface receives a transaction from the interchange and encryption is performed by a hardware security module (HSM) and the interchange interface is configured for triple DES encryption, the interface translates the PIN to single DES encryption under a single length intermediate key. The translated PIN and the encrypted intermediate key are placed in the STM/PSTM, and sent to the BASE24 Authorization process or the BASE24 Router/Authorization process.

When BASE24 acquires a transaction that must be sent to an interchange configured for triple DES encryption and the encryption of the external PIN is performed by an HSM ,the interchange interface translates the PIN from encryption under a single length key to triple DES encryption under a double length key. The translated PIN is be sent to the interchange.

Each interchange that supports dynamic key management sends a double length PIN encryption key in key change messages if double length keys and triple DES encryption are required. This key is sent either in a new field or in an expanded existing key field, depending on the interchange interface.

For more information about Triple DES, Contact your ACI Customer Manager.

BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to section 5.

General Enhancements

The following BASE24-atm enhancements are inherent in the release 6.0 offerings. These enhancements must be reviewed by all BASE24-atm users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

Several 6.0 general changes have been made to BASE24-atm modules:

These changes have minor impacts that users can learn more about in section 5 of this document.

- Three user fields were added to the following DDL layouts for future product enhancements, future regional enhancements, and future CSMs:
- BASE24-atm Terminal Data Dynamic File (ATDD1)
- BASE24-atm Terminal Data Static File (ATDS1)
- The NCR 5XXX Device Handler and CNFG5070 Load Group 24 support the Customer Selectable PIN state B for Encrypted PIN Block support. This enables the NCR 5XXX Device Handler to accomplish PIN change transactions, using encrypted PIN blocks, in a single request / response pair.

Device Handler Separation

The Device Handler Separation enhancement moved the Device Handler process-specific files and information to the Device Handler subvolumes. This reduces the amount of time needed to bring device enhancements to market. It also reduces the amount of effort needed to retrofit customer-specific changes. It helps insulate those BASE24 users not wanting to take advantage of new device enhancements when they first become available.

In the past, many device-specific requesters resided on the BASE24-atm AFT subvolume. These requesters have been moved to their respective device subvolumes. Now, when a change is made to a device-specific requester, only the device subvolume needs to be delivered. Subsequently, fewer subvolumes need to be recompiled after changes, since fewer Make files have each file as a dependency. Refer to the BASE24-atm General Enhancements in section 5 for additional information about this enhancement.

EMTs in BASE24 Classic

The EMTs in BASE24 Classic enhancement allows BASE24 processes to share extended memory tables, eliminating redundant copies of data in memory. Note that the BASE24-atm modules use EMTs in release 6.0 (e.g., APCFEMT, TRFEMT). Refer to the BASE24 Base Enhancements descriptions for more information about this enhancement.

Note: Not all BASE24-atm modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Multiple Logical Network Routing Indicator

This enhancement provides a new token for use by back office settlement and reconciliation systems to identify transactions that have been routed between two logical networks and logged to two TLFs. Refer to the BASE24 Base General Enhancements for more information about this enhancement.

TDF Expansion

The BASE24-atm Terminal Data File (TDF) is defined as an Enscribe relative file. The maximum length for a relative file is 4072 bytes. The file is at the maximum record length, and there are just a few user fields that can be redefined for future enhancements. Data for future enhancements requires a second TDF record to be defined for each terminal. If the TDF remained in the existing formats and all new data is added to a second TDF record, each transaction processed by the Device Handler process would require two reads of the TDF and two writes to the TDF (assuming the second TDF record contains data that is updated during the online transaction path).

The TDF Expansion Enhancement defines a method of storing additional TDF data without adversely impacting online transaction processing performance.

<u>TDF Record Layout</u>

The structure of the TDF is composed of four main components:

- Common configuration data for all Device Handlers.
- Device-specific configuration data. This data is displayed using the BASE24 AFT subsystem.

- Device dependent data specific for each device handler. This data is not displayed using the BASE24 AFT subsystem and is used primarily to monitor the status of the device.
- Last message area. The STM of the most recent transaction is stored in the TDF.

Three new terminal files were defined in release 6.0. These files are key-sequenced files. The BASE24-atm Terminal Definition files (ATD) are used by the following BASE24-atm enhanced Device Handler processes:

- Diebold 10XX/478X Device Handler
- NCR 5XXX Device Handler

Each terminal requires a record to exist in each ATD file. A record in the first ATD file contains dynamic data that is common for all Device Handlers (i.e., a fixed structure) and dynamic data that varies based on the device type. This record contains data that is updated during the online transaction path. It must be read from disk for each transaction and rewritten to disk upon completion of each transaction.

A record in the second ATD file contains static data. This record contains a fixed structure for static data that is common for all Device Handlers. The remainder of the static record contains device-specific data. This record contains configuration data that does not change during the course of online transaction processing. This data can be maintained in extended memory. The extended memory segment for the static data is not built using the EMT Build utility program. Because of the potential for a large terminal population and limited system resources at a customer site, the size of the static data extended memory segment is configurable. The extended memory segment is populated dynamically. The new TDF libraries, used for accessing both the ATD and PTD files, first check extended memory for a static record. If the record is not found in extended memory, the libraries read the static data from disk and put the static data record into extended memory. If there is not room in extended memory for the static record, an existing record in extended memory will be deleted to make room for the new static record. Note that the ATDD1 only stores selected STM fields that are required to generate a reversal.

A record in the third ATD file contains scratch-pad data for each ATM. This is an optional file for BASE24-atm users wanting to store token data in a file other than the TLF for reversal processing.

The new ATD records do not contain a hard-coded maximum size for device-specific data or last message area. Instead, the ATD records are defined based on the requirements of each device handler. The fixed structure portion of the dynamic record contains offsets and lengths of device-specific configuration data and device dependent data. This allows data for these areas to be defined as needed for the specific Device Handler process. For example, under the TDF structure a Device Handler process may need more than the 280 bytes of device-specific configuration data, but only need half of the device dependent data that is defined in the TDF record. By using offsets and lengths, a TDF record format can be defined for each device handler type.

Last STM and Token Data

Instead of maintaining the STM data and token data of the last transaction in the ATD, this data will be held in extended memory. When a response is received from the authorization entity, the Device Handler process puts the STM and token data into extended memory. If a transaction must be reversed, the Device Handler process uses the STM in extended memory to format the reversal message.

If the Device Handler process goes down before it is determined that a reversal must be generated, the STM data is lost. However, the ATD dynamic record has the TLF file name and relative byte address (RBA) of the last transaction. The Device Handler process reads the TLF, formats a STM reversal message, updates the ATD to back out the transaction impacts, and sends the message to the authorization entity.

In summary, the following benefits are a result of the TDF Expansion enhancement:

- The dynamic and static records have room to grow before the maximum record lengths are reached (i.e., there is a substantial amount of space available for future enhancements).
- Dynamic data was separated from the static data with the new ATD files, and therefore disk I/Os were held to a minimum.
- A new server object was written for the TDF and ATD files, so that the BASE24 AFT subsystem can present one view to the user.
- The method of saving token data has been modified, so that more token data can be retained for reversal processing.

Note: Not all BASE24 modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Transactions Allowed

Transactions Allowed parameters are now configured in a set of common files shared by multiple products. Note that there are product-specific files to configure with this enhancement (e.g., ATD). Refer to the BASE24 Base General Enhancements for more information about this enhancement

Note: Not all BASE24-atm modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

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Optional Enhancements

The following BASE24-atm Release 6.0 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

ACI Proactive Risk Manager Interface Components

The BASE24-atm Authorization process supports the ACI Proactive Risk Manager Token in the STM that is used to route the transaction to the ACI Proactive Risk Manager system if installed. Refer to the BASE24 Base Optional Enhancements descriptions for more information about this enhancement.

Dassault Device Handler

BASE24-atm has been enhanced to support Dassault ATMs. The BASE24-atm Dassault Device Handler module provides the interface between the device and BASE24-atm. This includes managing the communications and translating messages to and from the native format of the device.

Dial-up Support for Diebold 10XX/478X and NCR 5XXX Devices

The Diebold 10XX/478X and the NCR 5XXX Device Handler processes were enhanced to support all existing features and functionality in a dial-up environment. This enhancement relates to processing of ATM connections using a dial-up protocol. Existing processing involving dedicated leased line ATM connections has not been impacted by this enhancement.

The package extensions (Diebold 10XX/478X Dial-up Device Handler Extension and NCR 5XXX Dial-up Device Handler Extension) communicate with BASE24 using the Async (XN-PL215) or X.25 POSII (XN-PL070) protocols.

The surcharging functionality has been enhanced for the Dial-up Device Handler processes. Customers are able to configure, per dial-up terminal, whether surcharge transactions are processed as per standard product or with a non-interactive message sequence. The standard BASE24-atm product processes surcharge transactions with an interactive message sequence between the device and BASE24. In this environment, the customer selects the withdrawal transaction, and it is sent to the Authorization process, which determines the surcharge fee. This fee is sent back to the device to allow the customer to agree to the fee before continuing. The transaction is sent again to the Authorization process for approval. This results in long dial-up times for the institution. This enhancement adds the ability to set a maximum surcharge amount that can be charged for the transaction, which is displayed to the customer during the initial withdrawal transaction sequence. The actual amount the customer is charged for the transaction may be less than the maximum amount he or she was shown during the transaction selection, but the dial-up time is shorter.

Additionally, new NCS 9500 commands were added to the Device Handler process for dial-up ATM support. These commands cannot be performed from the DCT, but can be issued from the NCS.

A new file, the BASE24-atm Dial-up Command File (ADCF), contains the ATM dial-up command queue. The dial-up commands stored in the ADCF are normally executed using the FIFO (First in First Out) method. Exceptions to this are priority commands generated programmatically (e.g., initial loads started by the Device Handler process as a result of receiving ATM unsolicited power failure statuses). There is a configurable limit defined for the maximum number of these dial-up terminal load commands a Device Handler process can process at a time. Once this limit is reached, the Device Handler process does not send any queued commands to terminals until the number is below the maximum. A counter is increased any time a queued load message is sent from the Device Handler process and decreased when the command has been processed.

Note: Not all BASE24-atm Device Handler processes have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

EMV Support

BASE24-atm acquirer and issuer processing has been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV '96.

In addition, the NCR 5XXX Device Handler supports the NCR EMV CAM II Version 2.11 firmware.

Refer to "EMV Support" in the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all BASE24-atm modules have been modified to support this enhancement. Contact your ACI Customer Manager for further details.

Format 2 Files

Format 2 File Support, was released in Compaq's NSK D46 operating system. Format 2 File Support is available for the TLF in BASE24 Release 6.0, for D46 and higher Compaq NSK operating systems.

Format 2 File Support gives BASE24-atm Release 6.0 users the ability to format Enscribe disk file (TLF and ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain more data than the existing format 1 files.

Refer to "Format 2 File Support" in the BASE24 Base Optional Enhancements Section for more information.

Interactive Response Notification Fee Processing and Negative Surcharge Support

BASE24-atm fee processing has been enhanced to support interactive response notification of fees at an ATM. This differs from the original interactive **request** notification surcharge processing, in that the original interactive request notification surcharge processing performed an interactive request prior to the transaction going to the issuer. The cardholder account was impacted after the cardholder was notified of, and accepted, the fee.

The interactive **response** notification of fees, however, routes the transactions for authorization without notifying the cardholder first. The cardholder account is impacted before the cardholder accepts the fee, and the transaction may need to be reversed if the cardholder does not accept the fee displayed at the ATM.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) for only acquirer fees, and does not maintain information in its database on issuer fees. The setting of issuer fees must be done at a host system or an interchange whose interchange interface on BASE24 has been enhanced to carry the issuer fee information in a new BASE24 Issuer Fee/Rebate token.

With this enhancement, BASE24-atm customers have the option of which type of surcharge processing to perform by setting of a new LCONF param. Current interactive request notification surcharge processing is the default processing performed, unless the customer sets the param to indicate that they want to perform the enhanced interactive response notification processing of fees.

Standard product interactive request notification surcharge processing has been enhanced to allow a negative fee amount to provide a cardholder with an incentive to use the ATM device. This type of incentive is also available with the interactive response notification fee processing.

The following BASE24 modules were modified to support these enhancements:

- BASE24-atm Diebold 10XX/478X Device Handler
- BASE24-atm Diebold 10XX/478X Dial-up Device Handler Extension
- BASE24-atm NCR 5XXX Device Handler
- BASE24-atm NCR 5XXX Dial-up Device Handler Extension
- BASE24-atm BIC ISO Interface
- BASE24-atm ISO Host Interface (HISO)
- BASE24-atm Daily Terminal Settlement Report (SETL02-B24)
- BASE24-atm Transaction Log File Perusal
- BASE24-atm Receipt File (RCPT) File Maintenance
- BASE24 Base Application Utilities (BAUTILS)
- BASE24 Base Surcharge File (SURF) File Maintenance
- BASE24 Base Token Conversion Utilities

These enhancements are compliant with the UK LINK Switch Interchange Standard (LIS5), Version 5.1, and can be used by the following UK product module as well.

- UK LINK ISO Interchange (LIS5) Interface

Refer to “Interactive Response Notification Fee Processing and Negative Surcharge Support” in the BASE24-atm Optional Enhancements in Section 5 for more information about these enhancements.

Multiple Account Select By Qualifier Support

BASE24-atm has been enhanced to enable those BASE24-atm customers who are members of the MAC network (or any other network with similar requirements) to offer transactions using multiple accounts as outlined by the MAC Network.

MAC Multiple Account Select By Qualifier support is a multiple account selection system originally developed by the MAC switch network (Electronic Payment Systems - EPS) and now also supported through the NYCE switch network. Multiple Account Select By Qualifier support is used as a replacement for standard multiple account selection (often referred to in the industry as Open Account Relationship – OAR) through participating switch networks. In addition, Multiple Account Select By Qualifier support includes support for an additional account type, ‘Other’. As implemented in BASE24-atm, instead of using the standard BASE24-atm message set to select accounts, Multiple Account Select By Qualifier support selects accounts via fixed menus at the ATM, which are then passed to the Device Handler for mapping into a modified BASE24-atm message. This message is then recognized by Authorization, ISO Host Interface, the BIC ISO Interchange Interface, the MAC ISO Interchange Interface, and the NYCE ISO Interchange Interface.

Multiple Currency Support

BASE24-atm processing has been enhanced to allow institutions to offer accounts in different currencies, by defining each account with its own currency. In addition, cardholders can perform transactions in different currencies at different ATM devices, all associated with a single financial institution.

The following issues should be considered when configuring the Multiple Currency Support enhancement in BASE24-atm.

- Since each currency of an ATM hopper may not be unique, no ATMs connected to BASE24 may currently dispense in more than one currency. All of the ATMs are still considered single currency machines.
- There is no support for ATM device print or display of balances in more than one currency with the BASE24-atm Multiple Currency Support enhancement.
- The following transactions are not supported in BASE24-atm when the Multiple Currency Support enhancement is used:
  - Deposit with Cash Back Transactions
  - Generic Balance Inquiry Transactions (i.e., balance inquiry for two different account types in one transaction)
  - Split Deposit Transactions
- The Self-Service Banking add-on product has not been enhanced to support Multiple Currency functionality.
- Some loss of accuracy due to rounding will occur when the BASE24-atm limits, which are expressed in whole currency units, are converted to another currency. It is assumed that such rounding is acceptable for limit processing.

Refer to “Multiple Currency Support” in the BASE24 Base Optional Enhancements Section for more information about this enhancement.

Note: Not all BASE24-atm modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Tidel Device Handler

BASE24-atm has been enhanced to support Tidel ATMs. The BASE24-atm Tidel Device Handler module provides the interface between the device and front-end processor. This includes managing the communications and translating messages to and from the native format of the device.

The BASE24-atm Tidel Device Handler process supports the Tidel AnyCard-sc and AnyCard-td single cassette dial-up ATM devices. This new module supports Tidel's version 2.02 firmware.

The Tidel Device Handler transaction set includes withdrawals, balance inquiries and transfers for checking, savings and credit card applications. Support is also included for a print totals and administrative cutover transaction request (i.e., balance with standard cash). Features supported include ANSI PIN block, the Surcharge Application (BA-EN010) and inactivity timers for individual device monitoring. The Tidel Device Handler supports Tidel's Host Type #1 message format.

The control and monitoring of screens, receipts and device components is provided by the device. Tidel devices communicate with the Tidel Device Handler in a dial-up environment using the VISA2 Async and X.25 POS-2 protocol.

Transaction Security Services

The new BASE24 Transaction Security Services is now available in BASE24 Release 6.0 for managing all cryptographic functions such as PIN verification, message authentication, chip authentication and card verification. Transaction Security Services will need to be implemented to support full end-to-end triple DES and/or to support Atalla's new AKB key formats. This requires all keys be stored in a new database for all BASE24 modules that call hardware security modules. Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all PIN Verification methods are supported in TSS and not all ATM Device Handlers support triple DES encryption. Refer to the product-specific sections of this guide for more details about this enhancement.

Triple/Single DES Translation Support

BASE24-atm ISO Host Interface has been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 host interfaces and the host itself. Other BASE24 Interchange Interface modules have been enhanced to support triple DES. ACI will only support full end-to-end Triple DES processing in release 6.0. Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

Triton Device Handler

BASE24-atm has been enhanced to support Triton ATMs. The BASE24-atm Triton Device Handler module provides the interface between the device and the front-end processor. This includes managing the communications and translating messages to and from the native format of the device.

The BASE24-atm Triton Device Handler process supports the miniATM Script Terminal (model #9000) and the miniATM Cash Dispense for both single and multi-cassette (model #95xx and 96xx). It supports the following features: ANSI PIN block, the Surcharge Application (BA-EN010) and inactivity timers for individual device monitoring. The Device Handler process provides support for the following financial and administrative transactions: withdrawal, transfer, balance inquiry, host totals and return/clear host totals. The Triton device handler module also supports Non-Currency dispense functionality.

The control and monitoring of screens, receipts and device components is provided by the device. Triton devices can communicate with the Triton Device Handler in a dial-up environment using the VISA 2 asynchronous and X.25 POS-2 protocol.

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BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 offerings. These enhancements must be reviewed by all BASE24-pos users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

Several 6.0 general changes have been made to BASE24-pos modules:

These changes have minor impacts that users can learn more about within section 6 of this document.

- Three user fields were added to the following DDL layouts for future product enhancements, future regional enhancements, and future CSMs:
    - BASE24-pos Retailer Definition File (PRDF)
    - BASE24-pos Terminal Data Dynamic File (PTDD1)
    - BASE24-pos Terminal Data Static File (PTDS1)
- Several new optional ACI Standard POS Device Handler (SPDH) sub-fields were added as part of standard product processing. These SPDH sub-fields include:

FID	SFID	Picture	Length	Field Name	RQST	RESP
6	E	PIC 9(3)	3 bytes	Point of Service Entry Mode	✓	✓
6	I	PIC 9(3)	3 bytes	Transaction Currency Code	✓	✓
6	N	PIC 9(4)	4 bytes	Reason Online Code	✓	

- Preauthorization processing has been enhanced for the APACS 30/40 and SPDH Device Handlers, to allow preauthorization holds to be identified by either the transaction sequence number or additionally, by using the approval code generated for the preauthorization request.
- BASE24-mail can be licensed in conjunction with BASE24-pos and offers electronic mail capabilities between the POS device and BASE24. The SPDH supports the BASE24-mail functionality. No configuration changes or file conversions are required for release 6.0.
- BASE24-pos CRT Authorization provides a facility for telephone operators to authorize credit and debit card transactions, which enhances customer service levels to the merchant and cardholder. No configuration changes are required for release 6.0.

EMTs in BASE24 Classic

The EMTs in BASE24 Classic enhancement allows BASE24 processes to share extended memory tables, eliminating redundant copies of data in memory. Note that the BASE24-pos modules use EMTs in release 6.0 (e.g., APCFEMT). Refer to the BASE24 Base Enhancements for more information about this enhancement.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Multiple Logical Network Routing Indicator

This enhancement provides a new token for use by back office settlement and reconciliation systems to identify transactions that have been routed between two logical networks and logged to two PTLFs. Refer to the BASE24 Base General Enhancements for more information about this enhancement.

PTDF Expansion

The BASE24-pos Terminal Data File (PTDF) is defined as an Enscribe relative file. The maximum length for a relative file is 4072 bytes. The file is at the maximum record length, and there are just a few user fields that can be redefined for future enhancements. Data for future enhancements requires a second PTDF record to be defined for each terminal. If the PTDF remained in the existing formats and all new data is added to a second PTDF record, each transaction processed by the Device Handler process would require two reads of the PTDF and two writes to the PTDF (assuming the second PTDF record contains data that is updated during the online transaction path).

The PTDF Expansion Enhancement defines a method of storing additional PTDF data without adversely impacting online transaction processing performance.

<u>PTDF Record Layout</u>

The structure of the PTDF is composed of two main components:

- Common configuration data for all Device Handlers. This data is displayed using the BASE24 AFT subsystem.
- Device dependent data specific for each device handler. This data is not displayed using the BASE24 AFT subsystem and is used primarily to store PSTM data and monitor the status of the device. Note that unlike the TDF where the

entire STM is stored in the TDF record, the PTDF only stores selected PSTM fields that are required to generate a reversal.

Three new terminal files were defined in release 6.0. These files are key-sequenced files. The BASE24-pos Terminal Definition files (PTD) are used by the following BASE24-pos enhanced Device Handler modules:

- ACI Standard POS Device Handler module
- Hypercom Device Handler module

Each terminal requires a record to exist in each PTD file. A record in the first PTD file contains dynamic data that is common for all Device Handlers (i.e., a fixed structure) and dynamic data that varies based on the device type. This record contains data that is updated during the online transaction path. It must be read from disk for each transaction and rewritten to disk upon completion of each transaction.

A record in the second PTD file contains static data. This record contains a fixed structure for static data that is common for all Device Handlers. The remainder of the static record contains device-specific data. This record contains configuration data that does not change during the course of online transaction processing. This data can be maintained in extended memory. The extended memory segment for the static data is not built using the EMT Build utility program. Because of the potential for a large terminal population and limited system resources at a customer site, the size of the static data extended memory segment is configurable. The extended memory segment is populated dynamically. The new TDF libraries, used for accessing both the ATD and PTD files, first check extended memory for a static record. If the record is not found in extended memory, the libraries read the static data from disk and put the static data record into extended memory. If there is not room in extended memory for the static record, an existing record in extended memory will be deleted to make room for the new static record.

A record in the third PTD file contains scratch-pad data for each POS device. This is an optional file for BASE24-pos users wanting to store token data in a file other than the PTLF for reversal processing.

The new PTD records do not contain a hard-coded maximum size for device-specific data or last message area. Instead, the PTD records are defined based on the requirements of each device handler. The fixed structure portion of the dynamic record contains offsets and lengths of device-specific configuration data and device dependent data. This allows data for these areas to be defined as needed for the specific Device Handler module. For example, under the PTDF structure a Device Handler module may need more than the 280 bytes of device-specific configuration data, but only need half of the device dependent data that is defined in the PTDF record. By using offsets and lengths, a PTDF record format can be defined for each device handler type.

Last PSTM and Token Data

Instead of maintaining the PSTM data and token data of the last transaction in the PTD, this data will be held in extended memory. When a response is received from the authorization entity, the Device Handler process puts the PSTM and token data into extended memory. If a transaction must be reversed, the Device Handler module uses the PSTM in extended memory to format the reversal message.

If the Device Handler process goes down before it is determined that a reversal must be generated, the PSTM data is lost. However, the PATD dynamic record has the PTLF file name and relative byte address (RBA) of the last transaction. The Device Handler module reads the PTLF, formats a PSTM reversal message, updates the PTD to back out the transaction impacts, and sends the message to the authorization entity.

In summary, the following benefits are a result of the PTDF Expansion enhancement:

- The dynamic and static records have room to grow before the maximum record lengths are reached (i.e., there is a substantial amount of space available for future enhancements).
- Dynamic data was separated from the static data with the new PTD files, and therefore disk I/Os were held to a minimum.
- A new server object was written for the PTDF and PTD files, so that the BASE24 AFT subsystem can present one view to the user.
- The method of saving token data has been modified, so that more token data can be retained for reversal processing.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Transactions Allowed

Transactions Allowed parameters are now configured in a set of common files shared by multiple products. Note that there are product-specific files to configure with this enhancement (e.g., PTD, PRDF). Refer to the BASE24 Base General Enhancements for more information about this enhancement.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

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Optional Enhancements

The following BASE24-pos Release 6.0 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

ACI Proactive Risk Manager Interface Components

The BASE24-pos Router/Authorization module adds the new ACI Proactive Risk Manager token in the PSTM that is used to route the transaction to the ACI Proactive Risk Manager system if installed. Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

Derived Unique Key Per Transaction (DUKPT)

The BASE24-pos Release 6.0 Standard POS Device Handler (SPDH) has been enhanced to support the derived unique key per transaction (DUKPT) security management scheme. This is an uplift from the BASE24-pos Release 5.3.

SPDH was modified to support DUKPT for the encryption of PIN blocks for transactions originating at the terminals directly connected to BASE24-pos. Using the DUKPT method, each terminal security module (TSM) or PIN pad uses a derived unique key for each transaction.

The PTD PIN-ENCRYPT-TYP field supports a new value of “07” which indicates DUKPT usage. The SPDH message definition was modified to include a FID 6 subFID T, which is used to transmit the key serial number and descriptor from the terminal to BASE24. The SPDH uses the PTD PIN-ENCRPT field to determine if terminals are using DUKPT, and accesses the key derivation file, KEYD file, as needed, to process a DUKPT encrypted PIN block.

This enhancement is compatible only with the Atalla NSP and Thales security modules that support the commands necessary for this function.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Also, this enhancement is not yet supported with the Transaction Security Services enhancement. Contact your Customer Support Representative for further details.

BASE24-pos eCheck

BASE24-pos eCheck is a separately licensable add-on module to BASE24-pos which enables online positive balance check verification, guarantee or conversion services from the point-of-sale. With this solution, acquirers are able to offer full service electronic check processing and authorization of check transactions to merchant clients through their existing infrastructure. Issuers are able to perform check verification and check guarantee transactions in participation with current check verification or conversion programs now offered by several card associations and networks.

The BASE24-pos Router/Authorization module supports check verification and check guarantee transactions with an online-only authorization level, with or without a PAN. To facilitate eCheck, several BASE24 components provide the foundation for electronic check. With the eCheck Foundation, an issuer or acquirer is able to route and pass check verification and guarantee transactions through BASE24, taking advantage of additional check related tokens and routing improvements. Separate licensable add-on eCheck modules are required to allow BASE24-pos to authorize or stand-in for check verification and guarantee transactions or to receive and send check transactions through a network interface.

As part of the eCheck enhancement, the Check Authorization Routing Table (CART) was modified, and includes the following changes:

- A new alternate key of Institution ID Number (i.e., should match IDF alternate key value).
- A change to the primary key to include a FIID field.
- Modifications to the routing table to include the authorization level, authorization type (method) and a check SPF field.

Support for the Check Auth2 Token was added. Supported network interfaces or BASE24-pos Device Handler modules add the Check Auth2 Token and the pre-existing check tokens. The Router/Authorization module updates the token with the CART routing table authorization level, authorization type and check SPF fields.

PTLF perusal displays the check authorization transaction descriptions for the check verification and check authorization transactions. BASE24-pos reports use the new POS transaction table, which includes the transaction descriptions for check verification and check authorization transactions.

For BASE24-pos acquirers who want to acquire check verification and check guarantee transactions, the ACI Standard POS Device Handler (SPDH) provides additional information to the interchanges for processing.

Electronic Benefits Transfer (EBT)

With the release 6.0, the Electronic Benefits Transfer enhancement, previously supported as a channel-specific enhancement, has been fully incorporated into standard BASE24. This enhancement is specifically engineered for U.S. customers.

Electronic Benefits Transfer (EBT) provides retail customers with the functionality for acquiring and routing electronic benefits transactions. It also allows retailers to comply with federal (U.S.) mandates to electronically process government benefit transactions. EBT transactions allow consumers to access accounts for food stamps or cash benefits (e.g., Aide for Dependent Children).

EBT transactions are supported only through BASE24-pos, using SPDH as a BASE24 acquirer. If a transaction cannot be completed because the link to the authorizing entity is down, no stand-in processing is performed. There is no support for issuer traffic via an interchange interface.

The following interchange interfaces support EBT transactions:

- Alaska Option
- Deluxe ISO Gateway
- Money Payment Systems
- NPC ISO
- PayPoint ISO Gateway
- Star ISO

Note: Not all BASE24-pos modules have been modified to support this enhancement. Contact your ACI Customer Manager for further details.

EMV Support

BASE24-pos acquirer and issuer processing have been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV ’96. Note that the APACS, Hypercom and SPDH terminals have been enhanced to support this ICC standard.

Refer to “EMV Support” in the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Contact your ACI Customer Manager for further details.

Format 2 File Support

Format 2 File Support, was released in Compaq's NSK D46 operating system. Format 2 File Support is available for the PTLF in BASE24 Release 6.0, for D46 and higher Compaq NSK operating systems.

Format 2 File Support gives BASE24-pos Release 6.0 users the ability to format Enscribe disk file (PTLF and ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain more data than the existing format 1 files.

Refer to "Format 2 File Support" in the BASE24 Base Optional Enhancements Section for more information.

Multiple Currency Support

BASE24-pos processing has been enhanced to allow institutions and retailers to acquire transactions in different currencies at APACS and SPDH terminals. Functionality has been added to allow a single APACS or SPDH terminal to include the currency code with the transaction as it is sent to BASE24, thereby allowing a single terminal to accept transactions in many currencies. Institutions and retailers can also offer accounts in different currencies by defining a currency for each account.

The BASE24-pos CRT Authorization screens support multiple currency transactions and account processing. The PRF requester, PRF server, and MRF requester have been enhanced to include the alpha ISO code associated with each transaction.

The following issues should be considered when configuring the Multiple Currency Support enhancement in BASE24-pos.

- Parametric Authorization is not supported in the BASE24-pos Multiple Currency Support enhancement.
- Chargeback processing is not supported in the BASE24-pos Multiple Currency Support Enhancement.
- The PRF Report (B24POS30) has not been enhanced to support multiple currencies.
- Some loss of accuracy due to rounding will occur when BASE24-pos limits, which are expressed in whole currency units, are converted to another currency. It is assumed that such rounding is acceptable for limit processing.

Refer to "Multiple Currency Support" in the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement.

BASE24-pos Stored Value

BASE24-pos Release 6.0 has been enhanced to support Stored Value functionality. Previously an America's Channel product, Stored Value has been uplifted from release 5.3; and, as part of release 6.0, incorporated into the global product set.

The Stored Value pos module provides functionality to allow retailers to issue a stored value instrument in either a physical form; e.g. card, or as a virtual product over the Internet. All account balance information is stored on BASE24 as it is the database of record. Retailers can establish minimum and maximum replenishment amounts and configure the maximum amount of cash that the customer can receive back. Cards can be activated individually, or in batches, with options for setting expiration dates. Additional cards can be issued that tie back to the central funding account.

The Stored Value module only works in conjunction with the BASE24 Standard POS Device Handler (SPDH). Likewise, Stored Value only uses the Offline-Only authorization level and CAF/PBF authorization method. Stored Value transactions can be SPROUTE routed to a third party interface, provided that interface supports the required functionality.

The addition of Stored Value to BASE24-pos introduces four new transactions. They are Card Activation, Additional Card Activation, Replenishment and Full Redemption. A new Stored Value Token, "U2", is also being added to the Global Product. This token carries Stored Value data needed for transaction processing.

Note: Not all BASE24-pos modules have been modified to support this enhancement. Contact your ACI Customer Manager for further details.

Transaction Security Services

The new BASE24 Transaction Security Services is now available in BASE24 Release 6.0 for managing all cryptographic functions, such as PIN verification, message authentication, chip authentication, and card verification. Transaction Security Services will need to be implemented to support full end-to-end triple DES and/or to support Atalla’s new AKB key formats. This requires all keys be stored in a new database for all BASE24 modules that call hardware security modules. Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all PIN Verification methods are supported in TSS. DUKPT is not supported in TSS. APACS MACing is not supported in TSS. No BASE24-pos Device Handlers support triple DES encryption. Refer to the product-specific sections of this guide for more details about this enhancement.

Triple/Single DES Translation Support

BASE24-pos ISO Host Interface has been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 host interfaces and the host itself. Other BASE24 Interchange Interface modules have also been enhanced to support triple DES. ACI will only support full end-to-end Triple DES processing in release 6.0. Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

BASE24-teller Enhancements

The BASE24-teller product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-teller product, refer to Section 7.

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General Enhancements

There are no general enhancements inherent to this product in release 6.0.

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Optional Enhancements

There are no optional enhancements inherent to this product in release 6.0.

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BASE24 Kernel/IEP Enhancements

The BASE24 ITS Kernel/IEP product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 ITS Kernel/IEP product, refer to Section 8.

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General Enhancements

The following BASE24 Kernel/IEP enhancements are inherent in the release 6.0 offerings. These enhancements must be reviewed by all BASE24 Kernel/IEP users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

This 6.0 general change has been made to BASE24 Kernel/IEP modules:

This change has minor impacts that users can learn more about within section 8 of this document.

- Prior to release 6.0, BASE24 Kernel/IEP supported only one logical network (LN) per BASE24 AFT subsystem. In release 6.0, BASE24 Kernel/IEP has been enhanced so that multiple logical networks can share a single BASE24 AFT subsystem. Upon logon, users can chose the logical network database they want to access first by entering the desired logical network on the initial logon screen. They can then switch to another database at any time by entering the new logical network at the bottom of any screen.

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Optional Enhancements

There are no optional enhancements inherent to this product in release 6.0.

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BASE24-telebanking Enhancements

The BASE24-telebanking product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-telebanking product, refer to Section 9.

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General Enhancements

The following BASE24-telebanking enhancements are inherent in the release 6.0 offerings. These enhancements must be reviewed by all BASE24-telebanking users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

This 6.0 general change has been made to BASE24-telebanking modules:

This change has minor impacts that users can learn more about within section 9 of this document.

- Prior to release 6.0, BASE24-telebanking supported only one logical network (LN) per BASE24 AFT subsystem. In release 6.0, BASE24-telebanking has been enhanced so that multiple logical networks can share a single BASE24 AFT subsystem. Upon logon, users can chose the logical network database they want to access first by entering the desired logical network on the initial logon screen. They can then switch to another database at any time by entering the new logical network at the bottom of any screen.

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Optional Enhancements

The following BASE24-telebanking Release 6.0 enhancements are optional. BASE24-telebanking users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

Format 2 File Support

Format 2 File Support was released in Compaq's NSK D46 operating system. Format 2 File Support is available for the ITLF in BASE24-telebanking Release 6.0, for D46 and higher Compaq NSK operating systems.

Format 2 File Support gives BASE24-telebanking Release 6.0 customers the ability to format Enscribe disk file (ITLF) partitions to accommodate format 2 files. Files with format 2 partitions can contain more data than the existing Format 1 files.

Refer to "Format 2 File Support" in the BASE24 Base Optional Enhancements Section for more information about this enhancement.

Multiple Currency Support

BASE24-telebanking has not been enhanced to support multiple currency transactions. Changes have been made to BASE24-telebanking to ensure that BASE24-telebanking will run in a multiple currency system, including OLTP checks for unsupported currency codes and PBF Refresh file format changes. Refer to "Multiple Currency Support" in the BASE24 Base Optional Enhancements Section for more information about this enhancement.

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BASE24-billpay Enhancements

The BASE24-billpay product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-billpay product, refer to Section 10.

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General Enhancements

The following BASE24-billpay enhancements are inherent in the release 6.0 offerings. These enhancements must be reviewed by all BASE24-billpay users to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

6.0 General Changes

This 6.0 general change has been made to BASE24-billpay modules:

This change has minor impacts that users can learn more about within section 6 of this document.

- Prior to release 6.0, BASE24-billpay supported only one logical network (LN) per BASE24 AFT subsystem. In release 6.0, BASE24-billpay has been enhanced so that multiple logical networks can share a single BASE24 AFT subsystem. Upon logon, users can chose the logical network database they want to access first by entering the desired logical network on the initial logon screen. They can then switch to another database at any time by entering the new logical network at the bottom of any screen.

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Optional Enhancements

There are no optional enhancements inherent to this product in release 6.0.

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BASE24-customer service Enhancements

The BASE24-customer service product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-customer service product, refer to Section 11.

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General Enhancements

There are no general enhancements inherent to this product in release 6.0.

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Optional Enhancements

The following BASE24-customer service Release 6.0 enhancements are optional. BASE24-customer service users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

Format 2 File Support

Format 2 File Support was released in Compaq's NSK D46 operating system. Format 2 File Support gives BASE24-customer service user the ability to format Enscribe disk file (TLF and PTLF) partitions to accommodate format 2 files. Files with format 2 partitions can contain much more data than the existing format 1 files.

Refer to "Format 2 File Support" in the BASE24 Base Optional Enhancements Section for more information about this enhancement.

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BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 12.

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General Enhancements

Selected BASE24 Interchange Interface processes have been enhanced to support release 6.0 functionality within the Interchange Interface process or as a separate licensable module. The enhancements include all or some of the following functions: Transactions Allowed, EMV Support, Format 2 File Support, Triple/Single DES Translation Support and Transaction Security Services (TSS). Note that no specific modifications have been made to the Interchange Interface processes for the Multiple Currency Support enhancement.

The following table lists the enhancements available for selected Interchange Interface processes:

Interchange Interface	Catalog Number	EMV	Trans Allowed	Format 2 File Support	Triple/Single DES	TSS
Alaska Option ISO	SW-DS554					Yes
Alaska Option ISO EBT	SW-DS556					N/A
BASE24 Interface ANSI (BIC ANSI)	SW-DS015 SW-DS019					Yes
BASE24 Interface ISO (BIC ISO)	SW-DS004, SW-DS009	Yes	Yes	Yes	Yes	Yes
Banknet ISO	SW-DS764	Yes	Yes	Yes	Yes	Yes
CashStation ISO	SW-DS385					Yes
EPS-Net ISO	SW-DS860	Yes		Planned	Yes	Yes
EPS-Net IPF Refresh	SW-AE983	N/A	N/A	N/A	N/A	N/A
IPF Refresh - Cirrus	SW-AE001	N/A	N/A	N/A	N/A	N/A
IPF Refresh - Plus/Visa	SW-AE000	N/A	N/A	N/A	N/A	N/A
LINK ISO*	SW-DS969	Yes		Planned	Yes	Yes
MAC MASM	SW-DS792					Yes
MDS/Cirrus ISO	SW-DS133	Yes	Yes	Yes	Yes	Yes
MDS Maestro Gateway	SW-DS134	Yes	Yes	Yes	Yes	Yes
Moneystation	SW-DS172					Yes
MPS	SW-DS355					Yes
MPS EBT	SW-DS357					N/A
NEN	SW-DS300					Yes
NPC ISO	SW-DS290					Yes
Plus ISO – ATM	SW-DS063		Yes			Yes
Plus ISO/Interlink Gateway	SW-DS068		Yes			Yes
Publix	SW-DS310					Yes
Shazam (ITS) ISO	SW-Ds142					Yes
Switch Authorization Network (SWANI)	PS-DH987	Yes				

VisaNet ISO	SW-DS500	Yes	Yes	Yes	Yes	Yes
VisaNet Interlink Gateway	SW-DS509	N/A	Yes	Yes	Yes	Yes
VisaNet/PLUS Gateway	SW-DS508	N/A	Yes	Yes	Yes	Yes
VisaNet ISO IPF Refresh	SW-DS512	N/A	N/A	N/A	N/A	N/A

The Interchange Interface processes that have been enhanced with one or more of the above enhancements have been uplifted to a release 6.0 subvolume (SW60xxxx). As usual, BASE24 Release 6.0 Interchange Interface processes are backward compatible with previous versions of BASE24. All remaining supported Interchange Interfaces have been enhanced to be compatible with release 6.0. Compatible interfaces (i.e., those that have not been enhanced with release 6.0 functionality) maintain their current subvolume, either SW53xxxx or SW51xxxx, where xxxx is the Interchange Interface acronym name. The compatible interfaces have been modified to support release 6.0 and continue to support releases 5.3, 5.1.2, 5.0 or 4.0/3.4 of BASE24-atm and BASE24-pos.

Documentation on the Interchange Interface enhancements for release 6.0 and the modifications to the compatible Interchange Interface processes is available as informal documentation on the individual Interchange Interface module's subvolume. Each individual subvolume has a file with a file name of "XXXXMAN", where "XXXX" is the Interchange Interface acronym name. A formal manual is available on the ACI CD-ROM for the BIC ISO Interchange Interface and Europay EPS-Net Interchange Interface.

Customers need to consult with the applicable networks to determine if any certification testing is required, however ACI does recommend certification testing for any interface that has been enhanced for release 6.0.

EMTs in BASE24 Classic

The new EMTs in BASE24 Classic enhancement allows BASE24 processes to share extended memory tables, eliminating redundant copies of data in memory. Note that the enhanced Interchange Interface processes use the EMTs specified by the Transactions Allowed enhancement. Refer to the BASE24 Base Enhancements for more information about this enhancement.

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

Transactions Allowed

Transactions Allowed parameters are now configured in a set of common files shared by multiple products. Interchange Interfaces that have been enhanced with Transactions Allowed use the ICFE, APCFEMT, and IPCFEMT instead of the ICF. Refer to the BASE24 Base Enhancements for more information about this enhancement.

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Refer to the product-specific sections in this guide for more details about this enhancement. Contact your ACI Customer Manager for further details.

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Optional Enhancements

The following BASE24 Interchange Interface enhancements are optional. BASE24 Interchange Interface(s) users converting to release 6.0 should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment, before deciding to implement any of the changes.

- EMV Support
- Format 2 File Support
- Transaction Security Services
- Triple/Single DES Translation Support

BASE24 Release 6.0 Interchange Interface users should refer to the documentation provided for each Interchange Interface for implementation instructions. This documentation can be found on the release 6.0 Interchange Interface subvolume and the Interchange Interface add-on product subvolume with a file name of “XXXXMAN”, where “XXXX” is the Interchange Interface acronym name. All BASE24 Release 6.0 Interchange Interfaces that have been enhanced with the described general or optional enhancements are delivered on “SW60XXXX” subvolumes. A formal manual is available on the ACI CD-ROM for the BIC ISO Interchange Interface, Europay EPS-Net Interchange Interface, LINK (LIS5) Interchange Interface and Switch Authorization Network Interface (SWANI).

EMV Support

The following BASE24 Interchange Interface processes have been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV ’96. The following processes were enhanced in BASE24 Release 6.0.

- VisaNet ISO Interchange Interface
- EPS-Net Interchange Interface
- Banknet Interchange Interface
- LINK ISO Interchange Interface
- MDS ISO/MDS Maestro Interchange Interface
- BASE24 ISO (BIC) Interchange Interface
- Switch Authorisation Network Interface (SWANI)

Refer to “EMV Support” in the BASE24 Base Optional Enhancements for more information about this enhancement.

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Contact your ACI Customer Manager for further details.

Format 2 File Support

Format 2 File Support is available for selected Switch 6.0 interfaces in BASE24 Release 6.0 for D46 and higher Compaq NSK operating systems. Format 2 File Support gives BASE24 Release 6.0 users the ability to format Enscribe disk file (ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain more data than the existing format 1 files.

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Refer to the interface-specific subvolume of “XXXXMAN”, where “XXXX” is the Interchange Interface acronym, for more specific information related to this enhancement.

Transaction Security Services

The new BASE24 Transaction Security Services is available in BASE24 Release 6.0 for managing all cryptographic functions such as PIN verification, message authentication, chip authentication and card verification. Transaction Security Services will need to be implemented to support full end-to-end triple DES and/or to support Atalla’s new AKB key formats. The following BASE24 Release 6.0 Interchange Interfaces (Switch 6.0) have been enhanced to work with this new functionality. These BASE24 Release 6.0 Interchange Interfaces run in all releases of BASE24, but will only work with Transaction Security Services when running in a release 6.0 network.

- Alaska Option ISO Interchange Interface
- Banknet Interchange Interface
- BASE24 ISO (BIC ISO) Interchange Interface
- BASE24 ANSI (BIC ANSI) Interchange Interface
- Cash Station ISO Interchange Interface
- EPS-Net Interchange Interface
- LINK ISO Interchange Interface
- MAC MASM Interchange Interface
- MDS/Cirrus Interchange Interface
- MDS Maestro Interchange Interface
- Midwest Payment Systems Interchange Interface
- Moneystation Interchange Interface
- NEN Interchange Interface
- NPC ISO Interchange Interface
- PLUS ISO Interchange Interface

- Publix Interchange Interface
- Shazam (ITS) ISO Interchange Interface
- VisaNet ISO Interchange Interface

Refer to the BASE24 Base Optional Enhancements for more information about this enhancement.

Triple/Single DES Translation Support

Several BASE24 Release 6.0 Interchange Interfaces (Switch 6.0) have been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 interchange interfaces and the interchange itself. These BASE24 Release 6.0 interchange interfaces run in all releases of BASE24. No other BASE24 modules have been enhanced at this time to support triple DES. Note, ACI will support full end-to-end Triple DES processing in release 6.0 only. Because of interchange mandates, BASE24 interchange interfaces may be required to support triple DES encryption before customers can implement end-to-end support or when running BASE24 interchange interfaces in a release 5.3 or earlier network.

The interchange interfaces enhanced to support triple to single DES transactions are:

- BASE24 Interface ISO (BIC ISO) interchange interface
- Banknet ISO interchange interface
- MDS/Cirrus interchange interface
- MDS Maestro Gateway interchange interface
- VisaNet ISO interchange interface
- VisaNet Interlink Gateway interchange interface
- VisaNet/Plus Gateway interchange interface
- Link ISO interchange interface
- EPS-Net ISO interchange interface

Note: Not all BASE24 Interchange Interface modules have been modified to support this enhancement. Refer to the interface-specific subvolume of “XXXXMAN”, where “XXXX” is the Interchange Interface acronym, for more specific information related to this enhancement.

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BASE24 Value-Added Products Enhancements

A brief overview follows for each of the value-added products that have been enhanced in conjunction with other BASE24 product offerings. These value-added products have been enhanced to be compatible with release 6.0.

For more information about the enhancements in the BASE24 value-added products, refer to Section 13.

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BASE24-InfoBase

BASE24-InfoBase is the family name of a group of associated products, running on Compaq NSK hardware, which form a comprehensive solution for data collection and distribution, as well as transaction and file processing. The InfoBase products comprise a data transfer product, an EFT transaction processing application, support for the management of hot lists and a file version management system. The products in this family are frequently packaged together, as in the package BASE24 File Manager, which consists of InfoBase File Transfer and InfoBase Software Manager.

The entire suite of BASE24-InfoBase release 2.0.4 products is compatible with BASE24 Release 6.0. InfoBase release 2.5, which will be generally available soon after the release of BASE24 6.0, will also be compatible with BASE24 Release 6.0.

InfoBase File Manager

BASE24 File Manager is a software management and file transfer system that supports the management, transfer, and processing of diverse data across various and mixed networks. NonStop SQL and Compaq NSK's TMF products are used to manage the data flowing through the system and provide management processing.

BASE24 File Manager consists of two components: the Software Manager module and the File Transfer module. The Software Manager module is used to manage and report on the delivery of software and data files to remote devices. Files are created outside the BASE24 File Manager system, typically by the device vendor's own application. The files are presented to Software Manager, where they are associated with one or more remote devices. The user then releases them for delivery by the File Transfer delivery component.

InfoBase File Transfer

InfoBase File Transfer is the central component in the InfoBase product suite. All other InfoBase products essentially work in conjunction with File Transfer. File Transfer is the module that collects data from, and distributes data to, remote devices (e.g., ATM and POS devices).

InfoBase File Transfer operates independently of the business meaning of the data being handled, except where demanded by a particular EFT message format. File Transfer includes a comprehensive scheduling system for host initiated file transfer, as well as a monitoring system to provide a controlled environment for remote device-initiated transfers. File Transfer provides automated recovery processing for failed transfers to assure delivery.

Residing on the Compact NSK platform, InfoBase makes use of NonStop SQL database for all stored data and event logs.

InfoBase EFT

InfoBase EFT is the component of the InfoBase family that is used for the processing of EFT transactions. InfoBase EFT performs batch and transaction level validation on transactions that have been collected from a variety of terminals and presented to InfoBase EFT through several means, including input from the BASE24-pos Transaction Log File (PTLF). The EFT module also allows for the optional correction and input of data through the use of the TEAS subsystem.

All transactions are maintained in an SQL database for inquiry, control, audit and reporting functions. In addition, customers have the option of having output formats developed for delivery to their downstream processing hosts.

InfoBase Software Manager

InfoBase Software Manager is used to manage and report on the delivery of software and data files to remote devices. InfoBase Software Manager can process both object (software), textual (configuration parameters, table data) and graphics or video (marketing) files. These files are created outside InfoBase and are presented to InfoBase Software Manager, where they are associated with one or more remote devices. The user then releases them for delivery by InfoBase File Transfer. All activity is recorded, so the user can be sure that the correct files have been sent to the right devices at the right time, enabling the user to maintain version control information.

InfoBase HLMS

InfoBase HLMS is the InfoBase Hot Card List Management system. HLMS is a component that will receive and process hot lists (i.e., negative files of payment cards of checks) in a variety of formats from multiple sources. These hot lists are validated and merged to produce replacement or update files for distribution. The user defined destinations may be either terminals connected to the InfoBase File Transfer system or external systems such as third party processors.

BASE24 Retail Products

Retail customers who are making plans to upgrade to BASE24 Release 6.0 need to be aware that release 3.1 of BASE24-check auth, BASE24-frequent shopper, and BASE24-refunds are compatible with BASE24 Release 6.0. All retail products **prior to** release 3.1 will not be compatible with BASE24 Release 6.0. This means that when you uplift to BASE24-pos Release 6.0 you must also uplift the retail products to release 3.1.

The Retail Base 3.1 changes include the introduction of a Retail Configuration menu choice at the Virtual Menu that will reduce the number of steps required to access the user screens. In addition the creation of the Retail Configuration menu consolidates all the retail product configuration tables. With this change, minor modifications to the naming conventions of shared retail configuration tables have occurred.

BASE24-check auth

BASE24-check auth is a powerful tool for curbing check fraud and improving customer service. It provides a fast, convenient method for authorizing checks while limiting exposure to losses from check runners, organized fraud schemes, counterfeiters, forgers and customers who continually write non-sufficient funds (NSF) checks.

The BASE24-check auth system supports positive and negative information stored in the retailer or processor’s Compaq NSK database. These databases can be updated dynamically or through the batch process. This functionality uses positive data based on check-writing history with the retailer, negative data based upon returned-check information from the retailer or a third-party processor, and check validation based on the customer’s check velocity and dollar amounts.

BASE24-frequent shopper

BASE24-frequent shopper provides the ability to establish customized programs that collect valuable marketing and purchase data from the point of sale while rewarding customers for shopping. These programs include marketing campaigns, value accumulation and redemption.

BASE24-frequent shopper helps retailers build customer loyalty by providing a flexible way for retailers to learn about and understand customer behavior, thereby retaining and growing business. The system helps retailers increase sales through value-added services rather than through general price discounts and unfocused mail campaigns.

BASE24-refunds

BASE24-refunds gives retailers a potent defense against losses from reduced sales and increased shrinkage by combating fraudulent refunds schemes. The system validates returns online, providing powerful protection against refund fraud. The solution manages a systematic approach to handling refunds for both brick-and-mortar and virtual sites, allowing retailers to maintain a consistent and accurate refunds policy across retail channels.

Based on individual retailer parameters, refund requests can be declined online or issued as a configurable tender (check, voucher, or stored value). BASE24-refunds offers the ability to maintain information about customer receipts against which a refund is requested.

BASE24-card

BASE24 has been enhanced in release 6.0 to contain support for BASE24-card in standard product code. Previous releases of BASE24 supported BASE24-card as a Channel Specific Enhancement (CSE). BASE24-card is a Card Issue and Control System that enables institutions to control the issuing and maintenance of plastic cards on a Compaq NSK computer system. The application controls the generation of a plastic card files with data on the Cardholder Authorization File (CAF) and the Card Prefix File (CPF). In addition, BASE24-card provides the capability of maintaining the card database for online transaction authorization.

General Enhancements

All BASE24-card users should review the following enhancements to ensure that the correct configuration has been defined and that proper processing is invoked.

6.0 General Changes

Several 6.0 general changes have been made to BASE24-card modules. These changes have minor impacts that users can learn more about in section 13 of this document.

The release 6.0 CAF refresh format (CAFX) does not contain the alternate key fields as a result existing BASE24-card customers must to modify the host refresh process to no longer provide fields.

- The CAF refresh format currently includes ALPHA-KEY; however, this field is being removed from the CAF refresh format.
- The CAF GEN-KEY fields previously defined in the CAF Refresh are populated in the CAF database via an Issue Maintenance function with BASE24-card.

Customers may find that Host impacts could include:

- Hosts that process the OMF and/or ULF extract must be able to process new alternate key fields in the CAF record image.
- Hosts must be able to provide the new CAF Card segment in the refresh format and/or in the FHM ISO message.
- Hosts that process the OMF and/or ULF extract must be able to process the new Card segment in the CAF record image.

ACI Worldwide Inc.

ACI Host Link

ACI Host Link provides a CICS based interface between host applications and BASE24. It is compatible with IBM S/370, S/390 and z/Series mainframes running VSE, MVS or OS/390.

The product:

- Resides on the host and is compatible with IBMs CICS/ESA 4.1, CICS/TS 1.3, CICS/TS 2.1 and CICS/VSE 2.1.
- Provides support for both acquirer and issuer transaction processing for point-of-sale (POS), electronic mail, teller and automated teller machine (ATM) applications.
- Supports both ANSI and ISO message formats between BASE24 and the host processor.
- Accommodates either SNA/SDLC (LU2) or multi-point bisynchronous (BSC 3270) communication protocols between BASE24 and the host processor.
- Uses CICS command-level coding standards and conventions for ease of integration into the customer's host system.
- Features a table-driven set of processing parameters for ease of initial configuration, permitting timely, online changes to the operating environment. Since these processing parameters are memory resident, the need for disk I/O is virtually nonexistent.

ACI Host Link release 6.0 is compatible with BASE24 Release 6.0. ACI Host Link releases 5.1.2 and 5.3 are compatible with BASE24 Release 6.0 for customers that do not use the TeleBanking and/or From Host Maintenance products.

ACI Host Link release 6.0 is compatible with BASE24 Release 6.0 Triple/Single DES Translation Support.

ACI Host Link release 6.0 was made generally available in December 2001. Additional information on enhancements that will be included in ACI Host Link release 6.0 will be provided with the ACI Host Link release 6.0 documentation.

ACI Worldwide Inc.

ACI Plastic Card Control System

The ACI Plastic Card Control System offers a complete plastic card system for issuing cards, maintaining account information, tracking card usage, and providing customer service. ACI Plastic Card Control System supports multiple account types and allows online display and modification of pertinent account information.

This product supports all of the functions necessary to manage and service a cardholder base, from issuing a plastic card to tracking the card’s use. Some of the features of ACI Plastic Card Control System include multiple institution support, flexible maintenance, BASE24-atm transaction limits, BASE24-pos transaction limits, multiple application support, card/member/linked account relationships, special processing indicators, PIN processing, card production, online inquiries, and reporting.

ACI Plastic Card Control System Release 6.0 is compatible with BASE24 Release 6.0 for customers that do not use the following optional enhancements: EMV Support and Multiple Currency.

ACI Worldwide Inc.

ACI Claims Manager

ACI Claims Manager is a state-of-the-art claims and adjustment computer system that offers financial institutions a flexible solution that can enhance their customer server responsiveness and regulatory compliance while optimizing their staff utilization. ACI Claims Manager organizes and prioritizes the workflow that individual research analysts perform on a daily basis. The system helps ensure compliance with the appropriate regulations, improves operator productivity, and provides a vehicle to implement management policies.

The client application is developed primarily with PowerBuilder in a Windows environment. Clients have access to tasks and processing rules that are set up within ACI Claims Manager by a database administrator. Tasks and processing rules can be setup to ensure compliance with Reg E, MasterCard, regional networks, and institution specific requirements.

ACI Claims Manager Client/Server uses databases specific drivers to build, populate and access SQL tables that reside on the server hardware. This architecture allows the application to be server independent, providing optimum flexibility.

ACI Claims Manager Release 5.0 is compatible with BASE24 Release 6.0.

ACI Worldwide Inc.

ACI Proactive Risk Manager

ACI Proactive Risk Manager is a comprehensive client-server risk management/fraud control solution that analyzes EFT transactions and optionally uses a neural network to assign a fraud score based on past customer and institution transaction activity, and displays data for transactions which exceed a user-defined threshold for investigators to view. It combines the pattern recognition power of neural-network technology with statistical techniques to accurately score transactions.

ACI Proactive Risk Manager can operate with BASE24-atm and BASE24-pos in either a “near real-time” or “batch” mode to accommodate a variety of risk management strategies.

Alerts are generated for transactions scoring over the user-defined alert threshold, or by user-defined rules. The transactions are then queued for analysis and processing by the fraud investigators. A centralized Windows NT server delivers transactions to the reviewer workstation.

ACI Proactive Risk Manager is implemented in a distributed processing environment. ACI Proactive Risk Manager is designed to interoperate with existing authorization systems and runs on a wide-range of platforms, including Compaq NSK, IBM MVS and Windows NT.

BASE24 Release 6.0 is compatible with release 6.0 of ACI Proactive Risk Manager.

ACI Worldwide Inc.

ACI Settlement Manager

ACI Settlement Manager is a back office product providing creation of ACH or custom files for funds movement and cardholder posting; fee calculations, reporting and adjustment processing using the BASE24-atm Transaction Log File (TLF) and BASE24-pos Transaction Log File (PTLF). Reconciliation between BASE24 and external networks are also supported.

ACI Settlement Manager release 7.0 is compatible with BASE24 Release 6.0 for customers that do not use the following optional enhancements: EMV Support and Multi-Currency.

ACI Worldwide Inc.

ACI Payments Manager

ACI Payments Manager is an integrated software solution that manages customer accounts and automates the processing and settlement of electronic transactions. The solution is a database product that utilizes a centralized, online source of transaction updates in a near-real time environment. The solution's modules can be combined to provide a comprehensive electronic payments environment. The three-tier architectures: a user interface client, a business rules server and a data warehouse improves deployment and portability, streamlines installation, enhancements and functional updates.

ACI Payments Manager's core features integrate data and automate functions, including transactions storage and analysis; settlement and funds movement; and customer and transaction instrument management.

The integrated software solution consists of alerting, monitoring and executive analysis modules that enable users to analyze transaction information, as well as provide key business performance indicators and product performance statistics.

The settlement module provides comprehensive, automated processing for validation and settlement of electronic payment transactions, including reconciliation, funds management, fee calculation and exception processing for single or deferred settlement transactions.

The account management module provides a flexible parameter-driven electronic payments product issuance program that supports a comprehensive view of customer transaction and account activity.

Release 3.0 of ACI Payments Manager is compatible with BASE24 Release 6.0.

ACI Worldwide Inc.

ACI Smart Chip Manager

ACI Smart Chip Manager is a multifunction smart card lifecycle management solution designed to control every aspect of the lifecycle of a smart card and its applications. The solution supports the entire lifecycle of a card and the applications that reside on it, including the issuance, suspension/resumption and cancellation of cards and card applications.

The ACI Smart Chip Manager online updating module delivers online services to smart-card holders and enables issuers to add or update applications on cards. The module supports the connection between the front office, which provides the interface to the customer, and the back office, which handles the administration of the services. The online updating module simultaneously supports various services for different terminal types, as well as many cards used for the same or different services. The module's functionality includes identification and authentication for smart cards, downloading card applications, updating card applications, and processing transactions with a card application.

The ACI Smart Chip Manager card personalization module, a flexible smart card personalization tool, supports personalization of multifunctional smart cards, manages high volumes of cards, and allows card producers and their customers to turn smart card demand to their advantage. The card personalization module is suitable for use in large smart card personalization centers or by card issuers producing cards in-house. The module is highly configurable; the user selects and separately licenses modules based upon the personalization equipment to be used.

The ACI Smart Chip Manager customer management module, used primarily in small environments, supports cardholder enrollment and maintenance, including customer registration, card issuance, card-replacement and customer service functions, such as card and application blocking.

ACI Smart Chip Manager has been updated to accept CAF Updates from BASE24 Release 6.0.

ACI Worldwide, Inc.

Section 3

Release 6.0 Support Services

This section summarizes the various support services available from ACI for BASE24 Release 6.0. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; contact your ACI Customer Manager for more details of the services ACI can offer.

ACI Worldwide Inc.

Product Consulting And Project Management

No formal service programs have been created by ACI for upgrading customers from BASE24 Release 5.3/1.3 or Release 6.0 to latest version of BASE24 Release 6.0. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system to release 6.0.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

ACI Worldwide Inc.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system from release 5.3/1.3 to release 6.0. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

   One of the initial tasks for the release 6.0 upgrade is for the BASE24 user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

   An ACI sales representative can assist in determining whether additional hardware will be required for the release 6.0 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with Compaq NSK. Compaq NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

   If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

   If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE COMPAQ NSK SOFTWARE REQUIREMENTS

   ACI recommends that the BASE24 user be on a current Compaq NSK supported operating system to obtain maximum performance and support. Although release 6.0 is currently compatible with the Compaq NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 Release 6.0 requires TMF for several of the new files.

5. UPGRADE TO XPNET 3.0

   Customers should note that XPNET 3.0 is the minimum requirement for BASE24 Release 6.0.

6. PROJECT SCOPE MEETING

   ACI has available a BASE24 Release 5.3 to release 6.0 differences PSM if the customer chooses to have a Project Scope Meeting for their 6.0 upgrade project.

7. MIGRATION PLANNING

   With BASE24 Release 6.0 users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

   ACI does not believe this step is necessary for customers upgrading from BASE24 Release 5.3/1.3 to BASE24 Release 6.0.

9. DEFINE REQUIREMENTS DEFINITION

   The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new sync point. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 or if they are specific to the customer's environment and will need to be upgraded to the new release.

10. DEVELOP PROJECT PLAN

    The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new release, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 code. Customers implementing Triple/Single DES Translation or Transaction Security Services need to re-certify. Since the changes to the interface software for release 6.0 were minimal, except for Triple/Single DES Translation, Transaction Security Services or interchange interfaces also updated with new interchange mandates, customers not implementing these changes should not have to re-certify most interfaces.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 Release 6.0 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24 Release 6.0. A release 6.0 enhancement course is also available upon request. The course descriptions can be found in the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 Release 6.0 products. User/Operational impacts are noted in the detailed sections for each product within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 Release 6.0 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted to tape and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 software will need to be installed at the customer site. A tape containing the licensed release 6.0 software modules will be sent to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for BASE24 files between the BASE24 Release 5.3/1.3 products and BASE24 Release 6.0. There are a few that need to be converted. These files are discussed in the detail sections for each product.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 Release 5.3 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer’s project.

26. TURN OVER TO ACCOUNT MANAGEMENT

If an ACI project manager has handled the release 6.0 upgrade, a formal turnover meeting is held between the project manager and account manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 formal publications set has been updated to include the release 6.0 enhancements. The updated BASE24 release 6.0 documentation is available via CD-ROM.

The BASE24 6.0 CD-ROM also contains the previous release (5.3/1.3) documentation. The 5.3/1.3 version of a manual can be considered the most current if the manual is not listed under the 6.0 documentation menus. The XPNET 3.0 documentation is available on a separate CD-ROM titled NET24-XPNET.

BASE24 customers can order a copy of the BASE24 6.0 CD-ROM and/or XPNET 3.0 CD-ROM from their ACI Customer Manager.

ACI has discontinued hard copy distribution of manuals that are available on CD-ROM. Hard copies, however, can be printed at your site from the CD-ROM as necessary and in accordance with your license agreement.

ACI Worldwide Inc.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 Release 6.0 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Release 6.0 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

For a current education schedule, see the BASE24 Education Catalog available in hardcopy through your ACI Customer Manager or on the ACI Worldwide web site at \\www.aciworldwide.com.

ACI Worldwide Inc.

HELP24

Users migrating from BASE24 Release 5.3/1.3 to BASE24 Release 6.0 should use their existing procedures to report problems or ask questions. HELP24 will continue to use the same response and tracking methodology as they currently do today.

All HELP24 staff have received training in all methods of release upgrades. In addition, there is a complete library of migration documentation available for reference in quickly resolving upgrade issues.

HELP24 has implemented Remote Diagnostics or Access capabilities for assistance in troubleshooting complex problems. The service has grown because of the obvious benefits it offers. Information about Remote Diagnostics or Access is available from HELP24 or your ACI Customer Manager. HELP24 strongly recommends that customers start planning to take advantage of this service now, so it will be available and ready in a crisis situation.

Access to HELP24's bulletin board (in the EMEA this is provided through a web site) is still available as an additional method for communication of important support information. BASE24 users are assigned individual logons to use this facility and are able to peruse the information 24 hours a day. The system can be used to transfer files such as fix files, log files, audit files, and dump files to help expedite problem resolution.

ACI Worldwide Inc.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the release 6.0 upgrade. The number of staff weeks identified to accomplish these tasks are estimates and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new sync point level. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the release 6.0 upgrade project.

The Tech Lead will:

- Review the migration options and develop a detailed migration plan for the conversions project
- Review the Project Scope Document on behalf of the customer (if applicable)
- Review the customer's internal procedures to accommodate release 6.0 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code from release 5.3/1.3 to release 6.0. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

ACI Worldwide Inc.

Section 4

BASE24 Base Release 6.0

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 5.3/1.3 or a previous version of 6.0.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 Base modules changes that affect all BASE24 users, along with the associated conversion/implementation requirements that all BASE24 users must address.
- Optional enhancements describe the release 6.0 Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 13 to identify the changes that BASE24 users need to consider for upgrading to the latest version of release 6.0.

Standard Information Format

A standard format is used in sections 4 through 13 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given release 6.0 change or enhancement. This information includes Make defines that have been added, deleted, or modified with release 6.0.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given release 6.0 change or enhancement. This information includes assigns that have been added, deleted, or modified with release 6.0, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given release 6.0 change or enhancement. This information includes params that have been added, deleted, or modified with release 6.0, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given release 6.0 change or enhancement. This information includes file fields that have been added, deleted, or modified with release 6.0, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given release 6.0 change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given release 6.0 change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given release 6.0 change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given release 6.0 change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given release 6.0 change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given release 6.0 change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given release 6.0 change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given release 6.0 change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given release 6.0 change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given release 6.0 change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given release 6.0 change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given release 6.0 change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user’s host or co-network as a result of a given release 6.0 change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user’s internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given release 6.0 change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given release 6.0 change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given release 6.0 change or enhancement. BASE24 users can want to refer to these documents for additional information.

General Enhancements

This section describes the release 6.0 changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0.

- 6.0 General Changes
- EMTs in BASE24 Classic
- LCONF File Maintenance
- Multiple Logical Network Routing Indicator
- PI-TABLE Expansion
- Transactions Allowed

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. The BASE24 ITS Control Server name has changed to include the logical network name. This allows BASE24 ITS to have multiple logical networks under one BASE24 AFT subsystem.

2. A BASE24 Classic Control Server similar to the BASE24 ITS Control Server has been created. Several server objects (e.g., ATD, APCF) will be bound into one executable server, and therefore only one server will be defined in the PATHCONF for these new servers.

3. The Account Type Table (ATT) is used by BASE24 ITS and BASE24 Classic. The DDL source file for this table was moved from the OC13DDL subvolume to the BA60DDL subvolume.

4. The release 6.0 file conversion programs use a set of skeleton conversion files that are the basis for each file conversion program and necessary to Make the programs. These skeleton conversion source files are located on the BA10SC04 subvolume.

5. For release 6.0, the Make files (i.e., FM files, MM files, and M files) have been sequenced. When fixes are made to these files in the future, BASE24 uses will receive FIX files for these modifications instead of file replacements.

6. For release 6.0, ACI is introducing a new version of the Make utility (i.e., version 5). This new version handles the newly sequenced Make files. In addition, the new version is more efficient, because it runs native, runs High PIN, and handles memory better. For more information, refer to the MAKEHELP file on the MAKEV500 subvolume.

7. For release 6.0, release 4.0 formats for refresh, extract, and ISO messages used by ISO Host Interface, BIC ISO Interchange Interface, and From Host Maintenance processes are no longer supported.

8. For release 6.0, support for some modules has been dropped. Contact your ACI Customer Manager for the list of modules that have been dropped.

9. Three user fields were added to the following DDL layouts for future product enhancements, future regional enhancements, and future CSMs. These fields were added with the initial version of release 6.0, so that file conversions can be avoided with later enhancements and customer-specific changes.
   - ATDD1
   - ATDS1

- CAF
- CAFX (CAF refresh format)
- CPF
- ECF
- ERF
- FHSTM (BASE24-from host maintenance Standard Internal Message)
- FHM's ISO External Message
- ICFE
- IDF
- PBF
- PBFX (PBF refresh format)
- PTDD1
- PTDS1

10. Data fields were added to the following DDL layouts for CSEs that have already been done in a previous release, and may be supported in product in future releases. These fields were added with the initial version of release 6.0, so that file conversions can be avoided with later enhancements.
    - CAF
    - CAFX
    - CPF
    - ECF
    - FHSTM (BASE24-from host maintenance Standard Internal Message)
    - FHM's ISO External Message
    - KEYA (a redefine of VERIFY data area)
    - PRDF
    - PTDD1
    - PTDF
    - PTDS1

    Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The release 6.0 version 1 CAF refresh file message format contained the following BASE24-card fields: CARD-GENKEY and CARD-ALPHA-KEY. These fields have been removed from the release 6.0 version 2 CAF refresh file message format and the bytes were added to the field USER-FLD2. This is only a release 6.0 version 1 to release 6.0 version 2 consideration.

11. The Surcharge File (SURF) has been modified for the Multiple Currency Support enhancement to include a CURRENCY CODE field in the primary key. All BASE24 users who implement the surcharge feature, regardless of whether the Multiple Currency Support enhancement is being used, are required to configure this information. All surcharge users are required to run a SURF Conversion program to implement this modification.

12. The Extract Configuration File (ECF) has been modified for the Format 2 File Support enhancement to support larger values for the various extent fields in the ECF. The ECF has also been modified to allow the extract disk file to be buffered. In addition, the ECF supports the configuration of reading past the initially determined end-of-file or reading until to the initially determined end-of-file. With the modifications for Format 2 File Support, all BASE24 users, regardless of whether format 2 files are used in their BASE24 system, are required to run an ECF Conversion program.

    Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: A separate ECF conversion program was written, and must be run, for those BASE24 users converting from release 6.0 version 1 to release 6.0 version 2. If you are implementing format 2 files, refer to the Database Configuration subsection of the Format 2 File Support enhancement for further information.

13. The documentation files on the BASE24 Release 6.0 subvolumes are named to ensure that they are unique per product. If a change must be made to one of these files for a fix or an enhancement, the updated file is placed on the fix file-replacement subvolume (e.g., f05Xba60). This implies that the file name must be unique per product.

    Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The AAREADME files were renamed to ensure that the file name is unique per product. For example, the AAREAMDE file on the BA60UC04 subvolume has been renamed to AAREAD04.

CUSTMACS Changes

Added MYMACS and BACSE_CSEMACS hooks. If these are defined using -D on Make's command line, they will source in the appropriate file, otherwise, they point to files that do not exist.

Added defines for TRUE and FALSE to the top of the flags defined in this file to support the MYMACS and BACSE_CSEMACS hooks.

Renamed the pos_ckau_on flag to pos_echeck_on.

The following flags were deleted:

- apca_on
- atm_dh_c1000_sm_on
- atm_dh_c2300_on
- atm_dh_c906_on
- atm_dh_conv_c910_on flag
- atm_dh_n50_sm_on

- atm_dh_n1770_on
- atm_dh_omron_on
- badrp_on
- base_icc_on
- emea_on
- emea_fortify_on
- ebms_on
- fc_on
- interprocess_comm_on
- ipvd_vdps_refr_on
- message_interface_on
- message_interface_atm_on
- message_interface_pos_on
- plus_sm_on
- ssb_dh_on
- sm_on
- swapc_on
- swavs_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be modified in the LCONF to support this release. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>BASE24-REL</u>

A code indicating whether the BASE24 logical network is currently running release 3.4/4.0, release 5.x, or release 6.0. The release 6.0 Interchange Interface processes use this param to determine whether to use release 6.0 processing params or previous release processing parameters.

This param must be set to 60 if the BASE24 network being used is release 6.0.

Interchange Interface processes retrieve LCONF assigns according to how the BASE24-REL param is configured. If the BASE24-REL param contains a value other than one of the valid values or is not present in the LCONF, Interchange Interface processes default to the same release number as the subvolume on which the interchange interface code resides and perform processing according to the parameters of that release. Valid values are as follows:

- 40 = Release 3.4/4.0.
- 50 = Release 5.x. Default for Interchange Interfaces residing on SW5x subvolumes
- 60 = Release 6.0 or later. Default for interchange interfaces residing on SW60 subvolumes.

Finally, the EMF and KEYF servers read this param to determine which file to access. If the param equals 50 or 60, the EMF and KEYF files are accessed. Otherwise, the EMF5 and KEY5 files are accessed.

Database Configuration

New user fields were added to the CAF, CPF, ECF, ERF, ICFE, IDF, and PBF files for future product enhancements, future regional enhancements, and future CSMs. In addition, the new user fields were added to the CAF refresh format (CAFX) and PBF refresh format (PBFX). The new user fields are the following:

- USER-FLD-ACI is reserved for future BASE24 product use only. The designation of "product use only" provides ACI with the ability to deplete the number of bytes available within USER-FLD-ACI as product enhancements are introduced. When product enhancements require the addition of new fields within this file, the procedure to be followed is to deplete bytes from USER-FLD-ACI and use that number of bytes to define new fields. The new field definition(s) should precede the USER-FLD-ACI field.
- USER-FLD-REGN is reserved for ACI regional use only. Only ACI regions are allowed to use USER-FLD-REGN space.
- USER-FLD-CUST is reserved for BASE24 customer use only. Only customers are allowed to use USER-FLD-CUST space.

Note: The above fields allow space for future enhancements, and therefore they are not viewable using the BASE24 AFT subsystem. When a future enhancement uses a portion of any of the ACI reserved user fields, documentation on that enhancement will address the impacts to the BASE24 system.

<u>Extract Configuration File (ECF)</u>

Field Name: READ PAST INITIAL EOF
Screen: ECF Screen 1
DDL Name: ECF.BASE.READ-PAST-INITIAL-EOF

This field indicates whether extract reads past the initial end-of-file that was determined as the process began extracting records from that particular file. If additional records were added to the file after this initial end-of-file was determined, those records are not included in the extract unless extract was instructed to read past the initially determined end-of-file. Valid values are Y (i.e., Yes, extract reads past the initially determined end-of-file) and N (i.e., No, extract does not read past the initially determined end-of-file). The default is N.

When the last extract is performed on a transaction log file, the appropriate ECF record must have a value of Y.

Field Name: BUFFERED

Screen: ECF Screen 3
DDL Name: ECF.BASE.FILE-BUFFERED

This field indicates whether the Extract file should be created as a buffered file. Valid values are Y or N with a default of N.

Surcharge File (SURF)

Field Name: CURRENCY CODE
Screen: SURF Screen 2 and Screen 3
DDL Name: SURF.CPF.TXN-CRNCY-CDE
SURF.SPROUTE.TXN-CRNCY-CDE

Screen 2 and Screen 3 of the SURF have been modified to include a CURRENCY CODE field in the primary key. All surcharge users, regardless of whether the Multiple Currency Support enhancement is being used, are required to configure this information. All surcharge users are required to run a SURF Conversion program to implement this modification.

For additional information on the usage of this field, refer to the ***BASE24 Base Files Maintenance Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry was renamed in the SECTBL. DCT screen 32 was the Mirror Device Control screen, and is now the Olivetti 6000 Device Control screen.

Entries were deleted from the SECTBL for the IDF screen 45 (VisaCash Parameters), CNRF screens, DCT screen 42 (NCR 1700 Device Control), and DCT screen 46 (Omron Device Control).

SECTBL entries for ATT screens have been modified, so this file will appear on the BASE Product Menu.

PATHCONF Changes

Ensure that the MAXTERMDATA and MAXREPLY parameters for the TCPs defined in the PATHCONF are set to 125000 and 13312 respectively.

The BASE24 ITS Control Server name has changed from SERVER-ICTL to SRV-ICTL-<logical-network>. Thus, a BASE24 ITS Control Server will exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem.

A new server must be defined in the PATHCONF file for the Classic Control Server (i.e., CCTL Server). The CCTL Server can be added online through the PATHCOM utility, but it must also be added to the PATHCONF so that any future PATHCOLD operation still contains this server. The server name must be specified as SRV-CCTL-<logical network> in the PATHCONF. Thus, a BASE24 Classic Control Server will exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following files are required with other BASE24 Release 6.0 enhancements, and therefore the three new user fields and data fields added for CSEs will be initialized by those file conversions:

- CAF
- CPF
- ECF
- ICFE
- IDF

Extract Configuration File (ECF)

The ECF conversion program must be run to convert ECF records from the release 5.3 format to the release 6.0 format (i.e., the latest version of release 6.0). Running the conversion program is required for all users, not just those that are using the Format 2 File Support enhancement. Users must create the ECF file using the BADDLFUP file.

The ECF conversion program on the BA60UC04 subvolume can be used to convert the ECF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVECFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ecfcnv.t,name $ecfm,nowait/
cnvecfm –force –list –listreplace
```

Users must update the CNVECFR obey file on the BA60UC04 subvolume with the appropriate information. CNVECFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: A separate ECF conversion program must be run for those BASE24 users converting from release 6.0 version 1 to release 6.0 version 2. If you are implementing format 2 files, refer to the Database Configuration subsection of the Format 2 File Support enhancement for further information.

The ECF conversion program must be run to convert ECF records from the release 6.0 version 1 format to the release 6.0 version 2 format. Running the conversion program is required for all users using BASE24 Release 6.0 version 1, not just those that are using the Format 2 File Support enhancement. Users must create the ECF file using the BADDLFUP file.

The ECF conversion program on the BA60UC05 subvolume can be used to convert the ECF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVECFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ecfcnv.t,name $ecfm,nowait/
```

```
cnvecfm –force –list –listreplace
```

Users must update the CNVECFR obey file on the BA60UC05 subvolume with the appropriate information. CNVECFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD05 file on the BA60UC05 subvolume.

Refresh Group Control File (RGCF)

A RGCF conversion program that converts RGCF records from their release 5.3 format to the release 6.0 format is not supplied. Users must create the release 6.0 RGCF file using the BADDLFUP file.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Users must re-create the RGCF using the BADDLFUP file, since the RGCF has changed in length. The relative byte addresses (RBAs) saved in this file are 64-bits, not 32-bits. The 64-bit RBAs are required to read the format 2 transaction log files.

Surcharge File (SURF)

The SURF conversion program must be run to convert SURF records from the release 5.3 format to the release 6.0 format. Running the conversion program is required for any surcharge user, not just those that have the Multiple Currency Support enhancement. Users must create the SURF file using the BADDLFUP file.

The SURF conversion program on the BA60UC04 subvolume can be used to convert the SURF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVSURFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#surfcnv.t,name $surfm,nowait/
cnvsurfm –force –list –listreplace
```

Users must update the CNVSURFR obey file on the BA60UC04 subvolume with the appropriate information. CNVSURFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host impacts will be incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the From Host Maintenance, ISO Host Interface, Refresh, and Super Extract processes.

From Host Maintenance Process

The From Host Maintenance Process ISO message format allows the new user fields to be sent in the message from the host to BASE24 for this new information defined in the CAF or PBF. It also allows the new data fields added for CSEs to be sent in the message from the host to BASE24 for this new information defined in the CAF. Users utilizing release 5.*x* From Host Maintenance Process ISO message format will have the new fields defaulted to spaces.

ISO Host Interface Process

Host code changes are required for all BASE24 users that utilize the release 3.4/4.0 formats, since the release 3.4/4.0 format is no longer supported in the ISO Host Interface Process.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 3.4/4.0 formats, since the release 3.4/4.0 format is no longer supported in the Refresh Process.

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file message format to accept and process the new user fields and data fields for CSEs in the CAF file. Users utilizing the release 5.*x* CAF Refresh file message format will have the new fields defaulted to spaces.

Host code changes are required for all BASE24 users that utilize the release 6.0 PBF Refresh file message format to accept and process the new user field information in the PBF file. Users utilizing the release 5.*x* PBF Refresh file message format will have the new user fields defaulted to spaces.

Super Extract Process

Host code changes are required for all BASE24 users that utilize the release 3.4/4.0 formats, since the release 3.4/4.0 format is no longer supported in the Super Extract Process.

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process the new user fields and data fields for CSEs defined in the CAF, CPF, ECF, ERF, ICFE, IDF, and PBF.

User/Operational Impacts

Disk Impacts

The CAF, CPF, ECF, ICFE, IDF, and PBF require additional disk space for the three new user fields and the data fields added for CSEs. The total impact varies for each BASE24 user depending on the number of records defined in the BASE24 system. The conversion programs provided for these files initialize the new fields.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: During the transition between these versions, users cannot suspend and resume refresh since the RGCF must be re-created using the BADDLFUP file. After the RGCF is re-created, the fixes for the Format 2 File enhancement are applied, and an overall BASE24 Release 6.0 Make is run, the suspend and resume functionality of Refresh can be used.

User Impacts

The ECF has also been modified to allow the extract disk file to be buffered. The ECF conversion program sets the ECF base segment field FILE-BUFFERED to a value of N, and therefore BASE24 users must update those ECF records where the extract disk file is created with the buffered attribute.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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EMTs in BASE24 Classic

For performance reasons, components of the BASE24 ITS environment's Extended Memory Table files (EMTs) were utilized in BASE24 ITS applications (i.e., BASE24-customer service, BASE24 ITS Kernel/IEP, and BASE24-telebanking) to access critical Enscribe files (or SQL table data) from EMTs held in processor memory rather than from files or tables on disk. This saves time in the execution of time-critical operations. Some EMTs are shared by more than one object while others are used by a single object. The EMT Build utility provided with the BASE24 ITS environment was used to build shared extended memory tables from disk files or tables. EMTs that are not shared are built by the individual objects that use them. Once an EMT has been built or rebuilt, it must be allocated or reallocated before it can be used in processing.

This BASE24 Release 6.0 enhancement expands the use of the EMTs into BASE24 Classic, and BASE24 ITS serves as a model for BASE24 Classic implementation. Certain BASE24 Classic applications will use EMTs, and take advantage of the following EMT features:

- Data Verification Through Batch Processing

  Data verification is performed during construction of an EMT, which occurs in the background as a batch process. Batch processing enables resolution of problems encountered during EMT construction without disruption of transaction processing. EMT Build reports confirm the success of EMT construction.

- Conserved Initialization Time During Process Startup

  EMTs offer the significant advantage of greater efficiency during initialization and warmboot processing. Multiple I/Os of files, which can occur during initialization and warmboot, can be greatly reduced or avoided. The data in an EMT can be shared by multiple processes within a CPU.

- Greater Efficiency During Multiple-step Searches

  EMTs prove effective in cases where multiple searches are performed to retrieve specific records.

The BASE24 Classic modules in release 6.0 that utilize EMTs are the enhanced BASE24 Device Handlers, BASE24-atm Authorization, BASE24-pos Router/Authorization, and enhanced Interchange Interfaces. These modules already build exclusive extended memory tables upon initialization and warmboot. EMTs offer time and resource savings during initialization and warmboot processing, as well as standardization of data tables across applications.

Existing EMTs

BASE24 ITS Core applications use the Institution Definition File EMT (IDFEMT). This EMT has been expanded to accommodate BASE24-atm usage; the IDFEMT was modified to support additional product segments not used in BASE24 ITS. In addition, the EMT was moved to the new BASE24 Classic EMT Build subvolume (i.e., BA60EMTB) in order to facilitate sharing between BASE24 Classic and BASE24 ITS applications.

For this enhancement, the EMTB executable object, which includes the individual EMT Build modules for each product, is bound together into the new BASE24 Classic EMT Build Link object.

BASE24 Classic modules, which use the IDFEMT, support the requisite EMT functionalities. These include initialization of the EMTs, segment allocation, segment deallocation, and the ability to receive and interpret EMT warmboot commands.

It should be noted that, despite the aforementioned benefits of EMTs, some concerns exist about the impacts of EMT usage on performance. EMTs are not appropriate for excessively large tables and will not be used to access dynamic data (e.g., dates).

EMT Terminology

Extended Memory Table file (EMT): A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build): The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables.

custom-designed table: An extended memory table type that features custom construction. The table contains a header record and can be constructed with flat tables.

hash table: An extended memory table created by converting a record's primary key into a hashing function. Each key is mapped to a "cell" within the table. This mapping is referred to as "hashing." Conflicts occur when two keys hash to the same value during table construction. In order to produce the most efficient hash table (i.e., a table with the fewest possible conflicts), various hash algorithms are used; the most effective algorithm is used for final hash table construction. Hash tables are used to retrieve data with a specified key. Records are not maintained in a specific order.

radix table: A table created by sorting and resorting of data into bins. Radix tables are used to retrieve data with partial keys. In-order searches can be performed, as records are maintained in a specified order.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>ALT-COLL</u>

The fully qualified PPD name of the alternate EMS Collector process. This assign is used by the Extended Memory Table Build utility, SQL Refresh process, BASE24-telebanking Issuer Reporting program, and Event object.

If this assign is not present in the LCONF or contains an invalid value, the Extended Memory Table Build utility, SQL Refresh process, BASE24-telebanking Issuer Reporting program, and the processes that contain the Event object use a default value of $0.

<u>EMT-SWAPVOL</u>

The fully qualified file name of the swap volume in which the Extended Memory Table Build utility temporarily stores data when building extended memory tables.

If this assign is not present in the LCONF or contains an invalid value, the default volume is used and an error message is generated indicating that the assign needs to be corrected.

<u>IDFEMT</u>

The fully qualified file name of the Institution Definition File Extended Memory Table file (IDFEMT). The Extended Memory Table Build utility and the enhanced BASE24-atm Device Handlers use this assign.

If this assign is not present in the LCONF, the Extended Memory Table Build utility, the processes that contain the Financial Institution Access object, and the enhanced BASE24-atm Device Handlers do not initialize.

PRI-COLL

The fully qualified PPD name of the primary EMS Collector process. This assign is used by the Extended Memory Table Build utility, SQL Refresh process, BASE24-telebanking Issuer Reporting program, and Event object.

If this assign is not present in the LCONF or contains an invalid value, the process, programs, and object listed above use a default value of $0.

SPANCTL

The fully qualified file name of the Span Control File. This assign should point to the SPROUTE file.

The SPANCTL assign is used by the BASE24-atm Authorization process, BASE24-pos Device Handler/Router/Authorization process, XPNET, and the Extended Memory Table Build utility.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

ALT-TIM-LMT

The maximum time, in hundredths of seconds, the application waits for the call to the WRITE procedure to complete. This param is used when logging to the alternate EMS Collector process. Valid values are 0 through 2147483647. This param is used by the Extended Memory Table Build utility, BASE24-telebanking Issuer Reporting program, Event object, and SQL Refresh process.

CONSOLE-RPT

Indicates whether EMS event messages are printed on the operator console if the EMS compatibility distributor is used. This param is used by the Extended Memory Table Build utility, BASE24-telebanking Issuer Reporting program, Event object, Enscribe Refresh process, and SQL Refresh process.

DATE-DISPLAY-FORMAT

The format used to display dates on screens and print dates on reports. This param applies to fields that include only the month and day or month and year, as well as to fields that include all three portions of the date. This param is used by the Extended Memory Table Build utility and the ITS Transaction Log File Server.

IDFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the IDFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

IDFEMT-WARM*identifier*

A param containing the symbolic name of a process to be warmbooted for the IDFEMT. Processes that can be warmbooted for the IDFEMT are the enhanced BASE24-atm device handler processes. This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all IDFEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each IDFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another IDFEMT-WARM*identifier* param. For example, IDFEMT-WARM01 could contain the P1A^NCR^5XXX^01 process and IDFEMT-WARM02 could contain the P1A^NCR^5XXX^02 process.

PRI-TIM-LMT

The maximum time, in hundredths of seconds, the application waits for the call to the WRITE procedure to complete. This param is used when logging to the primary EMS Collector process. Valid values are 0 through 2147483647. This param is used by the Extended Memory Table Build utility, BASE24-telebanking Issuer Reporting program, Event object, and SQL Refresh process.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry was added to the SECTBL to support the new DCT screen 5 for EMT Warmboots.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new DCT screen 5 must be added through the SEC requester to the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The IDFEMT impacts disk requirements. The total impact varies for each BASE24-atm user depending on the number of financial institutions defined in the BASE24 system.

User Impacts

BASE24 users must understand the concept of EMT Builds and the procedure to create an Extended Memory Table file (EMT).

BASE24 users must understand the EMT Build Utility report that is generated by the EMT Build Utility.

BASE24 users must also understand the concept of EMT warmboots and the procedure to warmboot an EMT. While the BASE24 ITS applications do not re-read the LCONF on EMT warmboots, the BASE24 Classic applications do re-read the LCONF on EMT warmboots.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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LCONF File Maintenance

The LCONF File Maintenance enhancement changes the method for accessing the LCONF file for a BASE24 logical network. The change incorporates the LCONF file into the BASE24 AFT subsystem and provides an audit mechanism for changes to the file.

With this enhancement, the LCONF file will be accessed using the BASE24 AFT subsystem from the LNCF, a new selection on the Base menu.

This new LCONF Requester and Server perform most of the functions of the previous Entry program. The following functions are different:

- Online Maintenance File (OMF) auditing is available for LCONF database changes.
- It is possible for more than one person at a time to make changes to the file.
- The edit for file names on the Assigns screen has been changed. The file name is no longer required to match part of the template name.
- Standard BASE24 function keys have been used.
- It is not possible to create an empty LCONF by using a function key.
- It is not possible to read the previous record by using a function key.
- The LCONF Entry program ENTLCONF that maintained the LCONF files will no longer be supported.

When the LNCF Requester is started, the LCONF on the current logical network is the file that is opened. The first screen of the LNCF has a field where it is possible to enter a file name. This file name can be any file of type 2852 with the correct record length. This feature allows a LCONF for another logical network or a new logical network to be edited. For example, the current production LCONF can be copied to another location. The copied file name can be entered on screen 1. The assigns in the file name can then be edited to match the location of a new network.

It is not possible to start the BASE24 AFT subsystem without the existence of an LCONF. The LCONF must contain the required assigns and params (e.g., logical network name, location of the security file (SEC), location of the OMF, etc.). If an LCONF is not available with these assigns and params, the BASE24 user utilizes the FUP CREATE command to create the LCONF file and the LCONF Load Program to load the required assigns and params. Refer to the FUP input file LCONFFUP and the obey file LCONFLDR located on the BA60MISC subvolume to aid you in accomplishing these tasks.

The load program is executed through the use of the LCONFLDR obey file on $<volume>.BA60MISC. Refer to the ***BASE24 Logical Network File Maintenance Manual*** for more information on the LCONF Load program.

Usage codes (formerly Product codes) are used by the printing function to group assign or param records. In release 6.0, a table in LNCFTBL contains the abbreviations used as usage codes. This table can be modified by BASE24 users when a new usage code is needed. The LCONF DDL supports 256 usage codes per record. The first screen of the LNCF Requester displays only 105 usage codes. At a later time, if more are needed for a single record, a second screen will need to be added.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

<u>LNCF-RPT-LOC</u>

The fully qualified printer location for LCONF reports. The assign specified on LNCF screen 1 is not required, and the default is $S.#LNCF.

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

<u>Logical Network Configuration File (LCONF)</u>

Changes were made to the definition of LCONF records. The new definition provides support for 256 products to correspond to the changes made for the PI-TABLE changes. A new last file maintenance field was also added to the record to support OMF auditing.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The LNCF requester name was added to MEGATBL file, since RQMEGA communicates with the LNCF requester using the extended message format (i.e., use USER-CONTEXT-EXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the LNCF requester/server. LNCF is the four-character file abbreviation used to access the LCONF file.

PATHCONF Changes

The new LNCF (LCONF file) server must be defined in the PATHCONF file as SERVER-LNCF. Because the LCONF file must exist to start BASE24 AFT this server cannot be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new LNCF requester screens must be added through the SEC requester for the appropriate users.

LNCF is the four-character file abbreviation used to access the LCONF file.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Logical Network Configuration File (LCONF)

The LCONF conversion program must be run to convert LCONF records from the release 5.3 format to the release 6.0 format. New fields were added to the LCONF to support the standard BASE24 last file maintenance information and support 256 products in conjunction with the PI-TABLE expansion. The conversion program will set these new fields to zeroes.

The LCONF conversion program on the BA60UC04 subvolume can be used to convert the LCONF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVLNCFS file. Prior to converting the LCONF, verify that the OMF assign has a filename in the same format as the template filename. The new LNCF server expects the filename in the assign to be in AYYMMDDX format.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#lncfcnv.t,name $lncfm,nowait/
cnvlncfm –force –list –listreplace
```

Users must update the CNVLNCFR obey file on the BA60UC04 subvolume with the appropriate information. CNVLNCFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host impacts will be incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the Super Extract Process.

Super Extract Process

Host code changes are required for all BASE24 users that wish to utilize the OMF Extract Process to accept and process file maintenance made to the LCONF file using the BASE24 AFT subsystem.

User/Operational Impacts

Access to the LCONF using the LCONF Entry Program is obsolete. BASE24 users that wish to access the LCONF must use the new LNCF requester/server pair that is a part of BASE24 AFT subsystem.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Multiple Logical Network Routing Indicator

End users must have the ability to identify transactions that have been logged to two BASE24 logical networks transaction log files (TLFs or PTLFs). This would help eliminate the need for sorting and comparing the entire transaction log files to identify these transactions for settlement purposes.

Transactions are typically logged to two logical networks transaction log files when one logical network acquires the transaction, but has to send their (non-foreign) card to a second logical network for authorization. BASE24 needs to flag these transactions, so they can be identified. BASE24 uses this indicator when counting transactions for Transaction Based Pricing.

Most multiple logical network hops occur when the acquiring logical network does not own the card, but it has a link to the issuer logical network that will perform the authorization or route to the final authorization destination. For example, PRO1 acquires the transaction but cannot find the card prefix in PRO1's CPF. The authorization process will then perform a SPROUTE lookup. If PRO2 is the owner of the card, typically what is found in the PRECMD for PRO1 will be a copy of the CPF for PRO2. If the card matches a value in the CPF for PRO2 the destination for that transaction is typically an authorization process on the PRO2 logical network. When PRO2 receives this transaction, it either authorizes the transaction on BASE24 or forwards it using IDF routing to another destination for authorization. The response is returned to the PRO2 authorization process and logged to the PRO2 TLF/PTLF. The transaction is then sent back to the PRO1 authorization process and logged to the PRO1 TLF/PTLF. The PRO1 and PRO2 TLF/PTLF records are for the most part identical. There is not a way to consistently determine that a multiple logical network hop has occurred or whether the transaction log record is from the issuer or acquirer node.

SPROUTE lookup has been changed to indicate whether the file type was "IP" – IPF or "CP" – CPF. If authorization performs a SPROUTE lookup and a "CP" file type was used, the authorization would place its logical network (from the LCONF) into a token for logging and passing to the other logical network.

In the example given above, the authorization process on PRO1 would place PRO1 into the acquirer logical network field of the token (Token ID "BK") only if "CP" is indicated on the call from SPROUTE lookup and pass the transaction to the destination on PRO2. PRO2 would either authorize the transaction or send the transaction to the final authorization process. In both cases, token identified by a token ID of "BK" would be logged to the TLF/PTLF on both logical networks with the same value. When the end user wants to detect these transactions, they would compare the value in the token with the logical network they are processing for (i.e.

this is identified in the super extract header as PRO1 or PRO2 etc.). If the PRO1 transaction file is being processed and the value in the token matches then this is a potential duplicate logged transaction and the node was the acquirer of the transaction. If the value in the token doesn't match the logical networks log file then the node is the issuer. This processing would hold true for ATM and POS as well as all message types supported.

Current routing configurations should be reviewed and tested to ensure that transactions that cross multiple logical networks are properly identified. SPROUTE routing transactions from an acquirer logical network to another Switch Interface on another logical network might not be routed through an authorization process and will not be logged to a transaction log file on the second logical network. In this scenario, the transactions are not logged to two different logical network log files and are not reported on or counted twice.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

<u>MULT-LN-ACQ-TRAN-COUNTER</u>

Indicates that transactions routed between two logical networks using CPF SPROUTE routing should be included in the count for the reports. A value of Y indicates to count the corresponding transactions. A value of N indicates to not count the corresponding transactions. The default is Y.

Database Configuration

<u>Token File (TKN)</u>

The following BASE24 token has been added for the Multiple Logical Network Routing Indicator enhancement:

- The Multiple LN Token (Token ID “BK”) contains the logical network where the transaction was acquired.

If the new BASE24 token for this enhancement is not to be logged to the TLF or PTLF, the appropriate token records must be configured in the TKN.

If the new BASE24 token for this enhancement is to be extracted with the TLF or PTLF, the appropriate token records must be configured in the TKN.

If the destination is another BASE24 system, this token should not be configured to be sent in the ISO message using HISO or BIC ISO.

New Data Files

Not applicable

Template Changes

<u>COBNAMES</u>

New Multiple LN token names were added to the token table in the COBNAMES files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The newest version of SKEL must be used in order for this enhancement to be fully functional.

SCR #3000 SKELB - rel3^ver0^sutl13^020112

File Conversions

Not applicable

Host/Co-Network Impacts

BIC ISO Interchange Interface Process

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

Super Extract Process

If the new Multiple LN token for this enhancement is to be included in the TLF or PTLF, BASE24 users must consider this additional data in host programs that use the data from the log file.

ISO Host Interface Process Impacts

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

User/Operational Impacts

Disk Impacts

The TLF and PTLF require additional disk space because the new Multiple LN token for this enhancement can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and PTLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Transaction Based Pricing (TBP) Reports

The BASE24-atm and BASE24-pos TBP reports have been updated to use the new Multiple LN token. This allows the reports to properly count transactions that were SPROUTE routed using a CPF on another logical net. Users need to be on Version 7 or greater of the TBP Reports to take advantage of this change.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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PI-TABLE Expansion

The PI-TABLE is a file used by RQMEGA to build the BASE24 AFT Product Menu, give access to security records in the Security file, and pass information to other requesters about the products present in a BASE24 network. The PI-TABLE Expansion enhancement increases the number of products that can be supported by BASE24 from 32 to 256. It also clarifies the definition of segments used in BASE24 by creating a separate table to define segments. A maximum of 256 segments is supported as well. BASE24 has not been modified to utilize any of the new expanded indicators in release 6.0.

Refer to Appendix G, “PI-TABLE IMPLEMENTATION GUIDELINES”, for procedures to build modules to utilize new Product and Segment Ids under the new PI-TABLE methodology.

PI-TABLE File

In release 6.0, the PITABLE file on the BA60AFT subvolume contains five sections. The sections and their uses are:

- The section PI-TABLE is used by RQMEGA to build the Product Menu for the BASE24 Super/Super user. This table is a set of Y/N flags for all of the products defined for BASE24. Setting a flag to Y causes that product to appear on the Super/Super Product Menu. It also causes that product to be turned on in the message to the security server. The security server uses the information to allow access only to files in the Security Table (SECTBL) associated with products that are set to Y. This information is used to build the entries on screen 3 of the BASE24 AFT Security File (SEC) screens. The Product Menu and File Access information for all other operators is derived from the Security records created by the Super/Super user, not from direct access to the PI-Table.
- The section PRODUCT-DESC-TABLE contains a text description of each entry in the PI-TABLE. The Security Server utilizes the PRODUCT-DESC-TABLE to display product text descriptions on screen 3 of the BASE24 AFT Security File (SEC) screens.
- The section VIRTUAL-DESCR-PROD-TABLE contains a text description of each entry that can be displayed on the virtual menu. RQMEGA utilizes the VIRTUAL-DESCR-PROD-TABLE to display entries on the BASE24 AFT Product Menu. RQMEGA is the only module in BASE24 that accesses VIRTUAL-DESCR-PROD-TABLE. Previously, RQMEGA accessed an internal table titled WS-VIRT-DESC-PROD-TABLE for the Product Menu entry text descriptions.

- The section SEG-TABLE contains segments that are particular to products. The term “segment” refers to a segment that physically resides on a segmented data file (i.e., segmented file) in BASE24. Data contained in a segment is specific to a particular product or enhancement. The SEG-TABLE holds up to 256 segment entries.
- The section SEG-DESCR-TABLE contains a text description of each entry in the SEG-TABLE. The Institution Definition File requester utilizes the SEG-DESCR-TABLE to display segment text descriptions on screens 5 and 6 of the BASE24 AFT Institution Definition File (IDF) screens. The IDF File Requester is the only module in BASE24 that accesses the SEG-DESCR-TABLE. Previously, the IDF Requester accessed an internal table titled WS-FIID-DESC-PROD-TABLE for the segment text descriptions.

<u>Segmented File Changes</u>

In BASE24, segmented file records are not fixed length. Segments are added to records depending on the Y/N flag in the SEG-TABLE and the Y/N flags on screens 5 and 6 of the IDF Requester.

Prior to this enhancement, an application checked for the presence of a segment by checking the segment map in the BASE segment header. The BASE segment was always present in a segmented record. The segment map was a 32-position bit map. Each bit turned on in the segment map indicated that a particular product’s segment existed in that data record. The position of a bit in the segment map of the BASE segment header matched the position of the related product in the PI-TABLE.

The segment map could not reasonably be expanded to allow 256 positions in every record. Therefore a new segmentation scheme was developed. The segment map of the BASE segment is not used under the release 6.0 segmentation scheme and is set to zeroes. BASE24 applications do not check the segment map of the BASE segment to see if a particular segment exists in a segmented record. Instead, a segmented record is parsed by the new disassemble utility to find which segments are present. The utility builds a table of addresses for the segments that exist. BASE24 applications check whether a segment exists by interrogating the address table to see if the desired segment has a data address. If no address exists, the related segment does not exist.

New segmentation utilities have been developed to handle the new segmentation scheme. The new segmentation utilities are named the same as their old segmentation utility, except an “^EXT” tag has been appended to their names. The new segmentation utility DISASSEMBLE^SEG^REC^EXT converts segmented data records that are constructed under the old segmentation scheme to the new segmentation scheme using an online conversion when the record is disassembled.

The old segmentation utilities have been upgraded as part of this enhancement to deal with the fact that there are two segmentation schemes and the utilities recognize both. The old segmentation utility DISASSEMBLE^SEG^REC converts segmented data records that are constructed under the new segmentation scheme back to the old segmentation scheme using an online conversion. The online conversion that the old segmented utility DISASSEMBLE^SEG^REC performs will fail if a segment higher than 31 is present on the segmented data record. Any module that reads segmented data records that contain segments higher than 31 must use the new segmentation utilities. If a BASE24 network contains a mixture of processes using the old segmented utilities and the new segmented utilities, there can be many online conversions taking place between the two sets of segmentation utilities. This scenario harms performance to a degree. It is recommended that this mixture be avoided, if possible.

The following files are segmented files in BASE24:

- Cardholder Authorization File (CAF)
- Card Prefix File (CPF)
- Enhanced Interchange Configuration File (ICFE)
- Extract Configuration File (ECF)
- Host Configuration File (HCF)
- Interchange Configuration File (ICF)
- Institution Definition File (IDF)
- Negative Card File (NEG)
- Positive Balance File (PBF)
- Usage Accumulation File (UAF)

The following new segmentation utilities replace the functionality of the segmentation utilities that do not contain the “^EXT” suffix in their name. There are separate routines for handling data in extended memory, and these routines are noted below with the phrase “(Extended Memory Version)”.

- DISASSEMBLE^SEG^REC^EXT
- DISASSEMBLE^SEG^RECX^EXT (Extended Memory Version)
- REASSEMBLE^SEG^REC^EXT
- REASSEMBLE^SEG^RECX^EXT (Extended Memory Version)
- INITIALIZE^SEG^REC^EXT
- INITIALIZE^SEG^RECX^EXT (Extended Memory Version)
- DELETE^SEG^FROM^REC^EXT
- DELETE^SEG^FROM^RECX^EXT (Extended Memory Version)

- REPLACE^SEG^IN^REC^EXT
- REPLACE^SEG^IN^RECX^EXT (Extended Memory Version)
- RETRIEVE^SEG^FROM^REC^EXT
- RETRIEVE^SEG^FROM^RECX^EXT (Extended Memory Version)
- READ^SEGMENTED^FILE^EXT
- READ^SEGMENTED^FILEX^EXT (Extended Memory Version)
- WRITE^SEGMENTED^FILE^EXT
- WRITE^SEGMENTED^FILEX^EXT (Extended Memory Version)

With the new segmentation scheme described above, there were a number of modules that required changes to support the possibility of 256 segments in the BASE24 segmented data files (i.e., CAF, CPF, ECF, HCF, ICF, ICFE, IDF, NEG, PBF, UAF). Most references, outside of RQMEGA and the Security Requester/Server pair, to information derived from the PI-TABLE are not references for a product on the Product Menu. They are actually references to the presence or absence of a segment in a segmented data record. In many cases, the modifications required were to use the new segmentation utilities with a suffix of “^EXT” (e.g., DISASSEMBLE^SEG^REC^EXT). Also, many temporary storage fields also had to be expanded to allow for 256 segments.

The changes to modules to support the possibility of 256 segments are only covered as noted above. There are no product specific impacts, only internal code changes within product modules that enable the modules to deal with old and new segmentation schemes. Once the impacts in this section have been handled the other products will run accordingly.

<u>PATHWAY Changes – USER-CONTEXT</u>

In previous releases, the USER-CONTEXT message structure was used by RQMEGA to call subordinate requesters. It also carried data used by the subordinate requesters. The two fields inside of USER-CONTEXT of interest with this enhancement are USER-PROD-IND and SITE-PROD-IND. Each of the fields contained 32 positions. USER-PROD-IND was used to indicate which screens the operator can access based on how the user’s security was configured. USER-PROD-IND was filled in by the Security Server when the operator logged on to the BASE24 AFT subsystem. Each position within USER-PROD-IND matched the position of the related product in the PI-TABLE. The positions in USER-PROD-IND were set only if the particular operator had access to a product. The field SITE-PROD-IND was used by the IDF Requester to indicate which segment text descriptions appeared on screens 5 and 6 of the IDF screens in the BASE24 AFT subsystem. When the BASE24 user logs on, SITE-PROD-IND was filled based on the contents of the PI-TABLE.

With this enhancement, a new message structure titled USER-CONTEXT-EXT was created. The only difference between USER-CONTEXT-EXT and USER-CONTEXT are the fields PROD-IND and SITE-PROD-IND of USER-CONTEXT-EXT. These fields have been increased from 32 to 256 positions. SITE-PROD-IND is filled based on the contents of the SEG-TABLE. A separate message structure that supports 256 products and segments was added, because not all requesters were changed to support 256 positions and use the message structure USER-CONTEXT-EXT.

Only requesters that communicate with servers that update and add segmented data records were enhanced to use the message structure USER-CONTEXT-EXT. This allows segments greater than 31 to be added in the future. The servers that add and update segmented data records are already prepared to handle these segments. All future requester/server pairs will be developed using the message structure USER-CONTEXT-EXT even though they might not interact with segmented data records. All new release 6.0 requester/server pairs use the message structure USER-CONTEXT-EXT.

BASE24 supports communication with both USER-CONTEXT and USER-CONTEXT-EXT equally. To provide this message structure flexibility, the table PROD-RQ-EXT-FRMT-TBL was added in a new edit file titled MEGATBL. RQMEGA utilizes PROD-RQ-EXT-FRMT-TBL to decide which message structure to send to each requester. All requesters that communicate using USER-CONTEXT-EXT are listed in the PROD-RQ-EXT-FRMT-TBL table. In release 6.0, PROD-RQ-EXT-FRMT-TBL contains the necessary default values.

The edit file MEGATBLC is also provided as a part of release 6.0. The table CSE-RQ-EXT-FRMT-TBL within MEGATBLC provides a method for BASE24 users to add their custom requesters that they have enhanced to use the USER-CONTEXT-EXT message structure.

<u>PATHWAY Changes – MSG-FILE-MAINT-XXXX</u>

The MSG-FILE-MAINT-XXXX message structures are used to pass messages between requesters and servers. These messages are different lengths as indicated by the xxxx in the structure name. Each message structure contains the field SITE-PROD-IND as a part of the message header. SITE-PROD-IND is filled in from the SITE-PROD-IND of USER-CONTEXT in the requester before the message is sent to the server. In previous releases, SITE-PROD-IND consisted of 32 positions and each position within USER-PROD-IND matched the position of the related segment in the PI-TABLE.

New MSG-FILE-MAINT-XXXX-EXT message structures were created as part of this enhancement. The MSG-FILE-MAINT-XXXX-EXT message structures are functionally identical to their MSG-FILE-MAINT-XXXX message structure counterparts. The difference between the two message structures is the field SITE-PROD-IND that is part of the MSG-HEADER in all MSG-FILE-MAINT-XXXX and MSG-FILE-MAINT-XXX-EXT message structures. SITE-PROD-IND consists of 256 positions in the MSG-FILE-MAINT-XXXX-EXT message structures versus 32 positions in the MSG-FILE-MAINT-XXXX message structures. Requesters/server pairs that handle segmented data records were upgraded to use the new MSG-FILE-MAINT-XXXX-EXT message structures. This allows segments greater than 31 to be added in the future. The servers that add and update segmented data records are prepared to handle these segments. All future requester/server pairs will be developed using MSG-FILE-MAINT-XXXX-EXT message structures even though they might not interact with segmented data records. All new Release 6.0 requester/server pairs use the MSG-FILE-MAINT-XXXX-EXT message structures. BASE24 supports communication with both MSG-FILE-MAINT-XXXX and MSG-FILE-MAINT-XXXX-EXT equally.

To support the new message structures USER-CONTEXT-EXT and MSG-FILE-MAINT-XXXX-EXT, a number of new sections were added to BACOUTLS. The new sections are named like the old sections with a suffix of “-EXT”. The only exception is GENERAL-HEADER names; they were shortened to GENERAL-HDR due to the limit on the number of characters allowed.

<u>FIID-SEG-MAP Changes</u>

The FIID-SEG-MAP of the IDF is used to determine which segments are included in the authorization files (i.e., CAF, NEG, PBF, and UAF) for a financial institution. The user configures the FIID-SEG-MAP on screens 5 and 6 of the IDF screens in the BASE24 AFT subsystem. See the paragraph above that discusses the USER-CONTEXT message structure for an overview of how screens 5 and 6 are constructed. Physically, the FIID-SEG-MAP is a bitmap. Each bit represents a segment that exists in the authorization files for the financial institution. Servers that process segmented data records in any of the authorization files read the IDF to retrieve the financial institution’s FIID-SEG-MAP. The server checks the FIID-SEG-MAP to see whether the financial institution supports a specific segment before adding a corresponding segment to the segmented data record being added or updated.

With this enhancement, the FIID-SEG-MAP of the Institution Definition File increased from 32 bits to 256 bits. A field, FIID-SEG-MAP-EXT, was added adjacent to FIID-SEG-MAP. This allows financial institutions the ability to have 256 segments without changing all of the processes that read the current FIID-SEG-MAP. The functional purpose of the FIID-SEG-MAP has remained the same. Screen 6 has been added to the IDF Requester to accommodate the new size of the FIID-SEG-MAP. There is no functionality difference between screen 5 and the new screen 6 of the IDF in the BASE24 AFT subsystem. Screen 6 contains data only if more than 48 segments are turned on in the network. A total of 96 segments in a network can be displayed on the two screens. A conversion of the IDF is required to allow space for the new FIID-SEG-MAP-EXT field.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>Segmented Files</u>

In all segmented data records (i.e., CAF, CPF, ECF, HCF, ICF, ICFE, IDF, NEG, PBF, UAF), the segment map of the BASE Segment header is initialized to zeroes by the new segmentation utility DISASSEMBLE^SEG^REC^EXT. If an application writes the segmented record to disk after calling DISASSEMBLE^SEG^REC^EXT, the segment map is stored as zeroes on disk as well. If an application using the old segmentation utility DISASSEMBLE^SEG^REC dissembles a record with a segment map that is set to zeroes, the utility reconstructs the segment map into a meaningful segment indicator bitmap. If the application writes the segmented record to disk, the segment bitmap will not be all zeroes. Refer to the Segmented File Changes subsection earlier in this enhancement for additional information relating to segmentation schemes.

Institution Definition File

Changes were made to the definition of IDF records. The new field FIID-SEG-MAP-EXT adjacent to the existing field FIID-SEG-MAP provides support for 256 products to correspond to the changes made for the PI-TABLE changes.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The MEGATBL file has been added so RQMEGA can communicate with requesters that use the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT). MEGATBL is an edit file that contains a table titled PROD-RQ-EXT-FRMT-TBL. Each requester acronym listed in PROD-RQ-EXT-FRMT-TBL represents a requester that communicates with RQMEGA through the use of message structure USER-CONTEXT-EXT. RQMEGA uses PROD-RQ-EXT-FRMT-TBL to know which requesters should be sent the message structure USER-CONTEXT-EXT versus the message structure USER-CONTEXT. Refer to the PATHWAY Changes – USER-CONTEXT subsection earlier in this enhancement description for additional information relating to the functionality of USER-CONTEXT, USER-CONTEXT-EXT, and MEGATBL.

The MEGATBLC file is a new edit file with functionality similar to MEGATBL. It lists requesters used in Channel or Customer enhancements. BASE24 users can add requester suffix acronyms for enhanced requesters.

PI-Table Changes

A number of entries in the PI-TABLE have been moved, deleted, and consolidated as part of this enhancement. The following table illustrates the rearrangement of the effected products in the PI-TABLE. CSFC (Customer Service Fraud Control) product was broken apart. CUST SRVC (i.e., Customer Service) is a separate product, and FRAUD CNTL has been removed. A number of enhancements were a part of the old PI-TABLE so they would appear on screen 5 of the IDF and segments could be created. These enhancements have been moved to the Segment Table.

PI-TABLE

Position in Table	5.3 Product	6.0 Product
05	PCS	Undefined
08	KERNEL	KERNEL-IEP
10	Undefined	Undefined
11	BASE PS	Undefined
14	INTEGRATED END POINT	Undefined
15	Undefined	TELEBANKING
16	PBF CREDIT LINE	Undefined
17	PBF SHORT NAME	Undefined
18	SELF SERVICE BANKING BASE	Undefined
19	SELF SERVICE BANKING CHK	Undefined
20	ADDRESS VERIFICATION	Undefined
21	SMART CARD	Undefined
22	CUSTOMER SRVC/FRAUD CNTL	CUST-SRVC
23	PRE AUTH HOLD	Undefined
24	NON-CURRENCY DISPENSE	Undefined
25	Undefined	PRISM-COMPONENTS
26	DCT	RETAIL-CNFG
28	SWITCH FUNCTIONS	CHECK-DISB
29	FS	FREQ-SHOPPER
30	CHKA	CHECK-AUTH
31	RFND	REFUNDS
32	Undefined	Undefined

Note: In this table, the phrase “PRISM Components” is used even though other documentation refers to this product as “ACI Proactive Risk Manager.”

SEG-Table Changes

The Segment Table was added as a part of release 6.0. Only segments defined in one or more products are defined in the table. Only the first 32 segments are listed in the table. If a product is turned on in the Segment Table, it will appear on screen 5 of the IDF.

SEG-TABLE

Position in Table	6.0 Product
01	BASE
02	ATM
03	POS
04	TELLER
05 through 08	Undefined
09	FROM-HOST-MAINTENANCE
10	EMV
11	Undefined
12	MAIL
13	CARD-MANAGEMENT
14	Undefined
15	TELEBANKING
16	PBF-CREDIT-LINE
17	PBF-SHORT-NAME
18	SSB-BASE
19	SSB-CHECK
20	ADDR-VERIF
21	Undefined
22	CUST-SRVC
23	PRE-AUTH-HOLD
24	NON-CURRENCY-DISPENSE
25 through 31	Undefined
32	CAF-ACCOUNT/ INTERCHANGE SPECIFIC DATA

Note: Segment 32 is used for the CAF ACCOUNT Segment in the CAF and for Interchange Specific Data in the ICF and ICFE. This is the same as in previous releases of BASE24.

Security Table (SECTBL) Changes

The SECTBL has been modified. The fields in the Product Segment and Product Menu of table PROD-SCRN-ACCESS-TABLE-DATA are based on the Hexadecimal Number Base Format. Previously, these two fields were based on the Decimal Number Base Format. Also, the Product Segment for all non-segmented files has been set to 0.

SECTBL entries for ADMN, AST, CAPF, CHF, CRED, CRTA, CRUF, CSEC, MRF, NNF, OLT, PRF, and RTBL screens have been modified, so these files will appear on the POS Product Menu.

SECTBL entries for ICF and STF screens have been modified, so these files will appear on the BASE Product Menu.

SECTBL entries for DEST screens have been modified, so this file will appear on the KERNEL/IEP Product Menu.

The format of SECTBLC has been modified to correspond to the changes made to SECTBL.

PATHCONF Changes

Not applicable

CRT Access Security Changes

The Product entry BASE POS has been removed from the BASE24 AFT Product Menu. The screens previously under BASE POS included those for the ADMN, AST, CAPF, CHF, CRED, CRTA, CRUF, CSEC, MRF, NNF, OLT, PRF, and RTBL files. These screens were moved under the POS product entry. The security file and UPFR conversion programs allow operators to retain their previous security settings to these screens despite the transfer of the screens between Product entries. The conversion program for the UPFR file must be run before these screens will appear on the correct product entry. The transfer of screens will be transparent and will be handled jointly by the Security Server and RQMEGA.

The Product entry SWITCH FUNCTIONS has been removed from the BASE24 AFT Product Menu. The screens previously under SWITCH FUNCTIONS included those for the ICF and STF files. These screens were moved under the BASE product entry. The security file and UPFR conversion programs allow operators to retain their previous security settings to these screens despite the transfer of the screens between Product entries. The conversion program for the UPFR file must be run before these screens will appear on the correct product entry. The transfer of screens will be transparent and will be handled jointly by the Security Server and RQMEGA.

The Product entry INTEGRATED END POINT has been removed from the BASE24 AFT Product Menu. The screens previously under INTEGRATED END POINT included those for the DEST file. These screens were moved under the KERNEL/IEP product entry. The security file and UPFR conversion programs allow operators to retain their previous security settings to these screens despite the transfer of the screen between Product entries. The conversion program for the UPFR file must be run before these screens will appear under the correct product entry. The transfer of the screens will be transparent and will be handled jointly by the Security Server and RQMEGA.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

<u>Institution Definition File (IDF)</u>

The IDF conversion program must be run to convert IDF records from the release 5.3 format to the release 6.0 format. A new field titled IDF.IDFBASE.FIID-SEG-MAP-EXT has been added to the IDF file for this enhancement. The addition of the field IDF.IDFBASE.FIID-SEG-MAP-EXT allows individual institutions to support up to 256 segments. The Conversion Program also zero fills field IDF.IDFBASE.BASE-SEG.SEG-MAP. This initialization puts the IDF file under the new segmentation scheme.

The IDF conversion program on the BA60UC04 subvolume can be used to convert the IDF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVIDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#idfcnv.t,name $idfm,nowait/
cnvidfm –force –list –listreplace
```

Users must update the CNVIDFR obey file on the BA60UC04 subvolume with the appropriate information. CNVIDFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Security file (SEC)

The SEC File Conversion Program is needed for two purposes. One purpose pertains to the fact that the Product Indicator field of the SEC Base record has been increased from 32 product indicator entries to 256 product indicator entries. The second function is to clean up old products and segments in the Security file. Some products have been consolidated and others removed. The Conversion Program will remove all SEC screen access records that no longer correspond to products found in the PITABLE.

The SEC conversion program on the BA60UC04 subvolume can be used to convert the SEC. If modifications to the conversion program are necessary, the program can be modified by editing the CNVSECS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#seccnv.t,name $secm,nowait/
cnvsecm –force –list –listreplace
```

Users must update the CNVSECR obey file on the BA60UC04 subvolume with the appropriate information. CNVSECR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

User/Product File Relation file (UPFR)

The UPFR conversion program is needed because some products were consolidated and unsupported files must be removed from the UPFR.

The UPFR conversion program on the BA60UC04 subvolume can be used to convert the UPFR. If modifications to the conversion program are necessary, the program can be modified by editing the CNVUPFRS file.

Note: The UPFR File Conversion must be run after the SEC File Conversion.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#upfrcnv.t,name $upfrm,nowait/
cnvupfrm –force –list –listreplace
```

Users must update the CNVUPFRR obey file on the BA60UC04 subvolume with the appropriate information. CNVUPFRR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Hosts must no longer check the segment map of the BASE segment header on the ICF and IDF extract records to determine which segments are present. Hosts will now need to parse out the IDF, ICF, and ICFE extract records to verify which segments are present. The parsing mechanism a host will need to deploy should be similar to the code in segmentation utility DISASSEMBLE^SEG^REC^EXT. Refer to Segmented File Changes subsection for additional information relating to segmentation schemes.

User/Operational Impacts

For this enhancement, the table named WS-VIRT-DESC-PROD-TABLE of RQMEGA has been moved. It has been placed into the edit file PITABLE and renamed VIRTUAL-DESCR-PROD-TABLE. The virtual description product table contains text descriptions for product menus that will reside on the virtual menu. Refer to the PITABLE File subsection earlier in this enhancement for further information concerning VIRTUAL-DESCR-PROD-TABLE. A number of entries in VIRTUAL-DESCR-PROD-TABLE have been moved, renamed, deleted and consolidated as part of this enhancement.

The table on the following page shows the rearrangement of the affected product menu text descriptions in the VIRTUAL-DESCR-PROD-TABLE.

VIRTUAL-DESCR-PROD-TABLE

Position in Table	5.3 Product	6.0 Product
05	PCS	Undefined
08	KERNEL	KERNEL/IEP
10	Undefined	Undefined
11	BASE PS	Undefined
14	INTEGRATED END POINT	Undefined
15	Undefined	TELEBANKING
16	PBF CREDIT LINE	Undefined
17	PBF SHORT NAME	Undefined
18	SELF SERVICE BANKING BASE	Undefined
19	SELF SERVICE BANKING CHK	Undefined
20	ADDRESS VERIFICATION	Undefined
21	SMART CARD	Undefined
22	CUSTOMER SRVC/FRAUD CNTL	CUSTOMER SERVICE
23	PRE AUTH HOLD	Undefined
24	NON-CURRENCY DISPENSE	Undefined
25	Undefined	PRISM COMPONENTS
26	DCT	RETAIL CONFIGURATION
28	SWITCH FUNCTIONS	CHECK DISBURSEMENTS
29	FS	FREQUENT SHOPPER
30	CHKA	CHECK AUTHORIZATION
31	RFND	REFUNDS
32	Undefined	Undefined

Note: In this table, the phrase “PRISM Components” is used even though other documentation refers to this product as “ACI Proactive Risk Manager.”

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

Transactions Allowed

The Transactions Allowed enhancement changes the mechanism to configure allowed transactions for the BASE24 Authorization processes, the enhanced device handlers, and the enhanced interchange interfaces.

Transaction allowed tables, which were hard coded in several files (e.g., IDF, ICF, PTDF, TDF) throughout the BASE24 Classic products, are now configurable in a set of common files that will be shared by various BASE24 products, including BASE24-atm and BASE24-pos. These new files will be used to define the transactions allowed by transaction acquirers (e.g., ATMs, POS terminals, and interchanges) and card issuers. New configuration options are also supported at the processing code level.

<u>Transaction Acquirer</u>

The configuration for transactions allowed by the transaction acquirers will be stored in the new Acquirer Processing Code File (APCF). The APCF contains one record for each combination of acquirer transaction profile, message category, and ISO processing code that is supported by a BASE24-atm or BASE24-pos acquirer endpoint in the system (i.e., ATMs, POS terminals, and interchanges). The APCF also supports the use of wildcard characters for the acquirer transaction profile and message category, so one APCF record can cover many combinations. Each acquirer transaction profile defines a set of transactions supported for an individual acquiring terminal or group of terminals, or for an individual interchange or group of interchanges. For BASE24-pos, retailer and administrative card transaction profiles are also used to determine whether an administrative card is required to perform a transaction and whether a transaction is allowed for a specific administrative card.

Transaction profiles defined in the APCF are used in the following BASE24 files for an acquirer:

- Institution Definition File (IDF)

  The BASE24-atm and BASE24-pos acquirer transaction profiles define the default set of cardholder transactions supported for the acquiring institution (i.e., the acquiring terminal owner). For BASE24-pos, the IDF also contains a default retailer transaction profile and administrative card transaction profile. The retailer transaction profile defines the default set of transactions for which the retailer is required to use an administrative card. The administrative card transaction profile defines the default set of administrative transactions supported for administrative cards associated with the institution.

- Enhanced Interchange Configuration File (ICFE)

  The BASE24-atm and BASE24-pos acquirer transaction profiles define the set of cardholder transactions supported for inbound transactions from an acquiring interchange.

- BASE24-atm Terminal Data files (ATD)

  The acquirer transaction profile defines the not-on-us BASE24-atm cardholder transactions supported for an acquiring ATM. The acquirer transaction profile at this level overrides the default BASE24-atm acquirer transaction profile defined at the terminal owner level in the IDF.

- BASE24-pos Retailer Definition File (PRDF)

  The acquirer transaction profile defines the BASE24-pos cardholder transaction supported for an acquiring retailer. The acquirer transaction profile at this level overrides the default in the IDF. The retailer transaction profile defines the transactions for which the retailer is required to use an administrative card. The retailer transaction profile at this level overrides the default retailer transaction profile defined at the terminal owner level in the IDF. The retailer transaction profile defines the transactions for which the retailer is required to use an administrative card. The retailer transaction profile at this level overrides the default retailer transaction profile defined at the terminal owner level in the IDF.

- BASE24-pos Terminal Data files (PTD)

  The acquirer transaction profile defines the BASE24-pos cardholder transactions supported for an acquiring POS device. The acquirer transaction profile at this level overrides the BASE24-pos acquirer transaction profile defined at the retailer level in the PRDF and the terminal owner in the IDF.

- Administrative Card File (ADMN)

  The administrative card transaction profile defines the administrative transactions supported at POS devices for an administrative card. The administrative card transaction profile at this level overrides the default administrative card transaction profile defined at the terminal owner level in the IDF.

By manipulating the acquirer, retailer, and administrative card transaction profile values in multiple records, the BASE24 user can group the transactions allowed at different processing levels (e.g., interchange, terminal owner, or terminal) according to their business needs. For example, the user could use the same acquirer transaction profile in the ICFE for all interchanges or set up a different profile for each interchange record in the ICFE to meet the specific processing requirements of each interchange. If the user wants all their ATMs to allow the same transactions, the user could leave the acquirer transaction profile blank in the ATD, allowing it to default to the acquirer transaction profile defined in the IDF.

APCF records define the following for each acquirer, retailer, and administrative card transaction profile, message category (e.g., authorization, financial, administrative, etc.), and ISO processing code (i.e., the transaction code, from-account, and to-account) combination:

- An optional transaction description.
- A code indicating whether the transaction is allowed.
- An optional authorization destination that overrides the authorization process destination defined in the ATD, ICFE, or PTD. This allows for the direct routing of transactions from an endpoint to a destination other than a BASE24-atm Authorization process or a BASE24-pos Device Handler/Router/Authorization process. For example, this optional destination could be used to route bill payments entered at an ATM directly to a BASE24-telebanking Integrated Authorization Server process rather than to a BASE24-atm Authorization process).

Note: The processing codes used in this file are based on the ISO 8583:1993 standard. The internal BASE24 processing codes used on other BASE24 files should not be used here.

Information from the APCF is used by BASE24 processes in the Acquirer Processing Code File Extended Memory Table file (APCFEMT). Any time a change is made to the APCF, the APCFEMT should be rebuilt using the Extended Memory Table Build utility and reallocated (warmbooted) to processes that access the table using the EMT Control Commands screen or text commands entered from a network control facility.

The information in the APCF is verified against the data in the Account Type Table File (ATT), Processing Code Description File (PDF), and Transaction Code File (TCF). The ATT contains the account types for the ISO account types displayed in the ACCOUNT 1 TYPE and ACCOUNT 2 TYPE fields. The optional PDF contains the processing code descriptions for description tags used in the DESCR TAG field. The TCF contains the descriptions for ISO transactions codes displayed in the TRANSACTION CODE field.

The key to APCF records is a combination of the data entered in the ACQUIRER TRANSACTION PROFILE, MESSAGE CATEGORY, TRANSACTION CODE, ACCOUNT 1 TYPE, and ACCOUNT 2 TYPE fields.

Note: The non-enhanced device handlers and non-enhanced interchange interfaces will continue to use the transactions allowed tables in the TDF, PTDF, and ICF.

Card Issuer

The configuration for transactions allowed by the card issuers will be stored in the new Issuer Processing Code File (IPCF). The IPCF contains one record for each combination of issuer transaction profile, message category, and ISO processing code that is supported by a BASE24-atm or BASE24-pos card issuer in the BASE24 system. The IPCF also supports the use of wildcard characters for the issuer transaction profile and message category, so one IPCF record can cover many combinations. Each issuer transaction profile defines a set of cardholder transactions supported for an institution, card prefix, or individual cardholder account. For interchanges, each BASE24-atm or BASE24-pos issuer transaction profile defines a set of transactions allowed to be sent (outbound) to the issuer interchange.

Transaction profiles defined in the IPCF are used in the following BASE24 files for an issuer:

- Institution Definition File (IDF)

  The BASE24-atm and BASE24-pos issuer transaction profiles define the default set of cardholder transactions supported for the card-issuing institution. These profiles define the default set of transactions supported for cardholders associated with the institution.

- Card Prefix File (CPF)

  The BASE24-atm and BASE24-pos issuer transaction profiles define the set of cardholder transactions supported for a card prefix. The issuer transaction profiles at this level override the default issuer transaction profiles defined at the institution level in the IDF.

- Cardholder Authorization File (CAF)

  The BASE24-atm and BASE24-pos issuer transaction profiles define the transactions supported for an individual cardholder account. The issuer transaction profile at this level overrides the issuer transaction profile defined at the card prefix level in the CPF or at the card issuer level in the IDF.

- Enhanced Interchange Configuration File (ICFE)

  The BASE24-atm and BASE24-pos issuer transaction profiles define the set of cardholder transactions supported for outbound transactions to an issuer interchange.

By manipulating the issuer transaction profile values in multiple records, the BASE24 user can group the transactions allowed at different processing levels (i.e., card issuer, card prefix, cardholder account) according to your business needs. For example, the user could use the same issuer transaction profile in the CPF for all card prefixes or set up a different profile for each card prefix record in the CPF to meet the specific processing requirements of each. If the user wants the same transactions allowed for all of their cardholder accounts, the user could leave the issuer transaction profile blank in the CAF and the CPF, allowing it to default to the issuer transaction profile defined in the IDF.

IPCF records define the following for each issuer transaction profile, message category (e.g., authorization, financial, administrative, etc.), and ISO processing code (i.e., the transaction code, from-account, and to-account) combination:

- An optional transaction description.
- Codes indicating whether on-us (i.e., the institution’s cardholders using a terminal owned by the same institution) or not-on-us (i.e., the institution’s cardholders using a terminal not owned by the institution) transactions are allowed.
- A flag indicating whether a completion is required to be sent to the host for the transaction.

Note: The processing codes used in this file are based on the ISO 8583:1993 standard.

Information from the IPCF is used by BASE24 processes in the Issuer Processing Code File Extended Memory Table file (IPCFEMT). Any time a change is made to the IPCF, the IPCFEMT should be rebuilt using the Extended Memory Table Build utility and reallocated (warmbooted) to processes that access the table using the EMT Control Commands screen or text commands entered from a network control facility.

The information in the IPCF is verified against the data in the Account Type Table File (ATT), Processing Code Description File (PDF), and Transaction Code File (TCF). The ATT contains the account types for the ISO account types displayed in the ACCOUNT 1 TYPE and ACCOUNT 2 TYPE fields. The optional PDF contains the processing code descriptions for description tags used in the DESCR TAG field. The TCF contains the descriptions for ISO transactions codes displayed in the TRANSACTION CODE field.

The key to IPCF records is a combination of the data entered in the ISSUER TRANSACTION PROFILE, MESSAGE CATEGORY, TRANSACTION CODE, ACCOUNT 1 TYPE, and ACCOUNT 2 TYPE fields.

Note: The non-enhanced interchange interfaces will continue to use the transactions allowed tables in the ICF.

For more information about this enhancement, Refer to section 5 (ATM), section 6 (POS), and section 12 (BASE24 Interchange Interfaces).

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>APCF</u>

The fully qualified file name of the Acquirer Processing Code File. The APCF contains one record for each combination of acquirer transaction profile, message category, and ISO processing code that is supported by a BASE24-atm or BASE24-pos acquirer endpoint in the system. This is a required assign for the APCF Server object, APCFEMT Build utility object, and the BASE24-pos ACI Standard POS Device Handler Device Handler module (for downline loads).

<u>APCFEMT</u>

The fully qualified file name of the Acquirer Processing Code File Extended Memory Table file. This is a required assign for the APCFEMT Build utility object, enhanced BASE24-atm Device Handler processes, the enhanced BASE24-pos Device Handler modules, BASE24-pos Router/Authorization modules (for admin cards), and the enhanced Interchange Interface processes.

<u>ICFE</u>

The fully qualified file name of the Enhanced Interchange Configuration File. This is a required assign for the enhanced interchange interface processes.

<u>IPCF</u>

The fully qualified file name of the Issuer Processing Code File. The IPCF contains one record for each combination of issuer transaction profile, message category, and ISO processing code that is supported by a BASE24-atm or BASE24-pos card issuer in the BASE24 system. This is a required assign for the IPCF Server object and the IPCFEMT Build utility.

IPCFEMT

The fully qualified file name of the Issuer Processing Code File Extended Memory Table file. This is a required assign for the IPCFEMT Build utility object, the BASE24-atm Authorization module, BASE24-pos Router/Authorization module, and the enhanced Interchange Interface processes.

PDF

The fully qualified file name of the Processing Code Description File. The PDF contains one record for each processing code tag supported in the logical network. This is an optional file. This file can be used by issuers and acquirers to define long and short processing code descriptions based upon a description tag. This is a required assign for the PDF Server object.

TCF

The fully qualified file name of the Transaction Code File. The TCF contains one record for each transaction code supported in the logical network. The file is used to map ISO transaction codes to alphanumeric descriptions. This is a required assign for the TCF Server object.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

APCFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the APCFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

APCFEMT-WARM*identifier*

A param containing the symbolic name of a process to be warmbooted for the APCFEMT. Processes that can be warmbooted for the APCFEMT are the enhanced BASE24-atm Device Handler processes, enhanced BASE24-pos Device Handler modules, BASE24-pos Router/Authorization modules (for admin cards), enhanced Interchange Interface processes.

This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all APCFEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each APCFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another APCFEMT-WARM*identifier* param. For example, APCFEMT-WARM01 could contain the P1A^NCR^5XXX^01 process and APCFEMT-WARM02 could contain the P1A^NCR^5XXX^02 process.

IPCFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the IPCFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

IPCFEMT-WARM*identifier*

A param containing the symbolic name of a process to be warmbooted for the IPCFEMT. Processes that can be warmbooted for the IPCFEMT are the BASE24-atm Authorization processes, BASE24-pos Router/Authorization processes, and enhanced interchange interface processes. This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all IPCFEMT-WARM*identifer* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each IPCFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another IPCFEMT-WARM*identifier* param. For example, IPCFEMT-WARM01 could contain the P1A^ATAUTH^01 process and IPCFEMT-WARM02 could contain the P1A^ATAUTH^02 process.

Database Configuration

Token file (TKN)

The following new BASE24 Base tokens have been added for this enhancement:

- The Transaction Profile Token (Token ID “B8”) contains fields from the APCF and IPCF. This data is used for transactions allowed checking and determining if a reversal should be processed. The APCF data is added to the token by transaction acquirers, and the IPCF data is added to the token by transaction issuers.
- The Transaction Description Token (Token ID “B9”) contains the processing code descriptions from the APCF and IPCF files. This data is not used for transaction processing and is informational only. The APCF data is added to the token by transaction acquirers and the IPCF data is added to the token by transaction issuers.
- The Acquirer Routing Token (Token ID “BA”) contains the Authorization Destination and the Log Authorization Destination Response values from the APCF. The APCF data is added to the token by transaction acquirers.

If the new tokens for this enhancement are not to be logged to the PTLF, TLF or ILF, the appropriate token records must be configured. It is recommended that the Transaction Profile Token be configured to be logged to the ILF, since this reduces the Interchange Interface’s processing time for inbound reversal transactions.

If the new tokens for this enhancement are to be extracted with the PTLF, TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

This section describes the files that have added to support this enhancement.

Acquirer Processing Code File (APCF)

The APCF was created for storing processing code information based on the acquirer of the transaction. Through configuration of APCF records, acquirers define the transactions they allow. Acquirers can share common processing configurations through the specification of a common acquirer transaction profile.

ACI provides BASE24 users with a set of default APCF records at installation. The records are located in the APCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***. This file can be duplicated in its entirety to the appropriate data files subvolume or Tandem's FUP can be used to copy the default records to the user's APCF file. BASE24 users should then access the APCF records via standard BASE24 file maintenance and update the APCF records to reflect their own environment.

<u>Issuer Processing Code File (IPCF)</u>

The IPCF was created for storing processing code information based on the issuer of the transaction. Through configuration of IPCF records, institutions define the transactions allowed for on-us and not-on-us transactions. Issuers can share common processing configurations through specification of a common issuer transaction profile.

ACI provides BASE24 users with a set of default IPCF records at installation. The records are located in the IPCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***. This file can be duplicated in its entirety to the appropriate data files subvolume or Tandem's FUP can be used to copy the default records to the user's IPCF file. BASE24 users should then access the IPCF records via standard BASE24 file maintenance and update the IPCF records to reflect their own environment.

Note: Previously, there was one COMPLETION REQUIRED flag in the ATM segment of the IDF that pertained to all BASE24-atm transactions. There was also a COMPLETION REQUIRED flag for each BASE24-pos transaction code in the POS segment of the IDF. These flags in the IDF have been removed and replaced with a COMPLETION REQUIRED TO HOST flag in the IPCF. The flag in the IPCF allows institutions to define completions required at the processing code level.

<u>Processing Code Description File (PDF)</u>

The PDF was created for storing the description of a processing code. Refer to the LCONF Assign Changes subsection of this enhancement for further information about this file.

ACI provides BASE24 users with a set of default PDF records at installation. The records are located in the PDF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***. This file can be duplicated in its entirety to the appropriate data files subvolume or Tandem's FUP can be used to copy the default records to the PDF file. BASE24 users should then access the PDF records via standard BASE24 file maintenance and update the PDF records to reflect their own environment.

Transaction Code File (TCF)

The TCF was created for storing transaction code information. Refer to the LCONF Assign Changes subsection of this enhancement for further information about this file.

ACI provides BASE24 users with a set of default TCF records at installation. The records are located in the TCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***. This file can be duplicated in its entirety to the appropriate data files subvolume or Tandem's FUP can be used to copy the default records to the TCF file. BASE24 users should then access the TCF records via standard BASE24 file maintenance and update the TCF records to reflect their own environment.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The APCF requester name was added to MEGATBL file, since RQMEGA communicates with the APCF requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

The IPCF requester name was added to MEGATBL file, since RQMEGA communicates with the IPCF requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

The PDF requester name was added to MEGATBL file, since RQMEGA communicates with the PDF requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

The TCF requester name was added to MEGATBL file, since RQMEGA communicates with the TCF requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the APCF requester/server.

Entries were added to the SECTBL to support the screens for the IPCF requester/server.

An entry was added to the SECTBL to support the screen for the PDF requester/server.

An entry was added to the SECTBL to support the screen for the TCF requester/server.

PATHCONF Changes

The APCF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

The IPCF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

The PDF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

The TCF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

CRT Access Security Changes

Access to the new APCF requester screens must be added through the SEC requester to the appropriate users.

Access to the new IPCF requester screens must be added through the SEC requester to the appropriate users.

Access to the new PDF requester screens must be added through the SEC requester to the appropriate users.

Access to the new TCF requester screens must be added through the SEC requester to the appropriate users.

Routing Changes

If the AUTHORIZATION DESTINATION field in the APCF is not spaces, this symbolic name of an authorization destination overrides the authorization process destination defined in the ATD, ICFE, or PTD. This allows for the direct routing of transactions from an endpoint to an application process, other than a BASE24-atm Authorization process or a BASE24-pos Device Handler/Router/Authorization process, running in the XPNET system.

Note: When the AUTHORIZATION DESTINATION field in the APCF is not spaces, the LOG AUTH DEST RESPONSE field in the APCF must be set to a value of Y in order for the transaction to be written to the TLF/PTLF.

If the AUTHORIZATION DESTINATION field in the APCF is spaces, the symbolic name in the authorization process destination defined in the ATD, ICFE, or PTD is used, and normal transaction processing and routing occurs.

Network Configuration

Not applicable

File Conversions

<u>Account Type Table file (ATT)</u>

The ATT conversion program must be run to convert ATT records from the release 1.3 format to the release 6.0 format. Users must create the ATT file using the BADDLFUP file.

The ATT conversion program on the BA60UC04 subvolume can be used to convert the ATT. If modifications to the conversion program are necessary, the program can be modified by editing the CNVATTS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#attcnv.t,name $attm,nowait/
cnvattm –force –list –listreplace
```

Users must update the CNVATTR obey file on the BA60UC04 subvolume with the appropriate information. CNVATTR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Cardholders Authorization File (CAF)

The CAF conversion program must be run to convert CAF records from the release 5.3 format to the release 6.0 format. Users must create the CAF file using the BADDLFUP file.

The CAF conversion program on the BA60UC04 subvolume can be used to convert the CAF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCAFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cafcnv.t,name $cafm,nowait/
cnvcafm –force –list –listreplace
```

Users must update the CNVCAFR obey file on the BA60UC04 subvolume with the appropriate information. CNVCAFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Note: Even though a CAF conversion program is provided in release 6.0, it is recommended that BASE24 users convert the CAF using the Refresh process.

Card Prefix File (CPF)

The CPF conversion program must be run to convert CPF records from the release 5.3 format to the release 6.0 format. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/
cnvcpfm –force –list –listreplace
```

Users must update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. CNVCPFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Extract Configuration File (ECF)

The ECF conversion program must be run to convert ECF records from the release 5.3 format to the release 6.0 format (i.e., the latest version of release 6.0). Users must create the ECF file using the BADDLFUP file.

The ECF conversion program on the BA60UC04 subvolume can be used to convert the ECF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVECFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ecfcnv.t,name $ecfm,nowait/
cnvecfm –force –list –listreplace
```

Users must update the CNVECFR obey file on the BA60UC04 subvolume with the appropriate information. CNVECFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Institution Definition File (IDF)

The IDF conversion program must be run to convert IDF records from the release 5.3 format to the release 6.0 format. Users must create the IDF file using the BADDLFUP file.

The IDF conversion program on the BA60UC04 subvolume can be used to convert the IDF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVIDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#idfcnv.t,name $idfm,nowait/
cnvidfm –force –list –listreplace
```

Users must update the CNVIDFR obey file on the BA60UC04 subvolume with the appropriate information. CNVIDFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, From Host Maintenance, ISO Host Interface, Refresh, and Super Extract processes.

<u>BIC ISO Interchange Interface Process</u>

If the new tokens for this enhancement (i.e., Transaction Profile Token, Transaction Description Token, Acquirer Routing Token) are included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

<u>From Host Maintenance</u>

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF FHM ISO message format to accept and process the new information in the ATM or POS segment of the CAF file. Users using the 5.*x* CAF FHM ISO message format cannot specify values for the new information, so FHM defaults the fields to spaces.

<u>ISO Host Interface Process</u>

If the new tokens for this enhancement (i.e., Transaction Profile Token, Transaction Description Token, Acquirer Routing Token) are included in the ISO message to the host, this change to the message needs to be handled by the host programs.

<u>Refresh</u>

Host code changes are required for all BASE users that utilize the release 6.0 CAF Refresh file message format to initialize and process the new information in the ATM or POS segment of the CAF file. Users utilizing the release 5.*x* CAF Refresh file message format cannot specify values for the new information, so the Refresh process defaults the fields to spaces.

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process information in the APCF, CAF, CPF, IDF, IPCF, PDF, and TCF files.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF will require additional disk space as the new tokens for this enhancement (i.e., Profile Token, Transaction Description Token, and Acquirer Routing Token) can be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

The APCFEMT impacts disk requirements. The total impact varies for each BASE24 user depending on the number of APCF records defined in the BASE24 system.

The IPCFEMT impacts disk requirements. The total impact varies for each BASE24 user depending on the number of IPCF records defined in the BASE24 system.

Operational Impacts

BASE24 operators must be aware that the new files will appear on the BASE24 BASE Product Menu. The new BASE24 Base files for this enhancement are the APCF, IPCF, PDF, and TCF.

If the AUTHORIZATION DESTINATION field in the APCF is not spaces and the LOG AUTH DEST RESPONSE field in the APCF is Y (Yes), the AUTHORIZATION DESTINATION overrides the authorization process destination defined in the ATD, ICFE, or PTD, and therefore the BASE24-atm Authorization and BASE24-pos Authorization/Router processes will simply write the transaction to the TLF/PTLF.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF, PTLF, and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

- ISO 8583:1993 standard, Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions

Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion and implementation requirements. These enhancements are the following:

- ACI Proactive Risk Manager Components
- CAF Updates to the ACI Smart Chip Manager
- EMV Support
- Format 2 File Support
- Multiple Currency Support
- Transaction Security Services
- Triple/Single DES Translate Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 changes and conversion/implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 changes and conversion/implementation requirements for future reference.

ACI Worldwide Inc.

ACI Proactive Risk Manager Components

ACI Proactive Risk Manager is a fraud detection system that integrates with the BASE24-atm and BASE24-pos products. This enhancement provides the “hooks” that are required by the ACI Proactive Risk Manager product editions to support a “near real-time” interface between BASE24 and ACI Proactive Risk Manager. In this environment, the BASE24-atm Standard Internal Message (STM) or BASE24-pos Standard Internal Message (PSTM) is routed to the ACI Proactive Risk Manager Input Process (PIP). The Master File is updated using BASE24-from host maintenance and contains the cardholder specific data that is used by the PIP.

To support the ACI Proactive Risk Manager interface, changes have been made to the BASE24 Base, BASE24-atm, BASE24-pos, BASE24-from host maintenance, Extract and Refresh modules. The implementation of these product “hooks” are required for all BASE24 customers, whether or not they have ACI Proactive Risk Manager implemented in their environment.

Refer to the ACI Proactive Risk Manager Component Implementation Guide on the BASE24 system (i.e., $<volume>.PI60SRC.PRMUPDT) for information on implementing this enhancement into a BASE24 system.

ACI Worldwide Inc.

CAF Updates to the ACI Smart Chip Manager

This section describes the release 6.0 changes and conversion and implementation requirements associated with the CAF Updates to the ACI Smart Chip Manager.

Section 2 contains a functional overview of the CAF Updates to the ACI Smart Chip Manager enhancement.

The CAF Updates to the ACI Smart Chip Manager enhancement provides support for sending applicable CAF add, update, or delete messages from BASE24-from host maintenance to a host interface process and on to the ACI Smart Chip Manager. It also supports sending applicable CAF add, update or delete messages from the CAF requester/server to the ISO Host Interface process and on to the ACI Smart Chip Manager.

The ISO Host Interface logs the CAF add, update, and delete messages to the SAF when the ACI Smart Chip Manager is not available or if there is no online connection to the ACI Smart Chip Manager. The SAF file can be extracted and transferred to the ACI Smart Chip Manager.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Parameter Changes

This section describes the param records that must be modified in the LCONF to support this release. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>MONAD-NOTIFY-PROCESS</u>

This optional param contains the ACI Smart Chip Manager DPC number and ISO Host Interface process name. The format is “XXXXYYYYYYYYYYYYYYYY” where “XXXX” is the DPC number and “YYYYYYYYYYYYYYYY” is the ISO Host Interface process name.

The DPC number along with the process name are used to route to the correct ISO Host Interface process that is used to communicate with the ACI Smart Chip Manager application and to read the HCF.

Database Configuration

EMF

EMF records can be added for outbound BASE24-from host maintenance 0300 messages. EMF records can be added for inbound 0310 BASE24-from host maintenance messages.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

The new notification server must be defined in the PATHCONF file as SERVER-NTFY. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

A NCP user and password must be configured to support the deliver command from the Notify Server process. This user ID and password must be entered in the BA60AFT.SVNTFTBL file and will be sourced into the SVNTFYTS at MAKE time. If the SVNTFTBL is modified, the SVNTFYTS must be re-compiled.

The BA60AFT.SVNTFTBL file must be protected, since it contains a valid user ID and password (if configured). This can be accomplished using standard file protection or through a Compaq product such as Safeguard.

Routing Changes

Not applicable

Network Configuration

<u>Host Interface</u>

BASE24 users implementing this enhancement must configure an ISO Host Interface to communicate with the ACI Smart Chip Manager application. If no lines or stations are connected to the ACI Smart Chip Manager application, the ISO Host Interface logs the CAF add, update, and delete messages to the SAF. The ISO Host Interface process must include the BASE24-from host maintenance ISO Host Interface module.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The ISO Host Interface logs the CAF add, update, and delete messages to the SAF when the ACI Smart Chip Manager application is not available. This file must be set to a size large enough to accommodate the messages and must be extracted if the ACI Smart Chip Manager application is not available online.

User Impacts

EMF records can be added for outbound 0300 BASE24-from host maintenance messages. EMF records can be added for inbound 0310 BASE24-from host maintenance messages.

BASE24 users using the CAF Updates to the ACI Smart Chip Manager application enhancement need to enter the appropriate information in the LCONF.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

BASE24 users using the CAF Updates to the ACI Smart Chip Manager enhancement need to extract the SAF if the ACI Smart Chip Manager application is not available online.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

EMV Support

Europay, MasterCard, and Visa (EMV) have developed Integrated Circuit Card (ICC) credit and debit specifications to facilitate interoperability across competing payment systems. When these specifications are implemented, customers with EMV debit and credit cards can use any terminal that accepts EMV cards. The enhanced authentication processing supported by EMV cards is expected to reduce fraud, and also reduce telecommunications costs by allowing increased offline authorization.

EMV cards are produced with both a magnetic stripe and a chip. The chip includes an electronic version of the magnetic stripe data, as well as other data used for authentication and risk management. When used in an EMV-capable terminal, the card and terminal perform offline risk management to determine whether the transaction should be approved or declined offline, or sent online (via the acquirer) for authorization. If sent online, an Authorization Request Cryptogram (ARQC) is calculated using card and terminal data. The ARQC and component cryptogram data are sent online to the issuer with the authorization request.

The issuer checks the ARQC, using the cryptogram data and a secret DES key (kept in the KEY ICC File in BASE24), and calculates an Authorization Response Cryptogram (ARPC) using response data and the same key. The response cryptogram is returned to the card by the acquirer, which can then confirm that the issuer is genuine.

On successful completion of an online or offline transaction, another cryptogram, the Transaction Certificate (TC), is calculated by the card for use in verifying the offline clearing message.

It is possible for the issuer to provide limited updates to the chip through the use of scripts. The script is created by the issuer and sent to the card in the response message, along with the ARPC. Only two scripts are supported in BASE24 at this time. One allows the entire ICC on the card to be disabled. The other allows an update to the number of transactions that can be performed before a transaction must be authorized online.

The BASE24-atm and BASE24-pos products have been enhanced to support processing for both EMV and magnetic stripe transactions from a single terminal. The BASE24-atm EMV product is an add-on to the BASE24-atm product. The BASE24-pos EMV product is an add-on to the BASE24-pos product.

BASE24-atm or BASE24-pos Authorization can be configured to perform Card Authentication Method (CAM) authentication using an Atalla or Thales e-Security (Racal) Hardware Security Module (HSM). CAM checking is configurable at a prefix, rather than institution level, and is only performed on known prefixes.

When formatting a script command, BASE24 calculates a Message Authentication Code (MAC) based on the contents of the script. The MAC is appended to the script data.

When transactions enter a BASE24 system, they are converted to a standard internal message format recognized by all BASE24 processes for that product. Each BASE24 process that receives an internal message has the ability to receive a variable number of tokens with the message.

Five new tokens have been added to support EMV processing within BASE24:

- EMV Request Token (Token ID “B2”)—The EMV Request Data token contains the thirteen request data elements required for ARQC generation as defined by EMV. The Device Handler process or the Interchange Interface process creates this token and adds it to the P/STM before sending it to the Authorization process.
- EMV Discretionary Data Token (Token ID “B3”)—The EMV Discretionary Request Data token consists of EMV-related data that is not required for authorization. However, each data element is supported by more than one EMV-compliant interface, and therefore can be mapped between interfaces by BASE24.
- EMV Status Token (Token ID “B4”)—The EMV Status token holds data identifying the status of a transaction. Device Handler and Interchange Interface processes create this token and add it to the P/STM before sending it to the Authorization process. The token is added by the acquiring endpoint when the transaction originates from an EMV-capable terminal, regardless of whether or not the data relates to an EMV transaction.
- EMV Response Data Token (Token ID “B5”)—The EMV Response Data token contains the response cryptogram, data required to generate the response cryptogram, and flags used to identify the scripts to be returned to the acquirer. If the verification is performed on BASE24, the BASE24-atm or BASE24-pos Authorization process creates this token. If the transaction is routed to a switch for verification, an Interchange Interface creates this token.
- EMV Script Data Token (Token ID “B6”)—The EMV Script Data token holds EMV script data. The issuer process creates this token. In the context of EMV transactions, the issuer process can be an Interchange Interface process if the issuer is external to BASE24, or the Authorization Process if BASE24 is configured for offline or online/offline authorization. The token is added to the

internal message before returning the message to the acquiring process. This token is only present if an EMV transaction response contains script data.

The use of tokens allows support for EMV transactions in BASE24 ISO Host Interfaces, BASE24 ISO Interface Interchange (BIC ISO) and Extract processes without modification. Only a rebind of these modules to incorporate the new token conversion utilities is required.

The following BASE24-pos and BASE24-atm products have been enhanced to support EMV as an enhancement:

- BASE24-atm Authorization
- BASE24-atm NCR 5XXX Device Handler
- BASE24-atm Router/Authorization
- ACI Standard POS Device Handler
- APACS 30/40 POS Device Handler
- Banknet Interchange Interface (BASE24-pos)
- EPS-Net Interchange Interface (BASE24-atm and BASE24-pos)
- LINK ISO Interchange Interface (BASE24-atm)
- MasterCard Debit Switch Interchange Interface (BASE24-atm)
- Maestro Interchange Interface (BASE24-pos)
- VisaNet ISO Interchange Interface (BASE24-atm and BASE24-pos)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: EMV support for the NCR 5XXX ATM is available with BASE24-atm Release 6.0 version 2. EMV support for the Hypercom POS device is available with BASE24-pos Release 6.0 version 2.

<u>BASE24-atm Authorization and BASE24-pos Router/Authorization</u>

BASE24-atm Authorization and BASE24-pos Router/Authorization, referred to collectively as ‘Authorization’, have been modified to support acquirer or issuer functions for EMV. The EMV processing can be summarized as:

- Force Online Processing (POS only)
- Card and Issuer Authentication
- Application Transaction Counter (ATC) Limit Checking
- Status Field Verification
- Issuer Script Handling.

Most of the Authorization logic referred to above is performed within the utility source file, CAMUTILS. This is used to support the Card Authentication Method (CAM) and other EMV-specific functionality (e.g. CAM data verification and service code checking). The CAMUTILS file is shared by and bound into both POS and ATM Authorization modules. Standard Authorization code has been modified to support dummy entry-points for the CAM/EMV procedures. CAMUTILS performs CAM-specific security device input/output.

EMV authorization or issuer rules have been established, and where possible, are configurable at Prefix level in the Card Prefix File (CPF), at Institution level in the Institution Definition File (IDF) or on the Cardholder Authorization File (CAF). The Authorization logic has been modified to check for the presence of the EMV Request Token (B2). If present, and if configured on the CPF, the ARQC is checked and an ARPC generated.

<u>Transaction Acquirer</u>

When processing as an acquirer, the BASE24 EMV product operates in pass-through mode, routing the CAM data in the EMV Request Token to the issuer. The CAM data is logged to the TLF/PTLF in the appropriate tokens (i.e., EMV Request, EMV Status, EMV Discretionary Data, EMV Response, and EMV Script Data).

The EMV Request Token, EMV Status Token, and EMV Discretionary Data Token are added by Device Handler processes or the Interchange Interface processes, as appropriate. If the terminal is EMV capable, the EMV Status Token is added. If EMV data from the ICC is present in the incoming message, the EMV Request Token and EMV Discretionary Data Token are added. BASE24 optionally performs an EMV pre-screen check for all EMV requests using a configuration option on the CPF.

EMV transactions can follow the same routing logic as non-EMV transactions. However, BASE24 users utilizing an issuer host, issuing EMV cards in the same prefix ranges as non-EMV, are able to separately route the EMV transactions to a specific host interface or DPC by configuring the EMV Prefix Routing in the CPF. For BASE24-pos transactions using AST routing, it is possible to configure separate floor limits to determine whether a transaction should be routed online to a host or issuer system using the EMV Issuer param on the CPF.

In the event of host or issuer being unavailable or in the absence of a full cardholder database, Status Field Verification can be used by BASE24 to authorize, refer, or decline an EMV transaction. The Status Checking Action Index in the CPF tells which set of rules in the CVRTBL and TVRTBL will be used.

<u>Card Issuer</u>

For BASE24 processing as an issuer, the Authorization process (optionally) performs CAM authentication using an Atalla or Thales e-Security (Racal) Hardware Security Module (HSM).

CAM checking is configurable at a prefix, rather than institution level, and is only performed on known prefixes, i.e. those configured in the CPF. If a card has a CAF record, ATC checking can be enabled using the CPF.

All EMV Authorization processing is initiated by the presence of the EMV Status Token (B4) and the appropriate configuration data on the CPF, CAF and KEYI File (KEYI). Support for a pre-validated CAM (e.g., by Visa on the issuer's behalf) is also provided.

Script Data

Issuer script commands allow the issuer to alter the data or functionality of EMV cards already in circulation. If a script command is returned from the issuer in a response to an online authorization request, the device forwards the script to the card. The card uses the script data to re-configure itself.

BASE24 supports the following script commands:

- CARD BLOCK (an irreversible disabling of the ICC)
- PUT DATA (updates the card data)

Script data passed between the issuer and the card is secured with a MAC generated by an Atalla or Thales e-Security (Racal) HSM. Issuer Script Data is supported either in pass-through mode or as BASE24 originated response scripts.

For pass-through mode, BASE24 passes any script data received from an external authorizer to the acquirer without validation or modification. If the EMV Script Data Token (B6) is received from an issuer interface (host or interchange) in a response message, the token is passed to the acquirer end-point (interface or device handler).

BASE24 originates CARD BLOCK and PUT DATA script commands, based on an indicator in the CAF. Authorization supports the generation of only one script command in a response message. The Issuer script commands are stored by BASE24 as LCONF params. Refer to the LCONF Params subsection in this enhancement for more information.

Note: A PUT DATA script command is originated by BASE24 only if the card is configured for a positive authorization method.

CUSTMACS Changes

Several new flags have been added for the EMV Enhancement.

Set the following flag to true if any other EMV flags are set to true.

- emv_on

Set the following flags to true to add the EMV bindable pieces to the appropriate authorization modules:

- atm_emv_on
- pos_emv_on

Set each of the following flags to true if the module exists and support EMV transactions. This adds the bindable EMV extensions to the module.

- atm_dh_n50_emv_on
- pos_dh_apacs_emv_on
- pos_dh_spdh_emv_on
- bnet_emv_on
- euro_emv_on
- link_emv_on
- mds_maestro_emv_on
- mds_iso_emv_on
- swan_emv_on
- visa_emv_on

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>KEYI</u>

The fully qualified file name of the KEY ICC File (KEYI). The KEYI file contains the keys needed to validate and create the cryptograms used in EMV processing.

If this assign is not present in the LCONF, the KEYI file cannot be accessed by BASE24-atm Authorization and BASE24-pos Router/Authorization processes. This assign is required if the EMV extensions are present and initialization fails if the assign is missing.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

EMV-CARD-BLOCK-SCRIPT

Specifies the hexadecimal representation of binary data that is used to disable a chip card. It includes both the Script header and the Script command. This param is required if the EMV extensions are present and initialization fails if the param is missing.

EMV-PUT-DATA-SCRIPT

Specifies the hexadecimal representation of binary data that is used to reset the Lower Consecutive Offline Limit on a chip card. It includes both the Script header and the Script command. This param is required if the EMV extensions are present and initialization fails if the param is missing.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

Card Prefix File (CPF)

An EMV segment has been added to the CPF. For EMV authorization processing, screen 11 holds the CAM authentication control data for the card prefix. Set the fields on this screen according to the information in the ***BASE24-atm EMV Support Manual*** or ***BASE24-pos EMV Support Manual***.

Cardholder Authorization File (CAF)

An EMV segment has been added to the CAF. Screen 13 is specific to EMV authorization processing, and holds the CAM authentication control data for the ICC. This segment also contains the flags and values necessary to create an EMV Script Token to be sent to the card. Set the fields on this screen according to the information in the ***BASE24-atm EMV Support Manual*** or ***BASE24-pos EMV Support Manual***.

Institution Definition File (IDF)

Screen 5 of the IDF is used to set the segments to be created in authorization files. An EMV segment has been added to the CAF. In order to have this segment be a part of the CAF records, a Y must be placed next to EMV on screen 5. Refer to the SEG-Table subsection of this enhancement for further information needed to be sure EMV appears on IDF Screen 5/6.

Token file (TKN)

The following new BASE24 Base tokens have been added for this enhancement:

- The EMV Request Token (Token ID “B2”) contains the thirteen request data elements required for ARQC generation as defined by EMV. The Device Handler process or the Interchange Interface process creates this token and adds it to the STM/PSTM before sending it to the Authorization process.
- The EMV Discretionary Data Token (Token ID “B3”) consists of EMV-related data that is not required for authorization. However, each data element is supported by more than one EMV-compliant interface, and therefore can be mapped between interfaces by BASE24.
- The EMV Status Token (Token ID “B4”) holds data identifying the status of a transaction. Device Handler and Interchange Interface processes create this token and add it to the P/STM before sending it to the Authorization process. The token is added by the acquiring endpoint when the transaction originates from an EMV-capable terminal, regardless of whether or not the data relates to an EMV transaction.
- The EMV Response Data Token (Token ID “B5”) contains the response cryptogram, data required to generate the response cryptogram, and flags used to identify the scripts to be returned to the acquirer. If the verification is performed on BASE24, the BASE24-atm or BASE24-pos Authorization process creates this token. If the transaction is routed to a switch for verification, an Interchange Interface creates this token.
- The EMV Script Data Token (Token ID “B6”) holds EMV script data. The issuer process creates this token. In the context of EMV transactions, the issuer process can be an Interchange Interface process if the issuer is external to BASE24, or the Authorization Process if BASE24 is configured for offline or online/offline authorization. The token is added to the internal message before returning the message to the acquiring process. This token is only present if an EMV transaction response contains script data.

If the new tokens for this enhancement are not to be logged to the TLF, PTLF, or ILF, the appropriate token records must be configured.

If the new tokens for this enhancement are to be extracted with the TLF, PTLF, or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

KEY ICC File (KEYI)

The KEYI file contains the keys needed to validate and create the cryptograms used in EMV processing. The keys must be loaded using the KEYI requester.

Template Changes

New EMV related response codes have been added to BASE24-atm and BASE24-pos response code tables in the COBNAMES file.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The KEYI requester name was added to MEGATBL file, since RQMEGA communicates with the KEYI requester using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

The EMV segment must be set to Y in the SEG-TBL section of the PITABLE file. This causes EMV to appear on IDF screen 5. If EMV is set to Y in the IDF, it causes the EMV segments to be created in the CAF file.

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new screen for the CAF requester/server.

Entries were added to the SECTBL to support the new screen for the CPF requester/server.

Entries were added to the SECTBL to support the screen for the KEYI requester/server.

PATHCONF Changes

The new KEYI server must be defined in the PATHCONF file as SERVER-KEYI. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new CAF requester screen must be added through the SEC requester for the appropriate users.

Access to the new CPF requester screen must be added through the SEC requester for the appropriate users.

Access to the new KEYI requester screen must be added through the SEC requester for the appropriate users.

Routing Changes

The EMV transactions within a prefix can be routed differently than other cards in the prefix. To set up this function, the EMV prefix routing group number on screen 11 of the CPF can be used to define a routing group to be used for EMV transactions. Any routing group number specified in the CPF should have a corresponding group number entry in the IDF Routing Tables.

Network Configuration

In order to implement this enhancement as an issuer, BASE24 must have access to a hardware security module. The enhancement will support the use of an Atalla or Thales e-Security (Racal) security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

File Conversions

Cardholders Authorization File (CAF)

The new EMV segment can be added by updating the records using the BASE24 AFT subsystem. This should be done after running the CAF Conversion program provided for other release 6.0 enhancements. However, BASE24 users should add the new EMV segment using a full CAF Refresh.

Card Prefix File (CPF)

The new EMV segment is added by updating the records using BASE24 AFT subsystem. This should be done after running the CPF Conversion program provided for other release 6.0 enhancements.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the From Host Maintenance, Super Extract, ISO Host Interface Process, BIC ISO Interchange Interfacet, and Refresh processes.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF FHM ISO message format to accept and process the new EMV segment information in the CAF file. Users utilizing the Release 5.*x* CAF FHM ISO message format will not be able to perform file maintenance on this new segment.

Super Extract Process

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the KEYI files and changes to the CAF and CPF records.

ISO Host Interface Process

If the new tokens for this enhancement (i.e., EMV Status Token, EMV Request Token, EMV Discretionary Data Token, EMV Response Token and/or EMV Script Token) are included in the ISO message to the host, this change to the message needs to be handled by the host programs.

BIC ISO Interchange Interface Process

If the new tokens for this enhancement (i.e., EMV Status Token, EMV Request Token, EMV Discretionary Data Token, EMV Response Token and/or EMV Script Token) are included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file message format to accept and process the new EMV segment information in the CAF file. Users utilizing the release 5.*x* CAF Refresh file message format will not be able to refresh this new segment.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF require additional disk space as the new tokens for this enhancement (i.e., EMV Status Token, EMV Request Token, EMV Discretionary Data Token, EMV Response Token, EMV Script Token) can be included in the transaction records logged to these files.

The KEYI impacts disk requirements. The total impact varies for each BASE24 user depending on the number of KEYI records defined in the BASE24 system.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF, PTLF, and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Operational Impacts

BASE24 operators must be aware that the new KEYI file will appear on their BASE24 BASE Product Menu.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Vendor Impacts

In order to implement this enhancement as an issuer, BASE24 must have access to a hardware security module. The enhancement will support the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of software that is capable of supporting the EMV enhancement. For Atalla, the minimum version is 2.84. For the Thales e-Security (Racal), the minimum version is 5.05.

Reference Documents

- Compaq/Atalla Security Products, EMV/AMEX CSC Functional Description Document, Version 1.1, February 1,2000
- EMV ICC Application Specification for Payment Systems, Version 3.1.1, May 31, 1998
- Europay Integrated Circuit Card (ICC) Application Specification, Version 2.1, October 1999
- Europay International: Off-the-Shelf Card Profile Technical Implementation for Pay Now and Pay Later Cards, status Release 2.1, dated October 1999.
- MasterCard International: Business/Functional Requirements for Debit and Credit on Chip, version 1.0, dated 20th October 1997.
- MasterCard International: MasterCard Chip Recommended Specifications for Debit and Credit, version 2.1, dated November 1999.
- Thales e-Security (Racal): Host Security Module Standard Command Set to Support Issuer Authorization Functions for EMV Schemes, (Doc No: 1077-HS-0A), version 0A, February 2000
- Visa International: VIS ICC Specifications, version 1.3.2, dated July 1999.

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Format 2 File Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Format 2 File Support enhancement.

Section 2 contains a functional overview of the Format 2 File Support enhancement.

The Format 2 File Support enhancement provides format 2 file support for the CAF, NEG, PBF, TLF, PTLF, ITLF, and the ILFs for specific Interchange Interfaces. Format 2 files require a D46 or higher Compaq NSK Operating System. Files with format 2 partitions can contain more data than the existing format 1 files. While both file formats can have up to 16 partitions, entry-sequenced format 1 files are limited to four gigabytes, and key-sequenced format 1 files are limited to 32 gigabytes. Format 2 files are currently limited only by the physical disk size. Format 2 files have new block and label formats and can hold as much as 1024 gigabytes of data per partition. With key-sequenced files (i.e., CAF, NEG, and PBF) where more paths are needed to allow for more concurrent access to data within a single file, the BASE24 user can partition both format 1 files and format 2 files.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Parameter Changes

Not applicable

Database Configuration

<u>Extract Configuration File (ECF)</u>

Field Name: PRIMARY EXTENT
Screen: ECF Screen 3
DDL Name: ECF.BASE.FILE-CONF.PRI-EXT

The field size of the PRIMARY EXTENT field on screen 3 of the ECF has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: SECONDARY EXTENT
Screen: ECF Screen 3
DDL Name: ECF.BASE.FILE-CONF.SECONDARY-EXT

The field size of the SECONDARY EXTENT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: FILE FORMAT
Screen: ECF Screen 3
DDL Name: ECF.BASE.FILE-FRMT

This field indicates whether the Extract file should be created as a format 1 or a format 2 file. Valid values are 1 or 2. If the customer continues to have the Extract process create format 1 files, this field should be set to 1 (the default).

Field Name: PARTITION 1 PRI EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART1-PRI-EXT

The field size of the PARTITION 1 PRI EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 1 SEC EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART1-SECONDARY-EXT

The field size of the PARTITION 1 SEC EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 2 PRI EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART2-PRI-EXT

The field size of the PARTITION 2 PRI EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 2 SEC EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART2-SECONDARY-EXT

The field size of the PARTITION 2 SEC EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 3 PRI EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART3-PRI-EXT

The field size of the PARTITION 3 PRI EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 3 SEC EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART3-SECONDARY-EXT

The field size of the PARTITION 3 SEC EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 4 PRI EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART4-PRI-EXT

The field size of the PARTITION 4 PRI EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

Field Name: PARTITION 4 SEC EXT
Screen: ECF Screen 3
DDL Name: ECF.BASE.PART4-SECONDARY-EXT

The field size of the PARTITION 4 SEC EXT field has been increased by one character. The increased field size enables the Extract process to accommodate the larger transaction log files that can be extracted as a result of this enhancement.

For additional information on the usage of these fields, refer to the ***BASE24 Base Files Maintenance Manual.***

New Data Files

Not applicable

Template Changes

Using the Same File Formats for All Institutions

BASE24 users using format 2 CAF, NEG, and PBF files must create the format 2 templates before running refresh. The method that can be used to change to format 2 files is to rename the existing template, and then create the format 2 template files. When refresh is started, the CAF, NEG, and PBF files are created using the new format 2 template files.

The following is an example using FUP to create the format 2 CAF template file where the existing CAF template file name is CAF. The maximum values for the primary extent size and secondary extent size have changed with format 2 files, and therefore the user can also use the SET EXT command in the following creation of the CAF template files using FUP.

```
-SET LIKE CAF
-RENAME CAF, OLDCAF
-SET FORMAT 2
-SET REC 4040
-CREATE CAF
```

The format 2 CAF template file in the create command must have the same value as specified in the template file name field of the LCONF. For additional information on the usage of this assign, refer to the ***BASE24 Logical Network Configuration File Manual***.

Not Using the Same File Formats for All Institutions

If the user does not use the same file format for all of the institutions defined in the BASE24 system, the user must maintain at least two templates for these files. The following is an example using FUP to create the format 2 CAF template file where the existing CAF template file name is CAF (i.e., format 1 CAF template file). The maximum values for the primary extent size and secondary extent size have changed with format 2 files, and therefore the user can also use the SET EXT command in the following creation of the format 2 CAF template files using FUP.

```
-SET LIKE CAF
-SET FORMAT 2
-SET REC 4040
-CREATE CAFFRMT2
```

The appropriate CAF template file name must have the same value as specified in the template file name field of the LCONF. For additional information on the usage of this assign, refer to the ***BASE24 Logical Network Configuration File Manual***. Before the user performs a full-file CAF refresh using the format 1 CAF template file, the value in the LCONF must be CAF. Before the user performs a full-file CAF refresh using the format 2 CAF template file, the value in the LCONF must be CAFFRMT2.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The ECF conversion program must be run to convert ECF records from the release 5.3 format to the release 6.0 format (i.e., the latest version of release 6.0). Running the conversion program is required for all users, not just those that are using the Format 2 File Support enhancement. Users must create the ECF file using the BADDLFUP file.

The ECF conversion program on the BA60UC04 subvolume can be used to convert the ECF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVECFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ecfcnv.t,name $ecfm,nowait/
cnvecfm –force –list –listreplace
```

Users must update the CNVECFR obey file on the BA60UC04 subvolume with the appropriate information. CNVECFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: A separate ECF conversion program was written, and must be run for those BASE24 users converting from release 6.0 version 1 to release 6.0 version 2. If you are implementing format 2 files, refer to the Database Configuration subsection of the Format 2 File Support enhancement for further information.

The ECF conversion program must be run to convert ECF records from the release 6.0 version 1 format to the release 6.0 version 2 format. Running the conversion program is required for all users using BASE24 Release 6.0 version 1, not just those that are using the Format 2 File Support enhancement. Users must create the ECF file using the BADDLFUP file.

The ECF conversion program on the BA60UC05 subvolume can be used to convert the ECF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVECFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ecfcnv.t,name $ecfm,nowait/
cnvecfm –force –list –listreplace
```

Users must update the CNVECFR obey file on the BA60UC05 subvolume with the appropriate information. CNVECFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD05 file on the BA60UC05 subvolume.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the ISO Host Interface, BIC ISO Interchange Interface and Super Extract processes.

ISO Host Interface Process

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The TLF Token has changed in length, since the relative byte addresses for the records in a format 2 transaction log file is 64-bits, not 32-bits. If the TLF Token is included in the ISO message to the host, this change to the message needs to be handled by the host code.

BIC ISO Interchange Interface Process

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The TLF Token has changed in length, since the relative byte addresses for the records in a format 2 transaction log file is 64-bits, not 32-bits. If the TLF Token is included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

Super Extract Process

For those BASE24 users implementing this enhancement and extracting any of their data files to a format 2 file, the host code must process the new block format for the extract format 2 disk files. In addition, the host code can use the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The fourth byte of the user field in the Extract File Header is used to store this information.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The TLF Token has changed in length, since the relative byte addresses for the records in a format 2 transaction log file is 64-bits, not 32-bits. Since BASE24 users should not have configured this token to be extracted, this should not impact the host.

User/Operational Impacts

Disk Impacts

Any file converted to format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for format 2 files is increased. This is true when the primary file is format 2, regardless of whether the alternate key file(s) is format 2.

Since the maximum data area per block in format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a format 2 file. Users should check the length of entry sequenced records — TLF, PTLF, ITLF and ILF (for specific Interchange Interfaces modified for this enhancement) — and key sequenced, segmented file records — CAF and PBF — to see if fewer records will fit in the new format 2 maximum data area blocks. The format 2 maximum data area is 4048 for entry sequenced files and 4040 for key sequenced files. If fewer records will fit in the new format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry sequenced files against the TLFX, PTLFX, ITLFX and ILFX files, if format 2 versions of these are to be used when extracting to disk.

The NEG file is not segmented, and at its current size the same number of records could be written per block in both formats, so there should be no disk impacts for the NEG.

User Impacts

BASE24 users using format 2 transaction log files — TLF, PTLF, ITLF, and ILF — must modify their template files to be format 2 files.

BASE24 users using format 2 CAF, NEG, and PBF files must modify their Refresh templates to be format 2 files.

BASE24 users extracting to format 2 disk files must configure the appropriate information in the ECF.

BASE24 users who change to format 2 transaction log files must start at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files), and create the appropriate template files. At cutover, the transaction log files are created using the new template files. Refer to the Template Changes subsection in the product sections of this enhancement for further information.

The first time that the BASE24 user changes from using a format 1 CAF, NEG, or PBF file to a format 2 CAF, NEG, or PBF file, they can use the following methods:

- Using a full file refresh, the refresh process uses a template that is defined in the LCONF. Refer to the Template Changes subsection of this enhancement for further information.
- For a partial refresh, the appropriate file must be changed to a format 2 file format when the file is not in use. FUP can be used for this task. In the following example, the existing format 1 CAF template file name is CAF.

```
-SET LIKE CAF
-SET FORMAT 2
-SET REC 4040
-CREATE CAFFRMT2
```

  Use FUP load with the sorted feature to load the format 1 file records to the format 2 file.

```
-LOAD CAF,CAFFRMT2,SORTED
```

The sort feature of FUP LOAD will significantly cut down on the time, since the CAF is a key-sequenced file. Once the file format is set to a 2, all subsequent partial refreshes to this file will use the file format of 2.

Operational Impacts

New event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

Format 2 key-sequenced files allow a maximum logical record length of 4040 bytes. Before a BASE24 user changes their CAFs, NEGs, and/or PBFs to format 2 files, they should ensure these records do not exceed the 4040 byte limit. BASE24 processes utilizing format 1 key-sequenced files are not affected, since the DDLs for the CAF, PBF, and NEG were not changed to set the maximum record length to 4040 bytes.

BASE24 users must calculate the length of the records that are written to the transaction log files (i.e., TLF, PTLF, ITLF and ILF). Format 1 transaction log files allow a maximum logical record length of 4072 bytes. Format 2 transaction log files allow a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are written to the transaction log files, the record length of a transaction log file record could exceed the limit of 4048 bytes for format 2 files.

Before users change their transaction log files to format 2 files, they should ensure these records do not exceed the 4048 limit. BASE24 users utilizing format 1 are not affected, since the DDLs for the transaction log files were not changed to set the maximum record length to 4048.

Vendor Impacts

Not applicable

Reference Documents

- Tandem NonStop Kernel ***D-Series System Migration Planning*** for D46.00
- Tandem NonStop Kernel ***Enscribe Programmer's Guide*** for D46.00

Multiple Currency Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Multiple Currency Support enhancement.

Section 2 contains a functional overview of the Multiple Currency Support enhancement.

The Multiple Currency Support enhancement provides support for multiple currencies in BASE24-atm or BASE24-pos by managing the processing of a mixture of currencies involved in an individual transaction. It allows institutions to offer accounts in different currencies, with the ability to define each account with its own currency.

CUSTMACS Changes

Several new flags have been added for the Multiple Currency Support enhancement. Refer to the appropriate product section for any product-specific flags.

Set the following flag to true if any other Multiple Currency (MCU) flags are set to true.

- mcu_on

LCONF Assign Changes

<u>ERF</u>

The fully qualified name of the Exchange Rate File (ERF). The Extended Memory Table Build utility and BASE24 ERF File Maintenance use this assign. This assign is required if the MCU extensions are present. Additional information on this file can be found in the appropriate product ***Multiple Currency Support Manual.***

<u>ERFEMT</u>

The fully qualified file name of the Exchange Rate File Extended Memory Table file (ERFEMT). The ERF Extended Memory Table Build utility and Multiple Currency Conversion Utilities use this assign.

This assigned is required if the MCU extensions are present. If this assign is not present in the LCONF, the Extended Memory Table Build utility and the processes that use the ERFEMT will not initialize.

LCONF Parameter Changes

BASE-CURRENCY-CODE

This optional param contains a code indicating the base currency the logical network uses (for example, 840). If the param is not found, the default value of 840 (USD) is used. Each code consists of a three-digit numeric value. For the currency code values, refer to the ISO 4217:1995 standard, ***Codes for the Representation of Currencies and Funds***.

The currency code in this param must match the ERF Base currency code specified in the COBNAMES file.

CURRENCY-CODE

This existing param contains a code indicating the currency the logical network uses (for example, 840). Each code consists of a three-digit numeric value. For the currency code values, refer to the ISO 4217:1995 standard, ***Codes for the Representation of Currencies and Funds***. Prior to release 6.0, this param was used to determine the correct number of decimal positions for activity limits established in the Card Prefix File (CPF). However, with the release of the Multiple Currency Support enhancement, the CPF activity limits use the number of decimal positions of the currency code of the associated Institution from the IDF. This is true for all BASE24 users regardless of whether processing with the Multiple Currency Support enhancement.

The currency code in this param must match one of the values specified in the CURRENCY-CODE-TABLE in the COBNAMES file. If the param is not found, the default value of 840 (USD) is used.

ERFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the ERFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

ERFEMT-WARM*identifier*

This optional param contains the symbolic name of a process to be warmbooted for the ERFEMT. Processes that can be warmbooted for the ERFEMT include BASE24-atm Multiple Currency Authorization processes and the BASE24-pos Multiple Currency Router/Authorization processes. This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all ERFEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this EMT. A maximum of 500 processes can be warmbooted from the DCT.

Each ERFEMT-WARM param contains the name of one process. When multiple processes must be notified, this parameter can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another ERFEMT-WARMidentifier param. For example, ERFEMT-WARM01 could contain the P1A^AUTH1 process and ERFEMT-WARM02 could contain the P1A^AUTH2 process.

Database Configuration

Institution Definition File (IDF)

IDF screen 3 already contains the CURRENCY CODE field indicating the currency of the financial institution. If the Multiple Currency Support enhancement is being used, each FIID can have its own different currency code which must be one of those specified from the list in the CURRENCY-CODE-TABLE in the COBNAMES file. The value of the CURRENCY CODE does not have to be the same for all IDF records in the same refresh group.

For Acquiring Interchange Interfaces (i.e., those which deliver transactions to BASE24), an IDF record including the CURRENCY CODE must be configured for each FIID specified in the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE). This is required in order for BASE24 to establish the Acquiring Institution currency for a transaction.

An IDF record including the CURRENCY CODE must be configured for the FIID associated with each Acquiring Interchange Interface. This is required in order to establish the Acquiring Institution currency.

For additional information on the usage of this field, refer to the ***BASE24 Base Files Maintenance Manual***.

Positive Balance File (PBF)

PBF screen 1 has a new CURRENCY CODE field indicating the currency of the account. All BASE24 users, irrelevant of whether the Multiple Currency Support enhancement is being used, will be required to configure this information.

If a BASE24 user does not have the Multiple Currency Support enhancement, this field will automatically default to the first entry in the CURRENCY-CODE-TABLE in the COBNAMES file if a record is updated (without specifying the CURRENCY CODE) using the BASE24 file maintenance function.

If a BASE24 user does have the Multiple Currency Support enhancement, this field will also automatically default to the first entry in the CURRENCY-CODE-TABLE in the COBNAMES file, but it can also be set to any other valid entry within that table.

If this field is being updated using the FHM or Refresh processes, the user can specify the value if using 6.0 message formats. Otherwise, the value will be defaulted to the CURRENCY-CODE specified in the corresponding IDF record if using a 5.x message format.

For additional information on the usage of this field, refer to the ***BASE24 Base Files Maintenance Manual***.

Surcharge File (SURF)

SURF screens 2 and 3 have been modified to include a CURRENCY CODE field in the primary key. All BASE24 users who implement the surcharge feature, irrelevant of whether the Multiple Currency Support enhancement is being used, will be required to configure this information. All surcharge BASE24 users will be required to run a SURF Conversion program to implement this modification. Refer to the 6.0 General Changes section for additional information.

For additional information on the usage of this field, refer to the ***BASE24 Base Files Maintenance Manual.***

Token File (TKN)

The following new BASE24 Base token has been added for this enhancement:

- Multiple Currency Token (Token ID “BD”) contains information accumulated during the processing of a transaction relating to the currencies used in processing that transaction.

If the new token for this enhancement is not to be logged to the appropriate product Transaction Log File (TLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the new token for this enhancement is to be extracted with the TLF, PTLF, or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured in the TKN.

<u>CAF/CPF/UAF</u>

No data fields have been added to the Base segment of the CAF, CPF, or UAF for the Multiple Currency Support enhancement. Since these files contain amount fields that need to have a currency code associated with them, if the Multiple Currency Support enhancement is being used, the amount fields are therefore assumed to be in the currency of the corresponding Institution Definition File (IDF) record.

New Data Files

<u>Exchange Rate File (ERF)</u>

A new Base file, the Exchange Rate File (ERF), is used to contain the exchange rate information that is required in a Multiple Currency environment. Only users that have the Multiple Currency Support enhancement are required to enter information into this file. There is one ERF per Logical Network. Refer to the LCONF Assign Changes subsection of this enhancement for further information about this file. Refer to the appropriate product ***Multiple Currency Support Manual*** for additional information on the ERF.

Template Changes

<u>COBNAMES - CURRENCY-CODE-TABLE</u>

The following is an excerpt of the CURRENCY-CODE-TABLE from the COBNAMES file on the BA60SRC subvolume that needs to be modified if the Multiple Currency Support enhancement is being used. The highlighted text indicates the values that must be set according to the customer's configuration. The values that are used here are for explanation purposes only, and are not to be considered the intended configuration. This example shows the table with ten different currency entries.

After the modification of this table, the BASE24 Pathway requesters and servers must be recompiled.

```
*   Customers should alter the one line about currency to
*   reflect their currency. The values between the
*   quotes is the part to change. If the customers has a
*   currency other than US dollars change 840 to the number
*   of the currency. If the monetary system has 3 decimals,
*   change the 2 to a 3 for a 3 decimal monetary system. If
*   the monetary system has no fractional part, change the 2 to
*   a 0 to indicate no decimal place. The US can be changed to
*   2 letters to describe the currency.
```

```
*
*   At such time as a multiple currency code package is
*   developed and implemented by a customer, the
*   individual customers can add additional entries
*   to the table. At such time that more entries need to be
*   added to the table change the value to the correct number
*   of table entries. For example, if you have 2 currencies,
*   change value +1 to value +2.
*
  01  NUM-CRNCY-CDES             PIC S9(2)    VALUE +10.
  01  CRNCY-CDES.
*   To add more currency codes table entries, do it here
*   at the 05 level.

      05  UNITED-STATES          PIC X(12) VALUE "8408402(USD)".
      05  UNITED-KINGDOM         PIC X(12) VALUE "8268262(GBP)".
      05  EUROPEAN-CURRENCY      PIC X(12) VALUE "9789782(EUR)".
      05  GERMANY                PIC X(12) VALUE "2762762(DEM)".
      05  ITALY                  PIC X(12) VALUE "3803800(ITL)".
      05  GREECE                 PIC X(12) VALUE "3003000(GRD)".
      05  CANADA                 PIC X(12) VALUE "1241242(CAD)".
      05  MEXICO                 PIC X(12) VALUE "4844842(MXN)".
      05  BAHRAIN                PIC X(12) VALUE "0480483(BHD)".
      05  KUWAITI                PIC X(12) VALUE "4144143(KWD)".

    01  CRNCY-CDE-TABLE REDEFINES CRNCY-CDES.

  *   If you add country and currency codes to the table you
  *   also have to change the occurs number on this next line.

      05  CURRENCY-CODES         OCCURS  10 TIMES.
          10  CNTRY-CODE         PIC X(3).
          10  C-CODE             PIC X(3).
          10  DECIMAL            PIC X.
          10  L-PAREN            PIC X.
          10  COUNTRY-ABB        PIC X(3).
          10  R-PAREN            PIC X.
```

COBNAMES - BASE-ERF-CURRENCY-CODE

The following is an excerpt from the COBNAMES file on the BA60SRC subvolume of the BASE-ERF-CURRENCY-CODE that needs to be modified if the Multiple Currency Support enhancement is being used, and the customer wishes a different value than the default of 840 (USD) to be used as the Base currency in their system. The highlight indicates the value that must be set according to the customer's configuration.

After the modification of this table, the BASE24 ERF requester/server must be recompiled.

```
* Customers should alter the ERF-BASE-CRNCY-CDE if they are using
* the Multiple Currency Support enhancement and they want to have a
* currency other than US dollars. If a Base Currency other than
* US dollars is required, modify the 840 to the ISO numeric
* currency code as specified in the ISO booklet "Codes for the
* Representation of Currencies and Funds", ISO 4217-1995.
*
* The ERF-BASE-CRNCY-CDE is the single currency against which
* exchange rates for all other currencies are expressed.

     01 ERF-BASE-CRNCY-CDE       PIC X(3)   VALUE "840".
```

CUSTCNST

The following is an excerpt from the CUSTCNST file on the BA60DDL subvolume of a constant that needs to be modified if the Multiple Currency Support enhancement is being used. The highlight indicates the value that must be set according to the customer's configuration.

After the modification of this table, the BASE24 DDLs and Multiple Currency Support enhancement code must be recompiled.

```
*#  This field contains the maximum number of differing currency
*#  types in the Multiple Currency array in the Multiple Currency
*#  Token for a particular customer.
*#
*#  If the customer has more or less currency types being used in
*#  their system, they may modify the value here and recompile the
*#  Base DDLs. This value is ignored in a single currency system
```

```
*#  and does not need to be modified by those customers. This value
*#  defaults to 5 in a multiple currency system, but cannot be more
*#  than the BASE24 system maximum, which is currently defined at
*#  10.

constant cust-max-crncy-entries-l       value 5.
```

The value in this constant dictates the number of differing currency **types**, not necessarily the number of differing currency codes, that a customer's given logical network will contain in the Multiple Currency token. It is imperative that a customer consider the number of different currency types with differing currency codes in their system, and the transaction traffic that can flow from one logical network to another. On a given logical network, the Multiple Currency Support enhancement only processes transactions containing a Multiple Currency token with the number specified in the above constant.

The use of this constant allows a customer to tailor the size of the Multiple Currency token to its actual customer usage, thereby eliminating some possibly wasted space in the Multiple Currency token. The value in this constant should never be set to more than 10, as that is the maximum number of different currency types, excluding the Transaction currency, that BASE24 can handle in a single transaction.

For more information on currency types and the Multiple Currency token, refer to the appropriate product ***Multiple Currency Support Manual***.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The ERF requester name was added to the MEGATBL file, since RQMEGA communicates with the ERF requester using the extended message format (i.e., uses USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

An entry was added to the SECTBL to support the screen for the ERF requester/server.

PATHCONF Changes

The new ERF server must be defined in the PATHCONF file as SERVER-ERF. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new ERF requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

<u>Positive Balance File (PBF)</u>

A PBF file conversion can be performed using inline code through the BASE24 AFT subsystem and is transparent to the user. If a value has not been entered in the CURRENCY CODE field on PBF screen 1, it will automatically default to the first entry of the CURRENCY-CODE-TABLE in the COBNAMES file. However, BASE24 users can perform a PBF refresh or PBF from host maintenance transaction to initialize the new CURRENCY CODE field in the PBF to a value that they want.

<u>Surcharge File (SURF)</u>

The SURF conversion program must be run to convert SURF records from the release 5.3 format to the release 6.0 format. Running the conversion program is required for any BASE24 surcharge user, not just those that have the Multiple Currency Support enhancement. Users must create the SURF file using the BADDLFUP file.

The SURF conversion program on the BA60UC04 subvolume can be used to convert the SURF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVSURFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#surfcnv.t,name $surfm,nowait/
cnvsurfm –force –list –listreplace
```

Users must update the CNVSURFR obey file on the BA60UC04 subvolume with the appropriate information. CNVSURFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, From Host Maintenance, ISO Host Interface, Refresh, and Super Extract processes.

BIC ISO Interchange Interface Process

While the BIC ISO Interchange Interface Process is not specifically impacted by the Multiple Currency Support enhancement, the following are implications of Multiple Currency support:

- Issuer co-networks can receive authorization requests in different currencies, as BASE24 now is capable of delivering transactions acquired in multiple currencies if the Multiple Currency Support enhancement is being used. This includes receiving any of the new Multiple Currency tokens.
- Acquirer co-networks can deliver authorization requests to BASE24 in different currencies, as BASE24 now is capable of receiving transactions acquired in multiple currencies if the Multiple Currency Support enhancement is being used. This includes delivering any of the new Multiple Currency tokens.

The amount and currency fields in the BASE24 Standard External Message (SEM) conform to the ISO definitions and processing rules. The BASE24 defined Transaction currency is used for all standard amounts in requests, responses, reversals, and all other transaction messages.

From Host Maintenance Process

Host code changes are required for all BASE24 users that utilize the Positive Balance File (PBF) From Host Maintenance (FHM) process. BAS24 users utilizing the 6.0 FHM message format can specify an appropriate value for the CURRENCY CODE in the PBF. BASE24 users utilizing the 5.*x* FHM message format cannot specify a value for the CURRENCY CODE in the PBF, so FHM defaults this field to the CURRENCY CODE of the corresponding institution as specified in the IDF.

ISO Host Interface Process

The host can use of any of the Multiple Currency Support enhancement tokens to extend the functionality in a Multiple Currency environment.

While the standard ISO Host Interface (HISO) module is not specifically impacted by the Multiple Currency Support enhancement, host systems must be aware of the following implications of Multiple Currency support:

- Issuer hosts can receive authorization requests in different currencies, as BASE24 now is capable of delivering transactions acquired in multiple currencies if the Multiple Currency Support enhancement is being used. This includes receiving the new Multiple Currency tokens.
- Acquirer hosts can deliver authorization requests to BASE24 in different currencies, as BASE24 now is capable of receiving transactions acquired in multiple currencies if the Multiple Currency Support enhancement is being used.

The amount and currency fields in the BASE24 Standard External Message (SEM) conform to the ISO definitions and processing rules. The BASE24 defined Transaction currency is used for all standard amounts in requests, responses, reversals, and all other transaction messages.

Refresh Process

Host code changes are required for all BASE24 users that utilize the release 6.0 PBF Refresh file format to accept and process information in the PBF file. Users utilizing the Release 5.*x* PBF Refresh file format have the PBF CURRENCY CODE defaulted to the corresponding institution CURRENCY CODE from the IDF. A single PBF Refresh file can contain amounts specified in multiple currencies.

Note: The amount field in the PBF Refresh batch trailer record (RBT.AMT) represents a verification amount only, and should reflect the sum total of PBF ledger balances, regardless of currency, specified in the batch.

There are no explicit host code changes required for BASE24 users that utilize the release 6.0 CAF Refresh file message format. However, the amounts in the input file are assumed to be in the currency of the institution that issued the card. The value of the Institution currency does not have to be the same for all institutions belonging to the same refresh group. This means that a single CAF Refresh file can contain amounts specified in multiple currencies.

Super Extract Process

Host code changes are required for all BASE24 users that utilize OMF Extract to accept and process information in the ERF, PBF, and SURF files.

If the new token (i.e., Multiple Currency Token) for this enhancement is to be included in the TLF, PTLF, and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF require additional disk space as the new token for this enhancement (i.e., Multiple Currency Token) can be included in the transaction records logged to these files.

User Impacts

BASE24 users utilizing the Multiple Currency Support enhancement must enter exchange rate information in the Base Exchange Rate File (ERF).

BASE24 users utilizing the Multiple Currency Support enhancement must enter the appropriate currency code information in the LCONF, COBNAMES, and CUSTCNST files.

BASE24 users implementing the Multiple Currency Support enhancement must understand the concept of EMT Build and the procedure to create the Exchange Rate File Extended Memory Table file (ERFEMT).

BASE24 users must also understand the concept of EMT warmboot and the procedure to warmboot the ERFEMT.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Vendor Impacts

Not applicable

Reference Documents

- ISO 4217 :1995 standard, Codes for the Representation of Currencies and Funds
- European Council Regulation (EC) No 1103/97
- European Union Council Regulation (EC) No 2866/98

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Transaction Security Services

The Transaction Security Services (TSS) process, built using ACI's Enterprise Services architecture, is a new process created to serve as an interface between the security utilities bound into BASE24 processes and hardware security modules. The Transaction Security Services process was created to isolate and minimize modifications required to existing BASE24 database files and software for those customers who want to implement full end-to-end triple DES or to support the Atalla Key Block (AKB) key formats. Customers who do not want to implement these processing options are not required to implement Transaction Security Services. Transaction Security Services is only supported in BASE24-atm, BASE24-pos and BASE24 interchange interfaces.

The database impacts and the new/modified calls to the HSMs have been encapsulated within the BASE24 utilities (i.e., HISWUTLS, SECUTILS, CAMUTILS and CVUTILS) and the new Transaction Security Services process. Transaction Security Services stores the longer keys required by triple DES encryption and the Atalla AKB key format. Impacts to existing BASE24 Interchange Interface processes are limited to support for double-length and triple-length keys passed to or from interchanges for dynamic key management. Impacts to existing BASE24 device handler modules are limited to support for downloading double-length or triple-length keys to devices.

When Transaction Security Services is implemented, all BASE24 processes must be set to use it. BASE24 users cannot have some processes in a BASE24 network using the new Transaction Security Services and others using the existing hardware security module connection.

In addition to the BASE24 module changes, customers who want to implement Transaction Security Services must obtain and install the Transaction Security Services and ACI Desktop Graphical User Interface (GUI) modules. Refer to the BASE24 Transaction Security Services Processing Guide and the BASE24 Classic 6.0 Installation Guide for further information.

A set of database files used by the Transaction Security Services process supports extended length keys for triple DES and the AKB key block format. These files are maintained using the ACI Desktop Graphical User Interface (GUI). The Transaction Security Services process supports all of the following transaction security functions currently supported in BASE24 systems:

- Full end-to-end PIN encryption using single DES ensures that PIN information is maintained in a confidential manner. PIN verification ensures that a valid cardholder is performing the transaction.

- Message authentication ensures that data remains tamper-free while passing between an EFT device and its authorization destination.
- Card verification ensures that a valid card is being used to initiate the transaction.
- Chip authentication ensures that a credit or debit card with an embedded microprocessor chip is valid.

Besides the existing security-related functions described above, the Transaction Security Services process provides the following security enhancements to a BASE24 system:

- Triple DES for stronger PIN protection and reduced potential fraud losses
- Longer keys for protection against key exhaustion attacks
- Stronger key storage techniques to protect against key compromise

The Transaction Security Services process runs as an application process under the XPNET process. Security utilities bound into BASE24 application processes communicate with the Transaction Security Services process using a direct write/read operation.

When using the Transaction Security Services process, hardware-encrypted key information is maintained in the Transaction Security Services database. Except for the KEYA and KEYI, the hardware-encrypted key fields in the BASE24 database are set to a value known as a *key locator* value. The Transaction Security Services process uses the key locator value as the primary key value to retrieve the actual hardware-encrypted key information from its database.

Key locator values are either set to the primary key value for the record (e.g., the terminal ID for hardware key fields in the BASE24-atm Terminal Data files and BASE24-pos Terminal Data files) or an index value (for the BASE24 Key File). If the Transaction Security Services database is partitioned across more than one logical network, a two-character hexadecimal partition identifier is used as the first two characters of the key locator. For key locators in the terminal data files, the first fourteen characters of the terminal ID are concatenated to the partition identifier to create the key locator.

When using the Transaction Security Services process, the KEYA and KEYI are replaced by Transaction Security Services files and are no longer accessed in processing for PIN verification, card verification or chip authentication in hardware security modules. If a customer is performing PIN verification in software or is using Diebold PIN verification or Atalla Identikey PIN verification, Transaction Security Services cannot be implemented. Transaction Security Services does not support software encryption/decryption and does not support Diebold PIN verification or Atalla Identikey PIN verification. Transaction Security Services does not support the Derived Unique Key Per Transaction (DUKPT) method of PIN encryption at this time.

Each Transaction Security Services process must be configured as an application process to the XPNET system. In addition, a Timer process and User Interface Server processes must also be configured. The Timer process sets timers indicating how long a previous key stored in the database accessed by the Transaction Security Services process can be used. The User Interface Server processes communicate with the ACI Desktop Graphical User Interface. Devices, lines, and stations also must be configured to support the GUI. For more information on configuring processes, devices, lines and stations for Transaction Security Services, refer to the BASE24 Classic 6.0 Installation Guide.

Use of the Transaction Security Services process by BASE24 processes is controlled through LCONF assigns and params. BASE24 processes send messages directly to the Transaction Security Services processes identified in the SECURE-DEV1 and SECURE-DEV2 assigns using a write/read operation when the SECURE-DEV-TYP param is set to a value of “TSS” in the LCONF. The SECURE-DEV1 assign specifies the primary Transaction Security Services process. The SECURE-DEV2 assign specifies the secondary Transaction Security Services process. A value of “TSS” for the SECURE-DEV-TYP param indicates that Transaction Security Services is used.

The security utilities bound into BASE24 application processes communicate with a Transaction Security Services process using a direct write/read operation when a security function requiring the use of a hardware security module is needed. Each BASE24 process communicates with a single Transaction Security Services process. However, all BASE24 processes are not required to communicate with the same TSS process. The Transaction Security Services process can be replicated as needed for load balancing and transaction throughput.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to or modified in the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

SECURE-DEV1, SECURE-DEV2

The PPD names of the Transaction Security Services processes instead of the PPD names for RAM or BOXCAR processes.

The SECURE-DEV1 assign specifies the PPD name of the primary Transaction Security Services process for a BASE24 application process.

The SECURE-DEV2 assign specifies the PPD name of the secondary Transaction Security Services process for a BASE24 application process.

Different Transaction Security Services processes can be assigned to different BASE24 application processes by configuring multiple SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF and assigning each to a specific symbolic process name in the READ BY field on the LNCF Assign screen. The USER FIELD for the SECURE-DEV1 and SECURE-DEV2 assigns on the LNCF Assign screen must be set to spaces.

LCONF Param Changes

This section describes the param records that must be added to or modified in the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

<u>LN-IND</u>

The LN-IND param indicates a two-character hexadecimal partition identifier used as the first two characters of the key locator when Transaction Security Services data sources are to be partitioned across multiple logical networks. For key locators in the terminal data files, the first fourteen characters of terminal ID will be concatenated to the partition identifier to create the key locator.

<u>RSM-TERM-MASTER-KEY</u>

The RSM-TERM-MASTER-KEY param is not used, because it has been replaced by the RSM_TERM_MASTER_KEY param set in the Transaction Security Services environment. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

<u>SECURE-DEV-CONF-CMD100</u>

The SECURE-DEV-CONF-CMD100 param is not used, because it is replaced by a parameter that Transaction Security Services obtains from its own configuration files. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

SECURE-DEV-CONF-TXT

The SECURE-DEV-CONF-TXT param is not used, because it is replaced by a parameter that Transaction Security Services obtains from its own configuration files. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

SECURE-DEV-FAIL-NOTIFY

The SECURE-DEV-FAIL-NOTIFY param indicates the interval between event messages that indicate a Transaction Security Services process is not responding. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

SECURE-DEV-MONITOR-THRESHOLD

The SECURE-DEV-MONITOR-THRESHOLD param indicates the number of transactions processed by the Transaction Security Services process defined in the SECURE-DEV1 LCONF assign before checking the Transaction Security Services process defined in the SECURE-DEV2 LCONF assign. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

SECURE-DEV-TIM-LMT

The SECURE-DEV-TIM-LMT param specifies how long BASE24 processes wait for a response from a Transaction Security Services process. The time specified should be slightly longer than the amount of time Transaction Security Services waits for a response from the HSM. Refer to the BASE24 Transaction Security Services Processing Guide for further information.

SECURE-DEV-TYP

The SECURE-DEV-TYP param indicates the hardware security device type. It must be set to a value of “TSS” to use the Transaction Security Services.

SOFTWARE-PIN-VERIFICATION-ALWD

This optional param indicates whether PIN encryption is allowed to be configured to be done in software or in the clear, anywhere within the BASE24 system, when Transaction Security Services is being used. If this is set to Y, the KEYA must exist. In this situation, the clear keys are required at some point in the transaction path. If this parameter is not present and Transaction Security Services has been implemented, no processing in software is allowed.

Database Configuration

CRDVD

A CRDVD record must exist for every KEYA record in the existing BASE24 network that is used for card verification. The CRDVD contains the Card Verification data from the KEYA records.

Refer to the File Conversions subsection of this enhancement to see how CRDVD records for each KEYA record can be created programmatically.

EMVSD

An EMVSD record must exist for every KEYI record in the existing BASE24 network. The EMVSD contains the EMV Verification data from the KEYI records.

Refer to the File Conversions subsection of this enhancement to see how EMVSD records for each KEYI record can be created programmatically.

IDESD

An IDESD record must exist for every KEYA record in the existing BASE24 network that is used for IBM DES Pin Verification. The IDESD contains the IBM DES PIN Verification data from the KEYA records.

Refer to the File Conversions subsection of this enhancement to see how IDESD records for each KEYA record can be created programmatically.

KEYA

With the implementation of Transaction Security Services the KEYA is no longer used by BASE24, except when the LCONF param SOFTWARE-PIN-VERIFICATION-ALWD is set to Y.

KEYI

With the implementation of Transaction Security Services, the KEYI is no longer used by BASE24.

VPVVD

A VPVVD record must exist for every KEYA record in the existing BASE24 network that is used for Visa PIN Verification. The VPVVD contains the Visa PIN Verification data from the KEYA records.

Refer to the File Conversions subsection of this enhancement to see how VPVVD records for each KEYA record can be created programmatically.

New Data Files

Several new data files are required that replace BASE24 files no longer accessed in processing for verification in hardware security modules. These files can be created by running conversion programs or records can be added, deleted or modified manually using the ACI Desktop Graphical User Interface.

- CRDVD - Card Verification
- EMVSD - EMV Authorization Key
- IDESD -IBM DES Verification
- VPVVD - VISA PVV Verification

The following table shows the relationship of existing BASE24 Base database files to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process.

BASE24 Database File	Transaction Security Services Datatabase File
KEYA IBM DES records	IDESD
KEYA Visa PVV records	VPVVD
KEYA Card Verify records	CRDVD
KEYI	EMVSD

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If all PINS are encrypted in hardware, access to the KEYA and KEYI should be disabled through the BASE24 Security requester/server.

Access to the Transaction Security Services equivalents of these files will be through the new ACI Desktop Graphical User Interface that is set up separately from the BASE24 Pathway CRT access system. Refer to the BASE24 Classic 6.0 Installation Guide for further information.

Routing Changes

Not applicable

Network Configuration

To implement this enhancement, BASE24 must have access to a hardware security module. The enhancement supports the use of an Atalla or Thales e-Security (Racal) security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

The Transaction Security Services process runs as a satellite process under the XPNET process. It runs under control of the XPNET process for the purpose of process restart and recovery only. For normal transactions the XPNET process does not send transaction processing messages to the Transaction Security Services process. The same XPNET process control mechanisms available to other BASE24 processes can be used on the Transaction Security Services process. The process can be replicated as needed and can be started, stopped, etc., from a network control facility. Security utilities bound into BASE24 application processes communicate with the Transaction Security Services process using a direct write/read operation. They do not communicate with Transaction Security Services through the XPNET process. Refer to the BASE24 Classic 6.0 Installation Guide for details.

User Interface Server processes are required to support requests from ACI Desktop Graphical User Interface clients running on user workstations. A TCP/IP device, lines, and stations are required to interface GUI clients with the User Interface Server processes. Multiple lines and stations are required for multiple users. Service-based routing is used to route messages from GUI stations to multiple User Interface Server processes. All of the above processes, devices, lines, and stations must be configured in the Network Environment File (NEF) for the XPNET node. See the BASE24 Classic 6.0 Installation Guide for details.

A Timer process is required to set the expiration times for old keys in the Transaction Security Services database. This process must also be configured in the NEF for the XPNET node. See the BASE24 Classic 6.0 Installation Guide for details.

File Conversions

File conversions for the following Base files are supplied with the Transaction Security Services enhancement. There are no structural changes in any files, but Transaction Security Services files can be created and are populated from BASE24 files. Transaction Security Services data can also be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- KEYA
- KEYI

Note: Release 5.3 to release 6.0 conversions must be run on BASE24 files before the Transaction Security Services conversion programs are run. Transaction Security Services conversion programs must be run on release 6.0 files.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Transaction Security Services conversion programs must be run on release 6.0 version 2 files. All other version 1 to version 2 conversion programs must be run before the Transaction Security Services conversion programs are run.

<u>Key Authorization File (KEYA)</u>

The KEYA conversion program can be run to create IDESD, CRDVD and VPVVD records and a new KEYA. Encrypted keys in the new KEYA will be set to zeroes. The original KEYA is not modified. Running the conversion program is not required. IDESD, CRDVD and VPVVD records can be entered manually using the ACI Desktop Graphical User Interface instead.

The KEYA conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEYAS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#keyatss.t,name $keyam,nowait/
tsskeyam –force –list –listreplace
```

Users must update the TSSKEYAR obey file on the BA60UC05 subvolume with the appropriate information. TSSKEYAR must then be obeyed to execute the conversion program.

Due to a limitation in C++ libraries the largest year allowed in the IDESD, CRDVD and VPVVD records is 2035. KEYA records with an ending date larger that 2035 or with an ending date of all *s will have the ending date changed to 203512 in the corresponding IDESD, CRDVD and VPVVD records.

For more information on this file conversion, refer to the AAREAD05 and TSSKEYAR files on the BA60UC05 subvolume.

ICC Key File (KEYI)

The KEYI conversion program can be run to create EMVSD records and a new KEYI. Encrypted keys in the new KEYI are set to zeroes. The original KEYI is not modified. Running the conversion program is not required. EMVSD records can be entered manually using the ACI Desktop Graphical User Interface instead.

The KEYI conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEYIS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#keyitss.t,name $keyim,nowait/
tsskeyim –force –list –listreplace
```

Users must update the TSSKEYIR obey file on the BA60UC05 subvolume with the appropriate information. TSSKEYIR must then be obeyed to execute the conversion program.

Due to a limitation in C++ libraries the largest year allowed in EMVSD records is

2035. KEYI records with an ending date larger that 2035 or with an ending date of all *s will have the ending date changed to 203512 in the corresponding EMVSD records.

For more information on this file conversion, refer to the AAREAD05 and TSSKEYIR files on the BA60UC05 subvolume.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the ISO Host Interface and BIC ISO Interchange Interface processes.

<u>ISO Host Interface Process</u>

Host systems implementing triple DES encryption using the Transaction Security Services enhancement must make the necessary changes to support double-length keys and triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

<u>BIC ISO Interchange Interface</u>

Co-Networks implementing triple DES encryption using the Transaction Security Services enhancement must make the necessary changes to support double-length keys and triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

<u>Disk Impacts</u>

The KEYI file is not required if Transaction Security Services is implemented.

The KEYA file is not required if Transaction Security Services is implemented and all encryption/decryption is performed by an HSM.

The Transaction Security Services process requires several files. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for more details.

<u>Operational Impacts</u>

Previously, initial PIN selection has only been done in the clear. With Transaction Security Services, initial PIN selection in the clear does not work without the new SOFTWARE-PIN-VERIFICATION-ALWD param set to Y.

Event messages generated by Transaction Security Services components running on the Compaq NonStop system are sent to an EMS collector process and are displayed on the EMS log. The ACI Desktop Graphical User Interface can be used to view or modify Transaction Security Services event messages. Operators must be trained how to view and modify these messages.

New BASE24 event messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Performance Impacts

An Inter-Process Communication (IPC) message is added for each call to an HSM.

For transactions routed to an interchange, an additional disk I/O is required, since the PIN keys are stored in the ISECD. These keys defined in the ISECD can be shared by multiple Transaction Security Services processes. Keys are read from disk due to the frequency of key changes that can be done by any Transaction Security Services process.

Vendor Impacts

To implement this enhancement, BASE24 must have access to a hardware security module. The enhancement supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of software that is capable of supporting the Transaction Security Services enhancement.

- For Atalla HSMs, version 2.9 software and AKB version 1.0 are supported.
- For Thales e-Security (Racal) HSMs, the minimum version is 5.05.

Atalla AKB 1.0 HSMs require that all keys be stored in the database in the following format:

Header (8 bytes)	Encrypted Key (48 bytes)	MAC (16 bytes)

The AKB format protects stored keys against known attacks. The Header field contains eight one-byte values describing the attributes of the key. Before a key in AKB format is used in an Atalla Network Security Processor, the content of the header block is validated. The Encrypted Key field contains the encrypted key block. Single-length and double-length keys are padded before encryption to hide their length. The MAC field contains a message authentication code (MAC) of the header and encrypted key. The MAC value represents a MAC of the eight-byte header and the 48-byte key and is generated by an Atalla hardware security module (HSM) when the key is generated. This MAC value cryptographically binds the header and the encrypted key together. The header ensures the proper use of the key, the padding of the encrypted key block disguises the length of the key, and the MAC value ensures that the key block has not been tampered with. Because Atalla does not support both the old key block format and the new key block format in a single HSM, all keys must be converted to the new key block format at the same time.

The impacts of the new Atalla key block format extend beyond converting the database to accommodate larger key sizes. The actual key data must be converted from the old key block format to the new key block format. This impact is not directly related to the Triple DES mandate. The new Atalla AKB 1.0 key block format will even impact BASE24 customers who continue to use single DES.

When a BASE24 customer upgrades to a new Atalla HSM that supports the new Atalla AKB 1.0 key block format, all keys within the BASE24 database must be converted from the old key block format to the new key block format at the same time. In addition, if a BASE24 customer chooses to convert from single DES to triple DES when implementing the new Atalla HSM, new double/triple length keys must be generated.

Reference Documents

- Host Security Module RG7000 Programmers Manual (1270A514 Issue 3)
- Banking Command Reference Manual (424645-002 May 2001)
- Atalla Key Block Banking Command Reference Manual (TP522901-001, October 2001)

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Triple/Single DES Translation Support

Several BASE24 Release 6.0 interchange interfaces have been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 interchange interface and the interchange. These BASE24 Release 6.0 interchange interfaces run in all releases of BASE24.

The interchange interfaces enhanced to support triple to single DES transactions are the BASE24 Interface ISO (BIC ISO), Banknet ISO, MDS/Cirrus, MDS Maestro Gateway, VisaNet ISO, VisaNet Interlink Gateway, VisaNet/Plus Gateway, Link ISO and EPS-Net ISO interchange interfaces.

The ISO Host Interface (HISO) has been enhanced to support triple to single DES translations as well, but only for release 6.0.

A process specific param, DES-TRIPLE-SINGLE, has been added to each Release 6.0 interchange interface, indicating that the interface translates PINs from single DES encrypted inside BASE24 to triple DES encrypted when sent to the interchange and translates PINs from triple DES encrypted when received from an interchange to single DES encrypted inside BASE24.

If the DES-TRIPLE-SINGLE param indicates that triple DES encryption (3DEA) must be supported, a new version of the KEYF file (KEY6) must be used. The KEY6 supports double-length PIN encryption keys. The Intermediate keys remain single-length keys. The external key length has been added to this file. A new KEY6 DDL, requester and server have been created to support the KEY6.

Triple/Single DES Translation support invokes a new type of PIN translation. PINs can be translated from encryption under a double-length key (3DEA) to encryption under a single-length key (DEA) or from DEA encryption to 3DEA encryption. The special translation is done in three situations:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware
- External PIN encrypted in hardware to internal PIN encrypted in hardware

When an interchange interface receives a transaction from the interchange and encryption is performed by a hardware security module (HSM), the interchange interface checks the DES-TRIPLE-SINGLE LCONF param. If this LCONF param indicates that triple-to-single encryption is required, the application must call a new translation routine that translates the PIN to single DES encryption under a single-length intermediate key. Keys passed to and used by the new translation routine are the double-length KPE and the single-length encrypted intermediate key. The translated PIN and the encrypted intermediate key are placed in the STM/PSTM and then sent to the BASE24-atm Authorization process or BASE24-pos Router/Authorization process

When BASE24 acquires a transaction that must be sent to an interchange, the interchange interface checks the DES-TRIPLE-SINGLE LCONF param. If the encryption of the external PIN is performed by an HSM and the LCONF param indicates that single-to-triple translation is required, a new routine is called to translate the PIN from encryption under a single-length key to triple DES encryption under a double-length key. The translated PIN is sent to the interchange. Keys passed to and used by the new translation routine are the single-length encryption key from the STM/PSTM or intermediate key (if PIN is in the clear in the STM/PSTM, and the double-length KPE from the KEY6.

Each interchange that supports dynamic key management sends a double-length PIN encryption key in key change messages if the DES-TRIPLE-SINGLE LCONF param indicates double-length keys and triple DES encryption are required. This key is sent either in a new field or in an expanded existing key field, depending on the interchange interface.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

<u>KEY6</u>

The fully qualified file name of the KEY6 File (KEY6). The KEY6 file contains the double-length keys needed to perform triple DES encryption on PINs and keys.

If this assign is not present in the LCONF, the KEY6 file cannot be accessed by interchange interfaces. This assign is required when the DES-TRIPLE-SINGLE LCONF param is present and indicates triple DES encryption is to be performed. Initialization fails if the assign is missing when DES-TRIPLE-SINGLE indicates triple DES encryption.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications will be provided in the LCONFBA file located on the BA60MISC subvolume.

DES-TRIPLE-SINGLE

This optional param indicates whether an interchange interface supports triple DES encryption. This param is required if triple DES encryption is required. Single DES encryption is used if the param is missing or contains an invalid value. The default value for this param is N.

Database Configuration

Not applicable

New Data Files

KEY6 File (KEY6)

The KEY6 file contains the double-length keys needed to encrypt PINs and keys for Triple DES encryption. Keys must be loaded using the KEY6 requester.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the KEY6 requester/server.

PATHCONF Changes

The new KEY6 server must be defined in the PATHCONF file as SERVER-KEY6. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new KEY6 requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

To implement this enhancement, BASE24 must have access to a hardware security module. The enhancement supports the use of an Atalla or Thales e-Security (Racal) security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

File Conversions

There is no KEYF to KEY6 conversion, since the KEYF contains single-length keys and the KEY6 contains double-length keys.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the Super Extract, ISO Host Interface and BIC ISO Interchange Interface processes.

<u>Super Extract Process</u>

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the KEY6 files.

ISO Host Interface Process

Host systems implementing the Triple/Single DES Translate enhancement must make the necessary changes to support double-length keys and triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

BIC ISO Interchange Interface

Co-Networks implementing the Triple/Single DES Translate enhancement must make the necessary changes to support double-length keys and triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

Disk Impacts

The KEY6 impacts disk requirements. The total impact varies for each BASE24 user depending on the number of KEY6 records defined in the BASE24 system.

Operational Impacts

BASE24 operators must be aware that the new KEY6 file will appear on their BASE24 BASE Product Menu.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Performance Impacts

For those BASE24 users receiving transaction requests and/or advices from the interchange and the interchange sending triple DES encrypted PIN blocks, an additional I/O is required to the security device to translate the triple DES encrypted PIN block in the external message to the single DES encrypted PIN block that is sent in the internal message.

For those BASE24 users receiving transaction requests and/or advices from the interchange and the interchange sending single DES encrypted PIN blocks, no I/O is performed on the security device. The encrypted PIN block is simply moved from the external message to the internal message.

Vendor Impacts

To implement this enhancement , BASE24 must have access to a hardware security module. The enhancement supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of software that is capable of supporting the 3DEA enhancement.

- For Atalla, only version 2.84 software is supported.
- For the Thales e-Security (Racal), the minimum version is 5.05.

Reference Documents

- Host Security Module RG7000 Programmers Manual (1270A514 Issue 3)
- Banking Command Reference Manual (424645-002 May 2001

Section 5

BASE24-atm Release 6.0

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 5.3 or a previous version of release 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24-atm users, along with the associated conversion requirements and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion requirements and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes
- Device Handler Separation
- EMTs in BASE24 Classic
- Multiple Logical Network Routing Indicator
- TDF Expansion
- Transactions Allowed

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. Three user fields were added to the following DDL layouts for future product enhancements, future regional enhancements, and future CSMs. These fields were added now, so file conversions can be avoided with later enhancements and customer specific changes.
   - ATDD1
   - ATDS1

   The user fields were added to the core and device-specific portions of the ATDD1 and ATDS1. Refer to both the ATD DDL layout on the AT60DDL subvolume and the device specific DDL layout on the enhanced device handler subvolume (e.g., AT60N50.N50DDLS).

2. References to the TDF have been changed to ATD when navigating to BASE24 AFT subsystem screens and in Security Table entries. This change is necessary even if no enhanced device handlers are present in the network.

3. The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the BASE24-atm Transaction Log File (TLF) or reading up to the initially determined end-of-file of the TLF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following user fields were added to the ATDD1 and ATDS1 for future product enhancements, future regional enhancements, and future CSMs:

- USER-FLD-ACI is reserved for future BASE24 product use only. The designation of "product use only" provides ACI with the ability to deplete the number of bytes available within USER-FLD-ACI as product enhancements are introduced. When product enhancements require the addition of new fields within this file, the procedure to be followed is to deplete bytes from USER-FLD-ACI and use that number of bytes to define new fields. The new field definition(s) should precede the USER-FLD-ACI field.

- USER-FLD-REGN is reserved for ACI regional use only. Only ACI regions are allowed to use USER-FLD-REGN space.
- USER-FLD-CUST is reserved for BASE24 customer use only. Only customers are allowed to use USER-FLD-CUST space.

Note: The above fields allow space for future enhancements, and therefore they are not viewable using the BASE24 AFT subsystem. When a future enhancement uses a portion of any of the ACI reserved user fields, documentation on that enhancement will address the impacts to the BASE24 system.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

The BASE24 AFT subsystem security records that gave access to the TDF screens must be changed to refer to the ATD to gain access to ATD screens. The ATD requester and server object handles processing for the TDF as well as the ATD file. If the Security file conversion program is run, it changes all TDF references to ATD in the individual Security records. Refer to the subsection for the PI-Table enhancement in Section 4 for more information on the Security file conversion program.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

The three new user fields in the following files are initialized to spaces by the file conversion program for the TDF provided as part of the TDF Expansion enhancement.

- ATDD1
- ATDS1

Host/Co-Network Impacts

For additional information on Host/Co-Network Impacts with the ATDD1 and ATDS1 files, refer to the Device Handler Separation enhancement and the TDF Expansion enhancement.

User/Operational Impacts

Disk Impacts

The user fields in the ATDD1 and ATDS1 impacts disk requirements. The total impact varies for each BASE24-atm user depending on the number of receipt records defined in the BASE24-atm system.

NCR 5XXX Device Handler PIN Change Support

The NCR 5XXX Device Handler PIN Change functionality for Encrypted PIN Blocks has changed. Load group 24 contains the required states for Encrypted PIN Block support. Previously with load group 24, PIN Change transactions (encrypted PIN blocks), used PIN Entry state ‘B’ to solicit from the customer new PIN entries. Using state ‘B’ in this way required multiple requests to the ATM for the new PINs. Load group 24 has been changed to use the more efficient Customer Selectable PIN state ‘b’ for new PIN solicitation, which allows PIN Change transactions to complete in a single request. The NCR 5XXX Device Handler has been changed to support these changes to load group 24.

Customers should realize that some ATMs with older versions of NCR firmware may not support Customer Selectable PIN state ‘b’. In such cases, it will be necessary to either upgrade the ATM firmware, or forgo implementing this change until the firmware can be upgraded.

New DCT Hopper Mapping Table

The Hopper Mapping Table requester (AT60N50.RQHOPRS) has been added. The Hopper Mapping Table must reflect the contents of the device’s four hoppers. Currency Type, at present, must be set to 1, reflecting support for a single currency only at the device. Hopper values must reflect the contents of each hopper. The device can use this information to determine the validity of a particular transaction by comparing the requested amount with hopper contents via entry settings for state G. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for further information about the Hopper Mapping Table.

User Impacts

BASE24-atm users that extract their TLF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a TLF file.

The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the last daily extract of the TLF must be updated to have the value of Y for this field.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Device Handler Separation

The Device Handler Separation enhancement moved the device handler specific files and information to the device handler subvolumes. The following changes were made in release 6.0 for this enhancement:

- DDL layouts that are device specific (i.e., Device Information and Device Dependent Data) are now found on the appropriate device handler subvolume. The Device Dependent Data DDL layouts were already found on the device handler subvolumes.
- Device specific OMF Extract objects were created to accommodate the presence of the device specific DDL layouts that are now found on the appropriate device handler subvolumes.
- A requester and server object for device specific data located in the TDF are now found on the appropriate device handler subvolume.
- A requester and server object for device specific data located in the ATD files are now found on the appropriate device handler subvolume.
- A DCT requester/server pair that are device specific are now found on the appropriate device handler subvolume. The device specific DCT servers were already found on the device handler subvolumes.
- Device specific SNA modules are now found on the appropriate device handler subvolume.
- Make files (i.e., FM, MM, and M files) were changed to support the above changes.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>User Impacts</u>

BASE24 users must understand the concept of a separate device specific server objects for the TDF/ATD server. In addition, there are separate device specific objects for OMF extract. This concept already existed for the DCT server.

<u>Operational Impacts</u>

The DCT Menu is built for each BASE24 operator based on their SEC records. ACI recommends that BASE24 users give access via SEC records to the device-specific DCT screens for the device handlers that the user has purchased and that access is turned off to any other device-specific DCT screens. When the user turns off access to a device-specific DCT screen, it does not appear on the DCT Menu. If the user does not turn off access to device handlers that have not been purchased, the devices appear on the DCT Menu, but there are no corresponding device-specific DCT screens. When a BASE24 operator uses the NEW PAGE function key to access a screen that does not exist in the BASE24 system, the DCT Menu continues to be displayed, and the operator receives the following error message:

```
ERROR ON THE REQUESTER CALL. TERMINATION STATUS:            0020
```

Vendor Impacts

Not applicable

Reference Documents

Not applicable

EMTs in BASE24 Classic

This enhancement is being supported by the enhanced BASE24-atm Device Handlers, BASE24-atm Authorization, and the enhanced Interchange Interface processes.

The following BASE24-atm enhanced Device Handler processes support this enhancement:

- Diebold 10XX/478X
- NCR 5XXX

For more information on this enhancement, refer to section 4 (Base).

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

<u>RCPTEMT</u>

The fully qualified file name of the BASE24-atm Receipt file Extended Memory Table file (RCPTEMT). This assign is used by the RCPTEMT Build utility object and the enhanced BASE24-atm Device Handlers.

If this assign is not present in the LCONF, the Extended Memory Table Build utility and the enhanced BASE24-atm Device Handlers do not initialize.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

RCPTEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the RCPTEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

RCPTEMT-WARM*identifier*

A param containing the symbolic name of a process to be warmbooted for the RCPTEMT. Processes that can be warmbooted for the RCPTEMT are the enhanced BASE24-atm device handler processes. This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all RCPTEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each RCPTEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another RCPTEMT-WARM*identifier* param. For example, RCPTEMT-WARM01 could contain the P1A^NCR^5XXX^01 process and RCPTEMT-WARM02 could contain the P1A^NCR^5XXX^02 process.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Disk Impacts

The RCPTEMT impacts disk requirements. The total impact varies for each BASE24-atm user depending on the number of receipt records defined in the BASE24-atm system.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multiple Logical Network Routing Indicator

This section describes the release 6.0 changes and implementation requirements associated with the Multiple Logical Network Routing Indicator enhancement.

These changes and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Multiple Logical Network Routing Indicator enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

MULT-LN-ACQ-TRAN-COUNTER

Indicates that transactions routed between two logical networks using CPF SPROUTE routing should be included in the count for the reports. A value of Y indicates to count the corresponding transactions. A value of N indicates to not count the corresponding transactions. The default is Y.

Database Configuration

Token File (TKN)

The following BASE24 token has been added for the Multiple Logical Network Routing Indicator enhancement:

- The Multiple LN Token (Token ID “BK”) contains the logical network where the transaction was acquired.

If the new BASE24 token for this enhancement is not to be logged to the TLF, the appropriate token records must be configured in the TKN.

If the new BASE24 token for this enhancement is to be extracted with the TLF, the appropriate token records must be configured in the TKN.

If the destination is another BASE24 system, this token should not be configured to be sent in the ISO message using HISO or BIC ISO.

New Data Files

Not applicable

Template Changes

COBNAMES

New Multiple LN token names were added to the token table in the COBNAMES files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The newest version of SKEL must be used in order for this enhancement to be fully functional.

SCR #3000 SKELB - rel3^ver0^sutl13^020112

File Conversions

Not applicable

Host/Co-Network Impacts

BIC ISO Interchange Interface Process

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

Super Extract Process

If the new Multiple LN token for this enhancement is to be included in the TLF, BASE24 users must consider this additional data in host programs that use the data from the log file.

ISO Host Interface Process Impacts

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

User/Operational Impacts

Disk Impacts

The TLF requires additional disk space because the new Multiple LN token for this enhancement can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Transaction Based Pricing (TBP) Reports

The ATM TBP reports have been updated to use the new Multiple LN token. This will allow the reports to properly count transactions that were SPROUTE routed using a CPF on another logical net. Users need to be on Version 7 or greater of the TBP Reports to take advantage of this change.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

TDF Expansion

This section describes the BASE24-atm Release 6.0 changes and conversion/migration requirements associated with the TDF Expansion enhancement.

Section 2 contains a functional overview of the TDF Expansion enhancement.

This enhancement is only being supported by the following enhanced BASE24-atm Device Handlers.

- Diebold 10XX/478X
- NCR 5XXX

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

<u>ATM-ATD-DYN-GNRL</u>

The fully qualified file name of the BASE24-atm Terminal Data Dynamic file - general data. This is a required assign for the enhanced device handlers.

<u>ATM-ATD-DYN-SCRATCH-PAD</u>

The fully qualified file name of the BASE24-atm Terminal Data Dynamic file – scratch pad data. This is an optional assign. If the BASE24-atm user wishes to store token data in a file other than the TLF for reversal processing, ensure that this assign is present in the LCONF. Otherwise, ensure that this assign is not present in the LCONF.

<u>ATM-ATD-STATIC-GNRL</u>

The fully qualified file name of the BASE24-atm Terminal Data Static file - general data. This is a required assign for the enhanced device handlers.

ATM-DH-ATD-EXTMEM-SWAPVOL

The fully qualified file name of the swap volume in which the device handlers temporary store data when building extended memory for a terminal's ATD records. This is an optional assign. If the assign is not found, Compaq NSK chooses the swap volume for the application.

TDF

This assign is not used by the enhanced device handlers. If the BASE24-atm user only supports these device handlers, delete this assign from the LCONF. If the BASE24-atm user still supports a BASE24-atm non-enhanced device handler, do not delete this assign from the LCONF.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-DH-DYN-TBL-RETN-PRD-DIAL-UP

This param specifies the number of seconds the device handler waits for the next leg of a multiple leg transaction before the dynamic data for a terminal connected to a dial-up line may be removed from the dynamic data table. A value of zero disables this processing.

If a customer performs a multiple leg transaction, the device handler retains the dynamic data for the terminal until all legs of the transaction are complete. For the NCR 5XXX Device Handler, the effected transactions are administrative, surcharging, NCD, statement print, and LINK surcharging. For the Diebold 10XX/478X Device Handler, the effected transactions are surcharging, NCD, and LINK surcharging. If the customer cancels the transaction, the device does not notify the device handler about the cancellation (i.e., normal processing), and the terminal's data will be left in extended memory. To avoid a Dynamic Data Table full errors, the device handlers will search for terminals in the Dynamic Data Table where dynamic data meets the following criteria: a multiple leg transaction with a timestamp that exceeds the allowable retention period. If a terminal's dynamic data exceeds this limit, the dynamic data for the terminal will be removed from extended memory and replaced with the dynamic data for the terminal whose transaction was just initiated.

This is an optional param. If the param is not found or has a value greater than 32677, the default value of 60 is used.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: If the BASE24-atm user supports one of the enhanced device handlers, add this param to the LCONF for each logical network.

ATM-DH-DYN-TBL-RETN-PRD-LEASE

This param specifies the number of seconds the device handler waits for the next leg of a multiple leg transaction before the dynamic data for a terminal connected to a leased line may be removed from the dynamic data table. A value of zero disables this processing.

If a customer performs a multiple leg transaction, the device handler retains the dynamic data for the terminal until all legs of the transaction are complete. For the NCR 5XXX Device Handler, the effected transactions are administrative, surcharging, NCD, statement print, and LINK surcharging. For the Diebold 10XX/478X Device Handler, the effected transactions are surcharging, NCD, and LINK surcharging. If the customer cancels the transaction, the device does not notify the device handler about the cancellation (i.e., normal processing), and the terminal's data will be left in extended memory. To avoid a Dynamic Data Table full errors, the device handlers will search for terminals in the Dynamic Data Table where dynamic data meets the following criteria: a multiple leg transaction with a timestamp that exceeds the allowable retention period. If a terminal's dynamic data exceeds this limit, the dynamic data for the terminal will be removed from extended memory and replaced with the dynamic data for the terminal whose transaction was just initiated.

This is an optional param. If the param is not found or has a value greater than 32677, the default value of 60 is used.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: If the BASE24-atm user supports one of the enhanced device handlers, add this param to the LCONF for each logical network.

ATM-DH-MAX-IN-CORE

This param is not used by the enhanced device handlers. If the BASE24-atm user only supports these device handlers, delete this param from the LCONF. If the BASE24-atm user still supports a BASE24-atm non-enhanced device handler, do not delete this param from the LCONF.

ATM-DH-MAX-TERMS

This param specifies the maximum number of ATD record keys (i.e., terminals) that can be saved in extended memory. Thus, this param indicates the maximum number of terminals each module can manage. This param has the same purpose as the

ATM-DH-MAX-TDFS param that was used by this device handler in previous releases.

This is an optional param. If the param is not found, has the value 0 or a value greater than 2147483647, the default value of 100 is used.

ATM-DH-MAX-TERMS-IN-DYN-TBL

This param specifies the maximum number of ATD records each device handler saves in their dynamic data table located in extended memory. More than one dynamic data record can exist for a terminal ID, and the terminal's last message data, including the token data, is saved in this dynamic data table. Thus, this param indicates the number of terminals expected to ever be processing transactions simultaneously. This param has the same purpose as the ATM-DH-MAX-IN-CORE param that was used by this device handler in previous releases.

This param is an optional param. If this param is not found, has the value 0 or a value greater than ATM-DH-MAX-TERMS, the default value is either the value of ATM-DH-MAX-TERMS or 25 (the lower of the two).

The maximum number of terminals in the dynamic data table cannot exceed the number of timers allowed in the device handler's timer table.

ATM-DH-MAX-TERMS-IN-STATIC-TBL

This param specifies the maximum number of ATD records each device handler saves in their static data table located in extended memory. More than one static data record can exist for a terminal ID.

This param is an optional param. If this param is not found, has the value 0 or a value greater than ATM-DH-MAX-TERMS, the default value is either the value of ATM-DH-MAX-TERMS or 25 (the lower of the two).

ATM-DH-MAX-TDFS

This param is not used by the enhanced device handlers. If the BASE24-atm user only supports these device handlers, delete this param from the LCONF. If the BASE24-atm user still supports a BASE24-atm non-enhanced device handler, do not delete this param from the LCONF.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

BASE24-atm Terminal Data files (ATD)

Field Name: TOKEN RETRIEVAL OPTION
Screen: ATD Screen 2
DDL Name: ATDS1.[atds1-core].TKN-RETRV-FLG

When the device handler is processing a reversal, this field indicates where the token data should be retrieved. Valid values are:

0 = None

No tokens included in the STM

1 = ATD

Token data retrieved from the BASE24-atm Terminal Data Dynamic file – scratch pad data, and appended to the STM

2 = TLF

Token data retrieved from the TLF, and appended to the STM

9 = Use IDF (this is the default value)

The token retrieval option is specified at the institution level. Refer to the BASE24-atm segment of the IDF.

Institution Definition File (IDF)

Field Name: TOKEN RETRIEVAL OPTION
Screen: IDF Screen 13
DDL Name: IDF.ATM.TKN-RETRV-FLG

When the device handler is processing a reversal, this field indicates where the token data should be retrieved. Valid values are:

0 = None

No tokens included in the STM

1 = ATD

Token data retrieved from the BASE24-atm Terminal Data Dynamic file – scratch pad data, and appended to the STM

2 = TLF (this is the default value)

Token data retrieved from the TLF, and appended to the STM

Token file (TKN)

The following new BASE token has been added for this enhancement:

- The TLF Token (Token ID “B7”) contains the TLF file name, the RBA, and alternate key data of the record written to the TLF. This data is used by device handlers and AUTH to build a reversal message when the last transaction data is no longer known by the device handler.

If the new token for this enhancement is not to be logged to the TLF or ILF, the appropriate token records must be configured.

If the new token for this enhancement is to be extracted with the TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

This section describes the files that have been added to support this enhancement.

BASE24-atm Terminal Data Dynamic file – general data (ATDD1)

The ATDD1 was created for storing dynamic data for each ATM. Refer to the LCONF Assign Changes and LCONF Param Changes subsections of this enhancement for further information about this file.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of ATD.

BASE24-atm Terminal Data Dynamic file – scratch pad data (ATDD2)

The ATDD2 was created for storing scratch pad data for each ATM. This is an optional file for BASE24-atm users wishing to store token data in a file other than the TLF for reversal processing. Refer to the LCONF Assign Changes subsection of this enhancement for further information about this file.

This file is only accessed by the device handlers, and it is not accessible via the BASE24 AFT subsystem.

BASE24-atm Terminal Data Static file – general data (ATDS1)

The ATDS1 was created for storing static data for each ATM. Refer to the LCONF Assign Changes and LCONF Param Changes subsections of this enhancement for further information about this file.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of ATD.

Template Changes

BASE24-atm Transaction Log File (TLF)

BASE24-atm supports two different alternate keys on the BASE24-atm Transaction Log File (i.e., terminal ID, primary account number). The user has the option to configure which alternate keys are required in their particular environment. A DDL compile is no longer used to create the FUP output for the TLF. ACI provides BASE24-atm users with a standard template file at installation. This template file name TPLTTLF is located on the AT60MISC subvolume. This file can be used as is or the BASE24 user can alter the file to meet their requirements.

If the BASE24-atm users elect not to use the primary account number alternate key on their TLFs, reversal processing is adversely impacted when the Token Retrieval Option of 2 is specified in the ATD or the IDF.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The ATD requester name was added to MEGATBL file, since RQMEGA communicates with the ATD requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

The TDF references in the SECTBL file were changed to reference the ATD.

PATHCONF Changes

Delete the TDF Server (i.e., SERVER-TDF) from the PATHCONF file. The TDF Server can be deleted online through the PATHCOM utility, but it must also be deleted from the PATHCONF, so that any future PATHCOLD operation would not contain the TDF Server.

The functionality of the TDF Server is now contained within the ATD Server object that is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

CRT Access Security Changes

The BASE24 AFT subsystem security records that gave access to the TDF screens were changed to refer to the ATD to gain access to ATD screens. The ATD requester and server handle processing for the TDF as well as the ATD files.

When the Security file conversion program is run, it changes all TDF references to ATD in the individual Security records. Refer to the subsection for the PI-Table Enhancement in Section 4 for more information on the Security file conversion program.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Data files (ATD)

The ATD conversion program must be run to convert TDF records that exist for the Diebold 10XX/478X and NCR 5XXX ATMs (i.e., terminal type equals “30” and “22” respectively) into the new ATD records. In addition, this same conversion program copies those TDF records that are not for the two ATM types into a new TDF file. Users must create the ATD files and a new TDF file using the ATDDLFUP file.

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVATDS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdcnv.t,name $atdm,nowait/
cnvatdm –force –list –listreplace
```

Users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. CNVATDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host/Co-Network impacts are incurred by those BASE24 users implementing the BASE24-atm TDF Expansion enhancement. Modifications to existing files and processing impacts the Super Extract, ISO Host Interface, and BIC ISO Interchange Interface processes.

Super Extract Process

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process information in the ATD files.

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process the new information defined in the IDF.

New fields were added to the IDF. If the BASE24-atm user performs processing on the extracted IDF, they must be aware of the record format changes.

If the new TLF Token is included in the TLF and ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

ISO Host Interface Process

If the new TLF Token is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

BIC ISO Interchange Interface Process

If the new TLF Token is included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space as the new token for this enhancement (i.e., TLF Token) can be included in the transaction records logged to these files. The total impact varies for each BASE24-atm user depending on the transaction volume in the BASE24-atm system.

The terminal data for the non-enhanced device handlers remains in the TDF. The record size for these terminals does not change. The terminal data for the enhanced device handlers is maintained in the new ATD files. The first ATD file (i.e., ATDD1) consists of records with dynamic data for each terminal. Another ATD file (i.e., ATDS1) consists of records with static data for each terminal. A third ATD ATD file (i.e., ATDD2) is optional. This file consists of token information from the last message the device handler processed, and is only used when the BASE24 user has configured BASE24-atm to use token data from the ATDD2 for reversal processing. The maximum length for these ATD records is 4062. The total impact varies for each user depending on the number of enhanced terminals defined in the BASE24-atm system. The terminal data for the enhanced device handlers was removed from the TDF.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Operational Impacts

BASE24 users must be aware that the TDF data for the non-enhanced devices is displayed along with the ATD data. The BASE24 AFT subsystem File Destination for both the ATD data and the TDF data is “ATD “, and the BASE24 AFT subsystem File Destination “TDF “is no longer supported.

BASE24 users must perform a warmboot of the ATD static data using the new BASE24 DELIVER PROCESS text command ‘9529ATD’ after an operator performs file maintenance on any ATD static field. This is necessary since static data can not be read from disk during online transaction processing. This command is only supported via the BASE24 AFT NCS screen or NCPCOM. For more details, see the 9529ATD command in the BAS24-atm Process Commands section of the ***BASE24 Text Command Reference Manual***.

BASE24 users can determine the current status of the ATD extended memory tables using the new BASE24 DELIVER PROCESS text command ‘STATUSEXTMEM’. This command is only supported via the BASE24 AFT NCS screen or NCPCOM. For more details, see the STATUSEXTMEM command in the BASE24-atm Process Commands section of the ***BASE24 Text Command Reference Manual***.

The BASE24 DELIVER PROCESS text command RESET TDF DATES was changed to be the BASE24 DELIVER PROCESS text command RESET ATD DATES command. This command changes the posting date in the ATDD1 records as well as the TDF to the date specified in the command. For more details, see to the RESETATDDATES command in the BASE24-atm Process Commands section of the ***BASE24 Text Command Reference Manual***.

BASE24 users must configure the ATD and/or IDF to indicate whether to save token data and, if so, from where the token data should be retrieved, ATDD2 or TLF. The token data is only retrieved from the ATDD2 or the TLF if this data is no longer in extended memory.

BASE24 users must be aware that if the token data is configured to be retrieved from the TLF, any token data not logged to the TLF will not exist in the STM on the reversal. This can adversely affect reversal processing.

The values for the new LCONF params ATM-DH-MAX-TERMS, ATM-DH-MAX-TERMS-IN-DYN-TBL, and ATM-DH-MAX-TERMS-IN-STATIC-TBL must be determined due to memory management requirements for this enhancement. Refer to the following calculation for the amount of memory that is used by each device handler:

```
8298 +
(ATM-DH-MAX-TERMS’s value * 138) +
(ATM-DH-MAX-TERMS-IN-DYN-TBL’s value * 8048)+
(ATM-DH-MAX-TERMS-IN-STATIC-TBL’s value * 4062)
```

If the ATDD2 file is utilized, increase the above sum by the following:

```
(ATM-DH-MAX-TERMS-IN-DYN-TBL’s value * 4062)
```

For example, if the values of the LCONF params ATM-DH-MAX-TERMS, ATM-DH-MAX-TERMS-IN-DYN-TBL, and ATM-DH-MAX-TERMS-IN-STATIC-TBL are 100, 50, and 90 respectively, the sum is 790078 (i.e., 8298+100*138+50*8048+90*4062). If the ATDD2 file is utilized, the sum is 993178 (i.e., 790078+50*4062).

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The structure and routines that maintain the terminals in extended memory has been enhanced, but the amount of memory required for these structures is larger than the version 1 structures.

Delete the LCONF TDF assign, ATM-DH-MAX-IN-CORE param, and ATM-DH-MAX-TDFS param if the BASE24-atm user has only the BASE24-atm enhanced device handlers.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Transactions Allowed

The Transactions Allowed enhancement changes the mechanism to configure allowed transactions for the BASE24 Authorization processes, the enhanced device handlers, and the enhanced interchange interfaces.

Transactions Allowed (Terminal Receipt Processing)

The BASE24-atm enhanced device handlers use information in a new file called the Terminal Receipt File (TRF). This file is used to define the approved and denied receipt print options for a specified Terminal Receipt Profile, Message Category, and ISO processing code. This file is accessed instead of the hard-coded receipt tables that currently exist in the TDF.

For more information on this enhancement, refer to section 4 (Base). In addition to the impacts identified in section 4 (Base), the following items affect BASE24-atm.

Note: This enhancement is only being supported in the following enhanced BASE24-atm Device Handlers.

- Diebold 10XX/478X
- NCR 5XXX

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-TRF

The fully qualified file name of the BASE24-atm Terminal Receipt File (TRF). This assign is used by the TRFEMT Build utility object.

If this assign is not present in the LCONF, the Extended Memory Table Build utility does not initialize.

ATM-TRFEMT

The fully qualified file name of the BASE24-atm Terminal Receipt File Extended Memory Table file (TRFEMT). This assign is used by the TRFEMTB Build utility object and the enhanced BASE24-atm Device Handlers.

If this assign is not present in the LCONF, the Extended Memory Table Build utility and the enhanced BASE24-atm Device Handlers do not initialize.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-TRFEMT-MAX-TBL-SIZE

The maximum size (in megabytes) of the TRFEMT. This param is used to set the amount of memory allocated by the Extended Memory Table Build utility for the table. The maximum size allowed for the table is the default value of 127 MB.

ATM-TRFEMT-WARM*identifier*

A param containing the symbolic name of a process to be warmbooted for the TRFEMT. Processes that can be warmbooted for the TRFTEMT are the enhanced BASE24-atm device handler processes. This param indicates a process that receives an EMT warmboot command from the Device Control Terminal (DCT). The Device Control Terminal Server process retrieves all TRFEMT-WARM*identifier* params when a network control operator issues a command to reinitialize this specific EMT. A maximum of 500 processes can be EMT warmbooted from the DCT.

Each ATM-TRFEMT-WARM param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number, but can be any unique set of characters that has not already been declared by another ATM-TRFEMT-WARM*identifier* param. For example, ATM-TRFEMT-WARM01 could contain the P1A^NCR^5XXX^01 process and ATM-TRFEMT-WARM02 could contain the P1A^NCR^5XXX^02 process.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

BASE24-atm Terminal Data Files (ATD)

Field Name: ACQUIRER TXN PROFILE
Screen: ATD Screen 2
DDL Name: ATDS1.[atds1-core].ACQ-TKN-PRFL

The transaction profile for the acquirer who initiated the transaction. The acquirer transaction profile allows one or more acquirers to share processing code definitions and transactions allowed information. The value in this field overrides the acquirer transaction profile defined at the institution level in the IDF. The value entered in this field must match a transaction profile named in the Acquirer Processing Code File (APCF).

Cardholder Authorization File (CAF)

Field Name: ISSUER TXN PROFILE
Screen: CAF Screen 8
DDL Name: CAF.ATM.ISS-TKN-PRFL

A code identifying a group of BASE24-atm issuer transaction processing codes allowed for this cardholder in the Issuer Processing Code File (IPCF). The value in this field overrides the issuer transaction profile defined at the card prefix level in the CPF or at the institution level in the IDF. The value entered in this field must match a transaction profile named in the Issuer Processing Code File (IPCF).

Card Prefix File (CPF)

Field Name: ISSUER TXN PROFILE
Screen: CPF Screen 4
DDL Name: CPF.ATM.ISS-TKN-PRFL

A code identifying a group of BASE24-atm issuer transaction processing codes allowed for this card prefix in the Issuer Processing Code File (IPCF). The value in this field overrides the issuer transaction profile defined at the institution level in the IDF. The value entered in this field must match a transaction profile named in the Issuer Processing Code File (IPCF).

Institution Definition File (IDF)

Field Name: ACQUIRER TXN PROFILE
Screen: IDF Screen 9
DDL Name: IDF.ATM.ACQ-TKN-PRFL

A code identifying a group of default BASE24-atm transaction processing codes allowed for not-on-us cardholders at ATMs owned by this institution. This profile applies only if your BASE24-atm system uses the Diebold 10XX/478X- or NCR 5XXX Device Handler processes. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files (ATD). The value entered in this field must match a transaction profile named in the Acquirer Processing Code File (APCF).

Field Name: ISSUER TXN PROFILE
Screen: IDF Screen 9
DDL Name: IDF.ATM.ISS-TKN-PRFL

A code identifying a group of default BASE24-atm issuer transaction processing codes allowed for this institution's cardholders. The value in this field can be overridden at the card prefix level in the CPF or at the cardholder level in the CAF. The value entered in this field must match a transaction profile named in the Issuer Processing Code File (IPCF).

New Data Files

This section describes the files that have been added to support this enhancement.

BASE24-atm Terminal Receipt File (TRF)

The TRF was created for storing the receipt profile, message category, and processing code combination for each ATM.

ACI provides BASE24 users with a set of default TRF records at installation. The records are located in the TRF file on the AT60MISC subvolume and are also documented in the ***BASE24-atm Files Maintenance Manual***. This file can be duplicated in its entirety to the appropriate data files subvolume or Compaq NSK's FUP can be used to copy the default records to the user's TRF file. BASE24 users should then access the TRF records via standard BASE24 file maintenance and update the TRF records to reflect their own environment.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of TRF.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The TRF requester name was added to MEGATBL file, since RQMEGA communicates with the TRF requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the TRF requester/server.

PATHCONF Changes

The TRF Server object is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

CRT Access Security Changes

Access to the new TRF requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impacts the Super Extract Process.

Super Extract Process

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process information in the TRF file.

If the new tokens for this enhancement (i.e., Transaction Profile Token, Transaction Description Token, Acquirer Routing Token) are to be included in the TLF and ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TRFEMT impacts disk requirements. The total impact varies for each BASE24-atm user depending on the number of receipt records defined in the BASE24-atm system.

Operational Impacts

BASE24 operators must be aware that the new TRF file appears on their BASE24-atm Product Menu.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with Release 6.0, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Dassault Device Handler
- Dial-Up ATM Support for Diebold 10XX/478X and NCR 5XXX Devices
- EMV Support
- Format 2 File Support
- Interactive Response Notification Fee Processing and Negative Surcharge Support
- Multiple Account Select by Qualifier
- Multiple Currency Support
- Tidel Device Handler
- Transaction Security Services
- Triple/Single DES Translate Support
- Triton Device Handler

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 changes, conversion requirements, and implementation requirements for future reference.

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Dassault Device Handler

BASE24-atm has been enhanced to support Dassault ATMs. The BASE24-atm Dassault Device Handler module provides the interface between the device and Compaq NSK. The Dassault ATMs are defined in the BASE24-atm Terminal Data File (TDF). This device handler was first introduced in BASE24-atm Release 4.0.

ACI Worldwide, Inc.

Dial-Up ATM Support for Diebold 10XX/478X and NCR 5XXX Devices

This section describes the release 6.0 changes, conversion, and implementation requirements associated with the Dial-Up ATM Support for Diebold 10XX/478X and NCR 5XXX Devices enhancement.

The Diebold 10XX/478X and the NCR 5XXX Device Handler processes were enhanced to support all existing features and functionality in a dial-up environment. This enhancement relates to processing of ATM connections using a dial-up protocol. Existing processing involving dedicated leased line ATM connections has not been impacted by this enhancement.

This enhancement allows BASE24 users to take advantage of the XPNET X.25 POSII end-to-end protocol support for Message Failure 106 (no ACK received). Minimum version requirements for the X25NI and POS2 XPNET protocol modules are XPNET 3.0 rel3^ver0^x2504^990707 and rel3^ver0^pos205^990707.

If the Async VISA II protocol is used, XPNET SCR #2786 is required for this enhancement. Minimum version requirement for the XPNET Async Visa protocol modules is XPNET 3.0 rel3^ver0^visa04^991006. This XPNET SCR allows suppression of the automatic disconnect performed by XPNET when a CP completion message is received with the Async Visa II protocol. Historically, a CP completion is the last message in a communication sequence and XPNET automatically disconnects the line. However the communications sequences in the dial-up ATM environment allow additional messages after the CP. For any dial-up ATM devices using the Async Visa II protocol, the TYPE attribute on the corresponding XPNET DEVICE object must be set to a value of 255. This suppresses the automatic disconnect for CP messages. Any other value causes the automatic disconnect to remain in effect.

These changes, conversion, and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Dial-Up ATM Support for Diebold 10XX/478X and NCR 5XXX Devices enhancement.

CUSTMACS Changes

The following Make flag was added to the CUSTMACS file that indicates whether the NCR 5XXX Device Handler supports the Dial-Up Support for Diebold 10XX/478X and NCR 5XXX Devices enhancement:

- atm_dh_n50_dlup_on

The following Make flag was added to the CUSTMACS file that indicates whether the Diebold 10XX/478X Device Handler supports the Dial-Up Support for Diebold 10XX/478X and NCR 5XXX Devices enhancement:

- atm_dh_c1000_dlup_on

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

<u>ATM-ADCF</u>

The fully qualified file name of the BASE24-atm Dial-up Command File. This assign is required if the BASE24-atm user desires dial-up processing for their ATMs.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

<u>ATM-DH-MAX-QUEUED-CMD</u>

A param used by the Diebold 10XX/478X and NCR 5XXX Device Handlers. This param specifies the maximum number of commands that can be queued to a terminal. This is an optional param. If the param is not found, has the value 0 or a value greater than 32767, the default value of 20 is used.

<u>ATM-DH-MAX-LOAD-CMD</u>

A param used by the Diebold 10XX/478X and NCR 5XXX Device Handlers. This param specifies the maximum number of ADCF (dial-up) LOAD commands that can be processed by the device handler at one time. This is an optional param. If the param is not found, has the value 0 or a value greater than 32767, the default value of 30 is used.

<u>ATM-FEE-INTRACTV-RESP-NTFY</u>

An optional param used by the Diebold 10XX/478X and NCR 5XXX Device Handlers. This param indicates whether interactive response notification fee processing or standard interactive request notification surcharge processing is being performed.

For more information on the interactive response notification fee enhancement, refer to the “Interactive Response Notification Fee Processing and Negative Surcharge Support” subsection in this section.

Database Configuration

This section describes the fields in the files that have been modified to support this enhancement. The settings of each field depends on the configuration desired by the institution.

BASE24-atm Terminal Data Files (ATD)

Field Name: DIAL-UP
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].DIAL-UP-FLG
ATDS1.[atds1-dev-info-ncr-5xxx].DIAL-UP-FLG

This field identifies whether the terminal is a dial-up ATM.

Field Name: MSG-FAIL
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].DIAL-UP-MSG-FAIL
ATDS1.[atds1-dev-info-ncr-5xxx].DIAL-UP-MSG-FAIL

This field specifies if the terminal requires a reversal to be generated for message failures (i.e., 106 – no ACK received). Usage occurs where message failures would normally result in a reversal.

Field Name: DOWNLOAD SURCHARGE AMT
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].SURCG-AMT-FLG
ATDS1.[atds1-dev-info-ncr-5xxx].SURCG-AMT-FLG

This field identifies the surcharge amount to be downloaded to the device that is displayed to the cardholder on a non-interactive surcharge transaction.

Field Name: INACTIVITY TIME-OUT (WORKDAY)
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].INTERTRANSACTION.
WRKDAY-PEAK-INTERVAL
ATDS1.[atds1-dev-info-ncr-5xxx]. INTERTRANSACTION.
WRKDAY-PEAK-INTERVAL

The amount of time, in minutes, to use when setting the inactivity timer during the workday peak time. For peak inactivity timer processing to occur, this field must contain a value greater than 000 when the WORKDAY PEAK TIME starting time field and the WORKDAY PEAK TIME ending time field both contain values greater than zero. Valid values are 000 through 999.

The input values for this field have been expanded from two positions to three positions.

Field Name: INACTIVITY TIME-OUT (WEEKEND/HOLIDAY)
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].INTERTRANSACTION.
WKND-PEAK-INTERVAL
ATDS1.[atds1-dev-info-ncr-5xxx]. INTERTRANSACTION.
WKD-PEAK-INTERVAL

The amount of time, in minutes, to use when setting the inactivity timer during the weekend/holiday peak time. The Device Handler process uses the holiday table from the Institution Definition File (IDF) record (screen 4) when setting and processing the inactivity timers for holiday periods.

The input values for this field have been expanded from two positions to three positions.

Field Name: NON-PEAK INACTIVITY TIME-OUT
Screen: ATD Screen 7
DDL Name: ATDS1.[atds1-dev-info-diebold-10xx].INTERTRANSACTION.
NON-PEAK-INTERVAL
ATDS1.[atds1-dev-info-ncr-5xxx]. INTERTRANSACTION.
NON-PEAK-INTERVAL

The amount of time, in minutes, to use when setting the inactivity timer during the non-peak time (not in either of the defined peak periods). This field is required when workday or weekend/holiday peak inactivity timer processing is configured. Valid values are 000 through 999.

The input values for this field have been expanded from two positions to three positions.

New Data Files

This section describes the files that have been added to support this enhancement.

BASE24-atm Dial-up Command File (ADCF)

The ADCF was created for this enhancement that contains the ATM dial-up command queue.

This file is only accessed by the device handlers, and it is not accessible via the BASE24 AFT subsystem.

Template Changes

Not applicable

Device Configuration

The CNFGIX (Configuration File for the Diebold 10XX/478X) and CNFG5070 (Configuration File for the NCR 5XXX) files were modified to contain new load groups for this enhancement.

Diebold 10XX/478X Configuration File

The load group (305) has been added to the CNFGIX file to be used for the dial-up ATM non-interactive surcharge processing for withdrawal transactions. The new “fee notice” screen is displayed when a withdrawal transaction is attempted and the terminal is configured for non-interactive surcharging.

The load group (306) has been added to the CNFGIX file to be used for the dial-up ATM non-interactive surcharge processing for fastcash transactions. This load group must be loaded on top of group 5 (Fastcash group).

The load group (310) has been added to the CNFGIX file to be used for the dial-up ATM interactive response notification fee processing. This load group is loaded on top of group 0. For additional information on the interactive fee response notification enhancement, refer to the “Interactive Response Notification Fee Processing and Negative Surcharge Support” subsection in this section.

NCR 5XXX Configuration File

The load group (305) has been added to the CNFG5070 file to be used for the dial-up ATM non-interactive surcharge processing for withdrawal transactions. The new “fee notice” screen is displayed when a withdrawal transaction is attempted and the terminal is configured for non-interactive surcharging.

The load group (306) has been added to the CNFG5070 file to be used for the dial-up ATM non-interactive surcharge processing for fastcash transactions. This load group must be loaded on top of load group 305.

The load group (307) has been added to the CNFG5070 file to be used for the dial-up ATM interactive surcharge processing.

The load group (310) has been added to the CNFG5070 file to be used for the dial-up ATM interactive response notification fee processing. This load group must be loaded on top of group 0. For additional information on the interactive response notification fee enhancement, refer to the “Interactive Response Notification Fee Processing and Negative Surcharge Support” subsection in this section.

The load group (501) has been added to the CNFG5070 file to be used for the dial-up NCD processing.

The load group (502) has been added to the CNFG5070 file to be used for the dial-up NCD non-interactive surcharging processing. This load group must be loaded on top of load group 501.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

X.25 Protocol Diagram

```
| |                | |           |                 | |
| |                | +-+   +-+ |                 |X|
|D| <--’call’ 1--> | | |   | | |                 |.| <--call 1-->ATM 1
| |                | | | X | | |                 |2|
|H|                | |P|   |X| |                 |5|
| | <--’call’ 2--> | |O| P |.| |                 | | <--call 2-->ATM 2
| |                | |S|   |2| |     (leased)    |N|
| |                | |2| N |5| | <--X.25 line--> |E|
| | <--’call’ 3--> | | |   | | |                 |T| <--call 3-->ATM 3
| |                | |P| E |P| |                 |W|
| |                | |R|   |R| |                 |O|
| | <--’call’ 4--> | |O| T |O| |                 |R| <--call 4-->ATM 4
| |                | |T|   |T| |                 |K|
| |                | | |   | | |                 | |
| | <--’call’ n--> | +-+   +-+ |                 | | <--call n-->ATM n
| |                |           |                 | |
| |                |           |                 | |
```

The X.25 protocol implementation consists of a LINE definition with the PROTOCOL set to X.25 and the ETE (end-to-end) attribute set to POS2 (VISA2).

The n calls are multiplexed over one dedicated leased line from the X.25 network. To the ATM and the device handler, this appears as a conventional POS2 ASYNC connection.

The number of X.25 lines is dependent on the bandwidth of the line and the number of transactions per hour.

XPNET X.25 Dial-Up Device Configuration

The following is an example of an X.25 configuration viewed under Compaq NSK's SCF (Subsystem Control Facility):

```
*CallingAddr.. ON       *Threshold.. 500      *CharacterSet. EBCDIC
 Debug........ OFF      *SYNCS...... 3        *DSRTimeout... 0:20.00
*Extformat.... OFF      *L3window... 2        *IdleTimeout.. 0:05:00
*Interface.... RS422    *L3mod...... 8        *PVCrange.....
*Netid........ X25XA    *Retries.... 20       *SVCrange..... ( 1,150)
*CUG..........       *Framemode.. DTE      *T1Timeout.... 0:03.00
*CUGType...... BASIC    *BCUG.......       *Transitdelay.
*Calls........ TWO WAY   PacketSize. 128       Type......... (61,63)
*Clockmode.... DCE      *Clockspeed. CLOCK1200
*SrcAddr...... 12345678
*Program...... $SYSTEM.CSS01.C9317P00
*RPOA.........
*Bpadparms....
*Cpadparms....
```

The following is an example of a dial-up device using a X.25 configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET DEVICE
SET DEVICE RECVLEN 650
SET DEVICE XMITLEN 650
SET DEVICE ADD DESTSTRING  1, P1A^N50,   0, "50NCR"
SET DEVICE ADD DESTSTRING  2, P1A^C1000, 0, "50.7"
SET DEVICE ADD DESTSTRING  3, P1A^VISA2, 0, "50"
SET DEVICE ADD DESTSTRING  4, P1A^VISA2, 0, "?"
SET DEVICE ADD PROTOCOLSTRING 1, VISA2, 0, "?"
ADD DEVICE DIALUP-DEV, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

The following is an example of a dial-up line using a X.25 configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET LINE
SET LINE ATIMER 30
SET LINE BTIMER 10
SET LINE DUPLEX HALF
SET LINE ENQ 400
SET LINE ERANALYSIS 30
SET LINE ETE POS2
SET LINE PORT $X25T1.#S1
SET LINE WFCALL ON
SET LINE PROTOCOL X.25
ADD LINE L1A^X25^1, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

The following is an example of a dial-up station using a X.25 configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET STATION
SET STATION DESTINATION P1A^N50
SET STATION ERRORACT IGNORE
SET STATION RETRY 3
SET STATION DEVICE DIALUP-DEV
SET STATION LINENAME \ACI.P1A^GATE.L1A^X25^1
ADD STATION S1AX25^1, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

This example is for the NCR 5XXX device handler, and therefore the “dest-name” for the XPNET 3.0 attribute “station destination” would be different for the Diebold 10XX/478X device handler.

For more information about adding Devices, Lines, Stations, and Processes to the XPNET system, refer to the ***NET24-XPNET Network Control Operations Guide*** or the ***NET24-XPNET Network Control Command Reference Manual***.

XPNET ASYNC Dial-Up Device Configuration

The following is an example of an ASYNC configuration viewed under Compaq NSK’s CMI (Communications Management Interface):

```
Autobaud OFF      Autodconn OFF      Autolf OFF        Backspace 0
Brkcount OFF      Brkenable OFF      Charsize 7        Crlfdelay 0.00
Ctsflow OFF         Defcr 013  %015  %H0D  " "
Deflf 010  %012  %H0A  " "              Duplex FULL         Echo OFF
Hdxturnchar 000  %000  %H00  " "
            000  %000  %H00  " "
Intchar 004  %004  %H04  " "
        005  %005  %H05  " "
        006  %006  %H06  " "
        021  %025  %H15  " "
Etxchar 003  %003  %H03  " "           Etxenable 1    Interface RS232
Intterm ON         Linechar 013  %015  %H0D  " "      Modem ON
Page OFF           Pagechar 013  %015  %H0D  " "      Parity EVEN
Paritychk ON       Pollchar 000  %000  %H00  " "      Recsize 256
Retries 13         Reversechn OFF      Siso OFF       Spacing POST
Specialmode RESET  Speed 1200          Stopbits 1     Tpause OFF
Type (6,0)         Vtabdelay 0.00      Xoffchar 000  %000  %H00  " "
Xonchar 000  %000  %H00  " "
```

The following is an example of a dial-up device using an ASYNC configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET DEVICE
SET DEVICE RECVLEN 650
SET DEVICE TYPE 255
SET DEVICE XMITLEN 650
SET DEVICE ADD DESTSTRING  1, P1A^N50,   0, "50NCR"
SET DEVICE ADD DESTSTRING  2, P1A^C1000, 0, "50.7"
SET DEVICE ADD DESTSTRING  3, P1A^VISA2, 0, "50"
SET DEVICE ADD DESTSTRING  4, P1A^VISA2, 0, "?"
SET DEVICE ADD PROTOCOLSTRING 1, VISA2, 0, "?"
ADD DEVICE DIALUP-DEV, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

The following is an example of a dial-up line using an ASYNC configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET LINE
SET LINE ERANALYSIS 30
SET LINE PORT \GLOBAL.$ACB3.#T14
SET LINE SPEED 1200
SET LINE TIMEOUT 500
SET LINE PROTOCOL VISA
SET LINE MODE ASYNC
ADD LINE L1A^ASYNC^1, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

The following is an example of a dial-up station using an ASYNC configuration under XPNET 3.0. This example is intended to aid BASE24 users in their initial setup and can be altered to fit specific environments.

```
RESET STATION
SET STATION DESTINATION P1A^N50
SET STATION RETRY 3
SET STATION DEVICE DIALUP-DEV
SET STATION LINENAME \ACI.P1A^GATE.L1A^ASYNC^1
ADD STATION S1AASYNC, UNDER SYSNAME \ACI, UNDER NODE P1A^NODE
```

This example is for the NCR 5XXX device handler, and therefore the "dest-name" for the XPNET 3.0 attribute "station destination" would be different for the Diebold 10XX/478X device handler.

For more information about adding Devices, Lines, Stations, and Processes to the XPNET system, refer to the ***NET24-XPNET Network Control Operations Guide*** or the ***NET24-XPNET Network Control Command Reference Manual***.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Performance

Dial/connect time does not impact network performance, but the dial/connect time increases the time the cardholder spends at the ATM. Institutions should use care in defining the modem initialization string of the DIALUP file for NCR 5XXX Dial-UP ATMs or the ADI.DAT file for Diebold 10XX/478X Dial-Up ATMs. A mismatch in baud rates between the ATM modem and the Compaq NSK modem results in longer calls due to modem training.

For NCR 5XXX Dial-Up ATMs, Network performance is affected by the addition of three dial-up specific messages: No-Op Commands (Command Code ‘G’), Disconnect Commands (Command Code ‘F’), and I’m Alive statuses. No-Op and Disconnect Commands in combination with the CC/CP in the ATM’s dial-up message header are used to negotiate line disconnects.

For Diebold 10XX/478X Dial-Up ATMs, the Network performance is affected by the addition of two dial-up specific messages: Signoff Request Commands (Command Code ‘F’), and I’m Alive statuses. In addition, the ATM controls call termination by issuing a line disconnect after sending a message with “CP” in the ATM’s dial-up message header. The network does not have control of call termination.

A record is added to the BASE24-atm Dial-Up Command File (ADCF) when a 9500 command is issued from the NCS/NCP. In addition, the ADCF is updated at the beginning and end of execution of each 9500 command.

Diebold 10XX/478X Visa II Dial Application – CONFIG-DAT

The following CONFIG.DAT file contains general configuration data for the Visa II Dial Application for the Diebold device. The standard location for CONFIG.DAT is the C:\IBOLD\VISA2\DAT directory of the Diebold ATM hard drive C. The data in the file can be viewed or edited using a text editor, such as the OS/2 System Editor (E.EXE).

The contents of this file is the responsibility of the BASE24-atm user and/or Diebold. It's inclusion here is for information purposes only.

```
# VISAII Dial Application Configuration File
#
# This data file contains all the configuration data required by the
# VISAII Dial application. Each field is terminated with a comma.
#
#   Unlimited comments can be placed in this file. A comment is
#   designated by a '#'.
#
#   NOTE: Double quote all entries that require a space (e.g.
#   "DD 123 ") and follow with a comma.
#

# 1) High side LE name.
VS2HISIDE,

# 2) Low side LE name.
VS2LOSIDE,

# 3) High side message categorization file.
c:\ibold\visa2\dat\hsidecat.dat,

# 4) Low side message categorization file.
c:\ibold\visa2\dat\lsidecat.dat,

# 5) Operating Mode Protocol to use.
#      The following options are available:
#       VISA2, VSE (proprietory)
#    Note: If left blank, then default is VISA2
#
VISA2,

# 6) Terminal ID (up to 12 characters)
DC2000,

# 7) Byte (specified in hex by 2 characters 00-7F) to be used as a
#    field separator (e.g. 1C) between the Terminal ID and Message
#    Type fields in the protocol Dialup Header.
1C,

# 8) Polling/I'm-Alive Timer - number of minutes since the last host
#    communication before initiating a poll to the host.
```

```
#
#    Enter 0 (zero) to disable “I’m Alive” message
030,

# 9) Default message to send to the host to indicate that the
#    Signoff Request Operation Command has been honored.
22.000..9,

# 10) Notification of Disconnect - This message is sent to TCS
#     after a communication session has been unexpectedly terminated.
#     The TCS response is then stored and later transmitted to the
#     host when communication has been restored.
#
#     Note: If this field is blank, no notification message will be
#           sent to TCS. Consequently, a predefined message will be
#           transmitted to the HOST upon restored communications.
#
#     Note: Not used by VSE error recovery.
#
1...3,

# 11) Message type for message received from TCS as a response to a
#     Notification of Disconnect (see Loside Message Categorization
#     file for type values).
#
#     NOTE: Value must be present but is not used for VSE!
4,

# 12) Message subtype for message received from TCS as a response to a
#     Notification of Disconnect (see Loside Message Categorization
#     file for subtype values).
#
#     NOTE: Value must be present but is not used for VSE!
8,

# 13) Number of protocol dependent header bytes to strip from
#     beginning of msg (if any).
0,

# 14) Bytes to append to message before sending to host (up to 20
#     bytes, each specified in hex by 2 characters 00-7F). e.g.
#     3F315A = “?1Z”
#
#     NOTE: Do not separate bytes with spaces or commas.
#
#     Enter NONE if no trailer bytes are to be appended.
NONE,

# 15) Debug Flag (1 = ON, 0 = OFF)
0,

# 16) SXA connected to Loside (1 = YES, 0 = NO)
0,

# 17) Auto-InService on signoff - flag that specifies if VISAII should
```

```
#     automatically send a STARTUP to TCS before acknowledging any
#     host request for signoff (1 = send STARTUP,
#     0 = do not send STARTUP).
#
#     NOTE: Enabling this feature can cause a communication timeout
#           when logging into Maintenance Manager. Once the host
#           gets the Maintenance logon message, it can send a
#           disconnect request ('F' Op Command). If Auto-Inservice is
#           enabled, PLA will send a STARTUP to TCS. However, TCS
#           will not respond to the STARTUP until Maintenance logoff.
#           Thus, the host can timeout waiting for a disconnect
#           acknowledgement.
0,

# 18) Error Recovery: 0 = Keep ATM OFFLINE until connection
#     re-established
1,

# 19) Maximum Message Size (including STX,ETX,LRC)
1920,

# 20) Message Response Maximum Time - This is a 'worst case' timer.
#        It is effective if no response is ever received from the
#        HISIDE of this application, i.e no response is ever received
#        from the ADI application. This timer will expire and prevent
#        the ATM from being 'hung' while waiting forever for a
#        response from the HISIDE (ADI) which can never occur.
#        Default is 60 seconds
60,

# 21) Error Recovery Idle Wait Time: The amount of time the
#          application will wait after taking TCS OFFLINE during
#          initial error recovery before beginning the re-connection
#          process.
60,

# 22) RESERVED
0,

# 23) Enable the receiving of SCM published events.
0,
```

Diebold 10XX/478X Application Dial Interface – ADI.DAT

The following ADI.DAT file contains data used by the Application Dial Interface for the Diebold device. The standard location for ADI.DAT is the C:\IBOLD\CSS directory of the Diebold ATM hard drive C. The data in the file can be viewed or edited using a text editor, such as the OS/2 System Editor (E.EXE).

The contents of this file is the responsibility of the BASE24-atm user and/or Diebold. Its inclusion here is for information purposes only.

```
;
; assign and define the dial port(s) for remote device traffic
;
BEGIN_DIAL_PORT_DEFINITION
    :DIAL_PORT_DEVICE:          ADILINK00
; Modem setup intended for BOCA 33.6
    :ORIGINATE_MODEM_SETUP:     AT&F0E0V1Q0&C1&K3&S1&D2%C0X4S0=0\N0&Q6
    :ANSWER_MODEM_SETUP:        AT&F0E0V1Q0&C1&K3&S1&D2%C0X4S0=1\N3&Q5
; Modem setup is intended for USR 33.6 & Internal 56k
;  :ORIGINATE_MODEM_SETUP:AT&F1E0V1Q0&C1&H1&R2&S1&D2&K0X4S0=0&M4&N0&B1
;  :ANSWER_MODEM_SETUP:   AT&F1E0V1Q0&C1&H1&R2&S1&D2&K0X4S0=1&M4&N0&B1
;  :CONNECT_TIMER:             60

END

;
; example log setup (logging enabled)
;

BEGIN_LOG
    :LOG_SIZE_KBYTES: 24
END

BEGIN_APPLICATION_DEFINITION
     :DEVICE:               ADITOVS2
     :PRIMARY_NUMBER:       9,1231234
     :SECONDARY_NUMBER:     9,1231234
     :TLA:                  000001
     :DIAL_RETRY:           0
     :PRIORITY:             0
     :MSG_HEADER:           50.777777
     :FIELD_SEPARATOR:      1C
     :REMOVE_HEADER_COUNT:  0
     :REMOVE_TRAILER_COUNT: 0
     :BAUD_RATE: 2400
     :DISCONNECT_TIMER: 65
     :DISCONNECT_NOTIFY:
     :7_BIT_EVEN_PARITY:
     :DEBIT_TYPE_TRANS:
 END
```

NCR 5XXX Dial-Up Configuration File – DIALUP

The DIALUP configuration file is a data file held on the NCR NDC+ ATM's OS/2 hard drive (C:\SYSTEM) and defines NCR NDC+ configuration options pertaining to VISA 2 dial-up. The file is built using an NCR supplied DOS based utility (TABU.EXE). The utility produces the DIALUP file from an ASCII text file (DIALUP.DEF).

The contents of this file is the responsibility of the BASE24-atm user and/or NCR. Its inclusion here is for information purposes only.

```
VISA 2 Dialup Configuration File
@OPT Line Options Table
[ 000, /* Reserved Option */
  005, /* Modem Baud Rate (5 = 1200, 6 = 2400, 7 = 4800) */
  001, /* Comm Port (1 - 4) */
  007, /* Data Bits (7 or 8) */
  002, /* Parity (0 - 4) */
  001, /* Stop Bits (1 or 2) */
  'NCR5XXXDNXXXXXX          ',  /* ID Field (25 bytes), pad spaces */
  'ATE0V1X4M1&M0&K0S37=5'       /* Modem Initialization 1200 */
]

@TIM Timers and Application Options Table
[ 030, /* Reserved Timer */
  060, /* Reserved Timer */
  005, /* Reserved Timer */
  009, /* Power Fail Redial (9 = 90 seconds, MAC=20) */
  003, /* I'm Alive (1 = 100 seconds)  */
  060, /* Modem Connect (60 = 60 seconds) */
  000, /* ENQ after ATM ACK option (000-off, 255-on) */
  255, /* Allow message after CP is sent option (000-off, 255-on) */
  000, /* No Transaction Compl Msg option (000-off, 255-on) */
  000  /* Host Message Header Included option (000-off, 255-on) */
]

@NUM Host Dial Numbers
[ '915551231234',0,0,0,0,0,0,0,0, /* Primary # 20 digits, pad nulls */
  '915551231234',0,0,0,0,0,0,0,0  /* Secondary # also 20, pad nulls */
]

@DBG Debug Flag
[ 000 ]
```

Non-Interactive Surcharge Processing

The device handlers have been modified to support a maximum surcharge amount the customer can be charged which is displayed to the customer on the initial transaction selection flow. This maximum surcharge value is loaded to the surcharge notification screen on any response message to the terminal if a new ATD Download Surcharge Amount is entered for the terminal. The maximum surcharge value is also be sent to the ATM during any load that contains the surcharge notification screen.

When the surcharge notification (9901) message is received from AUTH, the Device Handler determines if non-interactive surcharging is in-use (i.e., there is a non-zero ATD DOWNLOAD SURCHARGE AMT amount and it has been downloaded to the ATM). If non-interactive surcharging is in-use, and the actual surcharge amount is less than or equal to the maximum amount the cardholder agreed to, the request is sent to AUTH immediately without presenting a surcharge notification screen to the cardholder.

Diebold 10XX/478X Interactive Surcharge Processing

There are no changes to Interactive Surcharge Processing for dial-up Diebold 10XX/478X ATMs.

Diebold 10XX/478X Interactive Response Notification Fee Processing

Load group 310 is used for interactive response notification fee processing for both leased line and dial-up ATMs.

NCR 5XXX Interactive Surcharge Processing

Load group 304 is used for interactive surcharging with leased line ATMs. A new load group, 307, is used for interactive surcharging with dial-up ATMs. Processing is slightly different for dial-up ATMs when the cardholder decides not to continue the transaction after being presented the interactive surcharge notification screen (they press “No” or “Cancel”). The changes in the group 307 allow a “clean” disconnect sequence which avoids a long connection time and an error 140 when the ATM times out and disconnects the line.

NCR 5XXX Interactive NCD Processing

Load group 500 is used for interactive NCD transactions with leased line ATMs. A new load group, 501, is used for interactive NCD with dial-up ATMs. Processing is slightly different for dial-up ATMs when the cardholder decides not to continue the transaction after being presented the interactive NCD confirmation screen (they press “No” or “Cancel”).

The changes in the group 501 allow a “clean” disconnect sequence which avoids a long connection time and an error 140 when the ATM times out and disconnects the line.

Load group 502 was added to allow dial-up ATMs to use NCD with non-interactive surcharging. States in the group 502 connect the NCD state flow to the non-interactive surcharging state flow.

NCR 5XXX Interactive Response Notification Fee Processing

Load group 310 is used for interactive response notification fee processing for both leased line and dial-up ATMs.

New NCS 9500 Commands

The following new NCS 9500 commands have been added to the device handlers as part of this enhancement. They are used for ADCF queue maintenance.

LISTQ - The ADCF is read with a generic read using the ATD's terminal ID for all records that belong to the terminal. The records are written to EMS.

CLEARQ - The ADCF is read with a generic read using the ATD's terminal ID for all records that belong to the terminal. The records are written to EMS and then deleted.

The format of the NCS command is:

```
DELIVER PROCESS <process>, "9500LISTQ <termid>"
```

The LISTQ and CLEARQ commands can only be performed from the BASE24 AFT NCS screen or NCPCOM.

For more details, see the DELIVER command in the Object Management Commands section of the ***NET24-XPNET Network Control Command Reference Manual***.

ADCF Command processing

For dial-up terminals, the Device Handler processes commands from the ADCF queue as follows:

An event message is generated to show that the command is being processed. The event message is generated before the command is actually processed so that if an error occurs that prevents the command from being sent, the operator knows what command was being performed.

The queued command is processed by the Device Handler according to standard product for a 9500 command (i.e., the logic that existed before the enhancement for the 9500 commands).

If there is an error processing the command, it remains on the command queue. The operator must correct the condition causing the error or purge the command queue and re-enter the correct commands.

If a command completes successfully, it is deleted from the queue and the queue is checked for another command to execute.

Message Flows

The device handler now accepts dial-up calls from an ATM and handle message flows outside the traditional leased line poll/select environment. Since the device handler must wait for the ATM to initiate calls, ATM control is dependent upon BASE24 receiving a transaction, an I'm Alive status, or a Supervisor Mode notification sequence (or other solicited or unsolicited message). At that time queued ADCF commands is initiated. The functionality for leased-line ATMs has not been affected by this enhancement.

The dial-up ATMs are identified by their six-digit machine number in the dial-up header instead of by line address.

The device handler recognizes I'm Alive status messages received from dial-up ATMs.

Since the device handler cannot initiate a call to a dial-up ATM, commands are queued and stored in the ATM Dial-Up Command File.

The Diebold 10XX/478X Device Handlers must now support the Signoff Request Command (Command Code 'F') that is used in the dial-up environment to ask the ATM to disconnect the communication line. The Signoff Request Command (Command Code 'F') can be referred to as a "disconnect request" or a "request for disconnect".

The NCR 5XXX Device Handlers must now support the Disconnect Command (Command Code 'F') and the No-Op Command (Command Code 'G') which are used in the dial-up environment to negotiate line disconnects. The Disconnect Command (Command Code 'F') can be referred to as a "disconnect request" or a "request for disconnect", and the No-Op Command (Command Code 'G') can be referred to as a "continue command" or a "no-op continue command".

Message Flows for the Diebold 10XX/478X ATMs

The following partial audits (truncated to 68 bytes per message) illustrate some of the message flows between a dial-up Diebold 10XX/478X ATM and the device handler.

The ADCF queued command used in some of the following audits was "9500LOADGROUP <terminal-id>". The load group in the ATD Screen 7 is 304, which is a small load group. The LOADGROUP 0304 command is not a component of every load or startup sequence, rather its inclusion in examples below is for illustrative purposes only.

ATM in and out of supervisor mode:

```
50.777777DC2000      ?CC12?000??320?000000?
1???F?
50.777777DC2000      ?CP22?000??9?
50.777777DC2000      ?CC12?000??300?001011?
1???3?
50.777777DC2000      ?CC22?000??8?<20000000?B03481>00000000<?0=20083
1???1?
50.777777DC2000      ?CC22?000??9?
1???F?
50.777777DC2000      ?CP22?000??9?
```

ATM in and out of supervisor mode with a queued command in the ADCF:

```
50.777777DC2000      ?CC12?000??320?000000?
1???2?
50.777777DC2000      ?CC22?000??9?
3???13?2000000000000041000084000?000?0004801023020030309304093050230
50.777777DC2000      ?CC22?000??9?
1???3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???13?2000000000000041000084000000017?000?0004801023020030309304093
50.777777DC2000      ?CC22?000??9?
1???3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???11?113?L000??@MFEE NOTICE?AC(NAME OF INSTITUTION), THE OWNER?BDO
50.777777DC2000      ?CC22?000??9?
3???12?094I09110500000000010000000000?110D094223000000000032000000?113
50.777777DC2000      ?CC22?000??9?
3???32?15412122324510919016512 2?
50.777777DC2000      ?CC22?000??9?
3???16?2000?
50.777777DC2000      ?CC22?000??9?
1???1?
50.777777DC2000      ?CC22?000??9?
1???F?
50.777777DC2000      ?CC12?000??320?001001?
1???F?
50.777777DC2000      ?CC12?000??300?001011?
1???3?
```

```
50.777777DC2000      ?CC22?000??8?<20000000?B03481>00000000<?0=20083
1???1?
50.777777DC2000      ?CC22?000??9?
1???F?
50.777777DC2000      ?CP22?000??9?
```

Transaction (without OAR):

```
50.777777DC2000      ?RQ11?000???12?;2123456700000007=12991000996899
4?000??051?0000000000000000?0928H000?203?I490???’  DATE?3TIME  STATI
50.777777DC2000      ?CP22?000??B?
```

Transaction (with OAR):

```
50.777777DC2000      ?RQ11?000???13?;2123456700000007=12991000996899
3?000??21100111?074??122?CIFR     CHECKING   ?FE1> DDA 2     ?GE2> D
50.777777DC2000      ?RQ11?000???14??????1C??
3?000??21100111?074??122?CITO     SAVINGS    ?FE1> SAV 2     ?GE2> S
50.777777DC2000      ?RQ11?000???15??????2C??
4?000??051?0000000000000000?0929H000?503?I490???’  DATE?3TIME  STATI
50.777777DC2000      ?CP22?000??B?
```

Transaction (with OAR) with a queued command in the ADCF:

```
50.777777DC2000      ?RQ11?000???14?;2123456700000007=12991000996899
1???2?
50.777777DC2000      ?CC22?000??9?
3???13?200000000000004100008400?000?0004801023020030309304093050230
50.777777DC2000      ?CC22?000??9?
1???3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???13?2000000000000041000084000000017?000?0004801023020030309304093
50.777777DC2000      ?CC22?000??9?
1???3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???11?113?L000??@MFEE NOTICE?AC(NAME OF INSTITUTION), THE OWNER?BDO
50.777777DC2000      ?CC22?000??9?
3???12?094I091105000000001000000000?110D0942230000000000032000000?113
50.777777DC2000      ?CC22?000??9?
3???32?160166231243198185096198?
50.777777DC2000      ?CC22?000??9?
```

```
3???16?2000?
50.777777DC2000      ?CC22?000??9?
1???1?
50.777777DC2000      ?CC22?000??9?
1???F?
50.777777DC2000      ?CP22?000??9?
```

I’m Alive message from ATM with a queued command in the ADCF:

```
50.777777DC2000      ?CC22?000??8?<20000000?B03481>00000000<?0=20083
1???1?
50.777777DC2000      ?CC22?000??9?
1????2?
50.777777DC2000      ?CC22?000??9?
3???13?20000000000000410000840000?000?0004801023020030309304093050230
50.777777DC2000      ?CC22?000??9?
1????3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???13?2000000000000041000084000000017?000?00048010230200303093040093
50.777777DC2000      ?CC22?000??9?
1????3?
50.777777DC2000      ?CC22?000??8?<00000000?B03481>00000000<?0=20083
3???11?113?L000??@MFEE NOTICE?AC(NAME OF INSTITUTION), THE OWNER?BDO
50.777777DC2000      ?CC22?000??9?
3???12?094I0911050000000001000000000?110D094223000000000032000000?113
50.777777DC2000      ?CC22?000??9?
3???32?0600701230690482101411119?
50.777777DC2000      ?CC22?000??9?
3???16?2000?
50.777777DC2000      ?CC22?000??9?
1???1?
50.777777DC2000      ?CC22?000??9?
1???F?
50.777777DC2000      ?CP22?000??9?
```

Message Flows for the NCR 5XXX ATMs

The following partial audits (truncated to 68 bytes per message) illustrate some of the message flows between a dial-up NCR 5XXX ATM and the Device Handler. The “01?” is the XPNET drop line control message. This is translated by XPNET into an EOT followed by the actual modem disconnect.

The ADCF queued command used in some of the following audits was “9500LAGROUP <terminal-id> 0305”. This command was chosen because group 0305 is small; generating a manageable audit for inclusion here. The LAGROUP 0305 command is not a component of every load or startup sequence, rather its inclusion in examples below is for illustrative purposes only. The presence of multiple ADCF queued commands would result in a corresponding delay in the disconnect sequence.

ATM in and out of supervisor mode:

```
50NCR5XXXDN123456    ?CC12?000??P21?
1???F?
50NCR5XXXDN123456    ?CC12?000??R09?
1???F?
50NCR5XXXDN123456    ?CC12?000??P20?
1???21?
50NCR5XXXDN123456    ?CP22?000??9?
1???4?
50NCR5XXXDN123456    ?CC22?000??F?204640000225009990099800897010000
1???7?
50NCR5XXXDN123456    ?CC22?000??F?10135?0000004000000000000000?157F0
3???32?0890411390060141161O3222?
50NCR5XXXDN123456    ?CC22?000??9?
3???16?0135?
50NCR5XXXDN123456    ?CC22?000??9?
1???1?
50NCR5XXXDN123456    ?CC22?000??9?
1???F?
50NCR5XXXDN123456    ?CP22?000??9?
01?
```

ATM in and out of supervisor mode with a queued command in the ADCF:

```
50NCR5XXXDN123456    ?CC12?000??P21?
```

```
1???F?
50NCR5XXXDN123456    ?CC12?000??R09?
1???F?
50NCR5XXXDN123456    ?CC12?000??P20?
1???21?
50NCR5XXXDN123456    ?CP22?000??9?
1???4?
50NCR5XXXDN123456    ?CC22?000??F?2046400002250099900998008970100000
1????7?
50NCR5XXXDN123456    ?CC22?000??F?10135?000000400000000000000000?157F0
3???32?204024238195206035124154?
50NCR5XXXDN123456    ?CC22?000??9?
3???16?0135?
50NCR5XXXDN123456    ?CC22?000??9?
1???1?
50NCR5XXXDN123456    ?CC22?000??9?
1???21?
50NCR5XXXDN123456    ?CC22?000??9?
3???12?035W108108036000000108108108?036F036141137255108036255187?108
50NCR5XXXDN123456    ?CC22?000??9?
3???11?108?(1??078?@JFEE NOTICE         108?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000??9?
3???11?118?(1??075?@JFEE NOTICE         118?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000??9?
3???16?0135?
50NCR5XXXDN123456    ?CC22?000??9?
1???1?
50NCR5XXXDN123456    ?CC22?000??9?
1???F?
50NCR5XXXDN123456    ?CP22?000??9?
01?
```

Transaction (without OAR):

```
50NCR5XXXDN123456    ?RQ11?000???1:?;123456700000003=12113000998635?
4?000??074??04655074075?CI  12465354.92 ?GI  12364435.80 ?:00?
50NCR5XXXDN123456    ?CP22?000??B?
01?
```

Transaction (with OAR):

```
50NCR5XXXDN123456    ?RQ11?000???1;?;123456700000003=12113000998635?
3?000??21100111?074???186?BEFR  ?BJCHECKING   ?DDDDA 2     ?EDDDA 1
50NCR5XXXDN123456    ?RQ11?000???1<??????4C??
3?000??21100111?074???186?BEFR  ?BJCHECKING   ?DDDDA 2     ?EDDDA 1
50NCR5XXXDN123456    ?RQ11?000???1=??????1C??
4?000??074??04665074075?CI      2219.74 ?GI       130.79 ?=00?
50NCR5XXXDN123456    ?CP22?000??B?
01?
```

Transaction (with OAR) with a queued command in the ADCF:

```
50NCR5XXXDN123456    ?RQ11?000???1>?;123456700000003=12113000998635?
3?000??21100111?074???186?BEFR  ?BJCHECKING   ?DDDDA 2     ?EDDDA 1
50NCR5XXXDN123456    ?RQ11?000???1???????4C??
3?000??21100111?074???186?BEFR  ?BJCHECKING   ?DDDDA 2     ?EDDDA 1
50NCR5XXXDN123456    ?RQ11?000???11??????1C??
4?000??074??04675074075?CI      2219.74 ?GI       130.79 ?100?
50NCR5XXXDN123456    ?CP22?000??B?
1???21?
50NCR5XXXDN123456    ?CC22?000???9?
3???12?035W1081080360000010810810 8?036F036141137255108036255187?108
50NCR5XXXDN123456    ?CC22?000???9?
3???11?108?(1??078?@JFEE NOTICE         108?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000???9?
3???11?118?(1??075?@JFEE NOTICE         118?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000???9?
3???16?0135?
50NCR5XXXDN123456    ?CC22?000???9?
1???1?
50NCR5XXXDN123456    ?CC22?000???9?
1???F?
50NCR5XXXDN123456    ?CP22?000???9?
01?
```

I'm Alive message from ATM with a queued command in the ADCF:

```
50NCR5XXXDN123456    ?CC22?000??F?60135?
1???21?
50NCR5XXXDN123456    ?CC22?000??9?
3???12?035W1081080360000010810810 8?036F036141137255108036255187?108
50NCR5XXXDN123456    ?CC22?000??9?
3???11?108?(1??078?@JFEE NOTICE        108?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000??9?
3???11?118?(1??075?@JFEE NOTICE        118?A@(NAME OF INSTITUTION),
50NCR5XXXDN123456    ?CC22?000??9?
3???16?0135?
50NCR5XXXDN123456    ?CC22?000??9?
1???1?
50NCR5XXXDN123456    ?CC22?000??9?
1???F?
50NCR5XXXDN123456    ?CP22?000??9?
01?
```

Vendor Impacts

This document assumes that the reader is aware of the hardware differences between leased-line and dial-up solutions. Consult hardware vendors for specifics.

Reference Documents

Not applicable

EMV Support

BASE24 Base processing has been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV '96. In addition to standard transaction functionality (cash withdraw, balance inquiry, etc), the NCR 5XXX Device Handler EMV Extension has been enhanced to include standard Surcharge, PIN Change, Fast Cash, and non-Currency Dispense support (NCD support also requires the standard NCR 5XXX Device Handler NCD Extension). Refer to the ***BASE24-atm EMV Support Manual*** for additional information.

This section describes the release 6.0 changes and conversion and implementation requirements associated with the EMV Support enhancement.

These changes, conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the EMV Support enhancement.

For more detailed information on this enhancement, refer to section 4 (Base).

CUSTMACS Changes

Several new flags have been added for the EMV Support enhancement.

Set the following flag to true to add the EMV bindable pieces to the BASE24-atm authorization module:

- atm_emv_on

Set the following flag to true to add the EMV bindable pieces to the NCR 5XXX Device Handler module:

- atm_dh_n50_emv_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes fields within BASE24 files relevant to EMV support. Field settings depend on desired configuration.

BASE24-atm Terminal Data Files (ATD)

Screen Name: EMV Terminal Capabilities
Screen: ATD Screen 21
DDL Name: ATDS1.[atds1-core].EMV-TERM-CAP

Indicates the card data input, CVM, and security capabilities of the terminal using a bit map to represent possible options.

The ATD has a separate screen 21 to configure the EMV Terminal capabilities. Set the flags on this screen according to the information in the ***BASE24-atm EMV Support Manual*** to define the EMV related capabilities for each terminal.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

NCR 5XXX Configuration File, CNFG5070

The NCR 5XXX Configuration File, CNFG5070, has been modified to include four new EMV load groups:

- Load Group 600

  Load group 600 is the basic EMV load group. This load group is loaded on top of base load group 0.

- Load Group 601

  Load group 601 is the EMV Fast Cash load group. If Fast Cash is implemented, this group is loaded on top of EMV load group 600. This load group requires standard Fast Cash load group 1 screens.

- Load Group 604

  Load group 604 is the EMV Surcharge load group. If surcharging is implemented, this group is loaded on top of EMV load group 600. This load group requires standard Surcharge load group 304 screens.

- Load Group 605

  Load group 605 is the EMV Non-Currency Dispense (NCD) load group. If NCD has been configured into the system, this load group is loaded on top of EMV load group 600. This load group requires standard non-Currency Dispense load group 500 screens.

- Load Group 624

  Load group 624 is the EMV Encrypted PIN Block Support load group. If Encrypted PIN Block Support is used for PIN Change transactions, this load group is loaded on top of EMV load group 600.

Enhanced Configuration Parameters (ECP)

Three new Enhanced Configuration Parameters have been added for EMV support. They are Cards Accepted, Include EMV Error Info, and EMV Actioned. Refer to the ***BASE24-atm EMV Support Manual*** for additional information about these parameters.

- Cards Accepted

  The Cards Accepted parameter instructs the terminal on the type of cards to be accepted.

- Include EMV Error Information

  The Include EMV Error Information parameter instructs the terminal to include or not include the ICC command and response data in solicited or unsolicited messages. This ECP also instructs the terminal to send the Application Information Version data in response to a ‘send configuration data’ request.

- EMV Actioned

  The EMV Actioned parameter instructs the terminal that ICC commands are or are not actioned.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new ATD NCR screen for EMV (screen 21) must be added through the Security requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Operational Impacts:</u>

There are five new requesters accessed via page 45 of the DCT:

- AT60IN50.RQEMMENS

  The AT60IN50.RQEMMENS requester formats and displays the EMV requester menu. There are four new tables accessed through this menu: Currency Data Objects table, Transaction Data Objects table, Terminal Data Objects table, and Application Identifier table. Each table must be appropriately configured. Refer to the ***BASE24-atm EMV Support Manual*** for additional information about the screens used to access these tables.

- AT6MIN50.RQEMAIDS

  The AT6MIN50.RQEMAIDS requester formats and displays the Application Identifier table.

- AT6MIN50.RQEMCRNS

  The AT6MIN50.RQEMCRNS requester formats and displays the Currency Data Objects table.

- AT6MIN50.RQEMTRMS

  The AT6MIN50.RQEMTRMS requester formats and displays the Terminal Data Objects table.

- AT6MIN50.RQEMTXNS

  The AT6MIN50.RQEMTXNS requester formats and displays the Transaction Data Objects table.

New Event Messages:

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

New BASE24-atm EMV related response codes were defined for this enhancement. Operators must be trained to recognize the following new EMV related response codes on the TLF Perusal screen:

- 81 HSM parameter error
- 82 HSM failure
- 83 KEYI record not found
- 84 ATC check failure
- 85 CVR decline
- 86 TVR decline
- 87 ARQC failure
- 88 Fallback Decline

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

Format 2 File Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Format 2 File Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Format 2 File Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

The TLF template file can have up to two alternate keys. The number of ways a BASE24-atm user wants to peruse their TLF records and the token retrieval option indicates the number of alternate keys to create. Refer to the TDF Expansion enhancement for further information on the token retrieval option.

There are three different examples of TLF template files that the BASE24-atm user can use. These TLF template example files are located on AT60MISC. You can create the template on the template subvolume with these examples, by using the VOLUME command to move to the logical network template subvolume (e.g., VOLUME $<volume>.<logical network>TPLT), and using the FUP utility to copy a template (e.g., FUP/IN $<volume>.AT60MISC.TPLTTLF). The template examples are described below.

TPLTTLF

The TPLTTLF file contains the necessary FUP commands to create the TLF template files as format 1 files.

TLF21

The TLF21 file contains the necessary FUP commands to create the TLF template primary file as a format 2 file; however, the alternate key file is a format 1 file. This option can be used to improve performance as long as the alternate key file does not exceed the format 1 size limit of 4 gigabytes.

TLF22

The TLF22 file contains the necessary FUP commands to create the TLF template files as format 2 files. Both the primary file and the alternate key file are format 2 files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

For those BASE24-atm users implementing this enhancement and extracting the TLF to a format 2 file, the host code must process the new block format for the extract format 2 disk file. In addition, the host code can use the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The value 2 in this field denotes a format 2 file. The fourth byte of the user field in the Extract File Header was used to store this information.

User/Operational Impacts

Disk Impacts

Any file converted to format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for format 2 files is increased. This is true when the primary file is format 2, regardless of whether the alternate key file(s) is format 2.

Since the maximum data area per block in format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a format 2 file. Users should check the length of their TLF records to see if fewer TLF records will fit in the new format 2 maximum data area blocks. The format 2 maximum data area is 4048 for entry-sequenced files, like the TLF. If fewer TLF records will fit in the format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry-sequenced files against the TLFX file, if a format 2 version of this file is to be used when extracting to disk.

User Impacts

BASE24-atm using format 2 TLF files must modify their template files to be format 2 files. Refer to the Template Changes subsection of this enhancement for further information about TLF Template files.

BASE24-atm users extracting their TLF files to format 2 disk files must configure the appropriate information in the ECF.

BASE24-atm users using format 2 TLF files must start at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files) and create the appropriate template files. Refer to the Template Changes subsection of this enhancement for further information about TLF Template files. At cutover, the TLF is created using the new template files.

Operational Impacts

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24-atm users must calculate the length of the records that are written to the TLF. Format 2 TLF files allow a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are written to the TLF files, the record length of a TLF record could exceed the limit of 4048 bytes for format 2 files.

Before a BASE24-atm user changes their TLFs to format 2 files, they should ensure these records do not exceed the 4048 limit. BASE24-atm users utilizing format 1 are not affected, since the DDL for the TLF was not changed to set the maximum record length to 4048.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

Interactive Response Notification Fee Processing and Negative Surcharge Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements.

Section 2 contains a functional overview of the Interactive Response Notification Fee Processing and Negative Surcharge Support enhancements.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support the interactive response notification fee processing enhancement. These modifications are provided in the LCONFAT file located on the BA60MISC subvolume.

ATM-FEE-INTRACTV-RESP-NTFY

This optional param indicates whether interactive response notification fee processing or standard interactive request notification surcharge processing is being performed. This param is not required if standard interactive request notification fee processing is being performed, regardless of whether negative surcharges are supported or not.

A value of Y (Yes) indicates that interactive **response** notification fee processing is being performed.

A value of N (No) indicates that standard interactive **request** notification surcharge processing is being performed.

The default for this param is a value of N. If the param is not present, standard product interactive request notification surcharge processing is performed.

This param can be set at the process level if the use of the enhancement varies among BASE24-atm Device Handler processes.

Database Configuration

For the Interactive Response Notification Fee Enhancement:

BASE24-atm Receipt File (RCPT)

The RCPT File can be used to display issuer fee or rebate information on a receipt.

The IF/ENDIF directives use the ISSFEES or NOISSFEES condition values for issuer fee or rebate receipt information.

The S2 field type code is used for the surcharge amount, and the S3 field type code is used for the issuer fee or rebate amount.

The A1 field type code contains the withdrawal amount plus any surcharge or issuer fees or rebates.

BASE24 Token File (TKN)

The following new BASE24 Base token has been added for this enhancement:

- Issuer Fee Rebate token (Token ID “30”) contains data relating to the transaction issuer fee or rebate amount assessed. This token is used if issuer fees or rebates are being processed:

If the Issuer Fee Rebate token is not logged to the Transaction Log File (TLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the Issuer Fee Rebate token is to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface process, the appropriate token records must be configured in the TKN.

For the Negative Surcharge Enhancement:

BASE24 Surcharge File (SURF)

The following existing fields in the Surcharge File (SURF) must be configured when performing negative surcharges:

Field Name: FLAT FEE
PERCENT FEE
Screen: SURF screens 2 and 3
DDL Name: SURF.ATM.TRAN.FLAT-FEE
SURF.ATM.TRAN.PCNT-FEE

Screens 2 and 3 of the SURF contain the surcharge information for transactions routed using the CPF and SPROUTE File, respectively. Either the flat fee, in whole and fractional currency units, can be used as a surcharge, or the percent fee, in whole and fractional units, can be used when calculating the surcharge. A flat fee, percent fee, or both must be defined for each transaction code. When using negative surcharges, a negative value can be entered in these fields. For additional information on these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

The interactive response notification of fees can include both issuer fees and acquirer fees (i.e., surcharges). BASE24 maintains information in the Surcharge File (SURF) only for acquirer fees and does not maintain information in its database for issuer fees. The setting of issuer fees must be done at a host system or an interchange whose BASE24 interchange interface has been enhanced to carry the issuer fee information in the new BASE24 Issuer Fee Rebate token.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

The following files were modified to contain new load groups for this enhancement.

Diebold 10XX/478X Configuration File (CNFGIX)

The load group (310) has been added to the CNFGIX file to be used for Interactive Request Notification fee processing. Some of the states and screen numbers in this load group overlap those that are in load group 200 for the MAC multiple account selection (MAS) enhancement. Because of that, users of MAC MAS can not be able to use the interactive request notification fee processing enhancement and load group 310.

An additional ATM screen for Negative Surcharges has been added to the following standard product surcharge request notification load group:

- 304 (Surcharge request notification)

<u>NCR 5XXX Configuration File (CNFG5070)</u>

The load group (310) has been added to the CNFG5070 file to be used for Interactive Request Notification fee processing. Some of the states and screen numbers in this load group overlap those that are in load group 200 for the MAC multiple account selection enhancement. Because of that, users of MAC MAS can not be able to use the interactive request notification fee processing enhancement and load group 310.

An additional ATM screen for Negative Surcharges has been added to the following standard product surcharge request notification load groups:

- 304 (Surcharge request notification)
- 307 (Dial-up ATM interactive surcharge processing)

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

The host or co-network must be modified to accept and process the Issuer Fee Rebate token in the ISO message from the BASE24-atm ISO Host Interface (HISO) or BASE24-atm BIC ISO process.

If negative surcharges are being used, the host or co-network must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

Super Extract Process

If the new Issuer Fee Rebate token is to be included in the TLF or ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

If negative surcharges are being used, the host must be modified to accept and process negative amounts in the existing Surcharge Data token in the TLF or ILF.

User/Operational Impacts

Disk Impacts

The TLF and ILF can require additional disk space because the new token for this enhancement (i.e., Issuer Fee Rebate token) can be included in the transaction records logged to these files.

If the TLF and/or ILF is extracted to a disk file, the extract record size is increased. This can increase the disk/tape space required for the extract file.

Operational Impacts

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multiple Account Select by Qualifier Support

BASE24-atm Multiple Account Select by Qualifier is a separately licensable add-on module to BASE24-atm, which enables BASE24-atm customers who are members of the MAC network with functionality to handle multiple accounts as outlined by the MAC Network. MAC mandates this processing and considers a member “out of compliance” if they do not offer the proper screen flows and verbiage at their MAC branded ATMs. Customers can be financially sanctioned if they are not in compliance.

Multiple Account Select by Qualifier is a multiple account selection system originally developed by the MAC switch network (Electronic Payment Systems - EPS) and now also supported through the NYCE switch network. Multiple Account Select by Qualifier is used as a replacement for standard multiple account selection (often referred to in the industry as Open Account Relationship - OAR), allowing multiple account selection through the participating switch network. In addition, Multiple Account Select by Qualifier includes support for an additional account type, ‘Other’. As implemented in BASE24-atm, instead of using STM 9906 messages to select accounts, Multiple Account Select by Qualifier selects accounts via fixed menus at the ATM, which are then passed to the Device Handler for mapping into a STM Account Qualifier Token. This token is recognized by Authorization, the ISO Host / BIC Interfaces, the MACI Interface, and the NYCI Interface. When authorizing on BASE24, Authorization performs the CAF account lookup using the account type from the STM transaction code, and from the STM Account Qualifier Token, the third digit of the account qualifier.

CUSTMACS Changes

Previous releases of Multiple Account Select by Qualifier support included a MASMMACS file, which was used in place of the CUSTMACS file when making the BASE24 system with Multiple Account Select by Qualifier support. BASE24 Release 6.0 includes Multiple Account Select by Qualifier support within CUSTMACS.

The following flag was added to allow the Multiple Account Select by Qualifier Authorization changes to be included in the MAKE stream:

atm_mas_by_qual_on

The following flags were updated to reflect the appropriate naming conventions for the Multiple Account Select by Qualifier enhancement:

Previous Flag	New Flag
atm_dh_c910_cse_mac_mas_on	atm_dh_c910_mas_by_qual_on
atm_dh_c1000_cse_mac_mas_on	atm_dh_c1000_mas_by_qual_on
atm_dh_n50_cse_mac_mas_on	atm_dh_n50_mas_by_qual_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Account Type Table File (ATT)

The ATT file contains one record for each valid account type in the BASE24 network. Each record contains an ISO code in the Account Type field and a corresponding description in the Account Type Name field.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: A record for the new Account type (Other – 9M) must be added to this file and the value is listed below.

Account Type: 9M Account Type Name: Other

ACI provides BASE24 users with a set of default ATT records at installation. The records are located in the ATT file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Processing Code Description File (PDF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the PDF file in order for description tags to be recognized in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

Description Tag	Long Description
ISO019M00	CASH FROM OTHER
ISO21009M	DEPOSIT TO OTHER
ISO21209M	SPLIT DEPOSIT DDA/OTHER
ISO211020	SPLIT DEPOSIT SAV/DDA
ISO21109M	SPLIT DEPOSIT SAV/OTHER

Description Tag	Long Description
ISO219M20	SPLIT DEPOSIT OTHER/DDA
ISO219M10	SPLIT DEPOSIT OTHER/SAV
ISO219M9M	SPLIT DEPOSIT OTHER/OTHER
ISO28009M	DEPOSIT TO OTHER,CASH BACK
ISO309M00	AVAIL FUNDS INQUIRY OTHER
ISO349M00	STATEMENT PRINT OTHER
ISO40209M	TRANSFER DDA/OTHER
ISO40109M	TRANSFER SAV/OTHER
ISO403010	TRANSFER CR/SAV
ISO40309M	TRANSFER CR/OTHER
ISO409M20	TRANSFER OTHER/DDA
ISO409M10	TRANSFER OTHER/SAV
ISO409M9M	TRANSFER OTHER/OTHER
ISO503030	PAYMENT CR/CR
ISO509M30	PAYMENT OTHER/CR

ACI provides BASE24 users with a set of default PDF records at installation. The records are located in the PDF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Acquirer Processing Code File (APCF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the APCF file in order for the Multiple Account Select by Qualifier transactions to be recognized by the Device Handler. The Authorization Destination field must be spaces for all records and the Log Auth Dest Response field must be set to N for all records.

Acquirer Transaction Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	Transaction Allowed
ATM	*	01	9M	00	ISO019M00	4
ATM	*	21	00	9M	ISO21009M	4
ATM	*	21	20	9M	ISO21209M	4
ATM	*	21	10	20	ISO211020	4
ATM	*	21	10	9M	ISO21109M	4
ATM	*	21	9M	20	ISO219M20	4
ATM	*	21	9M	10	ISO219M10	4
ATM	*	21	9M	9M	ISO219M9M	4
ATM	*	28	00	9M	ISO28009M	4
ATM	*	30	9M	00	ISO309M00	4
ATM	*	34	9M	00	ISO349M00	4
ATM	*	40	20	9M	ISO40209M	4
ATM	*	40	10	9M	ISO40109M	4

Acquirer Transaction Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	Transaction Allowed
ATM	*	40	30	10	ISO403010	4
ATM	*	40	30	9M	ISO40309M	4
ATM	*	40	9M	20	ISO409M20	4
ATM	*	40	9M	10	ISO409M10	4
ATM	*	40	9M	9M	ISO409M9M	4
ATM	*	50	30	30	ISO503030	4
ATM	*	50	9M	30	ISO509M30	4

BASE24 users also must configure the APCF (transactions allowed file) with the new Multiple Account Select by Qualifier transaction codes.

ACI provides BASE24 users with a set of default APCF records at installation. The records are located in the APCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Issuer Processing Code File

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the IPCF file in order for the Multiple Account Select by Qualifier transactions to be recognized by the issuer interchanges. The Completion Required to Host field must be set to N for all records.

Issuer Trans Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	On-Us Trans Allowed	Not-On-Us Trans Allowed
ATM	*	01	9M	00	ISO019M00	4	4
ATM	*	21	00	9M	ISO21009M	4	4
ATM	*	21	20	9M	ISO21209M	4	4
ATM	*	21	10	20	ISO211020	4	4
ATM	*	21	10	9M	ISO21109M	4	4
ATM	*	21	9M	20	ISO219M20	4	4
ATM	*	21	9M	10	ISO219M10	4	4
ATM	*	21	9M	9M	ISO219M9M	4	4
ATM	*	28	00	9M	ISO28009M	4	4
ATM	*	30	9M	00	ISO309M00	4	4
ATM	*	34	9M	00	ISO349M00	4	4
ATM	*	40	20	9M	ISO40209M	4	4
ATM	*	40	10	9M	ISO10109M	4	4
ATM	*	40	30	10	ISO403010	4	4
ATM	*	40	30	9M	ISO40309M	4	4
ATM	*	40	9M	20	ISO409M20	4	4
ATM	*	40	9M	10	ISO409M10	4	4

Issuer Trans Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	On-Us Trans Allowed	Not-On-Us Trans Allowed
ATM	*	40	9M	9M	ISO409M9M	4	4
ATM	*	50	30	30	ISO503030	4	4
ATM	*	50	9M	30	ISO509M30	4	4

BASE24 users must configure the IPCF with the new Multiple Account Select by Qualifier transaction codes.

ACI provides BASE24 users with a set of default IPCF records at installation. The records are located in the IPCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Institution Definition File (IDF)

Field Name: OTHER ACCT PROCESSING
Screen: IDF Screen 2
DDL Name: IDF.OTHER-ACCT-TYP

This field indicates whether ‘other’ accounts will be processed as credit accounts or debit accounts.

On screen 9, an ACCT TYPE of ‘60’ is now allowed in the ATM ROUTING TABLE.

On screen 13, a FAST CASH ACCOUNT TYPE of ‘60’ (OTHER) is now allowed in the ATM PROCESSING CONTROL PARAMETERS.

Cardholder Authorization File (CAF)

Field Name: QUAL
Screen: CAF Screens 3 and 4.
DDL Name: CAF.ACCT[X].QUAL

This field is used to differentiate between various accounts, which have the same account type.

Positive Balance File (PBF)

On screen 1, an ACCOUNT TYPE of ‘60’ (OTHER) is now allowed.

Token File (TKN)

No new BASE24-atm tokens have been added for the Multiple Account Select by Qualifier enhancement. The enhancement will use the existing BASE24 Account Qualifier Token (Token ID “18”) to log the appropriate account qualifier.

If the BASE24 token for this enhancement is not to be logged to the TLF or ILF, the appropriate token records must be configured in the TKN.

If the new token for this enhancement is to be extracted with the TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

COBNAMES

New Multiple Account Select by Qualifier related transaction descriptions were added to the ATM transaction description table in the COBNAMES file.

Device Configuration

The following files were modified to contain new load groups for this enhancement.

Diebold 10XX/478X Configuration File (CNFGIX)

The load group (200) has been added to the CNFGIX file to be used for Multiple Account Select by Qualifier. Some of the states and screen numbers in this load group overlap those that are in load group 310 for the Interactive Request Notification fee processing enhancement. Because of that, users of the Interactive Request Notification fee processing enhancement will not be able to use the Multiple Account Select by Qualifier enhancement and load group 200.

The following Load Groups were added for the Multiple Account Select by Qualifier enhancement:

- 200 (Multiple Account Select by Qualifier)
- 212 (MAS by Qualifier and SSB DH. Must be loaded on top of groups 0, 12 and 200.)

- 220 (MAS by Qualifier and NCD. Must be loaded on top of groups 0, 500 and 200.)
- 232 (MAS by Qualifier and NCD and SSB DH. Must be loaded on top of groups 0, 200, 12, 212 and 500.)
- 252 (MAS by Qualifier and SSB Base Split Deposit. Must be loaded on top of groups 0, 12, 52 and 200.)

<u>NCR 5XXX Configuration File (CNFG5070)</u>

The load group (200) has been added to the CNFG5070 file to be used for Multiple Account Select by Qualifier. Some of the states and screen numbers in this load group overlap those that are in load group 310 for the Interactive Request Notification fee processing enhancement. Because of that, users of the Interactive Request Notification fee processing will not be able to use the Multiple Account Select by Qualifier enhancement and load group 200.

The following Load Groups were added for the Multiple Account Select by Qualifier enhancement:

- 200 (Multiple Account Select by Qualifier)
- 206 (MAS by Qualifier and SSB DH – Check Handling. Must be loaded on top of groups 0, 12, 200 and 106.)
- 210 (MAS by Qualifier and SSB Base – Split Deposit. Must be loaded on top of groups 0, 12, 200 and 110.)
- 211 (MAS by Qualifier and SSB Base – Super Teller. Must be loaded on top of groups 0, 12, 200 and 111.)
- 250 (MAS by Qualifier and NCD. Must be loaded on top of groups 0, 200 and 500.)
- 256 (MAS by Qualifier and NCD and SSB DH – Check Handling. Must be loaded on top of groups 0, 200, 500, 12 and 106.)
- 261 (MAS by Qualifier and NCD and SSB Base – Super Teller. Must be loaded on top of groups 0, 200, 500, 12 and 111.)

<u>Diebold 910 Configuration File (CNFG910)</u>

The following Load Groups were added for the Multiple Account Select by Qualifier enhancement:

- 200 (Multiple Account Select by Qualifier)
- 207 (MAS by Qualifier and MAS by Qualifier Multiple Language. Must be loaded on top of groups 0, 200 and 7.)

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, From Host Maintenance, ISO Host Interface, Refresh, and Super Extract processes.

BIC ISO Interchange Interface Process

If the Account Qualifier token for this enhancement is included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs. In addition, the co-network must be able to handle a new account type of '9M' (Other).

From Host Maintenance

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF FHM ISO message format to accept and process the new information in the ACCT segment of the CAF file.

Super Extract Process

New fields were added to the IDF by redefining existing user fields. If the BASE24-atm user performs processing on the extracted IDF, they must be aware of the record format changes.

Host code changes are required for all BASE24-atm users that utilize the OMF Extract Process to accept and process the new information defined in the IDF.

ISO Host Interface Process Impacts

If the Account Qualifier token for this enhancement is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Host systems implementing the Multiple Account Select by Qualifier enhancement must make the necessary changes to support the new account type of '9M' (Other).

Refresh

Host code changes are required for all BASE24 users that utilize the release 6.0 CAF Refresh file message format to initialize and process the new information in the ACCT segment of the CAF file.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space because the new Account Qualifier token for this enhancement can be included in the transaction records logged to these files.

Operational Impacts

BASE24 users must be aware of the new Multiple Account Select by Qualifier transaction codes that can be displayed via TLF perusal and included in the ATM reports.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Multiple Currency Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Multiple Currency Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Multiple Currency Support enhancement.

The Multiple Currency Support enhancement allows cardholders to perform transactions in currencies different from their account currencies at different ATMs. Financial institutions can offer their cardholders the ability to withdraw funds from any account (regardless of the currency), and receive those funds in another currency, the account can or can not be held in the same currency as that dispensed by the ATM. This enhancement allows a currency to be defined and configured for the ATM, and each ATM's primary terminal currency (but not hopper currency) can be unique. Refer to the ***BASE24-atm Multiple Currency Support Manual*** for additional information.

CUSTMACS Changes

Several new flags have been added for the Multiple Currency Support (MCU) enhancement. Refer to section 4 for general MCU flags that are required to be set.

Set the following flag to true to add the BASE24-atm MCU bindable pieces to the appropriate authorization module:

- atm_mcu_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24 Terminal Data Files (ATD)

Field Name: CURRENCY CODE
Screen: ATD Screen 4 and 5
DDL Name: ATDD1.[atdd1-core].HOPR.CRNCY-CDE
TDF.HOPR.CRNCY-CDE

Screen 4 and 5 of the ATD contain the currency code for each hopper. Even in an environment with the Multiple Currency Support enhancement, these currency codes should all be set to the same value as the enhancement does not support dispensing in more than one currency. For additional information on these fields, refer to the ***BASE24-atm Files Maintenance Manual***.

Field Name: AMT CURRENCY CODE
Screen: ATD Screen 6
DDL Name: ATDD1.[atdd1-core].AMT-CRNCY-CDE
TDF.AMT-CRNCY-CDE

Screen 6 of the ATD contains the terminal's currency code. This value should be set to the same value as the hopper currency codes, as the Multiple Currency Support enhancement does not support dispensing in more than one currency. However, this value can be different across multiple ATD records within the same logical network if the Multiple Currency Support enhancement is being used. For additional information on this field, refer to the ***BASE24-atm Files Maintenance Manual***.

Cardholder Authorization File (CAF)/Usage Accumulation File (UAF)

No new data fields have been added to the BASE24-atm segment of the CAF or UAF for the Multiple Currency Support enhancement. Since these files contain amount fields that need to have a currency code associated with them, if the Multiple Currency Support enhancement is being used, the amount fields are therefore assumed to be in the currency of the corresponding Institution Definition File (IDF) record.

Card Prefix File (CPF)

No new data fields have been added to the BASE24-atm segment of the CPF for the Multiple Currency Support enhancement. Since this file contains amount fields that need to have a currency code associated with them, if the Multiple Currency Support enhancement is being used, the amount fields are therefore assumed to be in the currency of the corresponding Institution Definition File (IDF) record.

The following existing field requires special consideration when the Multiple Currency Support enhancement is being used:

Field Name: STANDARD CASH ADV INCR
Screen: CPF Screen 4
DDL Name: CPF.ATMCPF.STD-CCA-INCR

Screen 4 of the CPF already contains the STANDARD CASH ADVANCE INCREMENT. With the Multiple Currency Support enhancement, all amount fields in the CPF are assumed to be in the currency of the associated FIID. The user can still configure the BASE24-atm standard increment amount for Cash Advance transactions, but these are successful only in a Multiple Currency environment if the Transaction currency is the same as the Issuer Institution currency. Cash Advance transactions performed in any other currency are denied (to avoid this possibility no standard increment value should be configured in the CPF.) For additional information on this field, refer to the ***BASE24 Files Maintenance Manual***.

Token File (TKN)

The following BASE24-atm token has been added for the Multiple Currency Support enhancement:

ATM Balances Token (token ID “AB”) contains data relating to cardholder account balances in both the Account currency and the Transaction currency.

If the ATM Balances Token is not to be logged to the Transaction Log File (TLF) or Interchange Log File (ILF), the appropriate token records must be configured in the TKN.

If the new BASE24-atm token for this enhancement is to be extracted with the TLF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured in the TKN.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

If deposit credit amount transactions are being supported between BASE24 and a host system using the BASE24-atm Standard ISO Host Interface (HISO) module, these are supplied in the currency indicated by the Transaction Currency Code.

There are no essential changes required to HISO for passing balance information between BASE24-atm and host systems. However, the host can make use of the ATM Balances Token to extend the functionality in a Multiple Currency environment.

The following options are valid for a host to supply balance information in a response message (0210) to BASE24-atm:

- The host continues to supply data element P-44 (Additional Response Data) which contains the usage indicator and available or ledger Balance in the currency of the original request amount field (P-4, as indicated by P-49).
- With this option, balances in the transaction currency only are made available for display at the device, unless BASE24 is configured to override the balances provided by the host with the balances held on the PBF.
- The host can send balances in the new ATM Balances Token that is carried in data element S-126 (BASE24-atm Additional Data) together with any other token data. If the ATM Balances Token is supplied by the host, then the following information is required in the token:
- It is mandatory to supply the balance(s) in Account currency together with the ACCT-CRNCY-CDE.
- Balances in Transaction currency are optional. If not supplied, then the TXN-CRNCY-CDE, CONV-RATE, TXN-AMT-2, and TXN-AMT-3 should be initialized to zeroes. BASE24 supplies the appropriate values using conversion rates held in the ERF.
- It is mandatory to supply values for the display option indicators.
- Any information that is carried in data element P-44 is superceded.

Super Extract Process

Standard TLF Extract processing is not specifically impacted by the BASE24-atm Multiple Currency Support enhancement, but host systems must be aware that TLF Extract files can contain records in different currencies, as the BASE24-atm system is capable of processing transactions (and settlement data) which are acquired in multiple, different currencies.

The BASE24 defined transaction currency is used for all standard amount and currency fields in the TLF and TLF Extract records.

If the new token (i.e., Multiple Currency Token) for this enhancement is to be included in the TLF and ILF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files. The Multiple Currency Token can therefore be used to convey the transaction amounts in all applicable currencies to host systems, if required for host processing.

The amount field in the TLF Extract trailer record (XT.AMT) represents a verification amount only, and should reflect the sum total of TLF Amount-1 fields, regardless of currency, specified in the file.

Refresh Process

While there are not host modifications required for PBF Refresh impacting, this function has been enhanced to ensure that transaction impacting from the TLF is carried out in the appropriate account currency. The Refresh process always uses the account currency from the PBF record to ensure that impacting from the TLF is correctly valued.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space because the new tokens for this enhancement (i.e., Multiple Currency Token and BASE24-atm Balances Token) can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Tidel Device Handler

BASE24-atm has been enhanced to support Tidel ATMs. The BASE24-atm Tidel Device Handler module provides the interface between the device and Compaq NSK. The Tidel ATMs are defined in the BASE24-atm Terminal Data File (TDF). This device handler was first introduced in BASE24-atm Release 5.3.

Refer to the BASE24-atm Tidel Device Handler manual on the BASE24-atm system (i.e., $<volume>.AT60TIDL.TIDLMAN) for information on implementing this device and device handler into a BASE24-atm user's system.

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Transaction Security Services

This section describes the release 6.0 changes and the conversion and implementation requirements associated with the Transaction Security Services enhancement for BASE24-atm.

Section 2 contains a functional overview of the Transaction Security Services enhancement.

For more detailed information on this enhancement see section 4 (Base).

Standard BASE24-atm modules have been modified where required to support Transaction Security Services. The Diebold 10XX/478X Device Handler, NCR 5XXX Device Handler, Tidel Device Handler, and BASE24-atm Authorization processes have been modified to support triple DES encryption/decryption.

Note: Customers using Atalla Identikey PIN verification, Diebold PIN verification methods, or non-ANSI PIN blocks cannot implement Transaction Security Services at this time.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>ATD/TDF</u>

All key fields in the ATD/TDF records must be updated to contain key locators. The key locators are used by Transaction Security Services to read CSECD records to retrieve the actual encryption/decryption keys used by the ATM device.

Refer to the File Conversions subsection of this enhancement to see how the key fields in the ATD/TDF can be updated programmatically, using the terminal ID as the default key locator for each ATD/TDF record, and corresponding CSECD records created.

The ATD/TDF requesters/servers have been modified to read the SECURE-DEV-TYP param in the LCONF. If the SECURE-DEV-TYP param indicates Transaction Security Services, the key fields maintained in the ATD/TDF files are automatically modified to contain the terminal ID when a new record is added. The terminal ID is the key locator for ATM device records in the CSECD.

If the Transaction Security Services database is partitioned across multiple logical networks, a two-character hexadecimal partition identifier, set by the LCONF param LN-IND, is used as the first two characters of the key locator. For key locators in the terminal data files, the first fourteen characters of terminal ID is concatenated to the partition identifier to create the key locator.

CSECD

A CSECD (Channel Security) record must exist for every set of encryption/decryption keys used by ATMs in the BASE24 network. Normally, every ATM uses unique encryption/decryption keys so there would be one CSECD record for every ATD/TDF record.

Refer to the File Conversions subsection of this enhancement to see how CSECD records for each ATD/TDF record can be created programmatically.

If an ATM device is configured in the BASE24-atm Terminal Data files to send PINs in the clear to any interchanges on the BASE24 system, a Channel Security record (CSECD) must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an interchange interface is configured to support clear PINs, a Channel Security record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

New Data Files

A new data file is required that contains encryption/decryption keys referenced by key locators during transaction processing. This file can be created by running conversion programs or records can be added, deleted or modified manually using GUI.

- CSECD Channel Security

The following table shows the relationship of existing BASE24-atm database files to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process to get the real encryption/decryption keys used during transaction processing.

BASE24 Database File	**Transaction Security Services Database File**
ATD	CSECD
TDF	CSECD

Template Changes

Not applicable

Device Configuration

ATM devices supporting Triple DES encryption must have new software/hardware configured to use the Triple DES encryption algorithm and double-length or triple-length keys.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the keys stored in the CSECD records is through the new ACI Desktop Graphical User Interface, which is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following BASE24-atm files are supplied with the Transaction Security Services enhancement. There are no structural changes in any files, but Transaction Security Services files can be created and data populated from BASE24-atm files. BASE24-atm files are updated with key locators to point BASE24-atm processes to the correct Transaction Security Services records. Transaction Security Services data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- ATD
- TDF

Note: Release 5.3 to release 6.0 conversions must be run on BASE24 files before the Transaction Security Services conversion programs are run. Transaction Security Services conversion programs must be run on release 6.0 files.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Transaction Security Services conversion programs must be run on release 6.0 version 2 files. All other version 1 to version 2 conversion programs must be run before the Transaction Security Services conversion programs are run.

<u>BASE24-atm Terminal Data file (ATD)</u>

The ATD conversion program can be run to create CSECD records while updating ATD records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface. Key fields in corresponding ATD records would have to be updated manually as well.

The ATD conversion program is on the AT60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSATDS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdtss.t,name $atdm,nowait/
tssatdm –force –list –listreplace
```

Users must update the TSSATDR obey file on the AT60UC05 subvolume with the appropriate information. TSSATDR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous ATD conversion for Transaction Security Services or will be an empty file the first time a terminal data file conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSATDR files on the AT60UC05 subvolume.

BASE24-atm Terminal Data File (TDF)

The TDF conversion program can be run to create CSECD records while updating TDF records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding TDF records would have to be updated manually as well.

The TDF conversion program on the AT60UC05 subvolume can be used to create or update the CSECD. If modifications to the conversion program are necessary, the program can be modified by editing the TSSTDFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#tdftss.t,name $tdfm,nowait/
tsstdfm –force –list –listreplace
```

Users must update the TSSTDFR obey file on the AT60UC05 subvolume with the appropriate information. TSSTDFR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous TDF conversion for Transaction Security Systems or will be an empty file the first time a terminal data file conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSTDFR files on the AT60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- ***Host Security Module RG7000 Programmers Manual*** (1270A514 Issue 3)
- ***Banking Command Reference Manual*** (424645-002 May 2001)
- ***Atalla Key Block Banking Command Reference Manual*** (TP522901-001, October 2001)

Triple/Single DES Translation Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Triple/Single DES Translation Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Triple/Single DES Translation Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impacts the ISO Host Interface Process.

External Interfaces

Host systems and co-networks implementing the Triple/Single DES Translate enhancement must make the necessary changes to support double length keys and Triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Triton Device Handler

BASE24-atm has been enhanced to support Triton ATMs. The BASE24-atm Triton Device Handler module provides the interface between the device and Compaq NSK. The Triton ATMs are defined in the BASE24-atm Terminal Data File (TDF). This device handler was first introduced in BASE24-atm Release 5.3.

Refer to the BASE24-atm Triton Device Handler manual on the BASE24-atm system (i.e., $<volume>.AT60TRIT.TRITMAN) for information on implementing this device and device handler into a BASE24-atm user's system.

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Section 6

BASE24-pos Release 6.0

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 5.3 or a previous version of release 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with section 4 to identify the changes that BASE24-pos users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes
- Device Handler Separation
- Multiple Logical Network Routing Indicator
- PTDF Expansion
- Transactions Allowed

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. After the General Release of BASE24 Release 5.3, the CART file's primary key was changed, and an alternate key was added. If a BASE24-pos user has this file on their system, ensure that this file has been converted with the BASE24-pos Release 5.3 CART conversion program prior to implementing release 6.0. Refer to the PS53UC03 subvolume.

2. Three user fields were added to the following DDL layouts for future product enhancements, future regional enhancements, and future CSMs. These fields were added now, so file conversions can be avoided with later enhancements and customer specific changes.
    - PRDF
    - PTDD1
    - PTDS1

3. Several new ACI Standard POS Device Handler (SPDH) sub-FIDs have been included as part of the standard product. These are documented in the Vendor Impacts section.

4. The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the BASE24-pos Transaction Log File (PTLF) or reading up to the initially determined end-of-file of the PTLF.

5. Pre-authorization processing has been enhanced to allow pre-authorization holds to be identified by either the transaction sequence number or additionally, by using the approval code generated for the preauthorization request for the APACS 30/40 and SPDH Device Handlers. When a pre-authorization request transaction is authorized, BASE24 currently adds a pre-authorization 'hold' to the appropriate record on a customer file using the transaction sequence number to identify the hold. The pre-authorization processing support defined in Level 13 of the APACS 40 Standard specifies that the approval code generated for the transaction must be used to identify the hold, instead of the sequence number. POS Router/Authorization has been enhanced to allow BASE24-pos to support pre-authorization transactions received from any terminal supporting the APACS 40 Standard.

    This enhancement allows pre-authorization holds to be identified by either the transaction sequence number (as at present), or the approval code generated for the pre-authorization request.

The PSTM field PRE-AUTH-SEQ-NUM carries the value used to identify a pre-authorization hold. If the approval code is used to identify the hold, a hard-coded value of 'APPROVALCODE' can be set in this field by a BASE24-pos Device Handler for a pre-authorization request. If the message protocol allows it (e.g., SPDH), the terminal itself can pass the value 'APPROVALCODE' to BASE24-pos. If the message protocol does not allow it (e.g., APACS 40), the Device Handler can set up the value. Either way, the PSTM.PRE-AUTH-SEQ-NUM contains 'APPROVALCODE' when passed in a pre-authorization request to POS Router/Authorization.

If the contents of the PSTM PRE-AUTH-SEQ-NUM field does not contain 'APPROVALCODE', standard processing using the transaction sequence number applies; otherwise, the approval code is used. The last two characters of a BASE24-generated approval code (i.e., a space character followed by a B or X character) are removed from the PSTM.PRE-AUTH-SEQ-NUM field before a pre-authorization hold is added in this case. The last 2 digits of a BASE24-generated approval code are typically not returned to a device, and are logged to the PTLF solely to indicate that BASE24 authorized the transaction. Therefore, the last two digits are not present when the corresponding pre-authorization completion message is received from the device, and so cannot be used to locate the pre-authorization hold.

The associated pre-authorization completion is subsequently be received from a POS terminal. The value used to identify the pre-authorization hold (either the sequence number or the approval code) is included in the request message. Whether this value is a sequence number or an approval code is irrelevant to the acquiring interface; it simply moves the value to PSTM.PRE-AUTH-SEQ-NUM and passes the pre-authorization completion message to BASE24-pos Router/Authorization. BASE24-pos Authorization processing is completely standard. It uses the value in PSTM.PRE-AUTH-SEQ-NUM to retrieve the appropriate pre-authorization hold.

The benefit of this solution is that it applies to many BASE24-pos delivery processes (e.g., SPDH, HISO) with no changes to the acquiring interfaces. For example, if there is a requirement to implement this functionality with SPDH terminals, then terminals can be enhanced to set the PRE-AUTH-SEQ-NUM field to 'APPROVALCODE' in the SPDH pre-authorization request message. The SPDH itself does not have to be modified to ensure that the approval code was used to identify the pre-authorization hold.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The following user fields were added to the PRDF, PTDD1, and PTDS1 for future product enhancements, future regional enhancements, and future CSMs:

- USER-FLD-ACI is reserved for future BASE24 product use only. The designation of "product use only" provides ACI with the ability to deplete the number of bytes available within USER-FLD-ACI as product enhancements are introduced. When product enhancements require the addition of new fields within this file, the procedure to be followed is to deplete bytes from USER-FLD-ACI and use that number of bytes to define new fields. The new field definition(s) should precede the USER-FLD-ACI field.
- USER-FLD-REGN is reserved for ACI regional use only. Only ACI regions are allowed to use USER-FLD-REGN space.
- USER-FLD-CUST is reserved for BASE24 customer use only. Only customers are allowed to use USER-FLD-CUST space.

Note: The above fields allow space for future enhancements, and therefore they are not viewable using the BASE24 AFT subsystem. When a future enhancement uses a portion of any of the ACI reserved user fields, documentation on that enhancement will address the impacts to the BASE24 system.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following files were already provided to BASE24-pos users with other BASE24-pos Release 6.0 enhancements. The three new user fields in the PRDF file are initialized to spaces by the file conversion program for the PRDF provided as part of the Transactions Allowed enhancement. The three new user fields in the PTDD1 and PTDS1 files are initialized to spaces by the file conversion program for the PTDF provided as part of the PTDF Expansion enhancement.

- PRDF
- PTDD1

- PTDS1

Host/Co-Network Impacts

For additional information on Host/Co-Network Impacts with the PRDF file, refer to the Transactions Allowed enhancement.

For additional information on Host/Co-Network Impacts with the PTDD1 and PTDS1 files, refer to the PTDF Expansion enhancement.

User/Operational Impacts

User Impacts

BASE24-pos users that extract their PTLF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a PTLF file.

The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the last daily extract of the PTLF must be updated to have the value of Y for this field.

Vendor Impacts

ACI Standard POS Device Handler (SPDH) Message Formats

SPDH optional data field 6 contains new subfields (sub-FIDs) that provide additional optional information.

The following sub-FIDs (SFID) are available for standard processing:

FID	SFID	Picture	Length	Field Name	RQST	RESP
6	E	PIC 9(3)	3 bytes	Point of Service Entry Mode	✓	✓
6	I	PIC 9(3)	3 bytes	Transaction Currency Code	✓	✓
6	N	PIC 9(4)	4 bytes	Reason Online Code	✓	

Refer to the ***ACI Standard POS Device Message Specifications Manual*** for additional information.

Reference Documents

Not applicable

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Device Handler Separation

The Device Handler Separation enhancement moved the device handler specific files and information to the device handler subvolumes. The following changes were made in release 6.0 for this enhancement:

A server object for device specific data located in the PTD files is found on the appropriate device handler subvolume.

Make files (i.e., FM, MM, and M files) were changed to support the above change.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

BASE24 users must understand the concept of a separate device specific server objects for the PTD server.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Multiple Logical Network Routing Indicator

This section describes the release 6.0 changes and implementation requirements associated with the Multiple Logical Network Routing Indicator enhancement.

These changes and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Multiple Logical Network Routing Indicator enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

<u>MULT-LN-ACQ-TRAN-COUNTER</u>

Indicates that transactions routed between two logical networks using CPF SPROUTE routing should be included in the count for the reports. A value of Y indicates to count the corresponding transactions. A value of N indicates to not count the corresponding transactions. The default is Y.

Database Configuration

<u>Token File (TKN)</u>

The following BASE24 token has been added for the Multiple Logical Network Routing Indicator enhancement:

- The Multiple LN Token (Token ID “BK”) contains the logical network where the transaction was acquired.

If the new BASE24 token for this enhancement is not to be logged to the PTLF, the appropriate token records must be configured in the TKN.

If the new BASE24 token for this enhancement is to be extracted with the PTLF, the appropriate token records must be configured in the TKN.

If the destination is another BASE24 system, this token should not be configured to be sent in the ISO message using HISO or BIC ISO.

New Data Files

Not applicable

Template Changes

<u>COBNAMES</u>

New Multiple LN token names were added to the token table in the COBNAMES files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The newest version of SKEL must be used in order for this enhancement to be fully functional.

SCR #3000 SKELB - rel3^ver0^sutl13^020112

File Conversions

Not applicable

Host/Co-Network Impacts

<u>BIC ISO Interchange Interface Process</u>

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

<u>Super Extract Process</u>

If the new Multiple LN token for this enhancement is to be included in the PTLF, BASE24 users must consider this additional data in host programs that use the data from the log file.

<u>ISO Host Interface Process Impacts</u>

If the destination is another BASE24 system, then the Multiple LN token should not be configured to be sent in the ISO message. This is to avoid confusion on the secondary/issuer BASE24 system.

User/Operational Impacts

Disk Impacts

The PTLF requires additional disk space because the new Multiple LN token for this enhancement can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Transaction Based Pricing (TBP) Reports

The POS TBP reports have been updated to use the new Multiple LN token. This will allow the reports to properly count transactions that were SPROUTE routed using a CPF on another logical net. Users need to be on Version 7 or greater of the TBP Reports to take advantage of this change.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

PTDF Expansion

This section describes the BASE24-pos Release 6.0 changes and conversion/migration requirements associated with the PTDF Expansion enhancement.

Section 2 contains a functional overview of the PTDF Expansion Support enhancement.

This enhancement is only supported by the following BASE24-pos device handlers:

- ACI Standard POS Device Handler
- Hypercom Device Handler

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

<u>POS-PTD-DYN-GNRL</u>

The fully qualified file name of the BASE24-pos Terminal Data Dynamic file - general data. This is a required assign for the enhanced device handlers.

<u>POS-PTD-DYN-SCRATCH-PAD</u>

The fully qualified file name of the BASE24-pos Terminal Data Dynamic file – scratch pad data. This is an optional assign. If the BASE24-pos user wishes to store token data in a file other than the PTLF for reversal processing, ensure that this assign is present in the LCONF. Otherwise, ensure that this assign is not present in the LCONF.

<u>POS-PTD-STATIC-GNRL</u>

The fully qualified file name of the BASE24-pos Terminal Data Static file - general data. This is a required assign for the enhanced device handlers.

POS-DH-PTD-EXTMEM-SWAPVOL

The fully qualified file name of the swap volume in which the device handlers temporary store data when building extended memory for a terminal's PTD records. This is an optional assign. If the assign is not found, the Tandem chooses the swap volume for the application.

POS-PTDF

This assign is not used by the enhanced device handlers. If the BASE24-pos user only supports these device handlers, delete this assign from the LCONF. If the BASE24 user still supports a BASE24-pos non-enhanced device handler, do not delete this assign from the LCONF.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

POS-DH-MAX-IN-CORE

This param is not used by the enhanced device handlers. If the BASE24-pos user only supports these device handlers, delete this param from the LCONF. If the BASE24 user still supports a BASE24-pos non-enhanced device handler, do not delete this param from the LCONF.

POS-DH-MAX-TERMS

This param specifies the maximum number of PTD record keys (i.e., terminals) that can be saved in extended memory. Thus, this param indicates the maximum number of terminals each module can manage. This param has the same purpose as the POS-DH-MAX-TDFS param that was used by this device handler in previous releases.

This is an optional param. If the param is not found, has the value 0 or a value greater than 2147483647, the default value of 100 is used.

POS-DH-MAX-TERMS-IN-DYN-TBL

This param specifies the maximum number of PTD records each device handler saves in their dynamic data table located in extended memory. More than one dynamic data record can exist for a terminal ID, and the terminal's last message data, including the token data, is saved in this dynamic data table. Thus, this param indicates the number of terminals expected to ever be processing transactions simultaneously. This param has the same purpose as the POS-DH-MAX-IN-CORE param that was used by this device handler in previous releases.

This param is an optional param. If this param is not found, has the value 0 or a value greater than POS-DH-MAX-TERMS, the default value is either the value of POS-DH-MAX-TERMS or 25 (the lower of the two).

The maximum number of terminals in the dynamic data table cannot exceed the number of timers allowed in the device handler's timer table.

POS-DH-MAX-TERMS-IN-STATIC-TBL

This param specifies the maximum number of PTD records each device handler saves in their static data table located in extended memory. More than one static data record can exist for a terminal ID.

This param is an optional param. If this param is not found, has the value 0 or a value greater than POS-DH-MAX-TERMS, the default value is either the value of POS-DH-MAX-TERMS or 25 (the lower of the two).

POS-DH-MAX-TDFS

This param is not used by the enhanced device handlers. If the BASE24-pos user only supports these device handlers, delete this param from the LCONF. If the BASE24-pos user still supports a BASE24-pos non-enhanced device handler, do not delete this param from the LCONF.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depends on the configuration desired by the institution.

BASE24-pos Terminal Data files (PTD)

Field Name: TOKEN RETRIEVAL OPTION
Screen: PTD Screen 6
DDL Name: PTDS1.[ptds1-core].TKN-RETRV-FLG

When the device handler is processing a reversal, this field indicates where the token data should be retrieved. Valid values are:

0 = None

No tokens included in the PSTM

1 = PTD

Token data retrieved from the BASE24-pos Terminal Data Dynamic file – scratch pad data, and appended to the PSTM

2 = PTLF

Token data retrieved from the PTLF, and appended to the PSTM

9 = Use IDF (this is the default value)

The token retrieval option is specified at the institution level. Refer to the BASE24-pos segment of the IDF.

Institution Definition File (IDF)

Field Name: TOKEN RETRIEVAL OPTION
Screen: IDF Screen 19
DDL Name: IDF.POS.TKN-RETRV-FLG

When the device handler is processing a reversal, this field indicates where the token data should be retrieved. Valid values are:

0 = None

No tokens included in the PSTM

1 = PTD

Token data retrieved from the BASE24-pos Terminal Data Dynamic file – scratch pad data, and appended to the PSTM

2 = PTLF (this is the default value)

Token data retrieved from the PTLF, and appended to the PSTM

Token file (TKN)

If the new token for this enhancement (i.e., TLF Token) is not to be logged to the PTLF or ILF, the appropriate token records must be configured.

If the new token for this enhancement (i.e., TLF Token) is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

This section describes the files that have been added to support this enhancement.

BASE24-pos Terminal Data Dynamic file – general data (PTDD1)

The PTDD1 was created for storing dynamic data for each terminal ID. Refer to the LCONF Assign Changes and LCONF Param Changes subsections of this enhancement for further information about this file.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of PTD.

BASE24-pos Terminal Data Dynamic file – scratch pad data (PTDD2)

The PTDD2 was created for storing scratch pad data for each terminal ID. This is an optional file for BASE24-pos users wishing to store token data in a file other than the PTLF for reversal processing. Refer to the LCONF Assign Changes subsection of this enhancement for further information about this file.

This file is accessed only by the device handlers, and it is not accessible via the BASE24 AFT subsystem.

BASE24-pos Terminal Data Static file – general data (PTDS1)

The PTDS1 was created for storing static data for each terminal ID. Refer to the LCONF Assign Changes and LCONF Param Changes subsections of this enhancement for further information about this file.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of PTD.

Template Changes

BASE24-pos Transaction Log File (PTLF)

BASE24-pos supports two different alternate keys on the BASE24-pos Transaction Log File (i.e., terminal ID, primary account number). The user has the option to configure which alternate keys are required in their particular environment. A DDL compile is no longer used to create the FUP output for the PTLF. ACI provides BASE24-pos users with a set of standard template file at installation. Eight PTLF template files are located on the PS60MISC subvolume. These files can be used as is or the BASE24 user can alter the file to meet their requirements.

If the BASE24-pos users elect not to use the primary account number alternate key on their PTLFs, reversal processing is adversely impacted when the Token Retrieval Option of 2 is specified in the PTD or the IDF.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The PTD requester name was added to MEGATBL file, since RQMEGA communicates with the PTD requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

The PTDF references in the SECTBL file were changed to reference the PTD.

PATHCONF Changes

Delete the PTD Server (i.e., SERVER-PTD) from the PATHCONF file. This server was named SERVER-PTD, not SERVER-PTDF, in the previous releases of BASE24-pos). The PTD Server can be deleted online through the PATHCOM utility, but it must also be deleted from the PATHCONF, so that any future PATHCOLD operation would not contain the PTD Server.

The functionality of the PTD Server is now contained within the PTD Server object that is bound into the Classic Combined AFM Link Object (i.e., CCAFMLO).

CRT Access Security Changes

The BASE24 AFT subsystem security records that gave access to the PTDF screens were changed to refer to the PTD to gain access to ATD screens. The PTD requester and server handle processing for the PTDF as well as the PTD files.

When the Security file conversion program is run, it changes all PTDF references to PTD in the individual Security records. Refer to the subsection for the PI-Table Enhancement in Section 4 for more information on the Security file conversion program.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

<u>BASE24-pos Terminal Data files (PTD)</u>

The PTD conversion program must be run to convert PTDF records that exist for the Hypercom devices (i.e., terminal type equals “D0”, “D6”, “D7”, and “D8”) and the ASPDH devices (i.e., terminal type equals “41”) into the new PTD records. In addition, this same conversion program copies those TDF records that are not for the two TERMINAL ID types into a new TDF file. Users must create the PTD files and a new PTDF file using the PSDDLFUP file.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTDF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVPTDS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdcnv.t,name $ptdm,nowait/
cnvptdm –force –list –listreplace
```

Users must update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. CNVPTDR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing the BASE24-pos PTDF Expansion enhancement. Modifications to existing files and processing impact the Super Extract, ISO Host Interface, and BIC ISO Interchange Interface processes.

<u>Super Extract Process</u>

Host code changes are required for all BASE24-pos users that utilize the OMF Extract Process to accept and process information in the PTD files.

Host code changes are required for all BASE24-pos users that utilize the OMF Extract Process to accept and process the new information defined in the IDF.

New fields were added to the IDF. If the BASE24-pos user performs processing on the extracted IDF, they must be aware of the record format changes.

If the new TLF Token is included in the PTLF and ILF Extracts, BASE24-pos user must consider this additional data in host programs that use the data from log files.

<u>ISO Host Interface Process</u>

If the new TLF Token is included in the ISO message to the host, this change to the message needs to be handled by the host programs.

BIC ISO Interchange Interface Process

If the new TLF Token is included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space as the new token for this enhancement (i.e., TLF Token) can be included in the transaction records logged to these files. The total impact varies for each BASE24-pos user depending on the transaction volume in the BASE24-pos system.

The terminal data for the non-enhanced device handlers remains in the PTDF. The record size for these terminals does not change. The terminal data for the enhanced device handlers is maintained in the new PTD files. The first PTD file (i.e., PTDD1) consists of records with dynamic data for each terminal. Another PTD file (i.e., PTDS1) consists of records with static data for each terminal. A third PTD PTD file (i.e., PTDD2) is optional. This file consists of token information from the last message the device handler processed, and is only used when the BASE24 user has configured BASE24-pos to use token data from the PTDD2 for reversal processing. The maximum length for these PTD records is 4062. The total impact varies for each user depending on the number of enhanced terminals defined in the BASE24-pos system. The terminal data for the enhanced device handlers was removed from the TDF.

Operational Impacts

BASE24 users must be aware that the PTDF data for the non-enhanced devices is displayed along with the PTD data. The BASE24 AFT subsystem File Destination for both the PTD data and the TDF data is “PTD “, and the BASE24 AFT subsystem File Destination “PTDF “is no longer supported.

BASE24 users must perform a warmboot of the PTD static data using the new BASE24 DELIVER PROCESS text command ‘9529PTD’ after an operator performs file maintenance on any PTD static field. This is necessary since static data may not be read from disk during online transaction processing. This command is only supported via the BASE24 AFT NCS screen or NCPCOM. For more details, see the 9529PTD command in the BASE24-pos Process Commands section of the ***BASE24 Text Command Reference Manual***.

BASE24 users can determine the current status of the PTD extended memory tables using the new BASE24 DELIVER PROCESS text command 'STATUSEXTMEM'. This command is only supported via the BASE24 AFT NCS screen or NCPCOM. For more details, see the STATUSEXTMEM command in the BASE24-pos Process Commands section of the ***BASE24 Text Command Reference Manual***.

The BASE24 DELIVER PROCESS text command RESET TDF DATES was changed to be the BASE24 DELIVER PROCESS text command RESET PTD DATES command. This command changes the posting date in the PTDD1 records as well as the PTDF to the date specified in the command. For more details, see the RESETPTDDATES command in the BASE24-pos Process Commands section of the ***BASE24 Text Command Reference Manual***.

BASE24 users must configure the PTD and/or IDF to indicate whether to save token data and, if so, from where the token data should be retrieved, PTDD2 or PTLF. The token data is only retrieved from the PTDD2 or the PTLF if this data is no longer in extended memory.

BASE24 users must be aware that if the token data is configured to be retrieved from the PTLF, any token data not logged to the PTLF will not exist in the PSTM on the reversal. This can adversely affect reversal processing.

The values for the new LCONF params POS-DH-MAX-TERMS, POS-DH-MAX-TERMS-IN-DYN-TBL, and POS-DH-MAX-TERMS-IN-STATIC-TBL must be determined due to memory management requirements for this enhancement. Refer to the following calculation for the amount of memory that is used by each device handler:

```
8298 +
(POS-DH-MAX-TERMS's value * 138) +
(POS-DH-MAX-TERMS-IN-DYN-TBL's value * 8048)+
(POS-DH-MAX-TERMS-IN-STATIC-TBL's value * 4062)
```

If the PTDD2 file is utilized, increase the above sum by the following:

```
(POS-DH-MAX-TERMS-IN-DYN-TBL's value * 4062)
```

For example, if the values of the LCONF params POS-DH-MAX-TERMS, POS-DH-MAX-TERMS-IN-DYN-`TBL, and POS-DH-MAX-TERMS-IN-STATIC-TBL are 100, 50, and 90 respectively, the sum is 790078 (i.e., 8298+100*138+50*8048+90*4062). If the PTDD2 file is utilized, the sum is 993178 (i.e., 790078+50*4062).

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The structure and routines that maintain the terminals in extended memory has been enhanced, but the amount of memory required for these structures is larger than the version 1 structures.

Delete the LCONF POS-PTDF assign, POS-DH-MAX-IN-CORE param and POS-DH-MAX-TDFS param if the BASE24-pos user has only the BASE24-pos enhanced device handlers.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Transactions Allowed

The Transactions Allowed enhancement changes the mechanism to configure allowed transactions for the BASE24 Authorization processes, the enhanced device handlers, and the enhanced interchange interfaces.

This enhancement is only supported by the following BASE24-pos device handlers:

- ACI Standard POS Device Handler
- Hypercom Device Handler

For more information on this enhancement, refer to section 4 (Base).

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depends on the configuration desired by the institution.

Administrative Card (ADMN)

Field Name: ADMN TXN PROFILE
Screen: ADMN Screen 1
DDL Name: ADMN.ADMIN-TXN-PRFL

A code identifying a group of BASE24-pos transaction processing codes allowed for this administrative card. The value in this field overrides the default administrative card transaction profile defined at the institution level in the Institution Definition File (IDF). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

Cardholder Authorization File (CAF)

Field Name: ISSUER TXN PROFILE
Screen: CAF Screen 10
DDL Name: CAF.POS.ISS-TKN-PRFL

A code identifying a group of BASE24-pos issuer transaction processing codes allowed for this cardholder in the Issuer Processing Code File (IPCF). The value in this field overrides the issuer transaction profile defined at the card prefix level in the CPF or at the institution level in the IDF. The value entered in this field must match a transaction profile in the Issuer Processing Code File (IPCF).

Card Prefix File (CPF)

Field Name: ISSUER TXN PROFILE
Screen: CPF Screen 6
DDL Name: CPF.POS.ISS-TKN-PRFL

A code identifying a group of BASE24-pos issuer transaction processing codes allowed for this card prefix in the Issuer Processing Code File (IPCF). The value in this field overrides the issuer transaction profile defined at the institution level in the IDF. The value entered in this field must match a transaction profile in the Issuer Processing Code File (IPCF).

Institution Definition File (IDF)

Field Name: ACQUIRER TXN PROFILE
Screen: IDF Screen 19
DDL Name: IDF.POS.ACQ-TKN-PRFL

A code identifying a group of default BASE24-pos cardholder transaction processing codes allowed at POS terminals owned by this institution. This profile applies only to Hypercom and POS Standard Device Handler modules. The value in this field can be overridden at the retailer level in the BASE24-pos Retailer Definition File (PRDF) or at the terminal level in the BASE24-pos Terminal Data files (PTD). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

Field Name: ISSUER TXN PROFILE
Screen: IDF Screen 19
DDL Name: IDF.POS.ISS-TKN-PRFL

A code identifying a group of default BASE24-pos issuer transaction processing codes allowed for this institution's cardholders. The value in this field can be overridden at the card prefix level in the CPF or at the cardholder level in the CAF. The value entered in this field must match a transaction profile in the Issuer Processing Code File (IPCF).

Field Name: RETAILER TXN PROFILE
Screen: IDF Screen 19
DDL Name: IDF.POS.RTLR-TXN-PRFL

A code identifying a group of default BASE24-pos transaction processing codes allowed for retailers associated with this institution for transactions that require an administrative card. The value in this field can be overridden at the retailer level in the BASE24-pos Retailer Definition File (PRDF). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

Field Name: ADMN TXN PROFILE
Screen: IDF Screen 19
DDL Name: IDF.POS.ADMIN-TXN-PRFL

A code identifying a group of default BASE24-pos transaction processing codes allowed for administrative cards associated with this institution. The value in this field can be overridden at the administrative card level in the Administrative Card File (ADMN). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

<u>BASE24-pos Retailer Definition File (PRDF)</u>

Field Name: ACQUIRER TXN PROFILE
Screen: PRDF Screen 4
DDL Name: PRDF.ACQ-TKN-PRFL

A code identifying a group of BASE24-pos cardholder transaction processing codes allowed at POS terminals for this retailer. The value in this field overrides the default acquirer transaction profile at the institution level in the Institution Definition File (IDF). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

Field Name: RETAILER TXN PROFILE
Screen: PRDF Screen 4
DDL Name: PRDF.RTLR-TKN-PRFL

A code identifying a group of BASE24-pos administrative transaction processing codes allowed for this retailer that require employees to use an administrative card. The value in this field overrides the default retailer transaction profile defined at the institution level in the Institution Definition File (IDF). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

BASE24-pos Terminal Data files (PTD)

Field Name: ACQUIRER TXN PROFILE
Screen: PTD Screen 6
DDL Name: PTDS1.[ptds1-core].ACQ-TKN-PRFL

A code identifying a group of BASE24-pos cardholder transaction processing codes allowed at this POS terminal. The value in this field overrides the value at the retailer level in the POS Retailer Definition File (PRDF) and the value at the institution level in the Institution Definition File (IDF). The value entered in this field must match a transaction profile in the Acquirer Processing Code File (APCF).

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Administrative Card (ADMN)

The ADMN conversion program must be run to convert ADMN records from the release 5.3 format to the release 6.0 format. Users must create the ADMN file using the BADDLFUP file.

The ADMN conversion program on the PS60UC04 subvolume can be used to convert the ADMN. If modifications to the conversion program are necessary, the program can be modified by editing the CNVADMNS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#admncnv.t,name $admnm,nowait/
cnvadmnm –force –list –listreplace
```

Users must update the CNVADMNR obey file on the PS60UC04 subvolume with the appropriate information. CNVADMNR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

BASE24-pos Retailer Definition File (PRDF)

The PRDF conversion program must be run to convert PRDF records from the release 5.3 format to the release 6.0 format. Users must create the PRDF file using the BADDLFUP file.

The PRDF conversion program on the PS60UC04 subvolume can be used to convert the PRDF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVPRDFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#prdfcnv.t,name $prdfm,nowait/
cnvprdfm –force –list –listreplace
```

Users must update the CNVPRDFR obey file on the PS60UC04 subvolume with the appropriate information. CNVPRDFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the Super Extract process.

Super Extract Process

Host code changes are required for all BASE24-pos users that utilize the OMF Extract Process to accept and process the new information defined in the PRDF.

If the new tokens for this enhancement (i.e., Transaction Profile Token, Transaction Description Token, Acquirer Routing Token) are to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Derived Unique Key Per Transaction (DUKPT)
- eCheck Support
- Electronic Benefits Transfer (EBT)
- EMV Support
- Format 2 File Support
- Multiple Currency Support
- Stored Value Support
- Transaction Security Services
- Triple/Single DES Translation Support

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 changes, conversion requirements, and implementation requirements for future reference.

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Derived Unique Key Per Transaction (DUKPT) Support

This section describes the release 6.0 changes, conversion, and implementation requirements associated with the Derived Unique Key Per Transaction (DUKPT) enhancement.

These changes, conversion, and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Derived Unique Key Per Transaction Support enhancement.

The SPDH supports DUKPT for the encryption of PIN blocks for transactions originating at the terminals directly connected to BASE24-pos.

Using the DUKPT method, each terminal security module (TSM), or PIN pad, uses a derived unique key for each transaction. However, the TSM never contains any information that could lead to the determination of any past transaction keys used by the device, or of any past or future keys used by any other TSM. The originating TSM must derive the current transaction key from an initial key, loaded into the TSM on initialization. The receiving BASE24-pos security module (Atalla or Racal) then determines the current transaction key by using a key held on BASE24-pos, and non-secret information contained in the transaction message.

The PTD PIN-ENCRYPT-TYP field supports a new value of 07 (DUKPT). The Derivation Key File (KEYD) stores the derivation keys. The BASE24 AFT facility is used to maintain the new data in these files.

The Standard POS Device Handler (SPDH) message definition can optionally include FID 6 sub-FID T, which is used to transmit the Key Serial Number (KSN) and KSN descriptor from the terminal to BASE24.

Upon receipt of a terminal request message, the Standard POS Device Handler (SPDH) accesses the BASE24-pos Terminal Data file (PTD). The SPDH uses the PTD PIN-ENCRYPT-TYP field to determine whether the terminal is using DUKPT. The PTD RETAILER-ID, KEYD-GROUP and TERM-ID fields are used as the key to access the KEYD.

If the transaction contains a PIN and the PIN block is in the DUKPT format, the SPDH module converts the PIN block from encryption under the DUKPT to encryption under an intermediate PIN block encryption key. The intermediate PIN block encryption key is stored in the PTD M-KEY field (encrypted under variant 1 of the MFK for Atalla NSP Security Modules or LMK pair 06-07 for Racal Security

Modules). Encryption under the intermediate key allows standard BASE24 PIN processing to be used in Router/Auth and issuer interfaces. The PIN block translation is performed using the Security Utilities (SECUTILS) module.

The new PIN block and the intermediate key are loaded in the PSTM and forwarded to the Router/Auth process. The POS Standard Internal Message (PSTM) is used to transport DUKPT data between BASE24 processes using a token.

The configuration of a TSM can be converted to DUKPT encryption by adding the corresponding KEYD record, setting the PTD KEYD-GRP field, setting the LCONF param POS-DH-DUKPT-UPDATE-METHOD to a “Y” and changing out the TSM. If the KSN is present in the message (FID 6 sub-FID T), the PTD PIN-ENCRYPT-TYP field does not equal 07 (DUKPT) and the LCONF param POS-DH-DUKPT-UPDATE-METHOD does not equal N, the device handler automatically updates the PTD PIN-ENCRYPT-TYP field to 07 (DUKPT), generates an intermediate key for the PTD M-KEY field, and processes the transaction.

Note: DUKPT is not currently supported with the Transaction Security Services enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

<u>KEYD</u>

The fully qualified file name of the BASE24 Derivation Key File. This is a optional assign for the enhanced device handlers.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

<u>POS-DH-KSN-DESCR</u>

This optional param contains the default KSN descriptor needed for the Thallas

Security Module to perform a translate command when DUKPT is being used at the terminal to encrypt the PIN block. It is a three-byte numeric value. It contains the sum of the individual digits of the KSN minus the counter. A KSN that is sixteen bytes in length would have a KSN descriptor whose individual digits when added together would equal eleven (sixteen actual length of KSN – transaction counter). The KSN descriptor for a sixteen byte long KSN would be 803 (8 + 0 + 3 = 11).

The default for this param is a value of 644.

<u>POS-DH-KEYD-READ-FROM-DISK</u>

This optional param indicates whether the KEYD should be read from disk or placed into memory.

A value of Y (Yes) indicates the record is read from disk.

A value of N (No) indicates the record is read from memory.

The default value for this param is N. If the param is not present, the record is read from memory.

<u>POS-DH-DUKPT-UPDATE-METHOD</u>

This optional param specifies whether the SPDH module can automatically update the PTD PIN-ENCRYPT-TYP field when FID 6 sub-FID T is received for the first time and a KEYD record exists.

A value of Y (Yes) indicates the PTD PIN-ENCRYPT-TYP field is updated.

A value of N (No) indicates the PTD PIN-ENCRYPT-TYP field is not updated.

The default value for this param is N. If the param is not present, the PTD PIN-ENCRYPT-TYP field is not updated.

<u>SECURE-DEV-CONF-CMD-100</u>

This required param contains the 100 command that enables features for NSP devices.

This command enables DUKPT. Contact the security device vendor for the syntax of the command. This may be billable.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

BASE24-pos Terminal Data files (PTD)

The SPDH Device Handler had been enhanced to use the PTD in Release 6.0. For more information about how these files work, refer to the POS section on the PTDF Expansion.

Screen Name: PIN Encryption Type
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].ENCR-PIN.PIN-ENCRYPT-TYP

Indicates the PIN encryption type supported for the terminal. A value of 07 was added with this enhancement and indicates DUKPT.

Screen Name: PIN Master Key
Screen: PTD Screen 7
DDL Name: PTDD1.[ptdd1-core].ENCR-PIN.ENCR-KEYS.M-KEY

Contains a KPE encrypted under variant 1 of the MFK for Atalla NSP Security Modules or LMK pair 06-07 for Racal Security Modules for terminals supporting DUKPT. This is used for translation of the PIN block from DUKPT pin encryption to master/session pin encryption.

Screen Name: Derivation Key Group
Screen: PTD Screen 7
DDL Name: PTDS1.[ptds1-core].KEYD-GRP

Indicates the KEYD group used to retrieve the KEYD record.

ACI Standard Device Configuration File (ACNF)

FID 6 should be included in the request Field Map records of the ACNF for DUKPT driven terminals when a PIN is present.

Token File (TKN)

The following BASE24-pos tokens have been added for the Derived Unique Key Per Transaction Support enhancement:

- The DUKPT Data token (Token ID "CA") contains the KSN and KSN descriptor.

If the new BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the new BASE24-pos token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured in the TKN.

New Data Files

Derivation Key File (KEYD)

The Derivation Key File (KEYD) contains derivation keys required for SPDH-driven terminals supporting the encryption type of DUKPT. The Derivation Key File is a file created by the customer/operator which contains the 32-byte derivation key from the terminals encrypted under variant 8 of the MFK (Atalla)/LMK pair 28/29 (Racal).

The KEYD is read by the SPDH into memory on initialization/warmboot or from disk on every pinned transaction using DUKPT encryption(depending on the POS-DH-KEYD-READ-FROM-DISK LCONF parameter) as follows:

Matching Sequence	Retailer ID	KEYD Group	Terminal ID
1.	Exact	Exact	Exact
2.	Exact	Exact	****************
3.	Exact	****	****************
4.	****************	****	****************

Template Changes

Not applicable

Device Configuration

SPDH FID 6 sub-FID T is used to pass the key serial number (20 bytes) and the key serial number descriptor (3 bytes used for Racal security modules only).

NSP security devices must have DUKPT enabled (see SECURE-DEV-CONF-CMD100 LCONF parameter above)

Mega Table (MEGATBL) Changes

The KEYD requester name was added to MEGATBL file, since RQMEGA communicates with the KEYD requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

SECTBL entries for KEYD screens have been modified, so these files will appear on the BASE Product Menu.

PATHCONF Changes

The new KEYD server must be defined in the PATHCONF file as SERVER-KEYD. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new KEYD requester screen must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Since the KEYD derivation key field is a 32-byte key (double length) the MFK for the applicable security devices must also be a double-length key.

File Conversion

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interface and Super Extract processes.

External Interfaces

There are no essential changes required to HISO for passing the KSN and KSN descriptor information between BASE24-pos and host systems. However, the host can make use of the KSN to potentially detect fraud with a given PIN pad (the KSN should be sequential).

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by the BASE24-pos Derived Unique Key Per Transaction Support enhancement but host systems must be aware that PTLF.AUTH.ORIGINATOR field can contain a new value of “9” – DUKPT driven terminal using a Racal security module.

If the new token (i.e., DUKPT Data Token) for this enhancement is to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

Host code changes are required for all BASE24 users that wish to utilize the OMF Extract Process to accept and process file maintenance made to the KEYD file using the BASE24 AFT subsystem.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the new token for this enhancement (i.e., DUKPT Data Token) can be included in the transaction records logged to these files.

Disk space is required to accommodate the new KEYD file.

Operational Impacts

If the TSM (PIN PAD) is swapped out at the retailer, the network operator must update the PTD with the correct KEYD group to insure the correct derivation key is used (if applicable).

BASE24 operators must be aware that the new file will appear on their BASE24 BASE product menu.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

An additional I/O is required to the security device to translate the PIN from DUKPT encryption to using master/session encryption.

Additional I/O is required to either read the KEYD into memory on initialization/warmboot or to read the KEYD from disk on each applicable transaction. This can require several disk reads to find the correct KEYD record.

Vendor Impacts

The terminal vendor must encrypt the PIN block using DUKPT encryption. This is passed in FID b or FID c (admin). The clear derivation key used in the terminal must be encrypted and placed into the corresponding KEYD record.

SPDH FID 6 sub-FID T is used to pass the key serial number (20 bytes) and the key serial number descriptor (3 bytes used for Racal security modules only).

NSP security devices must have DUKPT enabled. See the SECURE-DEV-CONF-CMD100 LCONF parameter above.

Reference Documents

Not applicable

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eCheck Support

BASE24-pos eCheck is a separately licensable add-on module to BASE24-pos which enables online positive balance check verification, guarantee or conversion services from the point-of-sale. With this solution, acquirers are able to offer full service electronic check processing and authorization of check transactions to merchant clients through their existing infrastructure. Issuers are able to perform check verification and check guarantee transactions in participation with current check verification or conversion programs now offered by several card associations and networks.

The BASE24-pos Router/Authorization module supports check verification and check guarantee transactions with an online-only authorization level, with or without a PAN. To facilitate eCheck, several BASE24 components provide the foundation for electronic check. With the eCheck Foundation, an issuer or acquirer is able to route and pass check verification and guarantee transactions through BASE24, taking advantage of additional check related tokens and routing improvements. Separate licensable add-on eCheck modules are required to allow BASE24-pos to authorize or stand-in for check verification and guarantee transactions or to receive and send check transactions through a network interface.

Refer to the BASE24-pos eCheck manual on the BASE24-pos system (i.e., $<volume>.PS60ERTA.ERTAMAN) for information on implementing this enhancement into a BASE24-pos system.

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Electronic Benefits Transfer Support

This section describes the release 6.0 changes, conversion, and implementation requirements associated with the Electronic Benefits Transfer enhancement.

These changes, conversion, and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Electronic Benefits Transfer Support enhancement.

Electronic Benefits Transfer (EBT) enables retail customers to acquire and route electronic benefits transactions. EBT transactions are transactions which allow consumers to access accounts for Food Stamps or Cash Benefits (Aide For Dependent Children).

EBT transactions are supported only through BASE24-pos and only as BASE24 acquirer traffic; therefore if a transaction cannot be completed because the link to the authorizing entity is down, no stand-in processing is performed. No support exists for BASE24 issuer traffic via a switch interface acquirer. EBT transactions also can be run from other point-of-service (POS) devices attached to BASE24, but no modifications are required to support this functionality. These POS transactions are treated as withdrawals from checking to the corresponding EBT switch interface.

Typically a customer swipes the card through the SPDH terminal, enters the PIN and identifies if the funds for the transaction come from their Food Stamps or Cash Benefits account, if necessary. The request is sent to BASE24 using the SPDH message format. The Router module determines if this is a valid transaction, and standard AST/SPROUTE routing is used to determine the switch interface where the authorization request should be sent. Additional balance information present in a switch response is stored in the PSTM EBT Available Balance Token which is returned to the Device Handler module and mapped into the SPDH FID 8, sub-FID b of the response message for printing on the receipt at the device.

In the event that the BASE24 switch interface link to the authorizing entity is unavailable, the retailer can accept an EBT transaction authorized by voice authorization using a manual voucher form. The terminal operator is required to call into the authorization system for approval and an approval number. The SPDH terminal operator then forwards the transaction to the authorizing system electronically as a force post transaction with the approval code received from the authorizer and the voucher number from the manual voucher. The SPDH FID 8 sub-FID a is used to store the new voucher number in the request from the terminal. The Device Handler then places the voucher number in a PSTM EBT Voucher

Number Token. The transaction is SAFed by the switch interface until the link to the authorizing entity is reestablished.

CUSTMACS Changes

Previous releases of Electronic Benefits Transfer (EBT) support included an EBTMACS file, which was used in place of the CUSTMACS file when making the BASE24 system with EBT support. BASE24 Release 6.0 includes EBT support with no changes to CUSTMACS.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>ACI Standard Device Configuration File (ACNF)</u>

FID 8 should be included in the request and response Field Map records of the ACNF for EBT transactions.

<u>Account Type Table File</u>

The ATT file contains one record for each valid account type in the BASE24 network. Each record contains an ISO code in the Account Type field and a corresponding description in the Account Type Name field.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Records for the two new Account types (Cash Benefit – 96, Food Stamps – 98) must be added to this file and the values are listed below.

Account Type: 96 Account Type Name: CshBnf

Account Type: 98 Account Type Name: FdStmp

ACI provides BASE24 users with a set of default ATT records at installation. The records are located in the ATT file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Processing Code Description File (PDF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the PDF file in order for description tags to be recognized in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

Description Tag	Long Description
ISO009600	NORMAL PURCHASE CSHBNF
ISO009800	NORMAL PURCHASE FDSTMP
ISO1C9600	PRE-AUTH PURCHASE CSHBNF
ISO189600	PRE-AUTH COMPL CSHBNF
ISO209600	MERCHANDISE RETURN CSHBNF
ISO209800	MERCHANDISE RETURN FDSTMP
ISO019600	CASH FROM CSHBNF
ISO309600	AVAIL FUNDS INQUIRY CSHBNF
ISO309800	AVAIL FUNDS INQUIRY FDSTMP
ISO099600	PURCHASE CASH BACK CSHBNF

ACI provides BASE24 users with a set of default PDF records at installation. The records are located in the PDF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Acquirer Processing Code File (APCF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the APCF file in order for the EBT transactions to be recognized by the Device Handler. The Authorization Destination field must be spaces for all records and the Log Auth Dest Response field must be set to N for all records.

Acquirer Transaction Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	Transaction Allowed
POS	*	00	96	00	ISO009600	4
POS	*	00	98	00	ISO009800	4
POS	*	1C	96	00	ISO1C9600	4
POS	*	18	96	00	ISO189600	4
POS	*	20	96	00	ISO209600	4
POS	*	20	98	00	ISO209800	4
POS	*	01	96	00	ISO019600	4
POS	*	30	96	00	ISO309600	4
POS	*	30	98	00	ISO309800	4
POS	*	09	96	00	ISO099600	4

BASE24 users also must configure the APCF (transactions allowed file) with the new EBT transaction codes. This only impacts the endpoints (i.e., device handlers and interfaces), because BASE24 does not perform offline authorization for EBT transactions.

ACI provides BASE24 users with a set of default APCF records at installation. The records are located in the APCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Issuer Processing Code File

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the IPCF file in order for the EBT transactions to be recognized by the issuer interchanges. The Completion Required to Host field must be set to N for all records.

Issuer Trans Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	On-Us Trans Allowed	Not-On-Us Trans Allowed
POS	*	00	96	00	ISO009600	4	4
POS	*	00	98	00	ISO009800	4	4
POS	*	1C	96	00	ISO1C9600	4	4
POS	*	18	96	00	ISO189600	4	4
POS	*	20	96	00	ISO209600	4	4
POS	*	20	98	00	ISO209800	4	4
POS	*	01	96	00	ISO019600	4	4
POS	*	30	96	00	ISO309600	4	4
POS	*	30	98	00	ISO309800	4	4
POS	*	09	96	00	ISO099600	4	4

BASE24 users must configure the IPCF with the new EBT transaction codes. This only impacts the endpoints (i.e., device handlers and interfaces), because BASE24 does not perform offline authorization for EBT transactions.

ACI provides BASE24 users with a set of default IPCF records at installation. The records are located in the IPCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

POS Retailer Definition File (PRDF)

No new data fields have been added to the PRDF for the Electronic Benefits Transfer Support enhancement; however, many interchanges require that the Alternate Merchant ID field of Screen 2 of the PRDF be filled in accordingly for the EBT card types.

Token File (TKN)

The following BASE24-pos tokens have been added for the Electronic Benefits Transfer Support enhancement:

- The EBT Voucher Number Token (Token ID “U0”) contains the manual authorization voucher number for an EBT transaction.
- The EBT Available Balance Token (Token ID “U1”) contains available balances in a response message from an interchange.

If the new BASE24-pos tokens for this enhancement are not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the new BASE24-pos tokens for this enhancement are to be extracted with the PTLF or ILF, the appropriate token records must be configured in the TKN. EBT transactions cannot be IDF routed; therefore, they cannot be sent to a HISO process.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

SPDH FID 8 sub-FID a is used to pass the manual authorization voucher number for EBT transactions.

SPDH FID 8 sub-FID b is used to contain the available balances in response messages when applicable for an interchange authorized EBT transaction.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

EBT transactions can be AST or SPROUTE routed. Typically the card prefixes are not known to an acquiring entity of EBT transactions. In this case, transactions can be SPROUTE routed using the default destination for the terminal group as the EBT service provider.

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

<u>External Interfaces</u>

EBT transactions are not supported through the ISO Host Interface process and the BIC ISO Interchange Interface process.

<u>Super Extract Process</u>

Standard PTLF Extract processing is not specifically impacted by the BASE24-pos Electronic Benefits Transfer Support enhancement but host systems must be aware that PTLF Extract files (and any applicable ILFs) can contain records with the account type 96 (Cash Benefit) and 98 (Food Stamp), as the BASE24-pos system is now capable of processing Electronic Benefits Transfer transactions (and settlement data). The BASE24-pos Terminal Data files (PTD) totals include EBT transactions in the corresponding counts and amounts. These totals are logged to the PTLF.

If the new tokens (i.e., EBT Voucher Number and EBT Available Balance Tokens) for this enhancement are to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the new tokens for this enhancement (i.e., EBT Voucher Number and EBT Available Balance Token) can be included in the transaction records logged to these files.

Operational Impacts

BASE24 users must be aware of the new EBT transaction codes that can be displayed via PTLF perusal and included in the POS reports.

EBT transactions are included in the PTD totals kept by the SPDH Device Handler module.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

SPDH FID 8 sub-FID a is used to pass the manual authorization voucher number for EBT transactions.

SPDH FID 8 sub-FID b is used to contain the available balances in response messages when applicable for an interchange authorized EBT transaction.

Reference Documents

Not applicable

EMV Support

BASE24 Base processing has been enhanced to support the common Integrated Circuit Card (ICC) standard developed by Europay International, MasterCard International, and Visa International, commonly referred to as EMV '96. Refer to the ***BASE24-pos EMV Support Manual*** for additional information.

This section describes the release 6.0 changes and conversion and implementation requirements associated with the EMV Support enhancement.

These changes, conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the EMV Support enhancement.

For more detailed information on this enhancement, refer to section 4 (Base).

This enhancement is only supported by the following BASE24-pos device handlers:

- ACI Standard POS Device Handler
- APACS 30/40 Device Handler
- Hypercom Device Handler

CUSTMACS Changes

Set the following flag to true to add the EMV bindable pieces to the appropriate authorization modules:

- pos_emv_on

Set each of the following flags to true to add the EMV bindable pieces to the appropriate device handler modules.

- pos_dh_apacs_emv_on
- pos_dh_hpdh_emv_on
- pos_dh_spdh_emv_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

BASE24-pos Terminal Data files (PTD and PTDF)

The PTD screens are used to add data to the PTD or PTDF, depending on the device type. The SPDH Device Handler and Hypercom Device Handler have been enhanced to use the PTD in Release 6.0. The APACS Device Handler uses the PTDF. For more information about how these files work, refer to the POS section on the PTDF Expansion.

Screen Name:	EMV Terminal Capabilities
Screen:	PTD Screen 17
DDL Name:	PTDS1.[ptds1-core].EMV-TERM-CAP PTDF.EMV-TERM-CAP

Indicates the card data input, CVM, and security capabilities of the terminal using a bit map to represent possible options.

The PTD has a separate screen 17 to configure the EMV Terminal capabilities. Set the flags on this screen according to the information in the ***BASE24-pos EMV Support Manual*** to define the EMV related capabilities for each terminal.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new screen for the PTD requester/server.

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new PTD requester screen must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

New BASE24-pos EMV related response codes were defined with this enhancement. Operators must be trained to recognize the following new EMV related response codes on the PTLF Perusal screen:

- 130 ARQC failure
- 131 CVR Referral
- 132 TVR Referral
- 133 Reason On-line Referral
- 134 Fallback Referral
- 400 ARQC failure
- 401 HSM parameter error
- 402 HSM failure
- 403 KEYI Record not found
- 404 ATC check failure
- 405 CVR decline
- 406 TVR decline
- 407 Reason On-line decline
- 408 Fallback Decline
- 910 ARQC failure
- 911 CVR Capture
- 912 TVR Capture

Vendor Impacts

ACI Standard POS Device Handler (SPDH) Message Formats

SPDH optional data field 6 contains new subfields (sub-FIDs) that provide additional optional information.

The following sub-FIDs (SFID) are available for EMV processing:

FID	SFID	Picture	Length	Field Name	RQST	RESP
6	O	PIC X(128)	128 bytes	EMV Request Data	✓	
6	P	PIC X(64)	64 bytes	EMV Additional Request Data	✓	
6	Q	PIC X(64)	64 bytes	EMV Response Data		✓
6	R	PIC X(64)	64 bytes	EMV Additional Response Data		✓

Refer to the ***ACI Standard POS Device Message Specifications Manual*** for additional information.

Reference Documents

Not applicable

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Format 2 File Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Format 2 File Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Format 2 File Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

The PTLF template file can have up to five alternate keys. The number of ways a BASE24-pos user wants to peruse their PTLF records and the token retrieval option indicates the number of alternate keys to create. Refer to the PTDF Expansion enhancement for further information on the token retrieval option.

There are seventeen different examples of PTLF template files that the BASE24-pos user can use. These PTLF template example files are located on the PS60MISC subvolume. You can create the template on the template subvolume with these examples, by using the VOLUME command to move to the logical network template subvolume (e.g., VOLUME $<volume>.<logical network>TPLT), and using the FUP utility to copy a template (e.g., FUP/IN $<volume>.PS60MISC.TPLTPTL7). The template examples are described below.

PTLF217

The PTLF217 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME, TERMINAL/RETAILER, and RETAILER/CLERK. However, the alternate key files are format 1 files. This option can be used to improve performance as long as the alternate key file does not exceed the format 1 size limit of 4 gigabytes.

PTLF220

The PTLF220 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD and RETAILER. Both the primary file and the alternate key files are format 2 files.

PTLF221

The PTLF221 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER and TERMINAL ID/TIME. Both the primary file and the alternate key files are format 2 files.

PTLF222

The PTLF222 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER and TERMINAL/RETAILER. Both the primary file and the alternate key files are format 2 files.

PTLF223

The PTLF223 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER and RETAILER/CLERK. Both the primary file and the alternate key files are format 2 files.

PTLF224

The PTLF224 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME and TERMINAL/RETAILER. Both the primary file and the alternate key files are format 2 files.

PTLF225

The PLTF225 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME and RETAILER/CLERK. Both the primary file and the alternate key files are format 2 files.

PTLF226

The PTLF226 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER, TERMINAL/RETAILER, and RETAILER/CLERK. Both the primary file and the alternate key files are format 2 files.

PTLF227

The PTLF227 file contains the necessary FUP commands to create the PTLF template files as format 2 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME, TERMINAL/RETAILER, and RETAILER/CLERK. Both the primary file and the alternate key files are format 2 files.

TPLTPTL0

The TPLTPTL0 file contains the necessary FUP commands to create the PTLF template files as format 1 files using the alternate keys of CARD and RETAILER.

TPLTPTL1

The TPLTPTL1 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER and TERMINAL ID/TIME.

TPLTPTL2

The TPLTPTL2 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER and TERMINAL/RETAILER.

TPLTPTL3

The TPLTPTL3 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER and RETAILER/CLERK.

TPLTPTL4

The TPLTPTL4 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME and TERMINAL/RETAILER.

TPLTPTL5

The TPLTPTL5 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME and RETAILER/CLERK.

TPLTPTL6

The TPLTPTL6 file contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER, TERMINAL/RETAILER and RETAILER/CLERK.

TPLTPTL7

The TPLTPTL7 file is a FUP command file that contains the necessary FUP commands to create a PTLF template file as format 1 files using the alternate keys of CARD, RETAILER, TERMINAL ID/TIME, TERMINAL/RETAILER, and RETAILER/CLERK.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

For those BASE24-pos users implementing this enhancement and extracting the PTLF to a format 2 file, the host code must process the new block format for the extract format 2 disk file. In addition, the host code can use the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The value 2 in this field denotes a format 2 file. The fourth byte of the user field in the Extract File Header was used to store this information.

User/Operational Impacts

Disk Impacts

Any file converted to format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for format 2 files is increased. This is true when the primary file is format 2, regardless of whether the alternate key file(s) is format 2.

Since the maximum data area per block in format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a format 2 file. Users should check the length of their PTLF records to see if fewer PTLF records will fit in the new format 2 maximum data area blocks. The format 2 maximum data area is 4048 for entry-sequenced files, like the PTLF. If fewer PTLF records will fit in the new format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry-sequenced files against the PTLFX file, if a format 2 version of this file is to be used when extracting to disk.

User Impacts

BASE24-pos users using format 2 PTLF files must modify their template files to be format 2 files. Refer to the Template Changes subsection of this enhancement for further information about PTLF Template files.

BASE24-pos users extracting their PTLF files to format 2 disk files must configure the appropriate information in the ECF.

BASE24-pos users using format 2 PTLF files must start at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files) and create the format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the PTLF is created using the new format 2 template files.

Operational Impacts

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM or the BASE24 Documentation CD-ROM.

System Limitations

BASE24-pos users must calculate the length of the records that are written to the PTLF. Format 2 PTLF files allow for a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are written to the PTLF files, the record length of a PTLF record could exceed the limit of 4048 bytes for format 2 files.

Before BASE24-pos users change their PTLFs to format 2 files, they should ensure these records do not exceed the 4048 limit. BASE24-pos users using format 1 are not affected, since the DDL for the PTLF was not changed to set the maximum record length to 4048.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

ACI Worldwide Inc.

Multiple Currency Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Multiple Currency Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Multiple Currency Support enhancement.

The Multiple Currency add-on product allows cardholders to perform transactions in currencies different from their account currencies at different POS devices. In addition, this enhancement provides the ability for BASE24-pos to acquire transactions in different currencies from a single POS device for selected terminal types that allow transactions to be performed in different currencies. Refer to the ***BASE24-pos Multiple Currency Support Manual*** for additional information.

CUSTMACS Changes

Several new flags have been added for the Multiple Currency Support (MCU) enhancement. Refer to section 4 for general MCU flags that are required to be set.

Set the following flag to true to add the MCU bindable pieces to the appropriate authorization module:

- pos_mcu_on

Set each of the following flags to true to add the MCU bindable pieces to the appropriate device handler modules

- pos_dh_apacs_mcu_on
- pos_dh_spdh_mcu_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Authorization Selection Table (AST)

No new data fields have been added to the AST for the Multiple Currency Support enhancement. Since this file contains limit fields that need to have a currency code associated with them if the Multiple Currency Support enhancement is being used, the limit fields are assumed to be in the currency of the corresponding institution (IDF) record to which the routing information pertains. This can be an Issuer Institution or Acquirer Institution dependant upon configuration.

Cardholder Authorization File (CAF)/Usage Accumulation File (UAF) Segment

No new data fields have been added to the BASE24-pos segment of the CAF or UAF for the Multiple Currency Support enhancement. Since these files contain amount fields that need to have a currency code associated with them, if the Multiple Currency Support enhancement is being used, the amount fields are therefore assumed to be in the currency of the corresponding Institution Definition File (IDF) record.

BASE24-pos CRT Authorization Usage File (CRUF)

The CRT Authorization Device Handler has not been enhanced to support multiple currencies when processing using the CRUF. Therefore, CRUF records should not be configured in BASE24-pos systems supporting multiple currencies.

BASE24-pos Terminal Data files (PTD)

Field Name: CURRENCY CODE
Screen: PTD Screen 2
DDL Name: PTDD1.PTD1-CORE.CRNCY-CDE
PTDF.REC.CRNCY-CDE

Screen 2 of the PTD contains the terminal's currency code. With the BASE24-pos Multiple Currency Support enhancement, this value can be set to any currency code defined in the CURRENCY-CODE-TABLE within in the COBNAMES file. This value can be different across multiple PTD records within the same logical network if the Multiple Currency Support enhancement is being used.

Field Name: MC TOTALLING TYPE
Screen: PTD Screen 2
DDL Name: PTDS1.PTDS1-CORE.MULT-CRNCY-TTL
PTDF.REC.MULT-CRNCY-TTL

Screen 2 of the PTD has also been enhanced to contain the MULTIPLE CURRENCY TOTALLING TYPE field. This field indicates whether the terminal totals in a Multiple Currency environment are stored in the PTD as hash totals (i.e., transaction totals irrelevant of the Transaction currency) or stored as totals in the terminal's defined currency code for devices capable of supporting transactions in more than one currency. If the totals are to be stored in the terminal's currency code, the transaction amount can need to be converted into this currency before being added to the totals.

For additional information on these fields, refer to the ***BASE24-pos Files Maintenance Manual***.

Card Prefix File (CPF) – POS Segment

No new data fields have been added to the BASE24-pos segment of the CPF for the Multiple Currency Support enhancement. Since this file contains amount fields that need to have a currency code associated with them, if the Multiple Currency Support enhancement is being used, the amount fields are therefore assumed to be in the currency of the corresponding Institution Definition File (IDF) record.

The following existing field requires special consideration when the Multiple Currency Support enhancement is being used:

Field Name: STANDARD CASH ADV INCREMENT
Screen: CPF Screen 6
DDL Name: CPF.POSCPF.STD-CCA-INCR

Screen 6 of the CPF contains the STANDARD CASH ADVANCE INCREMENT. With the Multiple Currency Support enhancement, all amount fields in the CPF are assumed to be in the currency of the associated FIID. The user can still configure the BASE24-pos standard increment amount for Cash Advance transactions, but these are successful only in a Multiple Currency environment if the Transaction currency is the same as the Issuer Institution currency. Cash Advance transactions performed in any other currency are denied to avoid this possibility no standard increment value should be configured in the CPF. For additional information on this field, refer to the ***BASE24 Files Maintenance Manual***.

Token File (TKN)

The following BASE24-pos token has been added for the Multiple Currency Support enhancement:

- POS Balances Token (Token ID “CB”) contains data relating to cardholder account balances in both the Account currency and the Transaction currency.

If the new BASE24-pos token for this enhancement is not to be logged to the PTLF or ILF, the appropriate token records must be configured in the TKN.

If the new BASE24-pos token for this enhancement is to be extracted with the PTLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured in the TKN.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

External Interfaces

There are no essential changes required to HISO for passing balance information between BASE24-pos and host systems. However, the host can make use of the POS Balances Token to extend the functionality in a Multiple Currency environment.

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by the BASE24-pos Multiple Currency Support enhancement but host systems must be aware that PTLF Extract files can contain records in different currencies, as the BASE24-pos system is capable of processing transactions (and settlement data) which are acquired in multiple, different currencies.

The BASE24 defined transaction currency is used for all standard amount and currency fields in the PTLF and PTLF Extract records.

If the new token (i.e., Multiple Currency Token) for this enhancement is to be included in the PTLF and ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the data from the log files. The Multiple Currency Token can therefore be used to convey the transaction amounts in all applicable currencies to host systems, if required for host processing.

The amount field in the PTLF Extract trailer record (XT.AMT) represents a verification amount only, and should reflect the sum total of PTLF Amount-1 fields, regardless of currency, specified in the file.

Refresh Process

While there are not host modifications required for PBF Refresh impacting, this function has been enhanced to ensure that transaction impacting from the PTLF is carried out in the appropriate Account currency. The Refresh process always uses the Account currency from the PBF record to ensure that impacting from the PTLF is correctly valued.

User/Operational Impacts

Disk Impacts

The PTLF and ILF require additional disk space because the new tokens for this enhancement (i.e., Multiple Currency Token and BASE24-pos Balances token) can be included in the transaction records logged to these files.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Stored Value Support

This section describes the release 6.0 changes, conversion, and implementation requirements associated with the Stored Value enhancement.

These changes, conversion, and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Stored Value Support enhancement.

Stored Value allows retailers/processors to issue a stored value card. All account information is stored on BASE24 as it is the database of record. Retailers/processors are able to establish minimum and maximum replenishment amounts and configure the maximum amount of cash that can be received back by the customer.

The stored value card allows retailers to replace paper gift certificates, scrip, and/or promotions, merchandise vouchers and in-store vouchers with stored value cards. The concept can also be used for business incentive programs.

A retailer/processor location is assigned specific BINs for issuing of stored value cards. The customer then purchases a stored value card from the retailer. The clerk initiates a Card Activation transaction online using a modified POS device. This causes BASE24 to update the Cardholder Authorization File (CAF) record's card status for the card to be issued.

Once a card is activated, cardholders can initiate any standard POS transaction supported by the BASE24-pos SPDH device handler, provided the retailer/processor's terminals support those transactions. The BASE24-pos SPDH device handler is used to communicate with the modified POS device. It allows the retailer/processor to support all standard POS transactions as well as the following Stored Value transactions:

- Card Activation transaction

  A card activation transaction is initiated when a customer swipes the card at the POS device. That card's PAN and amount to be loaded (if it isn't pre-loaded) in the account, are provided to BASE24 and used to activate a CAF record and corresponding Positive Balance File (PBF) record.

- Additional Card Activation transaction

  The additional card activation transaction allows the customer to obtain additional cards that access the same account as their original card. To activate additional cards, the customer swipes an existing card and an additional card at the POS device. PBF balances are not impacted by an additional card activation transaction.

- Replenishment transaction

  A replenishment transaction is initiated when the customer requests additional funds to be loaded to an existing value card. The “maximum card balance value” restricts the amount that can be placed into an account. This transaction can only be accomplished with cards defined as re-loadable.

- Full Redemption transaction

  A full redemption transaction subtracts the remaining funds from an account. The card and account status are not altered. Redemption transactions are only allowed if there is no pre-authorization holds present and the account balance is not greater than the “maximum balance as cash” limit as specified on the CPF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the Assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

<u>POS-SVHF</u>

The fully qualified file name of the BASE24 Stored Value History File. This is required for Stored Value customers using the Stored Value History File.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFPS file located on the BA60MISC subvolume.

<u>POS-SVHF-ICHG-AUTH-LOG</u>

This param indicates whether interchange authorized transactions are logged to the SVHF. Valid values are Y and N with a default of Y.

POS-SVHF-NOTIFY-INTERVAL

The number of writes made to the Stored Value History File (SVHF) between calculating the percentage of file full. Each time a write is made to the SVHF, a counter is increased. When the counter is equal to the value in this param, the router module calculates how full the SVHF is. The counter is then reset to zero. When the file reaches the percent full specified in the POS-SVHF-NOTIFY-PERCENTAGE param, the router module generates an event message indicating that the file is becoming full. Until more file space is made available, the router module continues to generate event messages indicating that the file is becoming full at the interval specified in this param. Valid values are 10 to 32766, and the default is 5.

POS-SVHF-NOTIFY-PERCENTAGE

The percentage of file full at which the Router module of the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the Stored Value History File (SVHF) is becoming full. The Router module calculates the percentage of file full and continues to calculate the percentage at the interval indicated in the POS-SVHF-NOTIFY-INTERVAL param as described in the following paragraphs until more file space is made available. If the percentage has increased by one percent or more since the preceding calculation, the Router module generates an event message indicating that the file is becoming full. Valid values are 10 through 99, and the default is 70.

POS-SVHF-RET-PERIOD

The number of days worth of records, prior to the current transaction date, that are to remain on the system. If the transaction date plus the retention period is less than the current date, the record is deleted from the Stored Value History File (SVHF). The valid values are 3 through 99 and the default is 7. Used by the History File Cleanup module.

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field depend on the configuration desired by the institution.

ACI Standard Device Configuration File (ACNF)

Added Stored Value transactions to screen 8. The Stored Value transactions added are CARD ACTVT, ADNL CARD, REPLENISH, and FULL REDEM.

Stored Value required FIDs are as follows:

Request Transactions	Required FIDs
Card Activation	B (amount 1) and q (track2)
Additional Card Activation	B (amount 1), 6 (product subFIDs) and q (track2)
Replenishment	B (amount 1) and q (track2)
Full Redemption	q (track2)

ACI Standard Device Response File (ARSP)

Added Stored Value transaction descriptions to screen 5 of record number 04. The Stored Value transaction descriptions added are CARD ACTIVATION, ADNL CARD ACTIVATE, REPLENISHMENT, and FULL REDEMPTION.

Processing Code Description File (PDF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the PDF file in order for description tags to be recognized in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

Description Tag	Long Description
ISO600100	Replenishment
ISO610100	Full Redemption
ISO720100	Card Activation
ISO600000	Replenishment
ISO610000	Full Redemption
ISO720000	Card Activation

ACI provides BASE24 users with a set of default PDF records at installation. The records are located in the PDF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Acquirer Processing Code File (APCF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the APCF file in order for the Stored Value transactions to be recognized by the Device Handler. The Authorization Destination field must be spaces for all records and the Log Auth Dest Response field must be set to “N” for all records.

Acquirer Transaction Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	Transaction Allowed
POS	2	60	01	00	ISO600100	4
POS	2	61	01	00	ISO610100	4
POS	2	72	01	00	ISO720100	4
POS	2	60	00	00	ISO600000	4
POS	2	61	00	00	ISO610000	4
POS	2	72	00	00	ISO720100	4

BASE24 users must configure the APCF with the new Stored Value transaction codes.

ACI provides BASE24 users with a set of default APCF records at installation. The records are located in the APCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Issuer Processing Code File (IPCF)

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: The following entries must be added to the IPCF file in order for the Stored Value transactions to be recognized by the issuer interchanges. The Completion Required to Host field must be set to “N” for all records.

Issuer Trans Profile	Message Category	ISO Tran Code	Acct 1 Type	Acct 2 Type	Description Tag	On-Us Trans Allowed	Not-On-Us Trans Allowed
POS	2	60	01	00	ISO600100	4	0
POS	2	61	01	00	ISO610100	4	0
POS	2	72	01	00	ISO720100	4	0
POS	2	60	00	00	ISO600000	4	0
POS	2	61	00	00	ISO610000	4	0
POS	2	72	00	00	ISO720100	4	0

BASE24 users must configure the IPCF with the new Stored Value transaction codes.

ACI provides BASE24 users with a set of default IPCF records at installation. The records are located in the IPCF file on the BA60MISC subvolume and are also documented in the ***BASE24 Base Files Maintenance Manual***.

Institution Data File (IDF)

Field Name: PBF4
Screen: IDF Screen 1
DDL Name: IDF.BASE.PBF4-NAME

Screen 1 of the IDF contains the filename for PBF4. With the BASE24-pos Stored Value Support enhancement if this field is not blank then all Stored Value account type will be “09” and stored in PBF4. If this field is blank then Stored Value account type will be “01” and stored in PBF1.

Cardholder Account File (CAF)

No data fields were added to the CAF. However, the field name REFUND was changed to RFND/REPL for refunds and replenishments on screen 10.

Card Prefix File (CPF)

On screens 6 and 7 the field name REFUND was changed to REFUND/REPLENISH.

Field Name: MIN LOAD LIMIT
Screen: CPF Screen 10
DDL Name: CPF.SV.MIN-LOAD

The minimum amount that can be loaded on the Stored Value account by a Card Activation.

Field Name: MAX LOAD LIMIT
Screen: CPF Screen 10
DDL Name: CPF.SV.MAX-LOAD

The maximum amount that can be loaded on the Stored Value account by a Card Activation.

Field Name: MAX BALANCE AS CASH
Screen: CPF Screen 10
DDL Name: CPF.SV.MAX-AS-CASH

The maximum amount in the Stored Value account that can be returned as cash.

Field Name: MAX BALANCE AS CASH REDEMPTION TRAN
Screen: CPF Screen 10
DDL Name: CPF.SV.MAX-AS-CASH-REDEMPTION

The maximum amount in the Stored Value account that can be returned as cash in full redemption transactions.

Field Name: MAX CARD BALANCE
Screen: CPF Screen 10
DDL Name: CPF.SV.MAX-CRD-BAL

The maximum account balance that the Stored Value account can have.

Field Name: EXP DATE
Screen: CPF Screen 10
DDL Name: CPF.SV.CRD-EXP-DAT

The expiration date that will be placed on the CAF during a Card Activation.

Field Name: DURATION
Screen: CPF Screen 10
DDL Name: CPF.SV.CRD-DURATION

Contains the duration in months to add to the current month to set the expiration data in the CAF.

Field Name: RETAILER ID
Screen: CPF Screen 10
DDL Name: CPF.SV.RETAILER-ID

Retailer ID that is used by the Net Position report.

Field Name: CAF TEMPLATE LENGTH
Screen: CPF Screen 10
DDL Name: CPF.SV.CAF-TPLT-LGTH

Contains the number of leading digits of the PAN to use when reading a CAF template record during a Card Activation transaction.

Extract Configuration File (ECF)

Field Name: SVHF

Screen: ECF Screen 7
DDL Name: ECF.BASE.FILE-MAP.SVHF

The SVHF extract process flag. The Extract process uses this information to extract the records from the SVHF.

Field Name: START DATE
Screen: ECF Screen 7
DDL Name: ECF.BASE.FILE-MAP.SVHF-STRT-DAT

The SVHF extract start date. The Extract process uses this information to extract the records from the SVHF.

Field Name: START TIME
Screen: ECF Screen 7
DDL Name: ECF.BASE.FILE-MAP.SVHF-STRT-TIM

The SVHF extract start time. The Extract process uses this information to extract the records from the SVHF.

Field Name: END DATE
Screen: ECF Screen 7
DDL Name: ECF.BASE.FILE-MAP.SVHF-END-DAT

The SVHF extract end date. The Extract process uses this information to extract the records from the SVHF.

Field Name: END TIME
Screen: ECF Screen 7
DDL Name: ECF.BASE.FILE-MAP.SVHF-END-TIM

The SVHF extract end time. The Extract process uses this information to extract the records from the SVHF.

<u>Extract (EXTR)</u>

Field Name: SVHF
Screen: EXTR Screen 7
DDL Name: EXT-MSG.FILE-MAP.SVHF

The SVHF flag that controls the manual extract of the SVHF.

POS Transaction Log File (PTLF)

Field Name: ALT MRCHNT
Screen: PTLF Screen 4
DDL Name: STORED-VALUE-TKN.TRK2

The Track2 data from the Stored Value Token. This field is only displayed when the Track2 data is present in the Stored Value Token. If the transaction is not an Additional Card Activation or the Track2 data is empty, the new screen fields are not visible.

Token File (TKN)

The following BASE24-pos tokens have been added for the Stored Value Support enhancement:

- The Stored Value Token (Token ID “U2”) contains the Stored Value data needed for transaction processing.

If the new BASE24-pos token for this enhancement is not to be logged to the PTLF, the appropriate token records must be configured in the TKN.

If the new BASE24-pos token for this enhancement is to be extracted with the PTLF, the appropriate token records must be configured in the TKN.

New Data Files

Stored Value History File (SVHF)

The Stored Value History File (SVHF) was created and will be written to whenever a PTLF record is added for an approved, denied, and reversed Stored Value transaction (other than a balance inquiry). The SVHF records are referenced by the account number, so that all cardholders linked to a common account have read access to all applicable history data. Alternately, the records are keyed by the retailer ID and terminal ID, and therefore provides access to the information by retailer ID and terminal ID. A new user interface (SVHF) was added to allow read access to the SVHF file. The History File Cleanup Module is a stand alone module that deletes aged records from the SVHF.

Template Changes

COBNAMES

New Stored Value related card types have been added to BASE24-pos card type table in the COBNAMES file.

Device Configuration

SPDH FID 6 sub-FID S is used to contain the Balance As Cash Flag, Card Balance, Expiration Date and Additional Track2 for Stored Value transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Added an entry for Stored Value to the SEG-TABLE AND THE SEG-DESCR-TABLE.

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the new Stored Value screens for the ACNF, CPF, and Perusal of the Stored Value History File.

PATHCONF Changes

The new SVHF server must be defined in the PATHCONF file as SERVER-SVHF. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new ACNF, CPF Stored Value segment and SVHF requester screen must be added through the SEC requester for the appropriate users.

Routing Changes

Stored Value transactions are routed to the BASE24-pos router/authorization for authorization of the transaction using the authorization method CAF/PBF and the authorization level of offline. These transactions are logged to the PTLF and the SVHF (Stored Value History File).

The stored value cards can also be sproute routed to another interface process provided the given interface has been enhanced to support stored value transactions. If this occurs, logging to the SVHF is configurable since BASE24-pos is not the database of record.

Network Configuration

Not applicable

File Conversions

<u>Card Prefix File (CPF)</u>

There are three CPF conversion programs to choose from when converting the CPF to include the Stored Value segment. Only one of the three programs should be ran depending on the release currently being used.

The three conversion programs are:

- Convert a non-Stored Value CPF from Release 5.3 to Release 6.0 Version 2.
- Convert a non-Stored Value CPF from Release 6.0 Version 1 to Release 6.0 Version 2.
- Convert a Stored Value CPF from Release 5.3 to Release 6.0 Version 2.

This CPF conversion program must be run to convert non-Stored Value CPF records from the Release 5.3 format to the Release 6.0 Version 2 format. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/
cnvcpfm –force –list –listreplace
```

Users must update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. CNVCPFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: This CPF conversion program must be run to convert non-Stored Value CPF records from the Release 6.0 Version 1 format to the Release 6.0 Version 2 format. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC05 subvolume can be used to convert the CPF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/
cnvcpfm –force –list –listreplace
```

Users must update the CNVCPFR obey file on the BA60UC05 subvolume with the appropriate information. CNVCPFR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD05 file on the BA60UC05 subvolume.

This CPF conversion program must be run to convert Stored Value CPF records from the Release 5.3 format to the Release 6.0 Version 2 format. Users must create the CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVCPSVS file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cpsvcnv.t,name $cpfm,nowait/
cnvcpsvm –force –list –listreplace
```

Users must update the CNVCPSVR obey file on the BA60UC04 subvolume with the appropriate information. CNVCPSVR must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Super Extract Process

Standard PTLF Extract processing is not specifically impacted by the BASE24-pos Stored Value Support enhancement but host systems must be aware that PTLF Extract files can contain records with Stored Value transaction codes and card types, as the BASE24-pos system is now capable of processing Stored Value transactions (and settlement data). The BASE24-pos Terminal Data files (PTD) totals include Stored Value transactions in the corresponding counts and amounts. These totals are logged to the PTLF.

If the new token (i.e., Stored Value Token) for this enhancement is to be included in the PTLF, BASE24-pos users must consider this additional data in host programs that use the data from the log file.

The SVHF can be extracted to the host.

Refresh Process

While there are not host modifications required for PBF Refresh impacting, this function has been enhanced to ensure that the Stored Value accounts can be refreshed into a PBF4 file instead of included in PBF1 (if configured). If the PBF4 file is used, the account types for these records should be set to “09” – Stored Value.

User/Operational Impacts

Disk Impacts

The PTLF requires additional disk space because of the new token for this enhancement (i.e., Stored Value Token) can be included in the transaction records logged to this file.

The new SVHF impacts disk requirements. The total impact varies for each BASE24 user depending on the number of SVHF records defined in the BASE24 system. If extracting to disk, this would require additional disk space.

The CPF increased in length and could impact disk sizing.

PBF4 can now be configured to hold the stored value account records. This is a new file for Stored Value cards only. PBF1 can be used if PBF4 is left as spaces on the IDF screen 1. If PBF4 is used, the account types for those records should be set to "09" – Stored Value.

Operational Impacts

The PBF4 on the IDF can now be configured to contain the stored value account records. The account type for Stored Value Accounts in the PBF should be set to "09" – Stored Value if the PBF4 is used. This is required for the PBF server to access the records. If the name of the PBF4 file is the same as PBF1, PBF4 should be spaces and PBF1 will be used.

The SVHF can be extracted. This requires changes to the ECF and EXTR to extract the SVHF.

Clean up of the SVHF can be configured.

Clean up of the CAF and PBF files can be configured for stored value card types.

PTLF Perusal and BASE24-pos reports support the new Stored Value transactions. Operators should be trained to recognize the new transaction codes.

Stored Value transactions are included in the PTD totals kept by the SPDH Device Handler module.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Performance Impacts

Stored Value transactions that are logged to the SVHF require additional disk I/O.

A FUP reload is required occasionally to any CAF and PBF that supports stored value card types for disk fragmentation issues.

Vendor Impacts

SPDH FID 6 sub-FID S is used to contain the Balance As Cash Flag, Card Balance, Expiration Date and Additional Track2 for Stored Value transactions.

Reference Documents

Not applicable

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Transaction Security Services

This section describes the release 6.0 changes and the conversion and implementation requirements associated with the Transaction Security Services enhancement for BASE24-pos.

Section 2 contains a functional overview of the Transaction Security Services enhancement.

For more detailed information on this enhancement see section 4 (Base).

Modifications to Router/Auth and the PTD/PTDF requester/server are the only changes required to support Transaction Security Services in BASE24-pos. Router/Auth and the PTD/PTDF requester/server have been modified to use key locators instead of actual keys. No other changes have been made to support Transaction Security Services for BASE24-pos. When a POS device is modified to support triple DES encryption, changes for the associated device handler will need to be made.

Note: Customers using Derived Unique Key Per Transaction (DUKPT) with any POS device or who use Atalla Identikey PIN verification or Diebold PIN verification methods or who support non-ANSI PIN blocks cannot implement Transaction Security Services at this time. No BASE24-pos device handler has been enhanced to support triple DES encryption at this time.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

PTD/PTDF

All key fields in the PTD/PTDF records must be updated to contain key locators. The key locators are used by Transaction Security Services to read CSECD records to retrieve the actual encryption/decryption keys used by the POS device.

Refer to the File Conversions subsection of this enhancement to see how the key fields in the PTD/PTDF can be updated programmatically, using the terminal ID as the default key locator for each PTD/PTDF record, and corresponding CSECD records created.

The PTD/PTDF requesters/servers have been modified to read the SECURE-DEV-TYP param in the LCONF. If the SECURE-DEV-TYP param indicates Transaction Security Services, the key fields maintained in the PTD/PTDF files are automatically modified to contain the terminal ID when a new record is added. The terminal ID is the key locator for POS device records in the CSECD.

If the Transaction Security Services database is partitioned across multiple logical networks, a two-character hexadecimal partition identifier, set by the LCONF param LN-IND, is used as the first two characters of the key locator. For key locators in the terminal data files, the first fourteen characters of terminal ID is concatenated to the partition identifier to create the key locator.

CSECD

A CSECD (Channel Security) record must exist for every set of encryption/decryption keys used by POS devices in the BASE24 network. Normally, every POS device uses unique encryption/decryption keys, so there would be one CSECD record for every PTD/PTDF record.

Refer to the File Conversions subsection of this enhancement to see how CSECD records for each PTD/PTDF record can be created programmatically.

If a POS device is configured in the BASE24-pos terminal data files to send PINs in the clear to any interchanges on the BASE24 system, a Channel Security record (CSECD) must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an interchange interface is configured to support clear PINs, a Channel Security record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

New Data Files

A new data file is required that contains encryption/decryption keys referenced by key locators during transaction processing. This file can be created by running conversion programs or records can be added, deleted or modified manually using the ACI Desktop Graphical User Interface.

- CSECD Channel Security

The following table shows the relationship of existing BASE24-pos database files to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process to get the real encryption/decryption keys used during transaction processing.

BASE24 Database Files	**Transaction Security Services Database Files**
PTD	CSECD
PTDF	CSECD

Template Changes

Not applicable

Device Configuration

POS devices supporting Triple DES encryption must have new software/hardware configured to use the Triple DES encryption algorithm and double-length or triple-length keys.

Note: No BASE24-pos Device Handlers have been enhanced to support Triple DES for any POS devices at this time.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If all PINS will be encrypted in hardware, access to the keys stored in the CSECD that used to be stored in the PTD and PTDF is through the new ACI Desktop Graphical User Interface that is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 Classic 6.0 Installation Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable.

File Conversions

File conversions for the following BASE24-pos files are supplied with the Transaction Security Services enhancement. There are no structural changes in any files, but Transaction Security Services files can be created and data populated from BASE24 files. BASE24 files will be updated with locators to point BASE24 processes to the correct Transaction Security Services records. Transaction Security Services data can be entered manually using the ACI Desktop Graphical user Interface instead of using the conversion programs.

- PTD
- PTDF

Note: Release 5.3 to release 6.0 conversions must be run on BASE24 files before the Transaction Security Services conversion programs are run. Transaction Security Services conversion programs must be run on release 6.0 files.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Transaction Security Services conversion programs must be run on release 6.0 version 2 files. All other version 1 to version 2 conversion programs must be run before the Transaction Security Services conversion programs are run.

BASE24-pos Terminal Data files (PTD)

The PTD conversion program can be run to create CSECD records while updating PTD records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding PTD records would have to be updated manually as well

The PTD conversion program is on the PS60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSPTDS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdtss.t,name $ptdm,nowait/
tssptdm –force –list –listreplace
```

Users must update the TSSPTDR obey file on the PS60UC05 subvolume with the appropriate information. TSSPTDR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD must be created using the FUP utility. The old CSECD file will be the output of a previous PTD/PTDF conversion for Transaction Security Services or will be an empty file the first time a terminal data file conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSPTDR files on the PS60UC05 subvolume.

BASE24-pos Terminal Data File (PTDF)

The PTDF conversion program can be run to create CSECD records while updating PTDF records with key locators that point to the appropriate CSECD records. Running the conversion program is not required. CSECD records can be entered manually using the ACI Desktop Graphical User Interface instead. Key fields in corresponding PTDF records would have to be updated manually as well

The PTDF conversion program is on the PS60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSPTDFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#ptdftss.t,name $ptdfm,nowait/
tsstdfm –force –list –listreplace
```

Users must update the TSSPTDFR obey file on the PS60UC05 subvolume with the appropriate information. TSSPTDFR must then be obeyed to execute the conversion program.

Both a new and old CSECD file must exist before the conversion is run. The new CSECD must be created using the FUP utility. The old CSECD file will be the output of a previous PTD/PTDF conversion for Transaction Security Services or will be an empty file the first time a terminal data file conversion is run.

For more information on this file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSPTDFR files on the PS60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- ***Host Security Module RG7000 Programmers Manual*** (1270A514 Issue 3)
- ***Banking Command Reference Manual*** (424645-002 May 2001)
- ***Atalla Key Block Banking Command Reference Manual*** (TP522901-001, October 2001)

Triple/Single DES Translation Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Triple/Single DES Translation Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Triple/Single DES Translation Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the ISO Host Interface Process.

ISO Host Interface Process Impacts

Host systems implementing the Triple/Single DES Translate enhancement must make the necessary changes to support double length keys and Triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on $<volume>.BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Section 7

BASE24-teller Release 6.0

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 5.3 or a previous version of 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describes the release 6.0 changes that affect all BASE24-teller users, along with the associated conversion/implementation requirements that all BASE24-teller users must address.
- Optional enhancements describes the release 6.0 changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with section 4 to identify the changes that BASE24-teller users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- 6.0 General Enhancements

These areas are summarized below.

Section 2 contains a functional overview of the general enhancement described here.

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6.0 General Changes

6. BASE24-teller users that extract their TTLF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a TTLF file.

7. The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the last daily extract of the TTLF must be updated to have the value of Y for this field.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each. These enhancements include the following:

There are no optional enhancements inherent to BASE24-teller Release 6.0.

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Section 8

BASE24 Kernel/IEP Release 6.0

Section 8 discusses the changes and enhancements made to BASE24 Kernel/IEP in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24 Kernel/IEP users must consider when upgrading from release 5.3/1.3 or a previous version of 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24 Kernel/IEP users, along with the associated conversion/implementation requirements that all BASE24 Kernel/IEP users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24 Kernel/IEP users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24 users must address to implement the enhancements.

Section 8 must be used in conjunction with section 4 to identify the changes that BASE24 Kernel/IEP users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24 Kernel/IEP users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- 6.0 General Changes

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. The BASE24 ITS Control Server name has changed to include the logical network name. This allows BASE24 ITS to have multiple logical networks under one BASE24 AFT subsystem.

2. The BASE24 ITS EMS templates have been merged with the other BASE24 EMS templates on the SCRIBE subvolume. The ACITPL subvolume is no longer needed.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

The BASE24 ITS Control Server name has changed from SERVER-ICTL to SRV-ICTL-<logical-network>. Thus, a BASE24 ITS Control Server must exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Before BASE24 Release 6.0, the _EMS_TEMPLATES_ define for the BASE24 Kernel/IEP networks was required to point to the $<volume>.ACITPL subvolume. With this release, this define can point to the $<volume>.SCRIBE.EMSNRES file like the rest of BASE24 networks.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24 Kernel/IEP enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each.

There are no optional enhancements inherent to BASE24 Kernel/IEP Release 6.0.

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Section 9

BASE24-telebanking Release 6.0

Section 9 discusses the changes and enhancements made to BASE24-telebanking in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-telebanking users must consider when upgrading from release 1.3 or a previous version of 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24-telebanking users, along with the associated conversion/implementation requirements that all BASE24-telebanking users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24-telebanking users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24-telebanking users must address to implement the enhancements.

Section 9 must be used in conjunction with sections 4 and 8 to identify the changes that BASE24-telebanking users need to consider for upgrading to the latest version of release 6.0. Enhanced Telebanking users must also use section 10, BASE24-billpay, to identify all changes, as Enhanced Telebanking shares modules with BASE24-billpay.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24-telebanking users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. The BASE24 ITS Control Server name has changed to include the logical network name. This allows BASE24 ITS to have multiple logical networks under one BASE24 AFT subsystem. The Customer Service Requester (CSR) module is not impacted by this change.

2. The BASE24 ITS EMS templates have been merged with the other BASE24 EMS templates on the SCRIBE subvolume. The ACITPL subvolume is no longer needed.

3. The Account Type Table has been made a BASE24 Base file, and its DDL has been moved to the BA60DDL subvolume.

4. The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the BASE24-telebanking Transaction Log File (ITLF) or reading up to the initially determined end-of-file of the ITLF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

SECTBL entries for ATT screens have been modified, so this file will appear on the BASE Product Menu.

PATHCONF Changes

The BASE24 ITS Control Server name has changed from SERVER-ICTL to SRV-ICTL-<logical-network>. Thus, a BASE24 ITS Control Server must exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Before BASE24 Release 6.0, the _EMS_TEMPLATES_ define for the BASE24 Kernel/IEP networks was required to point to the $<volume>.ACITPL subvolume. With this release, this define can point to the $<volume>.SCRIBE.EMSNRES file like the rest of BASE24 networks.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

User Impacts

BASE24-telebanking users that extract their ITLF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a ITLF file.

The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the last daily extract of the ITLF must be updated to have the value of Y for this field.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-telebanking enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each.

- Format 2 File Support
- Multiple Currency Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 changes and conversion/implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 changes and conversion/implementation requirements for future reference.

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Format 2 File Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Format 2 File Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Format 2 File Support enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

For BASE24 users who wish to begin using format 2 ITLF files, the change to the new file format will take place at cutover. Before cutover, rename the existing ITLF template files (i.e., the primary file and all of its alternate key files), and create the format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the ITLF is created using the new format 2 template files.

Create the format 2 ITLF template file in the following manner using the FUP utility where the existing ITLF template file name is TLFMMDD, the existing ITLF template alternate key file name is TLFMMDD, and the existing alternate key file 1 name is TLFMMDD1. The maximum values for the primary extent size and secondary extent size has changed with format 2 files, and therefore the user can use the SET EXT command in the following creation of the ITLF template files (i.e., both the primary ITLF template file and the alternate key ITLF template files) using FUP.

```
-SET LIKE TLFMMDD
-RENAME TLFMMDD,OTLFMMDD
-RENAME TLFMMDD0,OTLFMMD0
-RENAME TLFMMDD1,OYLFMMD1
-SET FORMAT 2
-SET REC 4048
-CREATE TLFMMDD
```

The format 2 ITLF template file in the create command must have the same value as specified in the LCONF. When cutover occurs, the next day's ITLF will be created using the format 2 ITLF template file. The previous ITLF files can be format 1 files and other ITLF files can be format 2 files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

For those BASE24-telebanking users implementing this enhancement and extracting the ITLF to a format 2 file, the host code must process the new block format for the extract format 2 disk file. In addition, the host code can use the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The value 2 in this field denotes a format 2 file. The fourth byte of the user field in the Extract File Header was used to store this information.

User/Operational Impacts

Disk Impacts

Any file converted to format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for format 2 files is increased. This is true when the primary file is format 2, regardless of whether the alternate key file(s) is format 2.

Since the maximum data area per block in format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a format 2 file. Users should check the length of their ITLF records to see if fewer ITLF records will fit in the new format 2 maximum data area blocks. The format 2 maximum data area is 4048 for entry-sequenced files, like the ITLF. If fewer ITLF records will fit in the format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry-sequenced files against the ITLFX file, if a format 2 version of this file is to be used when extracting to disk.

User Impacts

BASE24-telebanking users wishing to use format 2 ITLF files must modify their template files to be format 2 files. Refer to the Template Changes subsection of this enhancement for further information about ITLF Template files.

BASE24-telebanking users extracting their ITLF files to format 2 disk files must configure the appropriate information in the ECF.

For BASE24-telebanking users who wish to begin using format 2 ITLF files, the change to the new file format will take place at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files) and create the appropriate template files. Refer to the Template Changes subsection of this enhancement for further information about ITLF Template files. At cutover, the ITLF is created using the new template files.

Operational Impacts

Not applicable

System Limitations

BASE24-telebanking users must calculate the length of the records that are written to the ITLF. Format 2 ITLF files allow a maximum logical record length of 4048 bytes.

Before a BASE24-telebanking user changes their ITLFs to format 2 files, they should ensure these records do not exceed the 4048-byte limit. BASE24-telebanking users utilizing format 1 are not affected, since the DDL for the ITLF was not changed to set the maximum record length to 4048 bytes.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Multiple Currency Support

The release 6.0 changes and conversion and implementation requirements associated with the Multiple Currency Support enhancement are described in section 4.

BASE24-telebanking has not been enhanced to provide full multiple currency support. However, it has been modified to deny transactions that are received with a transaction currency code that doesn’t match the currency code configured for the corresponding PBF record(s).

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Section 10

BASE24-billpay Release 6.0

Section 10 discusses the changes and enhancements made to BASE24-billpay in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-billpay users must consider when upgrading from release 1.3 or a previous version of 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24-billpay users, along with the associated conversion/implementation requirements that all BASE24-billpay users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24-billpay users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24-billpay users must address to implement the enhancements.

Section 10 must be used in conjunction with Sections 4 (Base), 8 (Kernel/IEP) and 9 (Telebanking) to identify the changes that BASE24-billpay users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24-billpay users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. The names of the BASE24-billpay Pathway servers have changed to include the logical network name. This allows BASE24-billpay to have multiple logical networks under one BASE24 AFT subsystem. The Customer Service Requester (CSR) module is not impacted by this change.

2. The BASE24-billpay EMS templates have been merged with the other BASE24 EMS templates on the SCRIBE subvolume. The ACITPL subvolume is no longer needed.

3. BASE24-billpay has been assigned an entry in the SEG-TABLE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

In order to allow the BASE24 AFT Subsystem to correctly detect the presence of the Billpay product, SEG-TABLE entry FILLER-7 has become the new Billpay entry.

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

The names of the BASE24-billpay Pathway servers have changed from SV-BP and SV-FM to SV-BP-<logical-network> and SV-FM-<logical-network>, respectively. Thus, the appropriate BASE24-billpay servers must exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Before BASE24 Release 6.0, the _EMS_TEMPLATES_ define for the BASE24-telebanking networks was required to point to the $<volume>.ACITPL subvolume. With this release, this define can point to the $<volume>.SCRIBE.EMSNRES file like the rest of BASE24 networks.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- BASE24-telebanking/billpay 6.0 Install Guide

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Optional Enhancements

This section identifies the BASE24-billpay enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-billpay in Release 6.0.

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Section 11

BASE24-customer service Release 6.0

Section 11 discusses the changes and enhancements made to BASE24-customer service in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24-customer service users must consider when upgrading from release 1.3 or a previous version of 6.0.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24-customer service users, along with the associated conversion/implementation requirements that all BASE24-customer service users must address.
- Optional enhancements describe the release 6.0 changes that affect only those BASE24-customer service users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24-customer service users must address to implement the enhancements.

Section 11 must be used in conjunction with sections 4 and 8 to identify the changes that BASE24-customer service users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24-customer service users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. The BASE24 ITS Control Server name has changed to include the logical network name. This allows BASE24 ITS to have multiple logical networks under one BASE24 AFT subsystem. This change does not apply to those servers that are specific to the BASE24-customer service product.

2. The BASE24 ITS EMS templates have been merged with the other BASE24 EMS templates on the SCRIBE subvolume. The ACITPL subvolume is no longer needed.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

The BASE24 ITS Control Server name has changed from SERVER-ICTL to SRV-ICTL-<logical-network>. Thus, a BASE24 ITS Control Server must exist in the PATHCONF for each logical network controlled by a particular BASE24 AFT subsystem. This change does not apply to those servers that are specific to the BASE24-customer service product.

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Before BASE24 Release 6.0, the _EMS_TEMPLATES_ define for the BASE24 Kernel/IEP networks was required to point to the $<volume>.ACITPL subvolume. With this release, this define can point to the $<volume>.SCRIBE.EMSNRES file like the rest of BASE24 networks.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-customer service enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each.

- Format 2 File Support

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 changes and conversion/implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 changes and conversion/implementation requirements for future reference.

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Format 2 File Support

This section describes the release 6.0 changes and conversion and implementation requirements associated with the Format 2 File Support enhancement.

These changes and conversion and implementation requirements are identified using a standard format defined in the introduction to section 4.

Section 2 contains a functional overview of the Format 2 File Support enhancement.

BASE24-customer service has been modified to read format 2 TLF and PTLF files as well as format 1 TLF and PTLF files. Other than these code changes, no other changes must be made to BASE24-customer service to support format 2 transaction log files.

For further information about the impacts to the TLF regarding format 2 files, refer to sections 4 (BASE) and 5 (ATM).

For further information about the impacts to the PTLF regarding format 2 files, refer to sections 4 (BASE) and 6 (POS).

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Section 12

BASE24 Interchange Interfaces Release 6.0

Section 12 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from Fortify.3 or a previous version of 6.0.

With the latest version of release 6.0, the following BASE24 Interchange Interfaces have been upgraded to SW60xxxx subvolumes:

- Alaska Option ISO Interchange Interface
- American Express Global Network Interchange Interface
- Banknet Interchange Interface
- BASE24 ANSI Interchange Interface
- BASE24 ISO Interchange Interface
- Cash Station ISO Interchange Interface
- Discover ISO Interchange Interface
- EPS-Net Interchange Interface
- LINK Interchange Interface
- MAC MASM Interchange Interface
- MagicLine ISO Interchange Interface
- MDS/Cirrus Interchange Interface
- MDS Maestro Interchange Interface
- Midwest Payment Systems Interchange Interface
- Moneystation Interchange Interface
- NBGC Interchange Interface
- NEN Interchange Interface
- NPC Interchange Interface
- NYCE ISO Interchange Interface
- Plus ISO Interchange Interface
- Publix Interchange Interface
- Shazam (ITS) ISO Interchange Interface
- Star System ISO Interchange Interface
- Switch Authorization Network Interface (SWANI)
- VisaNet ISO Interchange Interface

Although each of these release 6.0 Interchange Interfaces replace existing Fortify.3 or Fortify.1 Interchange Interfaces and can run in release 4.0, 5.0, 5.1, 5.1.2 and 5.3 BASE24 network environments, customers must implement a full release 6.0 network environment to take advantage of the release 6.0 enhancements available in the latest version of release 6.0 Interchange Interfaces.

All Fortify.3 and Fortify.1 BASE24 Interchange Interfaces not upgraded to release 6.0 can still run in release 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 BASE24 network environments. They do not support any of the release 6.0 enhancements described in this section. For information on how to implement Fortify.1 and Fortify.3 Interfaces in a 6.0 network environment, refer to the BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) – SW-DS000-04. This document is included as a Compaq NSK TGAL document and is shipped with the BASE24 Release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

This section is divided in two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the release 6.0 changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the release 6.0 Interchange Interface product changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 12 must be used in conjunction with section 4 to identify the changes that BASE24 Interchange Interface users need to consider for upgrading to release 6.0.

General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing Interchange Interface users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- 6.0 General Changes
- Transactions Allowed

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here. Further details may be found in the update guides and manuals on the individual Interchange Interface subvolumes.

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6.0 General Changes

1. A new Base token, the Original Currency 6.0 Token (Token ID “BE”) has been added and is used by the VisaNet and EPS-Net Interchange Interfaces. This token contains information relating to a currency conversion that has been performed by an external system (e.g., an interchange) prior to presenting the transaction to BASE24. No additional processing is performed within BASE24 on the data in this token. However, the token can be logged to the transaction log files and sent or extracted to a host or another logical network. Currently, this token is added on requests sent from an interchange into the VisaNet Interface or EPS-Net Interface irregardless of whether BASE24 is processing as a multiple currency system or not.

2. The Extract Configuration File (ECF) has been modified to support the configuration of reading past the initially determined end-of-file of the ILF or reading up to the initially determined end-of-file of the ILF.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>Token file (TKN)</u>

The following new BASE24 Base token has been added for this enhancement:

- The Original Currency 6.0 Token (Token ID “BE”) has been added and is used by the VisaNet and EPS-Net Interchange Interfaces. This token contains information relating to a currency conversion that has been performed by an external system (e.g., an interchange) prior to presenting the transaction to BASE24.

If the new token for this enhancement is not to be logged to the PTLF, TLF or ILF, the appropriate token records must be configured.

If the new token for this enhancement is to be extracted with the PTLF, TLF or ILF, sent to the host, or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interchange Interface, ISO Host Interface, and Super Extract processes.

BIC ISO Interchange Interface Process

If the new Original Currency 6.0 Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

ISO Host Interface Process

If the new Original Currency 6.0 Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

If the new Original Currency 6.0 Token is configured to be included in the TLF, PTLF, and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF, PTLF, and ILF require additional disk space if the new Original Currency 6.0 Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 user depending on the transaction volume in the BASE24 system.

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF, PTLF, and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system, logged to a log file, and passed to the host or co-network in an ISO External Message. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens may cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

User Impacts

BASE24 users that extract their ILF files must set the ECF Screen 1 field READ PAST INITIAL EOF to a value of Y for the ECF record that is used on the last daily extract of a ILF file.

The ECF conversion program sets the ECF base segment field READ-PAST-INITIAL-EOF to a value of N. The ECF record used for the ILF must be updated to have the value of Y for this field.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Transactions Allowed

The Transactions Allowed enhancement impacted the enhanced BASE24 Interchange Interfaces.

For more information on this enhancement, refer to section 4 (Base).

This enhancement is only supported for the following BASE24 Interchange Interface processes:

- BASE24 ISO Interchange Interface
- Banknet Interchange Interface
- MDS ISO/MDS Maestro Interchange Interface
- Plus ISO Interchange Interface
- VisaNet ISO Interchange Interface

These BASE24 Interchange Interfaces have been enhanced to support the Transactions Allowed Enhancement in their acquirer and issuer processing for BASE24-atm and BASE24-pos transactions. These Interchange Interfaces processes ensure that the inbound transactions are allowed from the interchange in the Acquirer Profile Configuration File (APCF) prior to forwarding the transaction to the authorization process. Similarly, the Interchange Interface processes ensure that the outbound transactions are allowed to be sent to the interchange in the Issuer Profile Configuration File (IPCF) prior to forwarding the transaction to interchange. Finally, the enhanced Interchange Interface processes were modified to utilize the Enhanced Interchange Configuration File (ICFE) instead of the Interchange Configuration File (ICF).

CUSTMACS Changes

Set each of the following flags to true for the appropriate enhanced Interchange Interface modules.

- bic_iso_enhanced_mode_on
- bnet_enhanced_mode_on
- mds_enhanced_mode_on
- plus_iso_enhanced_mode_on
- visa_enhanced_mode_on

If the enhanced Interchange Interface processes will be running in BASE24 Release 6.0 network environments, these flags must be set to true. Otherwise, these flags must be set to false.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that have been modified to support this enhancement. The settings of each field will depend on the configuration desired by the institution.

Enhanced Interchange Configuration File (ICFE)

Since the only difference between the ICF and ICFE is that the ICFE uses transaction profiles instead of the hard-coded tables to define the transactions allowed to and from the interchange, the new profile fields in the ICFE will be discussed in the section.

Field Name: ACQUIRER TXN PROFILE
Screen: ICFE Screen 8
DDL Name: ICFE.ATM.ACQ-TKN-PRFL

A code identifying a group of default BASE24-atm transaction processing codes allowed from this interchange. The value entered in this field must match a transaction profile named in the Acquirer Processing Code File (APCF).

Field Name: ISSUER TXN PROFILE
Screen: ICFE Screen 8
DDL Name: ICFE.ATM.ISS-TKN-PRFL

A code identifying a group of default BASE24-atm transaction processing codes allowed to be sent to this interchange. The value entered in this field must match a transaction profile named in the Issuer Processing Code File (IPCF).

Field Name: ACQUIRER TXN PROFILE
Screen: ICFE Screen 10
DDL Name: ICFE.POS.ACQ-TKN-PRFL

A code identifying a group of default BASE24-pos transaction processing codes allowed from this interchange. The value entered in this field must match a transaction profile named in the Acquirer Processing Code File (APCF).

Field Name: ISSUER TXN PROFILE
Screen: ICFE Screen 10
DDL Name: ICFE.POS.ISS-TKN-PRFL

A code identifying a group of default BASE24-pos transaction processing codes allowed to be sent to this interchange. The value entered in this field must match a transaction profile named in the Issuer Processing Code File (IPCF).

New Data Files

This section describes the files that have been added to support this enhancement.

Enhanced Interchange Configuration File (ICFE)

The ICFE was created for storing the parameters relevant to interchange transaction processing. Like the Interchange Configuration File (ICF), the ICFE is used by the institution to define institution and terminal interchange sharing, holidays, transaction handling in offline and online situations, and settlement handling. The difference between the ICF and ICFE is that the ICFE uses transaction profiles instead of the hard-coded tables to define the transactions allowed to and from the interchange.

This file is accessed via the BASE24 AFT subsystem by specifying a file destination of ICFE.

Note: In a BASE24 Release 6.0 network environment, the Interchange Interfaces not modified for this enhancement continue to use the ICF.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

The ICFE requester name was added to MEGATBL file, since RQMEGA communicates with the ICFE requesters using the extended message format (i.e., use USER-CONTEXT-EXT instead of USER-CONTEXT).

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the ICFE requester/server.

PATHCONF Changes

The new ICFE server must be defined in the PATHCONF file as SERVER-ICFE. This server can be added to the system online through the PATHCOM utility, but it must also be added to the PATHCONF, so that any future PATHCOLD operation would contain the ICFE Server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new ICFE requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Enhanced Interchange Configuration File (ICFE)

The ICFE conversion program must be run to convert ICF records that exist for the enhanced Interchange Interfaces into the new ICFE records. In addition, this same conversion program will copy those ICF records that are not for the enhanced Interchange Interfaces into a new ICF file. Users must create the ICFE file and a new ICF file using the BADDLFUP file.

The ICFE conversion program on the BA60UC04 subvolume can be used to convert the ICF. If modifications to the conversion program are necessary, the program can be modified by editing the CNVICFES file.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#icfecnv.t,name $icfem,nowait/
cnvicfem –force –list –listreplace
```

Users must update the CNVICFER obey file on the BA60UC04 subvolume with the appropriate information. CNVICFER must then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Host impacts will be incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing will result in impacts to the Super Extract process.

Super Extract Process

Host code changes are required for all BASE24 users that utilize the OMF Extract Process to accept and process information in the ICFE file.

If the new tokens for this enhancement (i.e., Transaction Profile Token, Transaction Description Token, Acquirer Routing Token) are to be included in the ILF Extract, BASE24 users must consider this additional data in host programs that use the ILF data.

User/Operational Impacts

<u>Disk Impacts</u>

The ICFE may impact BASE24 user's disk requirements. The total impact varies for each BASE24 user depending on the number of interchange interface records defined in the BASE24 system.

<u>User Impacts</u>

When upgrading from Fortify.3, BASE24 users must determine what mandates and enhancements that they are currently running. The user must run all conversion programs on the Interchange Interface 5.3 subvolumes with a date greater than the latest mandate or enhancement in their production Interchange Interface. The conversion programs must be run in sequential order.

<u>Operational Impacts</u>

BASE24 operators must be aware that the ICFE will appear on their BASE24 BASE product menu.

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the following event message documentation files:

- $<volume>.SW60BABI.BABILOGM for the BIC ISO Interchange Interface process
- $<volume>.SW60BNET.BNETLOGM for the Banknet Interchange Interface process
- $<volume>.SW60MDS.MDSLOGM for the MDS ISO/MDS Maestro Interchange Interface process
- $<volume>.SW60PISO.PISOLOGM for the Plus ISO Interchange Interface process
- $<volume>.SW60VISA.VISALOGM for the VisaNet Interchange Interface process
- $<volume>.BA60LOGM subvolume. This subvolume contains event message documentation files for BASE24 Classic modules, except interchange interface processes.

Vendor Impacts

BASE24 users need to evaluate the enhancements they are implementing and determine if re-certification is necessary for their environment.

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion and implementation requirements associated with each. These enhancements include the following:

- EMV Support
- Format 2 File Support
- Transaction Security Services
- Triple/Single DES Translation Support

BASE24 Interchange Interface users can choose whether or not to implement the above enhancement. Users who plan to implement this enhancement must address the corresponding release 6.0 changes and conversion and implementation requirements. Users who do not plan to implement this enhancement can simply review the corresponding Release 6.0 changes and conversion and implementation requirements for future reference.

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EMV Support

The EMV Support enhancement has been implemented in the following BASE24 Interchange Interfaces:

- Banknet Interchange Interface (BASE24-pos)
- BIC ISO Interchange Interface (BASE24-atm and BASE24-pos)
- EPS-Net Interchange Interface (BASE24-atm and BASE24-pos)
- LINK Interchange Interface (BASE24-atm)
- MasterCard Debit Switch Interchange Interface (BASE24-atm)
- Maestro Interchange Interface (BASE24-pos)
- Switch Authorisation Network Interface (BASE24-pos)
- VisaNet Interchange Interface (BASE24-atm and BASE24-pos)

Section 2 contains a functional overview of the EMV Support enhancement.

For more detailed information on this enhancement, refer to section 4.

For information about how to implement EMV within each Interchange Interface, refer to the Interchange Interface EMV manual located on the individual Interchange Interface EMV subvolume.

Note: The only changes to the BIC ISO modules were to map new BASE24-atm/pos EMV related response codes between the standard internal message and external ISO message. BIC ISO can pass EMV data in tokens using the standard TKN configuration options.

CUSTMACS Changes

Set each of the following flags to true to add the EMV bindable pieces to the appropriate interchange interface modules.

- bnet_emv_on
- euro_emv_on
- link_emv_on
- mds_maestro_emv_on
- mds_iso_emv_on
- swan_emv_on
- visa_emv_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- IBNTMAN (BankNet EMV Interface Manual – informal documentation) on $<volume>.SW60IBNT
- IMDSMAN (MasterCard Debit Switch ISO Interface EMV Manual – informal documentation) on $<volume>.SW60IMDS
- IVISMAN (VisaNet ISO Interface EMV Manual – informal documentation) on $<volume>.SW60IVIS
- BASE24 EPS-Net Interchange Interface Manual – informal documentation
- BASE24 LINK Interchange Interface User Guide – informal documentation
- BASE24 Switch Authorization Network Interface (SWANI) Manual

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Format 2 File Support

The Format 2 File Support enhancement has been implemented in the following BASE24 Interchange Interfaces:

- Banknet Interchange Interface (BASE24-pos)
- MasterCard Debit Switch Interchange Interface (BASE24-atm)
- Maestro Interchange Interface (BASE24-pos)
- Star ISO Interchange Interface(BASE24-atm and BASE24-pos)
- VisaNet Interchange Interface (BASE24-atm and BASE24-pos)

Section 2 contains a functional overview of the Format 2 File Support enhancement.

For more detailed information on this enhancement, refer to section 4.

For information about how to implement Format 2 File Support within each Interchange Interface, refer to the Interchange Interface manual located on the individual Interchange Interface subvolume.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

BASE24 users using format 2 ILF files must start at cutover. Before cutover, rename the existing template ILF files (i.e., the primary file and all of its alternate key files), and create the format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the ILF is created using the new format 2 template files.

Create the format 2 ILF template file in the following manner using the FUP utility where the existing ILF template file name is VAYYMMDD, the existing ILF template alternate key file name is VBYYMMDD. The maximum values for the primary extent size and secondary extent size has changed with format 2 files, and therefore the user can use the SET EXT command in the following creation of the ILF template files (i.e., both the primary ILF template file and the alternate key ILF template file) using FUP.

```
-SET LIKE VAYYMMDD
-RENAME VAYYMMDD,OPYYMMDD
-RENAME VBYYMMDD,OAYYMMDD
-SET FORMAT 2
-SET REC 4048
-CREATE VAYYMMDD
```

The format 2 ILF template file in the create command must have the same value as specified in the LCONF for this interchange interface's ILF. When the interchange interface cuts over, the next day's ILF will be created using the format 2 ILF template file. The previous ILF files can be format 1 files and other ILF files can be format 2 files.

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

For those BASE24 users implementing this enhancement and extracting the ILF to a format 2 file, the host code must process the new block format for the extract format 2 disk file. In addition, the host code may utilize the new file format indicator that is specified in the Extract File Header to decipher the file format of the extract disk file. The value 2 in this field denotes a format 2 file. The fourth byte of the user field in the Extract File Header was used to store this information.

User/Operational Impacts

Disk Impacts

Any file converted to format 2 with alternate key(s) requires more disk space for the alternate key file(s), as the key length for format 2 files is increased. This is true when the primary file is format 2, regardless of whether the alternate key file(s) is format 2.

Since the maximum data area per block in format 2 files is less than what is available in format 1 files, users may require more disk space for any file that is converted to a format 2 file. Users should check the length of their ILF records (for those Interchange Interfaces modified to support Format 2 File Support) to see if fewer ILF records will fit in the new format 2 maximum data area blocks. The format 2 maximum data area is 4048 for entry sequenced files, like the ILF. If fewer ILF records will fit in the new format 2 maximum data area per block, users will require more disk space for the same number of records.

The user should do the same check for entry sequenced files against the ILFX file, if a format 2 version of this file is to be used when extracting to disk.

<u>User Impacts</u>

BASE24 users using format 2 ILF files must modify their template files to be format 2 files. Refer to the Template Changes subsection of this enhancement for further information about ILF Template files.

BASE24 users extracting their ILF files to format 2 disk files must configure the appropriate information in the ECF.

BASE24 users using format 2 ILF files must start at cutover. Before cutover, rename the existing template (i.e., the primary file and all of its alternate key files), and create the format 2 template files (i.e., the primary file and all of its alternate key files). At cutover, the new ILF is created using the new format 2 template files.

<u>Operational Impacts</u>

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the following event message documentation files or the BASE24 Documentation CD-ROM:

- $<volume>.SW60BABI.BABILOGM for the BIC ISO Interchange Interface process
- $<volume>.SW60BNET.BNETLOGM for the Banknet Interchange Interface process
- $<volume>.SW60MDS.MDSLOGM for the MDS ISO/MDS Maestro Interchange Interface process
- $<volume>.SW60VISA.VISALOGM for the VisaNet Interchange Interface process
- $<volume>.BA60LOGM subvolume. This subvolume contains event message documentation files for BASE24 Classic modules, except interchange interface processes.

<u>Performance Impacts</u>

For those BASE24 users receiving transaction requests and/or advices from the interchange and the interchange sending triple DES encrypted PIN blocks, a call to the security box must be made to translate the triple DES encrypted PIN block in the external message to the single DES encrypted PIN block that is sent in the internal message.

For those BASE24 users receiving transaction requests and/or advices from the interchange and the interchange sending single DES encrypted PIN blocks, no call is made to the security box. The encrypted PIN block is simply moved from the external message to the internal message.

System Limitations

BASE24 users must calculate the length of the records that are written to the ILF. Format 2 Interchange log files allow a maximum logical record length of 4048 bytes. Depending on which, if any, tokens are written to the interchange log files, the record length of an interface log file record could exceed the limit the 4048 bytes.

Before you change ILFs to format 2 files, ensure these records do not exceed the 4048 limit. BASE24 users utilizing format 1 are not affected, since the DDLs for the interchange log files were not changed to set the maximum record length to 4048.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Transaction Security Services

This section describes the release 6.0 changes and the conversion and implementation requirements associated with the Transaction Security Services enhancement for interchange interfaces.

Section 2 contains a functional overview of the Transaction Security Services enhancement.

For more detailed information on this enhancement refer to Section 4 (Base).

Impacts to existing BASE24 interchange interfaces have been limited to support for keys passed to or from interchanges on Dynamic Key Exchanges. The interchange interfaces enhanced to support Transaction Security Services are as follows:

- Alaska Option ISO Interchange Interface
- Banknet Interchange Interface
- BIC ANSI Interchange Interface
- BIC ISO Interchange Interface
- Cash Station ISO Interchange Interface
- EPS-Net Interchange Interface
- LINK Interchange Interface
- MAC ICON Interchange Interface
- MasterCard Debit Switch Interchange Interface
- Midwest Payment Systems Interchange Interface
- Maestro Interchange Interface
- Moneystation Interchange Interface
- NEN Interchange Interface
- NPC ISO Interchange Interface
- PLUS ISO Interchange Interface
- Publix Interchange Interface
- Shazam ISO Interchange Interface
- VisaNet Interchange Interface

These Interchange Interfaces must run in a release 6.0 network in order to work with Transaction Security Services.

Note: The Triple/Single DES Translation enhancement is not required when Transaction Security Services is implemented.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>ISECD</u>

An ISECD (Interchange Security) record must exist for every set of encryption/decryption keys used by the interchange interfaces in the BASE24 network. Normally, every interchange interface uses unique encryption/decryption keys, so there is one ISECD record for every KEYF/KEY6 record.

Refer to the File Conversions subsection of this enhancement to see how ISECD records for each KEYF/KEY6 record can be created programmatically.

If an acquiring host or interface process is configured to send PINs in the clear to any interchange in the BASE24 system that supports encrypted PINs, an Interchange Security (ISECD) record must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an acquiring host or interface process encrypts PINs and an interchange interface supports clear PINs, an Interchange Security (ISECD) record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

<u>KEYF</u>

All key fields in the KEYF records must be updated to contain key locators. The key locators are used by Transaction Security Services to read ISECD records to retrieve the actual encryption/decryption keys used by the Interchange Interface.

Refer to the File Conversions subsection of this enhancement to see how the key fields in the KEYF can be updated and ISECD records created programmatically.

<u>KEY6</u>

A new KEYF record must be created for each KEY6 record and the key fields in the new KEYF records must be updated to contain key locators. The key locators are used by Transaction Security Services to read ISECD records to retrieve the actual encryption/decryption keys used by the Interchange Interface. The KEY6 records are no longer required when Transaction Security Services is implemented.

Refer to the File Conversions subsection of this enhancement to see how the new KEYF and ISECD records can be created for each KEY6 record programmatically.

New Data Files

A new data file is required that contains encryption/decryption keys referenced by key locators during transaction processing. This file can be created by running conversion programs or records can be added, deleted or modified manually using the ACI Desktop Graphical User Interface.

- ISECD Interchange Security

The following table shows the relationship of existing BASE24 Interchange Interface database files to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process to get the real encryption/decryption keys used during transaction processing.

BASE24 Database File	Transaction Security Services Database File
KEYF	ISECD
KEY6	ISECD

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

If all PINS are encrypted in hardware, access to the KEYF and KEY6 screens should be disabled through the BASE24 Security requester/server.

Access to the Transaction Security Services equivalents of these files is through the new ACI Desktop Graphical User Interface which is set up separately from the BASE24 Pathway CRT access system. Refer to the ***BASE24 6.0 Classic Install Guide*** for further information.

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

File conversions for the following BASE24 Interchange Interface files are supplied with the Transaction Security Services enhancement. There are no structural changes in any of these BASE24 files, but Transaction Security Services files can be created and populated from the BASE24 files. The BASE24 files are updated with locators to point BASE24 processes to the correct Transaction Security Services records. Transaction Security Services data can also be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

- KEYF
- KEY6

Note: Release 5.3 to release 6.0 conversions must be run on BASE24 files before the Transaction Security Services conversion programs are run. Transaction Security Services conversion programs must be run on release 6.0 files.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration: Transaction Security Services conversion programs must be run on release 6.0 version 2 files. All other version 1 to version 2 conversion programs must be run before the Transaction Security Services conversion programs are run.

Key File (KEYF)

The KEYF conversion program can be run to create ISECD records while updating KEYF records with key locators that point to the appropriate ISECD records. Running the conversion program is not required. ISECD records can be entered manually using the ACI Desktop Graphical User Interface. Key fields in corresponding KEYF records would have to be updated manually as well.

The KEYF conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEYFS file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#keyftss.t,name $keyfm,nowait/
tsskeyfm –force –list –listreplace
```

Users must update the TSSKEYFR obey file on the BA60UC05 subvolume with the appropriate information. TSSKEYFR must then be obeyed to execute the conversion program.

If both KEYF and KEY6 files are converted, the KEYF conversion must be run first.

For more information on this file conversion, refer to the AAREAD05 and TSSKEYFR files on the BA60UC05 subvolume.

Key6 File (KEY6)

The KEY6 conversion program can be run to create ISECD records and KEYF records with key locators that point to the appropriate ISECD records. Since the new ISECD can hold double-length and triple-length keys the KEY6 is no longer required. The KEY6 conversion program will copy KEY6 records to the KEYF. Running the conversion program is not required. ISECD records can be entered manually using the ACI Desktop Graphical User Interface. Corresponding KEYF records would have to be added manually.

The KEY6 conversion program is on the BA60UC05 subvolume. If modifications to the conversion program are necessary, the program can be modified by editing the TSSKEY6S file.

To compile the conversion program, enter the following command at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#key6tss.t,name $key6m,nowait/
tsskey6m –force –list –listreplace
```

Users must update the TSSKEY6R obey file on the BA60UC05 subvolume with the appropriate information. TSSKEY6R must then be obeyed to execute the conversion program.

For more information on this file conversion, refer to the AAREAD05 and TSSKEY6R files on the BA60UC05 subvolume.

Host/Co-Network Impacts

Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the ISO Host Interface and BIC ISO Interchange Interface processes.

<u>BIC ISO Interchange Interface</u>

Co-Networks implementing triple DES encryption using the Transaction Security Services enhancement must make the necessary changes to support double-length keys and triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

- ***Host Security Module RG7000 Programmers Manual*** (1270A514 Issue 3)
- ***Banking Command Reference Manual*** (424645-002 May 2001)
- ***Atalla Key Block Banking Command Reference Manual*** (TP522901-001, October 2001)

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Triple/Single DES Translation Support

Release 6.0 BASE24 interchange interfaces have been modified to support translation of PINs from triple to single DES encryption and vice versa, when processing transactions between the BASE24 interchange interfaces and the interchange itself. These release 6.0 BASE24 interchange interfaces will run in all releases of BASE24. No other BASE24 modules have been enhanced at this time to support triple DES.

The interchange interfaces enhanced to support triple to single DES transactions are as follows:

- Banknet Interchange Interface (BASE24-pos)
- BIC ISO Interchange Interface (BASE24-atm and BASE24-pos)
- EPS-Net Interchange Interface (BASE24-atm and BASE24-pos)
- LINK Interchange Interface (BASE24-atm)
- MAC MASM Interchange Interface (BASE24-atm and BASE24-pos)
- MasterCard Debit Switch Interchange Interface (BASE24-atm)
- Maestro Interchange Interface (BASE24-pos)
- VisaNet Interchange Interface (BASE24-atm and BASE24-pos)

A process specific param, DES-TRIPLE-SINGLE, has been added to each Release 6.0 interchange interface indicating that the interface translates PINs from being single DES encrypted inside BASE24 to triple DES encrypted when sent to the interchange and translates PINs from being triple DES encrypted when received from an interchange to being single DES encrypted inside BASE24.

If the DES-TRIPLE-SINGLE param indicates that triple DES encryption (3DEA) must be supported, a new version of the KEYF file (KEY6) must be used. The KEY6 supports double-length PIN encryption keys. The intermediate keys remain single length keys. The external key length has been added to this file. A new KEY6 DDL, requester and server have been created to support the KEY6.

Triple/Single DES Translation support invokes a new type of PIN translation. PINs are translated from encryption under a double-length key (3DEA) to encryption under a single-length key (DEA) or from DEA encryption to 3DEA encryption. The special translation will be done only in three situations:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware
- External PIN encrypted in hardware to internal PIN encrypted in hardware

When an interface receives a transaction from the interchange and encryption is performed by a hardware security module (HSM), the interface checks the DES-TRIPLE-SINGLE LCONF param. If this LCONF param indicates that triple-to-single encryption is required, the program must call a new translation procedure to translate the PIN to single DES encryption under a single length intermediate key. Keys passed to and used by the new translation procedure are the double-length KPE, and the single-length encrypted intermediate key. The translated PIN and the encrypted intermediate key are placed in the STM/PSTM that is then sent to the BASE24 Authorization process.

When BASE24 acquires a transaction that must be sent to an interchange, the interchange interface checks the DES-TRIPLE-SINGLE LCONF param. If the encryption of the external PIN is performed by an HSM and the LCONF param indicates that single-to-triple translation is required, a new procedure will be called to translate the PIN from encryption under a single-length key to triple DES encryption under a double-length key. The translated PIN is sent to the interchange. Keys passed to and used by the new translation procedure are the single-length encryption key from the STM/PSTM or the intermediate key if the PIN is in the clear in the STM/PSTM and the double-length KPE from the KEY6.

Each interchange that supports dynamic key management will send a double-length PIN encryption key in key change messages if the DES-TRIPLE-SINGLE LCONF param indicates double-length keys and triple DES encryption are required. This key will be sent either in a new field or in an expanded existing key field, depending on the interchange interface.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign records that must be added to the LCONF to support this enhancement. These modifications are provided in the LCONFBA file located on the BA60MISC subvolume.

<u>KEY6</u>

The fully qualified file name of the KEY6 File (KEY6). The KEY6 file contains the double-length keys needed to perform Triple DES encryption on PINs and keys.

If this assign is not present in the LCONF, the KEY6 file cannot be accessed by interchange interfaces. This assign is required if the DES-TRIPLE-SINGLE LCONF param is present and indicates Triple DES encryption is to be performed. Initialization fails if the assign is missing when DES-TRIPLE-SINGLE indicates Triple DES encryption.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support this enhancement.

DES-TRIPLE-SINGLE

Indicates whether an interchange interface will support Triple DES encryption. This param is required if Triple DES encryption is required. Single DES encryption will be used if the param is missing or contains an invalid value.

Database Configuration

Not applicable.

New Data Files

KEY6 File (KEY6)

The KEY6 file contains the double-length keys needed to encrypt PINs and keys for Triple DES encryption. Keys must be loaded using the KEY6 requester.

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Entries were added to the SECTBL to support the screens for the KEY6 requester/server.

PATHCONF Changes

The new KEY6 server must be defined in the PATHCONF file as SERVER-KEY6. This server can be added to the system online through the PATHCOM utility. After adding the server to the PATHCONF file, the BASE24 user must perform a PATHCOLD operation to actually gain access to the newly defined server. Refer to the PATHCONF file on the BA60CNTL subvolume for an example of this server.

CRT Access Security Changes

Access to the new KEY6 requester screens must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

In order to implement this enhancement, BASE24 must have access to a hardware security module. The enhancement will support the use of an Atalla or Thales e-Security (Racal) security module. Refer to the Vendor Impacts subsection of this enhancement for further information.

File Conversions

A KEYF to KEY6 conversion is unnecessary, since the KEYF contains single-length keys and the KEY6 contains double-length keys.

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing will result in impacts to the OMF Extract, ISO Host Interface Process and BIC ISO Interchange Interface processes.

Super Extract Process

Host code changes are required for all BASE24 users that use OMF Extract to accept and process information for the KEY6 files.

BIC ISO Interchange Interface Impacts

Co-Networks implementing the Triple/Single DES Translate enhancement must make the necessary changes to support double-length keys and Triple DES encryption. The MAC key length has been added in field 48 of the ISO message.

User/Operational Impacts

Disk Impacts

The KEY6 may impact disk requirements. The total impact varies for each BASE24 user depending on the number of KEY6 records defined in the BASE24 system.

Operational Impacts

BASE24 operators must be aware that the new KEY6 file may appear on their BASE24 Product Menu.

New event messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the following event message documentation files or the BASE24 Documentation CD-ROM.

- $<volume>.SW60BABI.BABILOGM for the BIC ISO Interchange Interface process
- $<volume>.SW60BNET.BNETLOGM for the Banknet Interchange Interface process
- $<volume>.SW60MDS.MDSLOGM for the MDS ISO/MDS Maestro Interchange Interface process
- $<volume>.SW60VISA.VISALOGM for the VisaNet Interchange Interface process

Vendor Impacts

To implement this enhancement, BASE24 must have access to a hardware security module. The enhancement supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of software that is capable of supporting the 3DEA enhancement.

- For Atalla, only version 2.84 software is supported.
- For the Thales e-Security (Racal), the minimum version is 5.05.

Reference Documents

- Host Security Module RG7000 Programmers Manual (1270A514 Issue 3)
- Banking Command Reference Manual (424645-002 May 2001)

Section 13

BASE24 Value-Added Products Release 6.0

Section 13 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 5.3 or a previous version of release 6.0.

The value-added products discussed in this section include the following:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager
- ACI Smart Chip Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describes the release 6.0 changes that affect all BASE24 value-added product users, along with the associated conversion/implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describes the release 6.0 changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion/implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 13 must be used in conjunction with sections 4 through 12 to identify the changes that BASE24 value-added product users need to consider for upgrading to the latest version of release 6.0.

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General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24 value-added product users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

This section describes the release 6.0 changes and conversion/implementation requirements that the following BASE24 value-added product users must address in order to upgrade to release 6.0:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Settlement Manager
- ACI Payments Manager
- ACI Smart Chip Manager

Section 2 contains a functional overview of the general enhancements made to the BASE24 value-added products in release 6.0.

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BASE24-InfoBase

This section describes the enhancements that impact the BASE24-InfoBase value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements affecting BASE24-InfoBase Release 6.0 users.

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BASE24 Retail Products

This section describes the enhancements that impact the value-added BASE24 Retail Products and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the users of the BASE24 Retail Products.

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BASE24-card

General Enhancements

This section describes the release 6.0 changes and conversion/implementation requirements that all existing BASE24-card users must address in order to upgrade to release 6.0.

Changes and requirements are identified using a standard format defined in the introduction to section 4.

- 6.0 General Changes

These areas are summarized below.

Section 2 contains a functional overview of the general enhancements described here.

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6.0 General Changes

1. Release 6.0 of BASE24 implements the support for the BASE24-card product into the standard product code. Previous releases of BASE24 supported the BASE24-card product as a CSE.

2. The release 6.0 CAF refresh format (CAFX) does not contain the alternate key fields. These fields are not to be updated by the Host.

Refer to the BASE24-card Reference Manual (CT-AE000-01) for details on implementing BASE24-card into a BASE24 system.

CUSTMACS Changes

Previous releases of BASE24-card included a CARDMACS file, which was used in place of the CUSTMACS file when making the BASE24 system with the BASE24-card product. BASE24-card release 6.0 only requires that the card_on flag be set to true in the CUSTMACS file if the customer wishes to make the BASE24-card product as part of the BASE24 system.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Cardholders Authorization File (CAF)

The CAF conversion program can be run to convert CAF records from the release 5.3 format to the release 6.0 format. This includes the CAF alternate key fields and the BASE24-card segment fields if the conversion program has been compiled with the card_on flag set to true and the BASE24-card segment exists in the release 5.3 CAF.

Note: Although it is recommended for performance purposes to convert the CAF using the Refresh process, the alternate key fields are not included in the 6.0 CAF Refresh file message format. This impacts existing BASE24-card customers that have run an Issue Maintenance function and wish to convert the CAF before running the Plastic Generation function. The CAF GEN-KEY alternate key field that was populated during the Issue Maintenance function will not be refreshed into the release 6.0 CAF. An Issue Maintenance function would have to be performed again on the 6.0 CAF file to reset the CAF GEN-KEY alternate key field. If the CAF conversion program is used to convert the CAF, the alternate keys will be converted into the release 6.0 CAF. For new BASE24-card customers, it is recommended that the Refresh process be used to convert the CAF.

Host/Co-Network Impacts

Host code changes are required for BASE24 users that utilize the release 6.0 CAF Refresh file message format because of the removal of the alternate keys from the message format. There is no support for the BASE24-card segment in the release 5.x CAF Refresh file message format.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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ACI Host Link

This section describes the enhancements that impact the ACI Host Link value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Host Link users.

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ACI Plastic Card Control System

This section describes the enhancements that impact the ACI Plastic Card Control System value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Plastic Card Control System users.

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ACI Claims Manager

This section describes the enhancements that impact the ACI Claims Manager value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Claims Manager users.

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ACI Proactive Risk Manager

This section describes the enhancements that impact the ACI Proactive Risk Manager value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Proactive Risk Manager users.

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ACI Settlement Manager

This section describes the enhancements that impact the ACI Settlement Manager value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Settlement Manager users.

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ACI Payments Manager

This section describes the enhancements that impact the ACI Payments Manager value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Payments Manager users.

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ACI Smart Chip Manager

This section describes the enhancements that impact the ACI Smart Chip Manager value-added product and the conversion/implementation requirements associated with the enhancements.

Enhancement Summary

There are no enhancements in release 6.0 affecting the ACI Smart Chip Manager users.

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 that have an impact on the following BASE24 value-added products, along with the changes and conversion/implementation requirements associated with them:

- BASE24-InfoBase
- BASE24 Retail Products
- BASE24-card
- ACI Host Link
- ACI Plastic Card Control System
- ACI Claims Manager
- ACI Proactive Risk Manager
- ACI Payments Manager
- ACI Settlement Manager
- ACI Smart Chip Manager

BASE24 users can choose whether or not to implement the enhancements described in this section. Users who plan to implement these enhancements must address the corresponding release 6.0 changes and conversion/implementation requirements. Users who do not plan to implement these enhancements can simply review the corresponding release 6.0 changes and conversion/implementation requirements for future reference.

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BASE24-InfoBase

This section describes the enhancement that impacts the BASE24-InfoBase value-added product.

There are no optional enhancements inherent to BASE24-InfoBase Release 6.0.

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BASE24 Retail Products

This section describes the enhancement that impacts the value-added BASE24 Retail Products.

There are no optional enhancements inherent to BASE24 Retail Products in release 6.0.

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BASE24-card

This section identifies the BASE24-card enhancements that can be enabled or disabled with release 6.0, along with the changes and conversion/implementation requirements associated with each.

There are no optional enhancements inherent to BASE24-card Release 6.0.

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ACI Host Link

This section describes the enhancement that impacts the ACI Host Link value-added product.

There are no optional enhancements inherent to ACI Host Link in release 6.0.

ACI Plastic Card Control System

This section describes the enhancement that impacts the ACI Plastic Card Control System value-added product.

There are no optional enhancements inherent to ACI Plastic Card Control System in release 6.0.

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ACI Claims Manager

This section describes the enhancement that impacts the ACI Claims Manager value-added product.

There are no optional enhancements inherent to ACI Claims Manager in release 6.0.

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ACI Proactive Risk Manager

This section describes the enhancement that impacts the ACI Proactive Risk Manager value-added product.

There are no optional enhancements inherent to ACI Proactive Risk Manager in release 6.0.

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ACI Payments Manager

This section describes the enhancement that impacts the ACI Payments Manager value-added product.

There are no optional enhancements inherent to ACI Payments Manager in release 6.0.

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ACI Settlement Manager

This section describes the enhancement that impacts the ACI Settlement Manager value-added product.

There are no optional enhancements inherent to ACI Settlement Manager in release 6.0.

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ACI Smart Chip Manager

This section describes the enhancements that impact ACI Smart Chip Manager value-added product.

- CAF Updates to the ACI Smart Chip Manager

The ACI Smart Chip Manager has been updated to accept CAF Updates from BASE24 via a SAF Extract tape. ACI Smart Chip Manager customers will need updated product information on the ACI Smart Chip Manager product to determine what needs to be done to uplift this product to be able to accept and process any CAF Updates from BASE24 Release 6.0.

Section 2 contains a functional overview of the functional changes made to uplift BASE24 Release 6.0 modules to send CAF Updates to the ACI Smart Chip Manager product.

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Appendix A

Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with Release 6.0 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0. Refer to the specified page numbers for explanations of how these assigns can or must be set.

TDF Expansion

Transactions Allowed:

Transaction Security Services:

Triple/Single DES Translation Support

PARAMS

The following params were added or modified for release 6.0. Refer to the specified page numbers for explanations of how these params can or must be set.

Multiple Currency Support:

Multiple Logical Network Routing Indicator:

PTDF Expansion

Stored Value Support:

TDF Expansion

Transactions Allowed:

Transaction Security Services:

Triple/Single DES Translation Support

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Appendix B

Host/CO-Network Impacts

Appendix B contains a list of release 6.0 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

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Appendix C

CUSTMACS File Changes

Appendix C contains a list of release 6.0 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 code.

ACI Worldwide Inc.

Enhancements

The following enhancements have CUSTMACS file changes:

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Appendix D

Re-Sequenced Files and New Files

Appendix D consists of the following tables:

- A table illustrating the files that have been re-sequenced in this release.
- A table illustrating the new files for the general enhancements.
- A table illustrating the new files for the optional ACI Proactive Risk Manager Interface Components enhancement.
- A table illustrating the new files for the optional ACI Smart Chip Manager enhancement.
- A table illustrating the new files for the optional BASE24-card product.
- A table illustrating the new files for the optional BASE24-atm Dassault Device Handler.
- A table illustrating the new files for the optional BASE24-pos Derived Unique Key Per Transaction (DUKPT) enhancement.
- A table illustrating the new files for the optional BASE24-atm Diebold 10XX Dial-Up Device Handler.
- A table illustrating the new files for the optional BASE24-pos eCheck Support enhancement.
- A table illustrating the new files for the optional EMV Support enhancement.
- A table illustrating the new files for the optional Format 2 File Support enhancement.
- A table illustrating the new files for the optional LINK Interchange Interface.
- A table illustrating the new files for the optional BASE24-atm Multiple Account Select by Qualifier Support Enhancement.
- A table illustrating the new files for the optional Multiple Currency Support enhancement.
- A table illustrating the new files for the optional BASE24-atm NCR 5XXX Dial-Up Device Handler.
- A table illustrating the new files for the optional BASE24-pos Stored Value Support enhancement.

- A table illustrating the new files for the optional BASE24-atm Tidel Device Handler.
- A table illustrating the new files for the optional Triple/Single DES Translate Support enhancement.
- A table illustrating the new files for the optional BASE24-atm Triton Device Handler.

The tables begin on the following page:

Re-Sequenced Files Table

Re-Sequenced File Location and File Name	**File Language Type**
at601000.c1000g	TAL
at601000.c1000s	TAL
at601000.c1000s	TAL
at601000.c100cses	TAL
at601000.c100nams	TAL
at601000.svd10xxs	TAL
at60b100.c100ssbg	TAL
at60b100.c100ssbs	TAL
at60bn50.n50ssbg	TAL
at60bn50.n50ssbs	TAL
at60c100.c100sscg	TAL
at60c100.c100sscs	TAL
at60cn50.n50sscg	TAL
at60cn50.n50sscs	TAL
at60d100.c100ssdg	TAL
at60d100.c100ssds	TAL
at60dct.svdcts	TAL
at60dn50.n50ssdg	TAL
at60dn50.n50ssds	TAL
at60dn50.ssdhnams	TAL
at60m100.c100mads	TAL
at60m100.c100mans	TAL
at60m100.c1000mas	TAL
at60m100.m100fm	MAKE
at60m100.m100mm	MAKE
at60m100.ma100m	MAKE
at60m100.mad100m	MAKE
at60m100.man100m	MAKE
at60mn50.madn50m	MAKE
at60mn50.man50m	MAKE
at60mn50.mann50m	MAKE
at60mn50.mn50fm	MAKE
at60mn50.mn50mm	MAKE
at60mn50.n50mads	TAL
at60mn50.n50mans	TAL
at60mn50.n50mas	TAL

Re-Sequenced File Location and File Name	File Language Type
at60n100.c100ncdg	TAL
at60n100.c100ncds	TAL
at60n100.n100nams	TAL
at60n50.n50cses	TAL
at60n50.n50g	TAL
at60n50.n50g	TAL
at60n50.n50nams	TAL
at60n50.n50ncdds	TAL
at60n50.n50rpqds	TAL
at60n50.n50s	TAL
at60n50.n50s	TAL
at60n50.n50smds	TAL
at60n50.n50ssbds	TAL
at60n50.svdn50s	TAL
at60nn50.n50ncdg	TAL
at60nn50.n50ncds	TAL
at60nn50.nn50nams	TAL
at60p100.c100dlug	TAL
at60p100.c100dlus	TAL
at60setl.setls	TAL
ba60aft.pitable	COBOL
ba60aft.svmhlps	COBOL
ba60dct.svdpcts	TAL
ba60ext.basexs	TAL
ba60ext.omfxg	TAL
ba60src.bacoutls	COBOL
ba60src.batkncvs	TAL
ba60src.bautils	TAL
ba60src.cobnames	COBOL
ba60src.hiswutls	TAL
ba60src.secutils	TAL
ps60ext.ptlfxs	TAL
ps60hpdh.hpdhg	TAL
ps60hpdh.hpdhs	TAL
ps60rtau.authlibs	TAL
ps60rtau.routers	TAL
ps60setl.setlg	TAL
ps60setl.setls	TAL

Re-Sequenced File Location and File Name	**File Language Type**
ps60spdh.aspdhg	TAL
ps60spdh.aspdhs	TAL
ps60spdh.spdhldrs	TAL
tb60ext.itlfxds	DDL

New Files for the General Enhancements Table

New File Location and File Name	File Language Type
acilibi.lconfld	Object
at601000.c100omxe	Generated during a compile
at601000.c100omxg	TAL
at601000.c100omxm	MAKE
at601000.c100omxo	Generated during a compile
at601000.c100omxs	TAL
at601000.rqd10xxm	MAKE
at601000.rqd10xxo	Generated during a compile
at601000.rqd10xxs	SCOBOL
at601000.rqt10xxm	MAKE
at601000.rqt10xxo	Generated during a compile
at601000.rqt10xxs	SCOBOL
at601000.scnd10xx	ACIAID
at601000.scrn10xx	ACIAID
at601000.sv10xxte	Generated during a compile
at601000.sv10xxtg	TAL
at601000.sv10xxtm	MAKE
at601000.sv10xxto	Generated during a compile
at601000.sv10xxts	TAL
at604730.at4730m	MAKE
at604730.i473cob	Generated during a compile
at604730.i473ddlm	MAKE
at604730.i473omxe	Generated during a compile
at604730.i473omxg	TAL
at604730.i473omxm	MAKE
at604730.i473omxo	Generated during a compile
at604730.i473omxs	TAL
at604730.sv4730te	Generated during a compile
at604730.sv4730tg	TAL
at604730.sv4730tm	MAKE
at604730.sv4730to	Generated during a compile
at604730.sv4730ts	TAL
at60aft.rqatdm	MAKE
at60aft.rqatdo	Generated during a compile
at60aft.rqatds	SCOBOL
at60aft.rqtrfss	ACIAID

New File Location and File Name	File Language Type
at60aft.rqtrfxm	MAKE
at60aft.rqtrfxo	Generated during a compile
at60aft.rqtrfxs	SCOBOL
at60aft.scrnatd1	ACIAID
at60aft.scrnatdb	ACIAID
at60aft.svatdlm	MAKE
at60aft.svatdlo	Generated during a compile
at60aft.svatdtbl	TAL
at60aft.svatdte	Generated during a compile
at60aft.svatdtg	TAL
at60aft.svatdtm	MAKE
at60aft.svatdto	Generated during a compile
at60aft.svatdts	TAL
at60aft.svtrfte	Generated during a compile
at60aft.svtrftg	TAL
at60aft.svtrftm	MAKE
at60aft.svtrfto	Generated during a compile
at60aft.svtrfts	TAL
at60aft.trfemtto	Generated during a compile
at60auth.authtbl	TAL
at60b650.atb650m	MAKE
at60b650.b650cob	Generated during a compile
at60b650.b650omxe	Generated during a compile
at60b650.b650omxg	TAL
at60b650.b650omxm	MAKE
at60b650.b650omxo	Generated during a compile
at60b650.b650omxs	TAL
at60b650.svb650te	TAL
at60b650.svb650tg	TAL
at60b650.svb650tm	MAKE
at60b650.svb650to	Generated during a compile
at60b650.svb650ts	TAL
at60c51.atc51m	MAKE
at60c51.c51cob	Generated during a compile
at60c51.c51ddlm	MAKE
at60c51.c51omfxe	Generated during a compile
at60c51.c51omfxg	TAL
at60c51.c51omfxm	MAKE

New File Location and File Name	File Language Type
at60c51.c51omfxo	Generated during a compile
at60c51.c51omfxs	TAL
at60c51.sv5100te	Generated during a compile
at60c51.sv5100tg	TAL
at60c51.sv5100tm	MAKE
at60c51.sv5100to	Generated during a compile
at60c51.sv5100ts	TAL
at60cse.atcsefm	MAKE
at60cse.atcsemm	MAKE
at60cse.atdctg	TAL
at60cse.tdfctg	TAL
at60dct.svdctg	TAL
at60ddl.ddlfatd	DDL
at60ddl.ddlftrf	DDL
at60ddl.ddlgatd	DDL
at60defs.atdefcob	Generated during a compile
at60defs.atdefsm	MAKE
at60defs.atdefsmm	MAKE
at60defs.atdeftal	Generated during a compile
at60defs.ddlgadef	DDL
at60defs.dictver	Generated during a compile
at60dieb.atdiebfm	MAKE
at60dieb.atdiebm	MAKE
at60dieb.atdiebmm	MAKE
at60dieb.dictver	Generated during a compile
at60dieb.diebcob	Generated during a compile
at60dieb.diebddlm	MAKE
at60dieb.diebddls	DDL
at60dieb.diebomxe	Generated during a compile
at60dieb.diebomxg	TAL
at60dieb.diebomxm	MAKE
at60dieb.diebomxo	Generated during a compile
at60dieb.diebomxs	TAL
at60dieb.diebTAL	Generated during a compile
at60dieb.svdiebte	Generated during a compile
at60dieb.svdiebtg	TAL
at60dieb.svdiebtm	MAKE
at60dieb.svdiebto	Generated during a compile

New File Location and File Name	File Language Type
at60dieb.svdiebts	TAL
at60emtb.aemtbfm	MAKE
at60emtb.aemtbm	MAKE
at60emtb.aemtbmm	MAKE
at60emtb.aemtdc	Generated during a compile
at60emtb.aemtdm	MAKE
at60emtb.aemtds	DDL
at60emtb.aemtdt	Generated during a compile
at60emtb.aemtdx	Generated during a compile
at60emtb.dictver	Generated during a compile
at60emtb.rcpemtte	Generated during a compile
at60emtb.rcpemttg	TAL
at60emtb.rcpemttm	MAKE
at60emtb.rcpemtto	Generated during a compile
at60emtb.rcpemtts	TAL
at60emtb.trfemtdc	Generated during a compile
at60emtb.trfemtdm	MAKE
at60emtb.trfemtds	DDL
at60emtb.trfemtdt	Generated during a compile
at60emtb.trfemtdx	Generated during a compile
at60emtb.trfemtte	Generated during a compile
at60emtb.trfemttg	TAL
at60emtb.trfemttm	MAKE
at60emtb.trfemtts	TAL
at60ibm.atibmfm	MAKE
at60ibm.atibmm	MAKE
at60ibm.atibmmm	MAKE
at60ibm.dictver	Generated during a compile
at60ibm.ibmcob	Generated during a compile
at60ibm.ibmddlm	MAKE
at60ibm.ibmddls	DDL
at60ibm.ibmomfxe	Generated during a compile
at60ibm.ibmomfxg	TAL
at60ibm.ibmomfxm	MAKE
at60ibm.ibmomfxo	Generated during a compile
at60ibm.ibmomfxs	TAL
at60ibm.ibmtal	Generated during a compile
at60ibm.sv36xxte	Generated during a compile

New File Location and File Name	File Language Type
at60ibm.sv36xxtg	TAL
at60ibm.sv36xxtm	MAKE
at60ibm.sv36xxto	Generated during a compile
at60ibm.sv36xxts	TAL
at60lib.albprops	TEMPLT
at60lib.atlibfm	MAKE
at60lib.atliblm	MAKE
at60lib.atliblo	Generated during a compile
at60lib.atlibm	MAKE
at60lib.atlibmm	MAKE
at60lib.atlibtbl	TAL
at60lib.atlibte	Generated during a compile
at60lib.atlibtg	TAL
at60lib.atlibtm	MAKE
at60lib.atlibto	Generated during a compile
at60lib.atlibts	TAL
at60misc.atlibman	DOC
at60misc.atlibtoc	DOC
at60misc.trf	Data
at60n50.n50omfxe	Generated during a compile
at60n50.n50omfxg	TAL
at60n50.n50omfxm	MAKE
at60n50.n50omfxo	Generated during a compile
at60n50.n50omfxs	TAL
at60n50.rqhoprm	MAKE
at60n50.rqhopro	Generated during a compile
at60n50.rqhoprs	TAL
at60n50.rqt5xxxm	MAKE
at60n50.rqt5xxxo	Generated during a compile
at60n50.rqt5xxxs	SCOBOL
at60n50.scnhopr	ACIAID
at60n50.scrn5xxx	ACIAID
at60n50.sv5xxxte	Generated during a compile
at60n50.sv5xxxtg	TAL
at60n50.sv5xxxtm	MAKE
at60n50.sv5xxxto	Generated during a compile
at60n50.sv5xxxts	TAL
at60p100.atp100fm	MAKE

New File Location and File Name	File Language Type
at60p100.atp100mm	MAKE
at60p100.c100dlue	Generated during a compile
at60p100.c100dlug	TAL
at60p100.c100dlum	MAKE
at60p100.c100dluo	Generated during a compile
at60p100.c100dlus	TAL
at60src.atdtg	TAL
at60src.atdxg	SCOBOL
at60src.tdftg	TAL
at60tidl.attidlm	MAKE
at60tidl.svtidlte	Generated during a compile
at60tidl.svtidltg	TAL
at60tidl.svtidltm	MAKE
at60tidl.svtidlto	Generated during a compile
at60tidl.svtidlts	TAL
at60tidl.tidlcob	Generated during a compile
at60tidl.tidlomxe	Generated during a compile
at60tidl.tidlomxg	TAL
at60tidl.tidlomxm	MAKE
at60tidl.tidlomxo	Generated during a compile
at60tidl.tidlomxs	TAL
at60uc04.aaread04	DOC
at60uc04.atuc04fm	MAKE
at60uc04.atuc04m	MAKE
at60uc04.atuc04mm	MAKE
at60uc04.cnvatd	Generated during a compile
at60uc04.cnvatde	Generated during a compile
at60uc04.cnvatdm	MAKE
at60uc04.cnvatdo	Generated during a compile
at60uc04.cnvatdr	Obey
at60uc04.cnvatds	TAL
ba10sc04.basc04fm	MAKE
ba10sc04.basc04m	MAKE
ba10sc04.basc04mm	MAKE
ba10sc04.cnve	Generated during a compile
ba10sc04.cnvg	TAL
ba10sc04.cnvlibe	Generated during a compile
ba10sc04.cnvlibo	Generated during a compile

New File Location and File Name	File Language Type
ba10sc04.cnvlibs	TAL
ba10sc04.cnvm	MAKE
ba10sc04.cnvo	Generated during a compile
ba10sc04.cnvr	Obey
ba10sc04.cnvs	TAL
ba60aft.ccafmlm	MAKE
ba60aft.ccafmlo	Generated during a compile
ba60aft.lncftbls	TAL
ba60aft.megatbl	COBOL
ba60aft.rqapcfss	ACIAID
ba60aft.rqapcfss	ACIAID
ba60aft.rqapcfxm	MAKE
ba60aft.rqapcfxm	MAKE
ba60aft.rqapcfxo	Generated during a compile
ba60aft.rqapcfxs	SCOBOL
ba60aft.rqapcfxs	SCOBOL
ba60aft.rqattxo	Generated during a compile
ba60aft.rqicfem	MAKE
ba60aft.rqicfeo	Generated during a compile
ba60aft.rqicfes	SCOBOL
ba60aft.rqipcfss	ACIAID
ba60aft.rqipcfss	ACIAID
ba60aft.rqipcfxm	MAKE
ba60aft.rqipcfxm	MAKE
ba60aft.rqipcfxo	Generated during a compile
ba60aft.rqipcfxs	SCOBOL
ba60aft.rqipcfxs	SCOBOL
ba60aft.rqkeyim	MAKE
ba60aft.rqkeyio	Generated during a compile
ba60aft.rqkeyis	SCOBOL
ba60aft.rqlncfss	ACIAID
ba60aft.rqlncfxm	MAKE
ba60aft.rqlncfxo	Generated during a compile
ba60aft.rqlncfxs	SCOBOL
ba60aft.rqpdfss	ACIAID
ba60aft.rqpdfxm	MAKE
ba60aft.rqpdfxo	Generated during a compile
ba60aft.rqpdfxs	SCOBOL

New File Location and File Name	File Language Type
ba60aft.rqtcfss	ACIAID
ba60aft.rqtcfxm	MAKE
ba60aft.rqtcfxo	Generated during a compile
ba60aft.rqtcfxs	SCOBOL
ba60aft.scnicfeo	Generated during a compile
ba60aft.scrnicfe	ACIAID
ba60aft.scrnkeyi	ACIAID
ba60aft.svapcfte	Generated during a compile
ba60aft.svapcftg	TAL
ba60aft.svapcftm	MAKE
ba60aft.svapcfto	Generated during a compile
ba60aft.svapcfts	TAL
ba60aft.svicfe	Generated during a compile
ba60aft.svicfem	MAKE
ba60aft.svicfes	COBOL
ba60aft.svipcfte	Generated during a compile
ba60aft.svipcftg	TAL
ba60aft.svipcftm	MAKE
ba60aft.svipcfto	Generated during a compile
ba60aft.svipcfts	TAL
ba60aft.svkeyi	Generated during compile
ba60aft.svkeyim	MAKE
ba60aft.svkeyis	COBOL
ba60aft.svlncfte	Generated during a compile
ba60aft.svlncftg	TAL
ba60aft.svlncftm	MAKE
ba60aft.svlncfto	Generated during a compile
ba60aft.svlncfts	TAL
ba60aft.svpdfte	Generated during a compile
ba60aft.svpdftg	TAL
ba60aft.svpdftm	MAKE
ba60aft.svpdfto	Generated during a compile
ba60aft.svpdfts	TAL
ba60aft.svtcfte	Generated during a compile
ba60aft.svtcftg	TAL
ba60aft.svtcftm	MAKE
ba60aft.svtcfts	TAL
ba60aft.svtdflte	Generated during a compile

New File Location and File Name	File Language Type
ba60aft.svtdfltg	TAL
ba60aft.svtdfltm	MAKE
ba60aft.svtdflto	Generated during a compile
ba60aft.svtdflts	TAL
ba60cse.desctbl	COBOL
ba60cse.megatblc	COBOL
ba60dct.ectscrns	ACIAID
ba60dct.rqdectm	MAKE
ba60dct.rqdecto	Generated during a compile
ba60dct.rqdects	SCOBOL
ba60ddl.custcnst	DDL
ba60ddl.ddlacnst	DDL
ba60ddl.ddlbcnst	DDL
ba60ddl.ddlfapcf	DDL
ba60ddl.ddlficfe	DDL
ba60ddl.ddlfipcf	DDL
ba60ddl.ddlfkeyi	DDL
ba60ddl.ddlfpdf	DDL
ba60ddl.ddlftcf	DDL
ba60ddl.ddlpcnst	DDL
ba60emtb.apcemtte	Generated during a compile
ba60emtb.apcemttg	TAL
ba60emtb.apcemttm	MAKE
ba60emtb.apcemtto	Generated during a compile
ba60emtb.apcemtts	TAL
ba60emtb.bemtbfm	MAKE
ba60emtb.bemtblm	MAKE
ba60emtb.bemtblo	Generated during a compile
ba60emtb.bemtblo	Generated during a compile
ba60emtb.bemtblte	Generated during a compile
ba60emtb.bemtbltg	TAL
ba60emtb.bemtbltm	MAKE
ba60emtb.bemtblto	Generated during a compile
ba60emtb.bemtblts	TAL
ba60emtb.bemtbm	MAKE
ba60emtb.bemtbmm	MAKE
ba60emtb.bemtdc	Generated during a compile
ba60emtb.bemtdm	MAKE

New File Location and File Name	File Language Type
ba60emtb.bemtds	DDL
ba60emtb.bemtdt	Generated during a compile
ba60emtb.bemtdx	Generated during a compile
ba60emtb.dictver	Generated during a compile
ba60emtb.idfemtdm	MAKE
ba60emtb.ipcemtte	Generated during a compile
ba60emtb.ipcemttg	TAL
ba60emtb.ipcemttm	MAKE
ba60emtb.ipcemtts	TAL
ba60ext.extrtbls	TAL
ba60ext.icfex	Generated during a compile
ba60ext.icfex6e	Generated during a compile
ba60ext.icfex6s	TAL
ba60ext.icfexe	Generated during a compile
ba60ext.icfexg	TAL
ba60ext.icfexm	MAKE
ba60ext.icfexs	TAL
ba60ext.idf5tal	TAL
ba60ext.idfx5e	Generated during a compile
ba60ext.idfx5s	TAL
ba60ext.idfx6e	Generated during a compile
ba60ext.idfx6s	TAL
ba60lib.balibdm	MAKE
ba60lib.balibfm	MAKE
ba60lib.balibgds	DDL
ba60lib.balibgdt	Generated during a compile
ba60lib.balibgdx	Generated during a compile
ba60lib.balibm	MAKE
ba60lib.balibmm	MAKE
ba60lib.balibtbl	TAL
ba60lib.balibte	Generated during a compile
ba60lib.balibtg	TAL
ba60lib.balibtm	MAKE
ba60lib.balibto	Generated during a compile
ba60lib.balibts	TAL
ba60lib.blbprops	TEMPLT
ba60lib.dictver	Generated during a compile
ba60misc.apcf	data

New File Location and File Name	File Language Type
ba60misc.att	Data
ba60misc.balibman	DOC
ba60misc.balibtoc	DOC
ba60misc.contents	DOC
ba60misc.ipcf	Data
ba60misc.lconfat	Data
ba60misc.lconfba	Data
ba60misc.lconfbpy	Data
ba60misc.lconfbcm	Data
ba60misc.lconfct	Data
ba60misc.lconffh	Data
ba60misc.lconfldr	Obey
ba60misc.lconfma	Data
ba60misc.lconfps	Data
ba60misc.lconftb	Data
ba60misc.lconftr	Data
ba60misc.pdf	Data
ba60misc.tcf	Data
ba60uc04.aaread04	DOC
ba60uc04.bauc04fm	MAKE
ba60uc04.bauc04m	MAKE
ba60uc04.bauc04mm	MAKE
ba60uc04.cnvatt	Generated during a compile
ba60uc04.cnvatte	Generated during a compile
ba60uc04.cnvattm	MAKE
ba60uc04.cnvatto	Generated during a compile
ba60uc04.cnvattr	obey
ba60uc04.cnvatts	TAL
ba60uc04.cnvcaf	Generated during a compile
ba60uc04.cnvcafe	Generated during a compile
ba60uc04.cnvcafm	MAKE
ba60uc04.cnvcafo	Generated during a compile
ba60uc04.cnvcafr	obey
ba60uc04.cnvcafs	TAL
ba60uc04.cnvcpf	Generated during a compile
ba60uc04.cnvcpfe	Generated during a compile
ba60uc04.cnvcpfm	MAKE
ba60uc04.cnvcpfo	Generated during a compile

New File Location and File Name	File Language Type
ba60uc04.cnvcpfr	obey
ba60uc04.cnvcpfs	TAL
ba60uc04.cnvecf	Generated during a compile
ba60uc04.cnvecfe	Generated during a compile
ba60uc04.cnvecfm	MAKE
ba60uc04.cnvecfo	Generated during a compile
ba60uc04.cnvecfr	obey
ba60uc04.cnvecfs	TAL
ba60uc04.cnvicfe	Generated during a compile
ba60uc04.cnvicfee	Generated during a compile
ba60uc04.cnvicfem	MAKE
ba60uc04.cnvicfeo	Generated during a compile
ba60uc04.cnvicfer	obey
ba60uc04.cnvicfes	TAL
ba60uc04.cnvidf	Generated during a compile
ba60uc04.cnvidfe	Generated during a compile
ba60uc04.cnvidfm	MAKE
ba60uc04.cnvidfo	Generated during a compile
ba60uc04.cnvidfr	obey
ba60uc04.cnvidfs	TAL
ba60uc04.cnvlncf	Generated during a compile
ba60uc04.cnvlncfe	Generated during a compile
ba60uc04.cnvlncfm	MAKE
ba60uc04.cnvlncfo	Generated during a compile
ba60uc04.cnvlncfr	obey
ba60uc04.cnvlncfs	TAL
ba60uc04.cnvsec	Generated during a compile
ba60uc04.cnvsece	Generated during a compile
ba60uc04.cnvsecm	MAKE
ba60uc04.cnvseco	Generated during a compile
ba60uc04.cnvsecr	obey
ba60uc04.cnvsecs	TAL
ba60uc04.cnvsurf	Generated during a compile
ba60uc04.cnvsurfe	Generated during a compile
ba60uc04.cnvsurfm	MAKE
ba60uc04.cnvsurfo	Generated during a compile
ba60uc04.cnvsurfr	obey
ba60uc04.cnvsurfs	TAL

New File Location and File Name	File Language Type
ba60uc04.cnvupfr	Generated during a compile
ba60uc04.cnvupfre	Generated during a compile
ba60uc04.cnvupfrm	MAKE
ba60uc04.cnvupfro	Generated during a compile
ba60uc04.cnvupfrr	obey
ba60uc04.cnvupfrs	TAL
ba60ud04.initpobj	FUP
ps60aft.svptdlm	MAKE
ps60aft.svptdlo	Generated during a compile
ps60aft.svptdtbl	TAL
ps60aft.svptdte	Generated during a compile
ps60aft.svptdtg	TAL
ps60aft.svptdtm	MAKE
ps60aft.svptdto	Generated during a compile
ps60aft.svptdts	TAL
ps60cse.ptdctg	TAL
ps60cse.ptdfctg	TAL
ps60ddl.ddlfptd	DDL
ps60ddl.ddlgptd	DDL
ps60ext.prdf5TAL	TAL
ps60ext.prdfx5e	Generated during a compile
ps60ext.prdfx5s	TAL
ps60ext.prdfx6e	Generated during a compile
ps60ext.prdfx6s	TAL
ps60hpdh.pshpdhm	MAKE
ps60hpdh.svhpdhte	Generated during a compile
ps60hpdh.svhpdhtg	TAL
ps60hpdh.svhpdhtm	MAKE
ps60hpdh.svhpdhto	Generated during a compile
ps60hpdh.svhpdhts	TAL
ps60lib.pslibfm	MAKE
ps60lib.psliblm	MAKE
ps60lib.psliblo	Generated during a compile
ps60lib.pslibm	MAKE
ps60lib.pslibmm	MAKE
ps60lib.pslibte	Generated during a compile
ps60lib.pslibtg	TAL
ps60lib.pslibtm	MAKE

New File Location and File Name	File Language Type
ps60lib.pslibto	Generated during a compile
ps60lib.pslibts	TAL
ps60misc.contents	DOC
ps60misc.pslibman	DOC
ps60misc.pslibtoc	DOC
ps60spdh.psspdhm	MAKE
ps60spdh.spdhcses	TAL
ps60spdh.spdhtbls	TAL
ps60spdh.svspdhte	Generated during a compile
ps60spdh.svspdhtg	TAL
ps60spdh.svspdhtm	MAKE
ps60spdh.svspdhto	Generated during a compile
ps60spdh.svspdhts	TAL
ps60src.ptdftg	TAL
ps60src.ptdtg	TAL
ps60src.ptdxg	COBOL
ps60uc04.aaread04	DOC
ps60uc04.cnvadmn	Generated during a compile
ps60uc04.cnvadmne	Generated during a compile
ps60uc04.cnvadmnm	MAKE
ps60uc04.cnvadmno	Generated during a compile
ps60uc04.cnvadmnr	obey
ps60uc04.cnvadmns	TAL
ps60uc04.cnvprdf	Generated during a compile
ps60uc04.cnvprdfe	Generated during a compile
ps60uc04.cnvprdfm	MAKE
ps60uc04.cnvprdfo	Generated during a compile
ps60uc04.cnvprdfr	obey
ps60uc04.cnvprdfs	TAL
ps60uc04.cnvptd	Generated during a compile
ps60uc04.cnvptde	Generated during a compile
ps60uc04.cnvptdm	MAKE
ps60uc04.cnvptdo	Generated during a compile
ps60uc04.cnvptdr	obey
ps60uc04.cnvptds	TAL
ps60uc04.psuc04fm	MAKE
ps60uc04.psuc04m	MAKE
ps60uc04.psuc04mm	MAKE

New File Location and File Name	File Language Type
rb60aft.rbmegas	COBOL
rb60aft.rbmhlps	COBOL
rb60aft.rbscrns	ACIAID
sw60cse.swcsefm	MAKE
sw60cse.swcsemm	MAKE
sw60euro.eurod	TAL
sw60mds.mdsd	TAL
tr60lib.trlibfm	MAKE
tr60lib.trliblm	MAKE
tr60lib.trliblo	Generated during a compile
tr60lib.trlibm	MAKE
tr60lib.trlibman	DOC
tr60lib.trlibmm	MAKE
tr60lib.trlibte	Generated during a compile
tr60lib.trlibtg	TAL
tr60lib.trlibtm	MAKE
tr60lib.trlibto	Generated during a compile
tr60lib.trlibtoc	DOC
tr60lib.trlibts	TAL
tr60uc04.aaread04	DOC
tr60uc04.truc04fm	MAKE
tr60uc04.truc04m	MAKE
tr60uc04.truc04mm	MAKE
tr60uc04.tssttdft	Generated during a compile
tr60uc04.tssttdfee	Generated during a compile
tr60uc04.tssttdfg	TAL
tr60uc04.tssttdfm	MAKE
tr60uc04.tssttdfo	Generated during a compile
tr60uc04.tssttdfr	obey
tr60uc04.tssttdfs	TAL

New Files for the Optional ACI Proactive Risk Manager Interface Components Enhancement Table

ACI Proactive Risk Manager Interface Components **New File Location and File Name**	**File Language Type**
at60path.authpie	Generated during a compile
at60path.authpig	TAL
at60path.authpim	MAKE
at60path.authpio	Generated during a compile
at60path.authpis	TAL
at60path.rtaupig	TAL
at60path.rtaupim	MAKE
at60path.rtaupio	Generated during a compile
at60path.rtaupis	TAL
pi60achd.achdclm	MAKE
pi60achd.achdcls	COBOL
pi60achd.piachdfm	MAKE
pi60achd.piachdmm	MAKE
pi60aft.piaftfm	MAKE
pi60aft.piaftm	MAKE
pi60aft.piaftmm	MAKE
pi60aft.rqptblm	MAKE
pi60aft.rqptblo	Generated during a compile
pi60aft.rqptbls	SCOBOL
pi60aft.rqsemfm	MAKE
pi60aft.rqsemfo	Generated during a compile
pi60aft.rqsemfs	SCOBOL
pi60aft.scrnptbl	ACIAID
pi60aft.scrnsemf	ACIAID
pi60aft.svptblm	MAKE
pi60aft.svptbls	COBOL
pi60aft.svsemfm	MAKE
pi60aft.svsemfs	COBOL
pi60ddl.ddlfachd	DDL
pi60ddl.ddlfptbl	DDL
pi60ddl.ddlfseif	DDL
pi60ddl.ddlfsemf	DDL
pi60ddl.ddlgsemx	DDL

ACI Proactive Risk Manager Interface Components New File Location and File Name	File Language Type
pi60ddl.piddlfm	MAKE
pi60ddl.piddlm	MAKE
pi60ddl.piddlmm	MAKE
pi60ext.piextfm	MAKE
pi60ext.piextm	MAKE
pi60ext.piextmm	MAKE
pi60ext.piomfxe	Generated during a compile
pi60ext.piomfxg	TAL
pi60ext.piomfxm	MAKE
pi60ext.piomfxs	TAL
pi60refr.pirefre	Generated during a compile
pi60refr.pirefrmm	MAKE
pi60refr.pirefrs	TAL
pi60seip.piseipfm	MAKE
pi60seip.piseipm	MAKE
pi60seip.piseipmm	MAKE
pi60seip.rqseipm	MAKE
pi60seip.rqseipo	Generated during a compile
pi60seip.rqseips	SCOBOL
pi60seip.scrnseip	ACIAID
pi60seip.seipcob	Generated during a compile
pi60seip.seipddlm	MAKE
pi60seip.seipddls	DDL
pi60seip.seipe	Generated during a compile
pi60seip.seipg	TAL
pi60seip.seiplogm	DOC
pi60seip.seipm	MAKE
pi60seip.seips	TAL
pi60seip.seiptal	Generated during a compile
pi60seip.semh	Generated during a compile
pi60src.achdh	Generated during a compile
pi60src.piddlcob	Generated during a compile
pi60src.piddlfup	Generated during a compile
pi60src.piddltal	Generated during a compile
pi60src.pisrcfm	MAKE
pi60src.pisrcmm	MAKE

ACI Proactive Risk Manager Interface Components New File Location and File Name	File Language Type
pi60src.pitplt	Generated during a compile
pi60src.prmfm	MAKE
pi60src.prmm	MAKE
pi60src.prmmm	MAKE
pi60src.prmupdt	DOC
pi60src.seifh	Generated during a compile
ps60prta.rtaupie	Generated during a compile
ps60rtau.rtaupid	TAL

New Files for the Optional ACI Smart Chip Manager Enhancement Table

ACI Smart Chip Manager New File Location and File Name	**File Language Type**
ba60aft.svntftbl	TAL
ba60aft.svntfyt	Generated during a compile
ba60aft.svntfyte	Generated during a compile
ba60aft.svntfytg	TAL
ba60aft.svntfytm	MAKE
ba60aft.svntfyto	Generated during a compile
ba60aft.svntfyts	TAL

New Files for the Optional BASE24-card Product Table

BASE24-card New File Location and File Name	File Language Type
ct60aft.ctaftfm	MAKE
ct60aft.ctaftm	MAKE
ct60aft.ctaftmm	MAKE
ct60aft.ctalgo	Generated during a compile
ct60aft.ctalgom	MAKE
ct60aft.ctalgos	TAL
ct60aft.rqcmfm	MAKE
ct60aft.rqcmfo	Generated during a compile
ct60aft.rqcmfs	SCOBOL
ct60aft.rqcofm	MAKE
ct60aft.rqcofo	Generated during a compile
ct60aft.rqcofs	SCOBOL
ct60aft.rqlstm	MAKE
ct60aft.rqlsto	Generated during a compile
ct60aft.rqlsts	SCOBOL
ct60aft.scrncmf	ACIAID
ct60aft.scrncof	ACIAID
ct60aft.scrnlst	ACIAID
ct60aft.svagen	Generated during a compile
ct60aft.svagenm	MAKE
ct60aft.svagens	COBOL
ct60aft.svcmf	Generated during a compile
ct60aft.svcmfm	MAKE
ct60aft.svcmfs	COBOL
ct60aft.svlst	Generated during a compile
ct60aft.svlstm	MAKE
ct60aft.svlsts	COBOL
ct60aft.svpcgen	Generated during a compile
ct60aft.svpcgenm	MAKE
ct60aft.svpcgens	COBOL
ct60ddl.ctddlfm	MAKE
ct60ddl.ctddlm	MAKE
ct60ddl.ctddlmm	MAKE
ct60ddl.ddlgcard	DDL
ct60src.cardfm	MAKE

BASE24-card New File Location and File Name	File Language Type
ct60src.cardm	MAKE
ct60src.cardmm	MAKE
ct60src.ctddlcob	Generated during a compile
ct60src.ctddltal	Generated during a compile
ct60src.ctsrcfm	MAKE
ct60src.ctsrcm	MAKE
ct60src.ctsrcmm	MAKE
ct60src.ctutil	Generated during a compile
ct60src.ctutile	Generated during a compile
ct60src.ctutilm	MAKE
ct60src.ctutilo	Generated during a compile
ct60src.ctutils	TAL
ct60src.soundex	Generated during a compile
ct60src.soundexm	MAKE
ct60src.soundexs	TAL

New Files for the Optional BASE24-atm Dassault Device Handler Table

Dassault **New File Location and File Name**	**File Language Type**
at60esd.atesdfm	MAKE
at60esd.atesdmm	MAKE
at60esd.atesdm	MAKE
at60esd.ddlesdm	MAKE
at60esd.esd	Generated during a compile
at60esd.esdddls	DDL
at60esd.esde	Generated during a compile
at60esd.esdm	MAKE
at60esd.esdo	Generated during a compile
at60esd.esds	TAL
at60esd.esdtal	Generated during a compile
at60esd.svdesdm	MAKE
at60esd.svdesdo	Generated during a compile
at60esd.svdesds	TAL
at60esd.rqdesdm	MAKE
at60esd.rqdesds	SCOBOL
at60esd.scndesd	ACIAID
at60esd.rqtesdm	MAKE
at60esd.rqtesds	SCOBOL
at60esd.scntesd	ACIAID

New Files for the Optional BASE24-pos DUKPT Support Enhancement Table

DUKPT **New File Location and File Name**	**File Language Type**
ba60ddl.ddlfkeyd	DDL
ba60aft.rqkeydm	MAKE
ba60aft.rqkeyds	SCOBOL
ba60aft.rqkeyd	Generated during a compile
ba60aft.svkeydm	MAKE
ba60aft.svkeyds	COBOL
ba60aft.svkeyd	Generated during a compile
ba60aft.scrnkeyd	ACIAID

New Files for the Optional BASE24-atm Diebold 10XX/476X Dial-Up Device Handler Table

Diebold 10XX/476X Dial-Up **New File Location and File Name**	**File Language Type**
at60p100.atp100fm	MAKE
at60p100.atp100mm	MAKE
at60p100.c100dlue	Generated during a compile
at60p100.c100dlug	TAL
at60p100.c100dlum	MAKE
at60p100.c100dluo	Generated during a compile
at60p100.c100dlus	TAL

New Files for the Optional BASE24-pos eCheck Support Enhancement Table

ECheck **New File Location and File Name**	**File Language Type**
ps60erta.ertaman	DOC
ps60erta.psertafm	MAKE
ps60erta.psertam	MAKE
ps60erta.psertamm	MAKE
ps60erta.rtauecke	Generated during a compile
ps60erta.rtaueckg	TAL
ps60erta.rtaueckm	MAKE
ps60erta.rtauecko	Generated during a compile
ps60erta.rtauecks	TAL
ps60rtau.rtaueckd	TAL

New Files for the Optional EMV Support Enhancement Table

EMV **New File Location and File Name**	**File Language Type**
at60auth.authemvd	TAL (Dummy Procs for EMV)
at60iath.atiathfm	MAKE
at60iath.atiathmm	MAKE
at60iath.authemve	Generated during a compile
at60iath.authemvg	TAL
at60iath.authemvm	MAKE
at60iath.authemvo	Generated during a compile
at60iath.authemvs	TAL
at60iath.cvrtbls	TAL
at60iath.tvrtbls	TAL
at60n50. n50emvde	TAL (Dummy Procs for EMV)
at60n50. n50emvds	TAL
at60in50.atin50fm	MAKE
at60in50.atin50mm	MAKE
at60in50.n50emve	Generated during a compile
at60in50. n50emvg	TAL
at60in50. n50emvm	MAKE
at60in50. n50emvo	Generated during a compile
at60in50. n50emvs	TAL
at60in50. rqemaidm	MAKE
at60in50. rqemaido	Generated during a compile
at60in50. rqemaids	SCOBOL
at60in50. rqemcrnm	MAKE
at60in50. rqemcrno	Generated during a compile
at60in50. rqemcrns	SCOBOL
at60in50. rqemmenm	MAKE
at60in50. rqemmeno	Generated during a compile
at60in50. rqemmens	SCOBOL
at60in50. rqemtrmm	MAKE
at60in50. rqemtrmo	Generated during a compile
at60in50. rqemtrms	SCOBOL
at60in50. rqemtxnm	MAKE
at60in50. rqemtxno	Generated during a compile
at60in50. rqemtxns	SCOBOL
at60in50. scnemaid	ACIAID

EMV New File Location and File Name	File Language Type
at60in50. scnemcrn	ACIAID
at60in50. scnemmen	ACIAID
at60in50. scnemtrm	ACIAID
at60in50. scnemtxn	ACIAID
ba60iemv.baiemvfm	MAKE
ba60iemv.baiemvm	MAKE
ba60iemv.baiemvmm	MAKE
ba60iemv.camutile	Generated during a compile
ba60iemv.camutilg	TAL
ba60iemv.camutilm	MAKE
ba60iemv.camutilo	Generated during a compile
ba60iemv.camutils	TAL
ps60apdh.apdhemvd	TAL (Dummy Procs for EMV)
ps60hpdh.hpdhemvd	TAL (Dummy Procs for EMV)
ps60iapd.apdhemve	Generated during a compile
ps60iapd.apdhemvg	TAL
ps60iapd.apdhemvm	MAKE
ps60iapd.apdhemvo	Generated during a compile
ps60iapd.apdhemvs	TAL
ps60iapd.psiapdfm	MAKE
ps60iapd.psiapdm	MAKE
ps60iapd.psiapdmm	MAKE
ps60ihpd.hpdhemve	Generated during a compile
ps60ihpd.hpdhemvg	TAL
ps60ihpd.hpdhemvm	MAKE
ps60ihpd.hpdhemvo	Generated during a compile
ps60ihpd.hpdhemvs	TAL
ps60ihpd.psihpdfm	MAKE
ps60ihpd.psihpdm	MAKE
ps60ihpd.psihpdmm	MAKE
ps60irta.cvrtbls	TAL
ps60irta.psirtafm	MAKE
ps60irta.psirtamm	MAKE
ps60irta.roctbls	TAL
ps60irta.rtauemve	Generated during a compile
ps60irta.rtauemvg	TAL
ps60irta.rtauemvm	MAKE

EMV New File Location and File Name	File Language Type
ps60irta.rtauemvo	Generated during a compile
ps60irta.rtauemvs	TAL
ps60irta.tvrtbls	TAL
ps60ispd.psispdfm	MAKE
ps60ispd.psispdm	MAKE
ps60ispd.psispdmm	MAKE
ps60ispd.spdhemve	Generated during a compile
ps60ispd.spdhemvg	TAL
ps60ispd.spdhemvm	MAKE
ps60ispd.spdhemvo	Generated during a compile
ps60ispd.spdhemvs	TAL
ps60rtau.rtauemvd	TAL (Dummy Procs for EMV)
ps60spdh.spdhemvd	TAL (Dummy Procs for EMV)
sw60ibnt.bnetemve	Generated during a compile
sw60ibnt.bnetemvm	MAKE
sw60ibnt.bnetemvo	Generated during a compile
sw60ibnt.bnetemvs	TAL
sw60ibnt.ibntfm	MAKE
sw60ibnt.ibntmm	MAKE
sw60ibnt.swibntm	MAKE
sw60ieur.euroemve	Generated during a compile
sw60ieur.euroemvm	MAKE
sw60ieur.euroemvo	Generated during a compile
sw60ieur.euroemvs	TAL
sw60ieur.ierumm	MAKE
sw60ieur.ieurfm	MAKE
sw60ieur.swieurm	MAKE
sw60ivis.ivisfm	MAKE
sw60imdm.imdmfm	MAKE
sw60imdm.imdmmm	MAKE
sw60imdm.mdsmemve	Generated during a compile
sw60imdm.mdsmemvm	MAKE
sw60imdm.mdsmemvo	Generated during a compile
sw60imdm.mdsmemvs	TAL
sw60imdm.swimdmm	MAKE
sw60imds.imdsfm	MAKE
sw60imds.imdsmm	MAKE

EMV New File Location and File Name	File Language Type
sw60imds.mdsemve	Generated during a compile
sw60imds.mdsemvm	MAKE
sw60imds.mdsemvo	Generated during a compile
sw60imds.mdsemvs	TAL
sw60imds.swimdsm	MAKE
sw60ilnk.ilnkfm	MAKE
sw60ilnk.ilnkmm	MAKE
sw60ilnk.linkemvm	MAKE
sw60ilnk.linkevme	Generated during a compile
sw60ilnk.linkemvo	Generated during a compile
sw60ilnk.linkemvs	TAL
sw60ilnk.swilnkm	MAKE
sw60ivis.ivismm	MAKE
sw60ivis.swvisim	MAKE
sw60ivis.visaemve	Generated during a compile
sw60ivis.visaemvm	MAKE
sw60ivis.visaemvo	Generated during a compile
sw60ivis.visaemvs	TAL

New Files for the Optional Format 2 File Support Enhancement Table

Format 2 File **New File Location and File Name**	**File Language Type**
at60misc.tlf22	FUP input file
at60misc.tlf21	FUP input file
ba60uc05.aaread05	DOC
ba60uc05.bauc05fm	MAKE
ba60uc05.bauc05m	MAKE
ba60uc05.bauc05mm	MAKE
ba60uc05.cnvecf	Generated during a compile
ba60uc05.cnvecfe	Generated during a compile
ba60uc05.cnvecfm	MAKE
ba60uc05.cnvecfo	Generated during a compile
ba60uc05.cnvecfr	obey
ps60misc.ptlf217	FUP input file
ps60misc.ptlf220	FUP input file
ps60misc.ptlf221	FUP input file
ps60misc.ptlf222	FUP input file
ps60misc.ptlf223	FUP input file
ps60misc.ptlf224	FUP input file
ps60misc.ptlf225	FUP input file
ps60misc.ptlf226	FUP input file
ps60misc.ptlf227	FUP input file
ps60misc.ptlf220	FUP input file
ps60misc.ptlf220	FUP input file

New Files for the Optional LINK Interchange Interface Table

LINK New File Location and File Name	File Language Type
sw60link.atlinke	Generated during a compile
sw60link.atlinkg	TAL
sw60link.atlinko	Generated during a compile
sw60link.atlinks	TAL
sw60link.balinkd	TAL
sw60link.balinke	Generated during a compile
sw60link.balinkg	TAL
sw60link.balinko	Generated during a compile
sw60link.balinks	TAL
sw60link.link	Generated during a compile
sw60link.linkcob	DDL output file
sw60link.linkddlm	MAKE
sw60link.linkddls	DDLs
sw60link.linkemvd	TAL
sw60link.linkfm	MAKE
sw60link.linkm	MAKE
sw60link.linkmm	MAKE
sw60link.linktal	DDL output file
sw60link.linktplt	FUP input file
sw60link.rp01lnk	Generated during a compile
sw60link.rp01lnkm	MAKE
sw60link.rp01lnks	COBOL
sw60link.rp09lnk	Generate during a compile
sw60link.rp09lnkm	MAKE
sw60link.rp09lnks	COBOL
sw60link.rqlinkm	MAKE
sw60link.rqlinko	Generated during a compile
sw60link.rqlinks	SCOBOL
sw60link.scrnlnk	ACIAID
sw60link.swlinkm	MAKE

New Files for the Optional BASE24-atm Multiple Account Select by Qualifier Support Enhancement

Multiple Account Select by Qualifier New File Location and File Name	**File Language Type**
at60math.atmathfm	MAKE
at60math.atmathmm	MAKE
at60math.authmase	Generated during a compile
at60math.authmasm	MAKE
at60math.authmaso	Generated during a compile
at60math.authmass	TAL

New Files for the Optional Multiple Currency Support Enhancement Table

Multiple Currency (MCU) New File Location and File Name	**File Language Type**
at60auth.authmcud	TAL (Dummy Procs for MCU)
at60uath.atuathfm	MAKE
at60uath.atuathmm	MAKE
at60uath.authmcue	Generated during a compile
at60uath.authmcug	TAL
at60uath.authmcum	MAKE
at60uath.authmcuo	Generated during a compile
at60uath.authmcus	TAL
at60umcu.atmcutl	Generated during a compile
at60umcu.atmcutle	Generated during a compile
at60umcu.atmcutlg	TAL
at60umcu.atmcutlm	MAKE
at60umcu.atmcutls	TAL
at60umcu.atumcufm	MAKE
at60umcu.atumcum	MAKE
at60umcu.atumcumm	MAKE
at60uref.atrfmcue	Generated during a compile
at60uref.atrfmcum	MAKE
at60uref.atrfmcuo	Generated during a compile
at60uref.atrfmcus	TAL
at60uref.atureffm	MAKE
at60uref.aturefm	MAKE
at60uref.aturefmm	MAKE
ba60aft.rqerfm	MAKE
ba60aft.rqerfo	Generated during a compile
ba60aft.rqerfs	TAL
ba60aft.scrnerf	ACIAID
ba60aft.sverf	Generated during a compile
ba60aft.sverfm	MAKE
ba60aft.sverfs	TAL
ba60ddl.ddlferf	DDL
ba60emtb.erfemtte	Generated during a compile
ba60emtb.erfemttg	TAL
ba60emtb.erfemttm	MAKE

Multiple Currency (MCU) New File Location and File Name	File Language Type
ba60emtb.erfemtto	Generated during a compile
ba60emtb.erfemtts	TAL
ba60logm.atatmcut	DOC
ba60logm.babamcut	DOC
ba60logm.pspsmcut	DOC
ba60refr.barfmcud	TAL (Dummy Procs for MCU)
ba60umcu.bamcutl	Generated during a compile
ba60umcu.bamcutle	Generated during a compile
ba60umcu.bamcutlg	TAL
ba60umcu.bamcutlm	MAKE
ba60umcu.bamcutls	TAL
ba60umcu.baumcufm	MAKE
ba60umcu.baumcum	MAKE
ba60umcu.baumcumm	MAKE
ba60uref.barfmcue	Generated during a compile
ba60uref.barfmcum	MAKE
ba60uref.barfmcuo	Generated during a compile
ba60uref.barfmcus	TAL
ba60uref.baureffm	MAKE
ba60uref.baurefm	MAKE
ba60uref.baurefmm	MAKE
ps60apdh.apdhmcud	TAL (Dummy Procs for MCU)
ps60refr.psrfmcud	TAL (Dummy Procs for MCU)
ps60rtau.rtaumcud	TAL (Dummy Procs for MCU)
ps60spdh.spdhmcud	TAL (Dummy Procs for MCU)
ps60uapd.apdhmcue	Generated during a compile
ps60uapd.apdhmcum	MAKE
ps60uapd.apdhmcuo	Generated during a compile
ps60uapd.apdhmcus	TAL
ps60uapd.psuapdfm	MAKE
ps60uapd.psuapdm	MAKE
ps60uapd.psuapdmm	MAKE
ps60umcd.psmcdhl	Generated during a compile
ps60umcd.psmcdhle	Generated during a compile
ps60umcd.psmcdhlg	TAL

Multiple Currency (MCU) New File Location and File Name	File Language Type
ps60umcd.psmcdhlm	MAKE
ps60umcd.psmcdhlo	Generated during a compile
ps60umcd.psmcdhls	TAL
ps60umcd.psumcdfm	MAKE
ps60umcd.psumcdm	MAKE
ps60umcd.psumcdmm	MAKE
ps60umcu.psmcutlg	TAL
ps60umcu.psmcutlm	MAKE
ps60umcu.psmcutls	TAL
ps60umcu.psumcufm	MAKE
ps60umcu.psumcum	MAKE
ps60umcu.psumcumm	MAKE
ps60uref.psrfmcue	Generated during a compile
ps60uref.psrfmcum	MAKE
ps60uref.psrfmcuo	Generated during a compile
ps60uref.psrfmcus	TAL
ps60uref.psureffm	MAKE
ps60uref.psurefm	MAKE
ps60uref.psurefmm	MAKE
ps60urta.psurtafm	MAKE
ps60urta.psurtamm	MAKE
ps60urta.rtaumcue	Generated during a compile
ps60urta.rtaumcug	TAL
ps60urta.rtaumcum	MAKE
ps60urta.rtaumcuo	Generated during a compile
ps60urta.rtaumcus	TAL
ps60uspd.psuspdfm	MAKE
ps60uspd.psuspdm	MAKE
ps60uspd.psuspdmm	MAKE
ps60uspd.spdhmcue	Generated during a compile
ps60uspd.spdhmcum	MAKE
ps60uspd.spdhmcuo	Generated during a compile
ps60uspd.spdhmcus	TAL

New Files for the Optional BASE24-atm NCR 5XXX Dial-Up Device Handler Table

NCR 5XXX Dial-Up New File Location and File Name	**File Language Type**
at60pn50.atpn50fm	MAKE
at60pn50.atpn50mm	MAKE
at60pn50.n50dlue	Generated during a compile
at60pn50.n50dlug	TAL
at60pn50.n50dlum	MAKE
at60pn50.n50dluo	Generated during a compile
at60pn50.n50dlus	TAL

New Files for the Optional BASE24-pos Stored Value Support Enhancement Table

Stored Value **New File Location and File Name**	**File Language Type**
ba60uc04.cnvcpsv	Generated during a compile
ba60uc04.cnvcpsve	Generated during a compile
ba60uc04.cnvcpsvm	MAKE
ba60us04.cnvcpsvo	Generated during a compile
ba60uc04.cnvcpsvr	Obey
ba60uc04.cnvcpsvs	TAL
ba60uc05.cnvcpf	Generated during a compile
ba60uc05.cnvcpfe	Generated during a compile
ba60uc05.cnvcpfs	TAL
ba60uc05.cnvcpfm	MAKE
ba60uc05.cnvcpfo	Generated during a compile
ba60uc05.cnvcpfr	Obey
ps60ddl.ddlfsvhf	DDL
ps60ddl.ddlgsvhx	DDL
ps60sv.scrnsvhf	ACIAID
ps60sv.rqsvhfm	MAKE
ps60sv.rqsvhfo	Generated during a compile
ps60sv.rqsvhfs	SCOBOL
ps60sv.svsvhfm	MAKE
ps60sv.svsvhf	Generated during a compile
ps60sv.svsvhfs	COBOL
ps60sv.cardcln	Generated during a compile
ps60sv.cardclne	Generated during a compile
ps60sv.cardclng	TAL
ps60sv.cardclnr	Obey
ps60sv.cardclno	Generated during a compile
ps60sv.cardclns	TAL
ps60sv.cardclnm	MAKE
ps60sv.hfclup	Generated during a compile
ps60sv.hfclupe	Generated during a compile
ps60sv.hfclupm	MAKE
ps60sv.hfclupg	TAL
ps60sv.hfclupo	Generated during a compile
ps60sv.hfclupr	Obey
ps60sv.hfclups	TAL

ps60sv.svhfx	Generated during a compile
ps60sv.svhfxe	Generated during a compile
ps60sv.svhfxm	MAKE
ps60sv.svhfxg	TAL
ps60sv.svhfxs	TAL
ps60sv.cursvhfs	TAL
ps60sv.cursvhfe	Generated during a compile
ps60sv.pssvfm	MAKE
ps60sv.pssvm	MAKE
ps60sv.pssvmm	MAKE

New Files for the Optional BASE24-atm Tidel Device Handler Table

Tidel **New File Location and File Name**	**File Language Type**
at60tidl.attidlfm	MAKE
at60tidl.attidlm	MAKE
at60tidl.attidlmm	MAKE
at60tidl.rqttidlm	MAKE
at60tidl.rqttidlo	Generated during a compile
at60tidl.rqttidls	SCOBOL
at60tidl.scrntidl	ACIAID
at60tidl.svdtidlm	MAKE
at60tidl.svdtidlo	Generated during a compile
at60tidl.svdtidls	TAL
at60tidl.svtidlte	Generated during a compile
at60tidl.svtidltg	TAL
at60tidl.svtidltm	MAKE
at60tidl.svtidlto	Generated during a compile
at60tidl.svtidlts	TAL
at60tidl.tidel	Generated during a compile
at60tidl.tidele	Generated during a compile
at60tidl.tidelg	TAL
at60tidl.tidelm	MAKE
at60tidl.tidels	TAL
at60tidl.tidelcob	Generated during a compile
at60tidl.tidlddls	DDL
at60tidl.tidlman	DOC
at60tidl.tidlnams	TAL
at60tidl.tidlomxe	Generated during a compile
at60tidl.tidlomxg	TAL
at60tidl.tidlomxm	MAKE
at60tidl.tidlomxo	Generated during a compile
at60tidl.tidlomxs	TAL
at60tidl.tidltal	Generated during a compile

New Files for the Optional Triple/Single DES Translation Support Enhancement Table

Triple/Single DES Translation Support New File Location and File Name	**File Language Type**
ba60aft.rqkey6m	MAKE
ba60aft.rqkey6o	Generated during a compile
ba60aft.rqkey6s	SCOBOL
ba60ddl.ddlfkey6	DDL
ba60aft.svkey6	Generated during a compile
ba60aft.svkey6m	MAKE
ba60aft.svkey6s	COBOL

New Files for the Optional BASE24-atm Triton Device Handler Table

Triton New File Location and File Name	File Language Type
at60ntrt.atntrtfm	MAKE
at60ntrt.atntrtmm	MAKE
at60ntrt.tritncde	Generated during a compile
at60ntrt.tritncdg	TAL
at60ntrt.tritncdm	MAKE
at60ntrt.tritncdo	Generated during a compile
at60ntrt.tritncds	TAL
at60trit.attritfm	MAKE
at60trit.attritm	MAKE
at60trit.attritmm	MAKE
at60trit.rqttritm	MAKE
at60trit.rqttrito	Generated during a compile
at60trit.rqttrits	SCOBOL
at60trit.scrntrit	ACIAID
at60trit.svdtritm	MAKE
at60trit.svdtrito	Generated during a compile
at60trit.svdtrits	TAL
at60trit.svtritte	Generated during a compile
at60trit.svtrittg	TAL
at60trit.svtrittm	MAKE
at60trit.svtritto	Generated during a compile
at60trit.svtritts	TAL
at60trit.tritcob	Generated during a compile
at60trit.tritddls	DDL
at60trit.tritman	DOC
at60trit.tritnams	TAL
at60trit.tritomxe	Generated during a compile
at60trit.tritomxg	TAL
at60trit.tritomxm	MAKE
at60trit.tritomxo	Generated during a compile
at60trit.tritomxs	TAL
at60trit.triton	Generated during a compile
at60trit.tritone	Generated during a compile
at60trit.tritong	TAL
at60trit.tritonm	MAKE

Triton **New File Location and File Name**	**File Language Type**
at60trit.tritons	TAL
at60trit.trittal	Generated during a compile
at60trit.tritupdt	DOC

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Appendix E

Moved, Renamed, Deleted Files

Appendix E contains a list of files that were moved, renamed, or deleted in BASE24 Release 6.0.

File Location and File Name	Action Taken	Additional Action Taken or Note
acitpl.*	Deleted	Merged with SCRIBE subvolume. Refer to BASE24 Install manual
at601000.svdc100m	Renamed as svd10xxm	
at601000.svdc100o	Renamed as svd10xxo	
at601000.svdc100s	Renamed as svd10xxs	
at604730.i4730fm	Renamed as at4730fm	
at604730.i4730mm	Renamed as at4730mm	
at60aft.rqt1770m	Deleted	Not Supported
at60aft.rqt1770o	Deleted	Not Supported
at60aft.rqt1770s	Deleted	Not Supported
at60aft.rqt2300m	Deleted	Not Supported
at60aft.rqt2300o	Deleted	Not Supported
at60aft.rqt2300s	Deleted	Not Supported
at60aft.rqt36xxm	Moved to subvol at60ibm	
at60aft.rqt36xxo	Moved to subvol at60ibm	
at60aft.rqt36xxs	Moved to subvol at60ibm	
at60aft.rqt4730m	Moved to subvol at604730	
at60aft.rqt4730o	Moved to subvol at604730	
at60aft.rqt4730s	Moved to subvol at604730	
at60aft.rqt5080m	Moved to subvol at60n50	Renamed as rqt5xxxm
at60aft.rqt5080o	Moved to subvol at60n50	Renamed as rqt5xxxo
at60aft.rqt5080s	Moved to subvol at60n50	Renamed as rqt5xxxs
at60aft.rqt5100m	Moved to subvol at60c51	
at60aft.rqt5100o	Moved to subvol at60c51	
at60aft.rqt5100s	Moved to subvol at60c51	
at60aft.rqtb650m	Moved to subvol at60b650	
at60aft.rqtb650o	Moved to subvol at60b650	
at60aft.rqtb650s	Moved to subvol at60b650	

File Location and File Name	Action Taken	Additional Action Taken or Note
at60aft.rqtdcspm	Moved to subvol at60dcsp	
at60aft.rqtdcspo	Moved to subvol at60dcsp	
at60aft.rqtdcsps	Moved to subvol at60dcsp	
at60aft.rqtdfm	Deleted	Not Supported
at60aft.rqtdfo	Deleted	Not Supported
at60aft.rqtdfs	Deleted	Not Supported
at60aft.rqtdiebm	Moved to subvol at60dieb	
at60aft.rqtdiebo	Moved to subvol at60dieb	
at60aft.rqtdiebs	Moved to subvol at60dieb	
at60aft.rqtfuj7m	Moved to subvol at60fuj7	
at60aft.rqtfuj7o	Moved to subvol at60fuj7	
at60aft.rqtfuj7s	Moved to subvol at60fuj7	
at60aft.rqtfujim	Moved to subvol at60fuj	
at60aft.rqtfujio	Moved to subvol at60fuj	
at60aft.rqtfujis	Moved to subvol at60fuj	
at60aft.rqtncshm	Moved to subvol at60ncsh	
at60aft.rqtncsho	Moved to subvol at60ncsh	
at60aft.rqtncshs	Moved to subvol at60ncsh	
at60aft.rqtoli6m	Moved to subvol at60oli6	
at60aft.rqtoli6o	Moved to subvol at60oli6	
at60aft.rqtoli6s	Moved to subvol at60oli6	
at60aft.rqtomrnm	Deleted	Not Supported
at60aft.rqtomrno	Deleted	Not Supported
at60aft.rqtomrns	Deleted	Not Supported
at60aft.rqttidlm	Moved to subvol at60tidl	
at60aft.rqttidlo	Moved to subvol at60tidl	
at60aft.rqttidls	Moved to subvol at60tidl	
at60aft.rqttritm	Moved to subvol at60trit	
at60aft.rqttrito	Moved to subvol at60trit	
at60aft.rqttrits	Moved to subvol at60trit	
at60aft.scrn1770	Deleted	Not Supported
at60aft.scrn2300	Deleted	Not Supported
at60aft.scrn36xx	Moved to subvol at60ibm	
at60aft.scrn4730	Moved to subvol at604730	
at60aft.scrn5080	Moved to subvol at60n50	Renamed as scrn5xxx
at60aft.scrn5100	Moved to subvol at60c51	
at60aft.scrnb650	Moved to subvol at60b650	

File Location and File Name	Action Taken	Additional Action Taken or Note
at60aft.scrndcsp	Moved to subvol at60dcsp	
at60aft.scrndieb	Moved to subvol at60dieb	
at60aft.scrnfuj7	Moved to subvol at60fuj7	
at60aft.scrnfuji	Moved to subvol at60fuj	
at60aft.scrnncsh	Moved to subvol at60ncsh	
at60aft.scrnoli6	Moved to subvol at60oli6	
at60aft.scrnomrn	Deleted	Not Supported
at60aft.scrntdf1	Deleted	Not Supported
at60aft.scrntdfb	Deleted	Not Supported
at60aft.scrntidl	Moved to subvol at60tidl	
at60aft.scrntrit	Moved to subvol at60trit	
at60aft.svtdf	Deleted	Not Supported
at60aft.svtdfm	Deleted	Not Supported
at60aft.svtdfs	Deleted	Not Supported
at60b650.b650fm	Renamed as at650fm	
at60b650.b650mm	Renamed as at650mm	
at60c51.c5100fm	Renamed as atc51fm	
at60c51.c5100m	Renamed as atc51m	
at60c51.c5100mm	Renamed as atc51mm	
at51dass.atdassfm	Moved to subvol at60esd	Renamed atesdfm
at51dass.atdassm	Moved to subvol at60esd	Renamed atesdm
at51dass.atdassmm	Moved to subvol at60esd	Renamed atesdmm
at51dass.ddlesdm	Moved to subvol at60esd	
at51dass.esdm	Moved to subvol at60esd	
at51dass.esds	Moved to subvol at60esd	
at51dass.svdesdm	Moved to subvol at60esd	
at51dass.svdesds	Moved to subvol at60esd	
at60dct.rqd36xxm	Moved to subvol at60ibm	
at60dct.rqd36xxo	Moved to subvol at60ibm	
at60dct.rqd36xxs	Moved to subvol at60ibm	
at60dct.rqd4730m	Moved to subvol at604730	
at60dct.rqd4730o	Moved to subvol at604730	
at60dct.rqd4730s	Moved to subvol at604730	
at60dct.rqd5100m	Moved to subvol at60c51	
at60dct.rqd5100o	Moved to subvol at60c51	
at60dct.rqd5100s	Moved to subvol at60c51	
at60dct.rqdb650m	Moved to subvol at60b650	

File Location and File Name	Action Taken	Additional Action Taken or Note
at60dct.rqdb650o	Moved to subvol at60b650	
at60dct.rqdb650s	Moved to subvol at60b650	
at60dct.rqddiebm	Moved to subvol at60dieb	
at60dct.rqddiebo	Moved to subvol at60dieb	
at60dct.rqddiebs	Moved to subvol at60dieb	
at60dct.rqdfujim	Moved to subvol at60fuj	
at60dct.rqdfujio	Moved to subvol at60fuj	
at60dct.rqdfujis	Moved to subvol at60fuj	
at60dct.rqdn50m	Moved to subvol at60n50	
at60dct.rqdn50o	Moved to subvol at60n50	
at60dct.rqdn50s	Moved to subvol at60n50	
at60dct.rqdoli6m	Moved to subvol at60oli6	
at60dct.rqdoli6o	Moved to subvol at60oli6	
at60dct.rqdoli6s	Moved to subvol at60oli6	
at60dct.scnd36xx	Moved to subvol at60ibm	
at60dct.scnd4730	Moved to subvol at604730	
at60dct.scnd5100	Moved to subvol at60c51	
at60dct.scndb650	Moved to subvol at60b650	
at60dct.scnddieb	Moved to subvol at60dieb	
at60dct.scndfuji	Moved to subvol at60fuj	
at60dct.scndn50	Moved to subvol at60n50	Renamed as scndn5xx
at60dct.scndoli6	Moved to subvol at60oli6	
at51eaft.rqtesdm	Moved to subvol at60esd	
at51eaft.rqtesds	Moved to subvol at60esd	
at51eaft.scrnesd	Moved to subvol at60esd	Renamed to scntesd
at51edct.rqdesdm	Moved to subvol at60esd	
at51edct.rqdesds	Moved to subvol at60esd	
at51edct.scndesd	Moved to subvol at60esd	
at60ext.atprvtal	Deleted	Not Supported
at60ext.prvtlfe	Deleted	Not Supported
at60ext.prvtlfs	Deleted	Not Supported
at60fuj.fujfm	Renamed as atfujfm	
at60fuj.fujmm	Renamed as atfujmm	
at60m100.m100m	Deleted	Moved into at601000.c1000m
at60mn50.mn50m	Deleted	Moved into at60n50.n50m
at60mmas.athisog	Deleted	Moved into at60hiso.athisog
at60mmas.athisos	Deleted	Moved into at60hiso.athisos

File Location and File Name	Action Taken	Additional Action Taken or Note
at60mmas.atrefrs	Deleted	Moved into at60refr.atrefrs
at60mmas.authg	Deleted	Moved into at60auth.authg
at60mmas.authlibs	Deleted	Moved into at60auth.authlibs
at60mmas.auths	Deleted	Moved into at60auth.auths
at60mmas.cn100doc	Deleted	Moved into at601000.cn100doc
at60mmas.cn910doc	Deleted	Moved into at60c910.cn910doc
at60mmas.cnfg5070	Deleted	Moved into at60n50.cnfg5070
at60mmas.cnfg910	Deleted	Moved into at60c910.cnfg910
at60mmas.cnfgix	Deleted	Moved into at601000.cnfgix
at60mmas.cnn50doc	Deleted	Moved into at60n50.cnn50doc
at60mmas.n50g	Deleted	Moved into at60n50.n50g
at60mmas.rqtlfs	Deleted	Moved into at60aft.rqtlfs
at60mmas.svtlfs	Deleted	Moved into at60aft.svtlfs
at60n50.n50smde	Deleted	Not Supported
at60n50.n50smds	Deleted	Not Supported
at60setl.setlfm	Renamed as atsetlfm	
at60setl.setlm	Renamed as atsetlm	
at60setl.setlmm	Renamed as atsetlmm	
at60src.snadiebs	Moved to subvol at60dieb	
at60src.snadocs	Moved to subvol at60c51	
at60src.snaibms	Moved to subvol at60ibm	
at60src.snaoli6s	Moved to subvol at60oli6	
at60tidl.tidelfm		Renamed as attidlfm
at60tidl.tidelmm	Renamed as attidlfm	
at60trit.tritonfm	Renamed as attritfm	
at60trit.tritonmm	Renamed as attritmm	
ba60aft.rqidfncm	Deleted	Not Supported
ba60aft.rqidfnco	Deleted	Not Supported
ba60aft.rqidfncs	Deleted	Not Supported
ba60aft.rqidfsmm	Deleted	Not Supported
ba60aft.rqidfsmo	Deleted	Not Supported
ba60aft.rqidfsms	Deleted	Not Supported
ba60card.ddlfcaf	Deleted	Moved into ba60ddl.ddlfcaf
ba60card.ddlgcafx	Deleted	Moved into ba60ddl.ddlgcafx
ba60card.omfxg	Deleted	Moved into ba60ext.omfxg
ba60card.omfxs	Deleted	Moved into ba60ext.omfxs
ba60card.refrg	Deleted	Moved into ba60refr.refrg

File Location and File Name	Action Taken	Additional Action Taken or Note
ba60card.refrs	Deleted	Moved into ba60refr.refrs
ba60card.rqcafs	Deleted	Moved into ba60aft.rqcafs.
ba60card.rqmegas	Deleted	Moved into ba60aft.rqmegas
ba60card.scrncaf	Deleted	Moved into ba60aft.scrncaf
ba60card.svcafs	Deleted	Moved into ba60aft.svcafs
ba60crdm.cardmacs	Deleted	Not Supported
Ba60crdm.cardupdt	Deleted	Not Supported
Ba60ebtm.ebtdoc	Deleted	Not Supported
Ba60ebtm.ebtmacs	Deleted	Not Supported
Ba60logm.secutils	Renamed as basecutl	Xprintba file updated
ba60masm.masmmacs	Deleted	Not Supported
ba60masm.mmasupdt	Deleted	Not Supported
ba60mmas.cobnames	Deleted	Moved into ba60src.cobnames
ba60mmas.curidfs	Deleted	Moved into ba60ext.idf6xs
ba60mmas.ddlfcaf	Deleted	Moved into ba60ddl.ddlfcaf
ba60mmas.ddlfidf	Deleted	Moved into ba60ddl.ddlfidf
ba60mmas.ddlgcafx	Deleted	Moved into ba60ddl.ddlgcafx
ba60mmas.ddlgiso	Deleted	Moved into ba60ddl.ddlgiso
ba60mmas.hiswutls	Deleted	Moved into ba60src.hiswutls
ba60mmas.omfxs	Deleted	Moved into ba60ext.omfxs
ba60mmas.refrs	Deleted	Moved into ba60refr.refrs
ba60mmas.rqcafs	Deleted	Moved into ba60aft.rqcafs
ba60mmas.rqidfs	Deleted	Moved into ba60aft.rqidfs
ba60mmas.rqpbfs	Deleted	Moved into ba60aft.rqpbfs
ba60mmas.scrncaf	Deleted	Moved into ba60aft.scrncaf
ba60mmas.scrnidf	Deleted	Moved into ba60aft.scrnidf
ba60mmas.svcafs	Deleted	Moved into ba60aft.svcafs
ba60mmas.svidfs	Deleted	Moved into ba60aft.svidfs
ba60mmas.svpbfs	Deleted	Moved into ba60aft.svpbfs
ba60sv.bacoutls	Deleted	Moved into ba60src.bacoutls
ba60sv.basvfm	Deleted	
ba60sv.basvm	Deleted	
ba60sv.basvmm	Deleted	
ba60sv.cardclng	Moved to subvol ps60sv	
ba60sv.cardclnm	Moved to subvol ps60sv	
ba60sv.cardclns	Moved to subvol ps60sv	
ba60sv.cobnames	Deleted	Moved into ba60src.cobnames

File Location and File Name	Action Taken	Additional Action Taken or Note
ba60sv.ddlfcaf	Deleted	
ba60sv.ddlfcpf	Deleted	Moved into ba60ddl.ddlfcpf
ba60sv.ddlfecf	Deleted	
ba60sv.ddlfidf	Deleted	
ba60sv.ddlgext	Deleted	Moved into ba60ddl.ddlgext
ba60sv.ddlgpstm	Deleted	Moved into ba60ddl.ddlgpstm
ba60sv.extrm	Deleted	Moved into ba60ext.extrm
ba60sv.omfxs	Deleted	Moved into ba60ext.omfxs
ba60sv.rqcpfs	Deleted	Moved into ba60aft.rqcpfs
ba60sv.rqecfs	Deleted	Moved into ba60aft.rqecfs
ba60sv.rqextrs	Deleted	
ba60sv.rqidfs	Deleted	Moved into ba60aft.rqidfs
ba60sv.scrncaf	Deleted	Moved into ba60aft.scrncaf
ba60sv.scrncpf	Deleted	Moved into ba60aft.scrncpf
ba60sv.scrnecf	Deleted	Moved into ba60aft.scrnecf
ba60sv.scrnextr	Deleted	
ba60sv.scrnidf	Deleted	Moved into ba60aft.scrnidf
ba60sv.sectbl	Deleted	Moved into ba60aft.sectbl
ba60sv.svcpfs	Deleted	Moved into ba60aft.svcpfs
ba60sv.svdoc	Deleted	
ba60sv.svidfs	Deleted	Moved into ba60aft.svidfs
ba60sv.svmhlps	Deleted	Moved into ba60aft.svmhlps
bp60crta.auprvtal	Deleted	Not Supported
fh60card.ddlfhstm	Deleted	Moved into fh60ddl.ddlfhstm
fh60card.fhmcafs	Deleted	Moved into fh60caf.fhmcafs
fh60card.fhmgbls	Deleted	Not Supported
fh60card.fhmrpts	Deleted	Moved into fh60rpt.fhmrpts
fh60card.fhms	Deleted	Moved into fh60fhm.fhms
fh60card.ulfxg	Deleted	Moved into fh60ext.ulfxg
fh60card.ulfxs	Deleted	Moved into fh60ext.ulfxs
fh60mmas.ddlfhstm	Deleted	Moved into fh60ddl.ddlfhstm
fh60mmas.fhmcafs	Deleted	Moved into fh60caf.fhmcafs
fh60mmas.fhmrpts	Deleted	Moved into fh60rpt.fhmrpts
fh60mmas.fhms	Deleted	Moved into fh60fhm.fhms
oc60afm.rqattso	Moved to subvol ba60aft	
oc60afm.rqattss	Moved to subvol ba60aft	
oc60afm.rqattxm	Moved to subvol ba60aft	

File Location and File Name	Action Taken	Additional Action Taken or Note
oc60afm.rqattxs	Moved to subvol ba60aft	
oc60afm.svattte	Moved to subvol ba60aft	
oc60afm.svatttm	Moved to subvol ba60aft	
oc60afm.svattts	Moved to subvol ba60aft	
oc60ddl.attds	Moved to subvol ba60ddl	
oc60fi.idfemtdc	Moved to subvol ba60emtb	
oc60fi.idfemtdm	Moved to subvol ba60emtb	
oc60fi.idfemtds	Moved to subvol ba60emtb	
oc60fi.idfemtdt	Moved to subvol ba60emtb	
oc60fi.idfemtte	Moved to subvol ba60emtb	
oc60fi.idfemttg	Moved to subvol ba60emtb	
oc60fi.idfemttm	Moved to subvol ba60emtb	
oc60fi.idfemtto	Moved to subvol ba60emtb	
oc60fi.idfemtts	Moved to subvol ba60emtb	
oc60obj.cpobjcod	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
oc60obj.cpobjdir	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
Oc60obj.cpobjsym	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
oe60obj.epobjcod	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
oe60obj.epobjdir	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
oe60obj.epobjsym	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
ok60ddl.confds	Deleted	Not Supported
ok60obj.kpobjcod	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g.,

File Location and File Name	Action Taken	Additional Action Taken or Note
		BA60OBJ.POBJ)
ok60obj.kpobjdir	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
ok60obj.kpobjsym	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
ps60ext.prvptlfe	Deleted	Not Supported
ps60ext.prvptlfs	Deleted	Not Supported
ps60ext.psprvtal	Deleted	Not Supported
ps60ext.rtlfx	Deleted	Not Supported
ps60ext.rtlfxe	Deleted	Not Supported
ps60ext.rtlfxg	Deleted	Not Supported
ps60ext.rtlfxm	Deleted	Not Supported
ps60ext.rtlfxs	Deleted	Not Supported
ps60hpdh.hpdhfm	Renamed as pshpdhfm	
ps60hpdh.hpdhmm	Renamed as pshpdhmm	
ps60spdh.spdhfm	Renamed as psspdhfm	
ps60spdh.spdhmm	Renamed as psspdhmm	
sw60sdf.sdprvtal	Deleted	Not Supported
tb60obj.tpobjcod	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
tb60obj.tpobjdir	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
tb60obj.tpobjsym	Deleted	Requester objects SCUP COPYed directly into DEST_POBJ (e.g., BA60OBJ.POBJ)
Tr60ext.asufsm	Deleted	Not Supported
tr60ext.asufx	Deleted	Not Supported
tr60ext.asufxe	Deleted	Not Supported
tr60ext.asufxg	Deleted	Not Supported
tr60ext.asufxs	Deleted	Not Supported
tr60ext.prvttlfe	Deleted	Not Supported

File Location and File Name	Action Taken	Additional Action Taken or Note
tr60ext.prvttlfs	Deleted	Not Supported
tr60ext.trprvtal	Deleted	Not Supported
tr60refr.trprvtal	Deleted	Not Supported

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Appendix F

Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 that are specific to each region.

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Appendix G

PI-Table Implementation Guidelines

Appendix G contains a summary of steps that explain how a developer can apply the new methodology associated with the PITABLE enhancement to a particular module. If a module is developed using the new PITABLE methodology as described in this appendix, your BASE24 network must include the PITABLE enhancement.

This appendix consists of two sections. The first section describes changes specific to pathway modules. The second section describes changes that relate to all processes.

It should be noted, this appendix is intended to provide an overview of changes that are generally common among modules. Some changes might not be applicable to certain modules.

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Changes And Requirements Specific To PATHWAY

To allow a Requester/Server to access a segment greater than 32, the following changes must be made. In addition, the Items listed under the section “Changes and Requirements relating to all Processes” will need to be implemented for Requester/Server pairs.

- Change MEGATBL (for product enhancements) or MEGATBLC (for Channel enhancements and Customer Enhancements) to recognize the new requester. MEGATBL and MEGATBLC are used by RQMEGA to create a list of requesters that have been upgraded. This informs RGMEGA to send USER-CONTEXT-EXT to this requester.

- Copy in the necessary “-EXT” sections from BADDLCOB and BACOUTLS.

- Change the FRMT-CDE value sent to the Security Server to reflect the correct extended message. This will mean changing an “A” value to a “C” and a “B” value to a “D”.

- Change all references to the MSG-FILE-MAINT-XXXX structures and USER-CONTEXT to add the “-EXT” suffix.

- Change all references to the segmentation utilities to use the new utilities with the “^EXT” suffix.

- For COBOL Servers working storage field LST-SEG-DEF-C must be added if the field does not presently exist. If the field does not exist but the server is required to handle higher numbered segments, the field must be updated. The value of the field must be equal to the segment number of the last segment available plus one. Servers utilize this field in routines that perform such tasks as deleting unneeded segments (i.e., Paragraph 315-DELETE-EXTRA-SEGS).

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Changes And Requirements Relating To All Processes

To allow a process to access a segment greater than 32, the following changes must be made.

- Change all references to the segmentation utilities to use the new utilities with the "^EXT" suffix.

- Change any working storage fields that store the SITE-PROD-IND from 32 to 256. Change any working storage fields that store the FIID-SEG-MAP from 32 to 256. Check all working storage fields with values of 32. Some of these may refer to the SITE-PROD-IND, FIID-SEG-MAP or other fields related to the segments or products.

- Any definitions for PTR-ARRAY (or other array variables) set for 32 should be evaluated. If they are used to store address to segments in files, they should be increased to 256.

- Any structure definitions for segmented files that are passed to the segmentation utilities must be modified to allow for 256 segment addresses.

- Before a segmented record can be added, the segment ID and B24-RSRVD field must be set in each segment's header. This initialization does not pertain to the BASE segment. Constants are available in the new DDLBCNST file for these values.

- Before a segmented record can be added, the SEG-MAP of the BASE segment header must be set to zeroes.

- Look at all uses of the SEG-MAP field in the BASE segment's header. This field will now contain zeroes. Any code that was testing the SEG-MAP to see if a segment is present in a record should now check the pointer array returned from the Disassemble Segmentation Utilities (e.g., DISASSEMBLE^SEG^REC^EXT). If the address is negative one, the segment does not exist.

- Examine the calls to BIT^TO^BYTE or BYTE^TO^BIT. Many of these manipulate the bits in the old SEG-MAP field and must be replaced. If a call to these routines still exists, check to see if the NUM-BITS parameter must be changed to 256.

- Look at the use of the IDF FIID-SEG-MAP field. This is now 256 bits long in two adjacent fields. If the field is defined as a pointer to the beginning of the field, it will access all 256 entries. If the code is reading or changing a particular bit, the ACI utilities (TESTBIT and SETBIT) may still be used. Be sure to allow for all 256 entries in the field.

Appendix H

Version 1 to Version 2 Considerations

Appendix H contains information about release 6.0 version 2 enhancements that users should consider when determining impacts for converting from release 6.0 version 1 to release 6.0 version 2.

Refer to the specified page numbers for more information about these enhancements.

This section is divided into two sections. The first lists the enhancements that are new to Version 2 and the page numbers where information about the functionality and the changes that were made to BASE24 for them can be found. The second section contains a list of considerations noted within the document specifically for customers converting from release 6.0 version 1 to release 6.0 version 2, and the page numbers within this document where users can find these considerations.

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Version 2 Enhancements

The following enhancements were added to Release 6.0 Version 2 and users that want to take advantage of these enhancements should review the following pages.

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Considerations

The following considerations need to be reviewed when converting from release 6.0 version 1 to release 6.0 version 2. Refer to the specified page numbers for more information about these considerations.

Release 6.0 Version 1 to Release 6.0 Version 2 Consideration:

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Appendix I

Converting from Release 5.1.2/1.1.2 to 6.0

The standard conversion programs provided with release 6.0 are designed to convert BASE24 database files from release 5.3/1.3 to release 6.0. In order to assist BASE24 Release 5.1.2/1.1.2 customers converting to 6.0, ACI is providing the necessary release 5.3/1.3 subvolumes to compile and run the standard 5.3/1.3 conversion programs. Customers may convert their release 5.1.2/1.1.2 database to a release 5.3/1.3 database using the release 5.3/1.3 conversion programs, and then convert the resulting 5.3/1.3 database to 6.0 using the standard 6.0 conversion programs. Customers can choose to write their own 5.1.2/1.1.2 to 6.0 conversion programs instead, possibly starting with and modifying the standard BASE24 release 6.0 conversion programs.

Along with the necessary release 5.3/1.3 subvolumes, special overall Make subvolumes are provided that are used to compile and build the standard 5.3/1.3 conversion programs. Customers are not sent all of the BASE24 Release 5.3/1.3 code, so the standard BASE24 Release 5.3/1.3 Make stream cannot be used. This special subvolume is named BA60UC03 for BASE24 classic customers and OC60UC03 for BASE24 Remote Banking customers. For more information about these special overall Make subvolumes, customers should refer to the AAREAD03 files on the subvolumes.

For more information about the BASE24 Release 5.3/1.3 conversion programs, refer to the AAREADME files on the following subvolumes as well as the BASE24 Release 5.3/1.3 Implementation Guide on the ACI Publications CD-ROM:

- BA53UC03
- AT53UC03
- BP53UC03
- PS53UC03
- OC13UC03

BASE24 Release 6.0 classic customers receive the following subvolumes to convert their release 5.1.2 database to release 5.3:

- AT53DDL
- AT53UC03
- BA10SKEL
- BA53CSE
- BA53DDL
- BA53DEFS
- BA53SRC
- BA53UC03
- BA60UC03
- BP53CRTA[1]
- BP53UC03[1]
- PS53DDL
- PS53UC03

Note 1: These subvolumes are only delivered for those customers who purchased the BASE24-pos CRT Authorization Device Control module for release 6.0.

BASE24 6.0 Remote Banking customers receive the following subvolumes to convert their release 1.1.2 database to release 1.3:

- BA10SKEL
- BA53CSE
- BA53DDL
- BA53DEFS
- BA53SRC
- OC13DDL
- OC13INTF
- OC13SRC
- OC13UC03
- OC60UC03
- OK13DDL
- OK13LIB
- OK13SRC

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